19-G-0309/19-G-0310 - 2-11-20 - Brooklyn Union Gas

CASE 19-G-0309 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service.

CASE 19-G-0310 - Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service

EVIDENTIARY HEARING

DATE: February 11, 2020

LOCATION: 19th Floor Hearing Room
Agency Building Three,
Empire State Plaza
Albany, New York

ALJ JAMES A. COSTELLO, DPS

ALJ MAUREEN F. LEARY, DPS

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Exhibit 617 916
1036 Request, two pages

Exhibit 618 920
Information request response to information request, D.P.S. 699, it was issued on June 26, 2019 and a response was submitted on July 8th, 2019. It contains eight attachments and is 64 pages.

Exhibit 619 1039
Staff info - response December 6th

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(On the record 9:00 a.m.)

THE REPORTER: On the record.

MR. GAVILONDO: Good morning, Your Honors, Carlos Gavilondo for the company and with me at counsel table are Patrick O'Brien, Kenneth Maloney of the law firm of Cullen and Dykman and Phillip Decicco.

A.L.J. COSTELLO: Good morning.

A.L.J. LEARY: Good morning.

MR. GOODRICH: Good morning, Your Honors. Brandon Goodrich for the Department of Public Service and with me are Nick Forst and Raquel Parks.

A.L.J. COSTELLO: Good morning.

A.L.J. LEARY: Good morning.

A.L.J. COSTELLO: The Utility Intervention Unit?

MS. O'HARE: Good morning, Your Honors. Kathleen O'Hare for the Utility Intervention Unit.

A.L.J. LEARY: Yeah, that's not going to work for us because I can't hear that microphone. So you really have to get close and you need to speak up.

MS. O'HARE: Okay.

A.L.J. COSTELLO: Okay. The Public Utility Law Project?

MR. BERKLEY: Good morning, Your Honors. Richard Berkley for the Public Utility Law Project.

MS. WHEELOCK: And Laurie Wheelock.

A.L.J. LEARY: I can hear you guys. Good, thanks.

A.L.J. COSTELLO: City of New York?

MR. FUNG: Good morning, Your Honors. Justin Fung for the City of New York and Bob Loughney is at the other table.

A.L.J. COSTELLO: Good morning. The Long Island Power Authority?

MR. PODOLNY: Good morning, the firm of Read and Laniado by Konstantin Podolny.

A.L.J. COSTELLO: Good morning. Environmental Defense Fund?

MS. MURPHEY: Erin Murphey, for the Environmental Defense Fund joined by Martin Rothfelder with Rothfelder Stern on behalf of E.D.F.

A.L.J. LEARY: Good morning.

A.L.J. COSTELLO: Yeah. Bob Wyman?

MR. WYMAN: Bob Wyman, for myself.

A.L.J. COSTELLO: Okay. I don't believe they're here, the Sane Energy Project? No. Okay, is there anyone who is making an appearance today who I have not called? Okay. Hearing none. We're going to get started.

I am going to -- just want to do a bit of housekeeping with respect to some exhibits that were left off the exhibit list. They are pre-marked exhibits by the companies and I just want to -- they were left off the exhibit list. They -- they're Appendix 1 to KEDNY SIR cost control efforts and that was to the testimony of the -- the SIR witness on behalf of the companies and that will be marked as Exhibit 613. And Appendix 1, KEDLI SIR cost control efforts and that will be marked as Exhibit 614 and both of them are 8 pages long.

A.L.J. LEARY: Do they have a date?

A.L.J. COSTELLO: I --

A.L.J. LEARY: Any date? Counsel, do you know the date on these documents?

MR. O'BRIEN: Be April 30th, 2019.

A.L.J. LEARY: My birthday. Thank you.

A.L.J. COSTELLO: Okay. And we're going to start with the Department of Public Service Staff Earnings Adjustment Mechanisms panel.

MR. BERKLEY: Your Honor, a question. We have an affidavit from who's been excused. How do you want to handle this?

A.L.J. COSTELLO: I can collect it from you in a -- when -- when we take a break, I'll collect it from you. And what we're going to do is probably sometime later on when we have all the -- all the affidavits that we need, we'll just admit everything at one time -- all that testimony at one time.

MR. BERKLEY: Okay. Just -- in the affidavit, to the truthfulness of a number of I.R. responses they provided, and the Company would be providing those as an exhibit at some point, but they're covered by the affidavit.

A.L.J. COSTELLO: Okay. Thank you.

A.L.J. LEARY: Hold on a second. This is -- these are I.R.s by a different witness that the company is going to cost but -- offer but the witness is not going to testify to?

MR. MALONEY: Sorry, Your Honor --.

A.L.J. LEARY: You've stipulated that these are --.

MR. MALONEY: Yes, I mean, Your Honor, these are I.R.s that were propounded by the company on New York City and that witness -- that witness responded. We're now going to put them into the record through the affidavit. We'll have copies of them this afternoon to admit as an -- mark as an exhibit and put it there.

A.L.J. LEARY: Okay.

A.L.J. COSTELLO: This was -- the exhibits that you needed so that you didn't have to cross?

MR. MALONEY: That's exactly right, Your Honor.

A.L.J. LEARY: Okay. Thanks.

MR. MALONEY: Yes. This is Ken Maloney for this -- for KEDNY and KEDLI.

A.L.J. COSTELLO: Okay. I'm going to ask each of you to state your name and business address for the record to -- one at a time. Thank you.

MR. CULLY: Robert Cully. 3 Empire State Plaza, Albany, New York.

MS. MAMMEN: Katy Mammen. 3 Empire State Plaza, Albany, New York.

MS. KOKKINIDES: Laurie Kokkinides. 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: And, may I ask you to stand and please raise your right hand? Do you swear or affirm that the testimony you will provide is the truth?

MR. CULLY: Yes.

MS. MAMMEN: Yes.

MS. KOKKINIDES: Yes.

Robert Cully; Sworn

Katy Mammen; Sworn

Laurie Kokkinides; Sworn

A.L.J. COSTELLO: Thank you.

A.L.J. LEARY: I'm just going to ask the court reporter. Do you have -- do you need spellings, or you have everything?

THE REPORTER: No. Because I got it right on the paper that you --

A.L.J. LEARY: Great. Thanks.

A.L.J. COSTELLO: Okay.

THE REPORTER: So, I know that I have everybody's names because I went around, but can you

just state your name before you talk so I know who is talking, so I know I'm getting the right name down.

A.L.J. LEARY: Yeah. Mr. Maloney is new.

THE REPORTER: I did grab his -- I grab -- I went around and I grabbed everybody's names. I just want to make sure when everybody's talking that I'm getting the right person to the right -- right face to the right name.

A.L.J. LEARY: Okay. Thanks. Everybody, got that? Okay.

A.L.J. COSTELLO: Mr. Goodrich?

MR. GOODRICH: Your -- Your Honors, this is Brandon Goodrich. May we have just one moment? We seem to be missing one of the members of the panel.

A.L.J. COSTELLO: Sure.

A.L.J. LEARY: Sure.

A.L.J. COSTELLO: We'll go off the record.

(Off the record 9:05 a.m.)

A.L.J. LEARY: So let's -- if you can swear that additional witness in on the record.

THE REPORTER: I don't know if I know

who so I can't swear --

A.L.J. LEARY: No -- no -- no, I'm asking you did you pick up on the record that we swore this fourth witness on the panel in --.

THE REPORTER: You're going to have to do that one again.

A.L.J. LEARY: Okay.

A.L.J. COSTELLO: Okay. Sorry about that.

THE REPORTER: That's okay.

A.L.J. COSTELLO: Please stand, state you name and business address.

MR. MAURIELLO: Davide Mauriello. 3 Empire State Plaza, Albany, New York 12223.

A.L.J. COSTELLO: Do you swear or affirm that the testimony you will give is the truth?

MR. MAURIELLO: I do.

Davide Mauriello; Sworn

A.L.J. COSTELLO: Okay. Thank you. You may be seated.

DIRECT EXAMINATION

BY MR. GOODRICH:

Q. Good morning, panel. For the record, this is Brandon Goodrich and I have a few

questions for you folks to -- to get us started off. So the Staff Earnings Adjustment Panel, before you is a document entitled Prepared Testimony of Staff Earnings Adjustment Panel consisting of a cover page and 77 pages of questions and answers dated August 2019. Was that testimony prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Also before you, is a document entitled Prepared Supplemental Testimony of Staff Earnings Adjustment Panel consisting of a cover page and 4 pages of questions and answers, dated January 2020. Excuse me. Was that testimony prepared by you or under your direct supervision?

A. (Cully) Yes. Just to clarify, it's the Staff Earnings Adjustment Mechanisms Panel. The Mechanisms didn't come in through the last two ones.

Q. Sorry. Do you have any changes or corrections to make to either of your prepared testimonies?

A. No.

Q. Do you affirm that the information contained in the panel's prepared

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testimonies is true and accurate to the best of your knowledge?

A. Yes.

Q. And if you are asked the same questions today under oath, would you answer them in the same way?

A. (Panel) Yes.

Q. Also before you, are exhibits identified at the time of filing as S.E.A.M.P.-1, S.E.A.M.P.-2, S.E.A.M.P.-3, S.E.A.M.P.-4, -5, -6 and -7. These have now been marked for identification purposes as Exhibits 416, 417, 418, 419, 420, 421 and 517 respectively. Were those 7 exhibits prepared by you or under your direct supervision?

A. (Cully) Yes.

Q. Do you affirm that the information contained in the 7 exhibits just identified is true to the best of your knowledge?

A. Yes.

MR. GOODRICH: Your Honors, I ask that the Prepared Initial and Supplemental Testimonies of the Staff Earnings Adjustment Mechanisms Panel be entered into the record as if given orally?

A.L.J. COSTELLO: That motion is

granted. And at this point in the record, the court reporter will be inserting D.P.S. Staff Earnings Adjustment Mechanisms Panel direct testimony and D.P.S. Staff Earnings Adjustment Mechanisms Panel supplemental testimony. #orally

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Earnings Adjustment
Mechanisms Panel

Robert Cully
Utility Engineering Specialist 3

Kathryn Mammen
Utility Analyst 3

Laurie Kokkinides
Utility Analyst 1
Office of Markets and Innovation

Davide Maioriello
Utility Engineering Specialist 3
Office of Electric, Gas and
Water

State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Staff Earned Adjustments Mechanisms Panel (Panel), please state your names, employer, and business address.

A. Our names are Robert Cully, Kathryn Mammen, Laurie Kokkinides and Davide Maioriello. We are employed by the New York State Department of Public Service (Department). Our business address is Three Empire State Plaza, Albany, NY 12223.

Q. Have each of the members of the Panel already discussed their credentials in testimony submitted in these proceedings?

A. Yes. Mr. Cully's, Ms. Mammen's, and Ms. Kokkinides' credentials are provided in the direct testimony of the Staff Efficiency and Sustainability Panel. Mr. Maioriello's credentials are provided in the direct testimony of the Staff Gas Supply Panel.

Q. What is the purpose of the Panel's testimony?

A. We will provide a summary and our recommendations regarding The Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan Gas East Corporation d/b/a National Grid's (KEDLI) (jointly, the Companies) proposed

Earnings Adjustment Mechanisms (EAMs), Platform Service Revenues (PSRs), KEDLI's EmPower Replacement Program Incentive Mechanism, and Non-Pipes Alternatives (NPA) process and associated Incentive Mechanism.

Q. In this testimony, will the Panel refer to, or otherwise rely upon, any information produced during the discovery phase of these proceedings?

A. Yes, we will refer to, and have relied upon, several responses to Department Staff and other party Information Requests (IRs), which are referred to by the Department or other parties' given designation (e.g., DPS-123). The IRs that we rely upon are included in Exhibit___(SEAMP-1).

Q. Is the Panel sponsoring any other exhibits with its testimony?

A. Yes, we are also sponsoring five additional exhibits with our testimony.

Q. Please describe these exhibits.

A. Exhibit___(SEAMP-2) provides an overview of our EAM proposals.

Exhibit___(SEAMP-3) shows how we developed our Share the Savings EAM metric.

Exhibit___(SEAMP-4) provides data regarding the Companies' proposed Affordability EAM metric. Exhibit___(SEAMP-5) compares our recommended Utility Energy Services Contracts Platform Service Revenue with the Companies' proposal. Exhibit___(SEAMP-6) provides data regarding the KEDLI EmPower Replacement Program.

Q. What is the Rate Year in these proceedings?

A. The Rate Year is the 12 months ending March 31, 2021. The Companies referred to Data Years as follows: Data Year 1 is the 12 months ending March 31, 2022; Data Year 2 is the 12 months ending March 31, 2023; and Data Year 3 is the 12 months ending March 31, 2024. For ease of reference, we will use the same references here as the Companies did in their testimony.

EARNINGS ADJUSTMENT MECHANISMS

Q. Provide a brief overview of EAMs.

A. EAMs are metrics with targets and associated earnings opportunities designed to align utility shareholder interests with New York State energy policy goals. If the utility meets the EAM targets, benefits are achieved, and utility

shareholders earn financial rewards. The Commission initially approved EAMs in the May 19, 2016 Order Adopting a Ratemaking and Utility Revenue Model Policy Framework in Case 14-M-0101, referred to as the Track Two Order.

Q. Provide a brief overview of the EAM requirements set forth in the Track Two Order.

A. Pages 59-60 of the Track Two Order explained the Commission's expectation that EAMs would play a transitional role until other forms of market-based revenues are available. The Track Two Order established the Commission's preference for: outcome-based EAMs, see pages 61-65; EAMs that are generally positive-only, see pages 66-67; a maximum amount of earnings totaling 100 basis points, see page 68; and establishing EAM awards on a sliding scale with the size of awards growing with greater achievement, see page 77. The Track Two Order also set the Commission's expectations for the types of EAM metrics it would consider and stated that rate proceedings are the proper venue for developing EAMs.

Q. What EAM metrics were discussed in the Track Two

Order?

A. In general, the Track Two Order discussed EAMs designed predominantly for electric utilities only. The Track Two Order implemented a System Efficiency EAM based on both peak load reduction and improvements in load factor, see pages 73-77; an Energy Efficiency EAM based on achieving greater energy efficiency program savings or cost savings and electric usage intensity across the utility's territory, see pages 82-83; and an Interconnection EAM based on an evaluation of application quality and satisfaction of applicants with the process, see pages 86-87. Although the Track Two Order states, on page 89, that no general EAM is necessary for Customer Engagement and Information Access, it did invite proposals for such an EAM based on customer uptake for opt-in innovative rates and initiatives related to electric vehicle adoption and ground source heat pumps, as explained on pages 89-90. Similarly, on pages 90-91, the Track Two Order invited proposals related to incentivizing greenhouse gas reductions.

Q. Have there been other proceedings since the

Track Two Order that indicate the way the Commission currently views EAMs?

A. Yes, there have been two other statewide policy proceedings, as well as further developments in recent rate proceedings, which demonstrate the Commission's current stance toward EAMs and our recommendations in these proceedings. Following the Track Two Order, EAMs have been developed within rate proceedings. The Commission has approved various EAMs in the rate proceedings for Consolidated Edison Company of New York, Inc. (Con Edison) in Case 16-E-0060, Niagara Mohawk Power Corporation d/b/a National Grid (Niagara Mohawk) in Case 17-E-0238, Central Hudson Gas & Electric Corporation (Central Hudson) in Case 17-E-0459, and Orange and Rockland Utilities, Inc. (O&R) in Case 18-E-0067.

Q. Are there any common features of the EAM portfolios approved by the Commission as part of these recent rate proceedings of other utilities?

A. Yes. The Commission approved an outcome-based electric Peak Reduction metric at Niagara

Mohawk, Central Hudson, and O&R. The Commission has approved programmatic energy efficiency metrics based on achieving greater energy savings through utility-run energy efficiency programs, most for both the utilities' electric and gas businesses, and electric-only outcome-based greenhouse gas reduction metrics based on increased adoption of carbon dioxide (CO2) reducing technologies for Con Edison, Niagara Mohawk, Central Hudson, and O&R. There are a number of other common features of EAM portfolios which have been approved by the Commission for Con Edison, Niagara Mohawk, Central Hudson, and O&R. However, since these metrics are electric-only, their relevance to the Companies' gas-only businesses in these proceedings is limited.

Q. Please describe the first statewide policy proceeding in which the Commission has further refined its position on EAMs.

A. On March 15, 2018 in case 15-M-0252, the Commission issued its Order Authorizing Utility-Administered Energy Efficiency Portfolio Budgets and Targets for 2019–2020 (March 2018 Energy

Efficiency Order). On pages 20-27 of this Order, the Commission required all programmatic EAMs proposed after March 15, 2018 to be calculated and awarded based on verified gross energy savings. On page 28 o the order, the Commission also required that the utilities file an Evaluation Measurement & Verification (EM&V) plan demonstrating when and how EM&V activities will be performed to support awarding EAM financial incentives based reported energy efficiency savings.

Q. Please define gross savings.

A. Gross savings are energy savings that result directly from program-related actions taken by participants in an energy efficiency program, regardless of why they participated. This topic is also discussed in the direct testimony of the Staff Efficiency and Sustainability Panel.

Q. Please define verified gross savings.

A. Verified gross savings are the values reported by an independent evaluator after the energy efficiency program activities and gross savings analysis are complete. Verified gross savings are distinct from gross savings in two ways.

First, verified gross savings are the product of a complete gross savings analysis using methods consistent with industry standard best practices. Second, verified gross savings are produced by an independent evaluator, not by the utility.

Q. Please define the process used to determine verified gross savings.

A. As part of the energy efficiency budgets, each utility has also been provided a budget for EM&V activities and studies. EM&V is a broadly inclusive description of any activities intended to determine energy efficiency program impacts, including quantifying the savings realized by the program. The ratio of the savings determined after verification to the gross savings is called the verified gross savings realization rate.

Q. Please describe the second statewide policy proceeding in which the Commission has further refined its position on EAMs since the Track Two Order?

A. On December 13, 2018 in case 18-M-0084, the Commission issued the Order Adopting Accelerated

Energy Efficiency Targets (December 2018 Energy Efficiency Order). In this Order, the Commission established a new paradigm in Energy Efficiency EAMs by requiring utilities to develop and propose new “share the savings” EAMs related to their energy efficiency programs. On page 68 of the December 2018 Energy Efficiency Order, the Commission states that EAMs for energy efficiency programs will continue to be developed in individual rate proceedings, using benchmarked dollars-per-lifetime-million British Thermal Unit (MMBTU) costs. In addition, on page 69 of the December 2018 Energy Efficiency Order, the Commission also required that such EAMs be designed to share any cost savings achieved below authorized budgets or by achieving significantly greater energy savings for the same budget. To encourage longer lived savings and discourage over-reliance on measures with shorter estimated useful lives, the Commission required programmatic energy efficiency EAMs to use a dollars-per-lifetime MMBTU basis. On pages 67 and 68 of the Order, the Commission expressed its preference for

"share the savings" energy efficiency EAMs, as they measure near term results commensurate with the length of three-year rate plans. As a corollary, the Commission explained its reduced reliance on Energy Intensity EAMs as they are better suited for measuring longer term trends than the length of three-year rate plans.

Q. Provide an overview of the Companies' EAM proposals.

A. KEDLI and KEDNY's EAM proposals are included in two documents. First, the Companies initially proposed a set of EAMs as part of the direct testimony of the Future of Heat Panel (FOH Panel). Second, the Companies proposed an update to the Incremental Energy Efficiency EAM to include targets based on gross savings as opposed to net savings of EAMs in the Corrections and Updates Testimony of The FOH Panel.

EAM Overview

Q. Provide an overview of the EAM proposals contained in the direct testimony of KEDLI and KEDNY's FOH Panel.

A. In its direct testimony, the FOH Panel describes the Companies EAM metric development, measurement, and target-setting methodologies for its proposed metrics. The Companies propose three categories of EAMs: (1) System Efficiency; (2) Energy Efficiency; and (3) Carbon Reduction. The Companies propose a total of six EAM metrics within these three categories. On page 97 of the FOH Panel's direct testimony, the Companies propose that each of these metrics be measured on a calendar year basis.

Q. What level of financial awards do the Companies propose related to these EAMs?

Q. KEDNY proposes to earn up to 78 basis points of financial incentives for 2020, and 63 basis points of financial incentives each for 2021, 2022, and 2023. KEDLI proposes to earn up to 77 basis points in financial incentives for 2020, 62 basis points in financial incentives for 2021, and 63 basis points in financial incentives for 2022 and 2023. As shown in the Companies' response to IR DPS-361, KEDNY and KEDLI propose to value basis points on a Rate Year basis, whereas EAM performance would be

measured on a calendar year basis.

Q. Did the Companies make any additional EAM proposals in their Corrections and Updates Testimony?

A. Yes, the FOH Panel's Corrections and Updates Testimony provided a correction to the Companies' Incremental Energy Efficiency metric targets, which we will describe later in this testimony. The Companies did not propose any further updates to any of their other EAM metrics or targets in the Corrections and Updates Testimony.

Q. Prior to discussing the individual EAM metrics proposed by the Companies, does the Panel have any preliminary recommendations regarding the Company's overall EAM proposals?

A. Yes, we have four overall recommendations.

Q. What is the Panel's first overall recommendation?

A. First, we recommend that the Commission reject KEDNY's proposal to earn up to 78 basis points and KEDLI's proposal to earn up to 77 basis points during the Rate Year at maximum achievement levels, and instead adopt, for each

Company, a maximum financial incentive of 10 basis points, plus the earnings related to the Share the Savings energy efficiency metric, which we will discuss later in our testimony. Our recommended financial incentives are shown in Exhibit__(SEAMP-2). We will also provide recommendations on the specific number of basis points for each metric as part of our recommendations for each of those metrics.

Q. What are the dollar amounts associated with one basis point during the Rate Year?

A. Based on Staff's recommended rate base and revenue requirement, the value of one basis point is $335,537 for KEDNY and $211,211 for KEDLI. The calculations of basis point values are provided in the direct testimony of the Staff Finance Panel.

Q. What is the Panel's second overall recommendation?

A. Second, we recommend that the Commission approve more aggressive targets than those proposed by the Companies, even at minimum EAM achievement levels. We will explain our recommended EAM achievement levels when we address each

individual EAM later in this testimony.

Q. Why should the Commission adopt more aggressive EAM achievement levels?

A. The Companies' target-setting methodology results in achievement levels that do not represent stretch goals indicative of exceptional utility performance. The Track Two Order twice provided guidance that EAM targets in general should be ambitious to encourage a portfolio approach beyond conventional programs, see page 75, and that utilities should not begin earning EAM incentives on programmatic metrics until such programs achieve their minimum targets, see page 82.

Q. What is the Panel's third overall recommendation?

A. Third, we propose that the EAMs ultimately be expressed in terms of absolute dollars instead of basis points. While we express our potential earnings in basis points in our testimony, we recommend the values be converted to a dollar amount once the capital structure and rate base are determined in these proceedings.

Q. Explain why the Panel recommends that EAMs be

awarded in terms of dollars and not in basis points.

A. Since the value of a basis point fluctuates over time, as it is directly related to the amount of plant in service, basing EAMs strictly on basis points could create a perverse incentive for the Companies to increase its net plant-in-service to inflate the value of a basis point. Instead, the basis points assigned to each EAM should be converted into absolute dollar figures based on the equity ratio and net plant-in-service targets ultimately approved by the Commission in these proceedings.

Q. What is the Panel’s fourth overall recommendation?

A. Fourth, in the event that a settlement is reached resulting in a multi-year rate plan, we recommend that the EAM metrics, targets, and financial incentive levels be set for the full term of the rate plan.

Q. Why should the EAM metrics, targets, and financial incentive levels be set for the full term of a multi-year rate plan?

A. A term of a multi-year rate plan strikes a

reasonable balance between providing a long-term incentive structure around which the Company will be able to plan its business and operations, and ensuring that the metrics, targets, and financial incentives remain reasonable and relevant. Furthermore, setting the EAM metric targets for multiple years helps to reduce a utility gaming opportunity whereby the Companies might have an incentive, if EAMs are reset each year based on performance during the prior year, to artificially depress its progress toward the metric targets if it can reasonably expect that metric targets would be reduced in a subsequent proceeding, thus requiring less effort for greater financial reward.

Q. Are there risks involved in setting multi-year EAM metrics, targets, and financial incentives?

A. Setting EAM metric targets for a multi-year term does present a risk that targets may be set either too strict, such that the utility is not able to achieve the targets regardless of its actions, or too lenient, such that the utility earns an incentive with very little effort.

However, it is our opinion that this risk is more than balanced given our process of developing targets that reflect achievable goals through utility intervention and efforts and the benefits of providing a stable and predictable incentive structure for KEDLI and KEDNY. This combination is beneficial for the Companies, their customers, and the State of New York.

Q. Are there ongoing statewide policy proceedings that may require reconsideration of the EAM targets during the term of a multi-year rate plan?

A. Yes, there are a number of ongoing statewide policy proceedings that may require reconsideration of certain EAM metrics or targets if the Commission authorizes new programs or significantly increases the budgets for existing programs or portfolios. In particular, the Commission is currently considering the Companies and other utilities' May 2019 energy efficiency proposals in Case 18-M-0084.

Q. What does the Panel recommend in the event there are developments in these other proceedings?

A. We recommend that the EAM targets be reconsidered as part of a collaborative among the Companies, Staff, and other stakeholders in the event that the Commission authorizes significant new programs or substantial increases to the budgets for existing programs or portfolios as a result of ongoing statewide policy proceedings. While the targets and metrics should be re-examined, the total number and value of basis points available to the Companies during the three-year period should remain as set in these proceedings.

System Efficiency EAMs

Q. Please describe the Companies' proposed System Efficiency EAM.

A. The proposed System Efficiency EAM comprises two metrics: Peak Reduction, and Supply Diversification.

Q. Provide an overview of the Companies' proposed Peak Reduction metric.

A. As described by the FOH Panel on page 98 of its direct testimony, the Companies' proposed Peak Reduction metric would measure reduction in gas

usage during peak events by averaging the reduction across all peak events as part of the Demand Response Demonstration Project.

Q. How do the Companies propose to set targets for their Peak Reduction metric?

A. On page 99-100 of the FOH Panel's direct testimony, the Companies explain that they set the minimum achievement level for each Company equal to that Company's respective forecast of peak load reductions achieved through their Demand Response programs on a Dekatherm (Dth) reduced per hour, per event, basis, averaged among all demand response events called during a given season. The Companies further propose to set the midpoint and maximum achievement levels 10% and 20% above the forecast levels, respectively. The Companies claim that their proposed midpoint and maximum targets reflect the incremental effort required to exceed the design of the expanded demonstration.

Q. How many basis points of financial awards do the Companies propose related to these targets?

A. KEDNY proposes to earn one basis point, three basis points, and five basis points of incentive

awards at the minimum, midpoint, and maximum achievement levels, respectively, for each calendar year from 2020 through 2023. KEDLI proposes to earn one basis point, two basis points, and three basis points of incentive awards at the minimum, midpoint, and maximum achievement levels, respectively, for each year from 2020 through 2023.

Q. Does the Panel agree with the Companies' proposed Peak Reduction metric?

A. While we agree with the need for an EAM metric designed to encourage the Companies to reduce gas peak demand, we disagree with the specifics of the metric as proposed by the Companies.

Q. What aspects of the Companies' proposed Peak Reduction metric does the Panel disagree with?

A. We disagree with the Companies' proposal to base an EAM metric solely on the results of their proposed Demand Response program, especially since, as the Companies describes it, such program is still in the demonstration phase. The Commission has repeatedly expressed its preference for outcome-based EAM metrics in general, as discussed previously, as well as for

outcome-based peak reduction metrics specifically.

Q. Where has the Commission expressed a desire for outcome-based Peak Reduction metrics specifically?

A. The Commission espoused outcome-based peak reduction metrics in the Track Two Order on page 75, and reiterated that position on page 40 of its Order Establishing Energy Storage Goal and Deployment Policy issued in Case 18-E-0130 on December 13, 2018. Although the Commission's Orders in those proceedings are related to electric Peak Reduction EAMs, the Commission's desire for outcome-based Peak Reduction metrics is similarly applicable to the gas business.

Q. How does the Panel recommend devising a peak reduction metric?

A. As a preliminary recommendation, we believe that such a metric should be based on peak-day usage reductions from firm customers on a per-Heating Degree Day basis, to be measured during the month of January or February when gas companies are most likely to experience the heating season's peak day.

Q. Does the Panel recommend reserving a level of financial incentives related to this metric, once developed?

A. Yes. We recommend that up to 10 basis points of gas incentives be reserved for this metric at maximum achievement levels beginning as soon as practicable, either in 2021 or 2022. As a preliminary recommendation, we believe that the minimum and midpoint incentives should be 2.5 basis points and five basis points, respectively.

Q. What process does the Panel envision for developing this metric and targets?

A. We invite the Companies and other parties to assist in this effort by recommending possible metrics and target development strategies as part of their respective rebuttal testimonies. We anticipate being able to develop targets based on available historical data and additional data obtained as needed during the 2019-2020 heating season. Once the metric and associated targets are developed, the Companies should petition the Commission for authorization to begin earning the reserved EAM financial

incentives.

Q. Provide an overview of the Companies' proposed Supply Diversification metric.

A. As proposed by the Companies on page 100 of the FOH Panel's direct testimony, the Supply Diversification metric would reward each Company for increasing annual deliveries of Renewable Natural Gas (RNG) onto the Companies' distribution system. Specifically, the Companies propose to measure the incremental RNG introduced into their respective natural gas networks each year.

Q. How did the Companies propose to set the minimum, midpoint, and maximum achievement levels for this metric?

A. The Companies used a two-step process for developing their respective minimum targets, described on pages 101 and 102 of the FOH Panel's direct testimony. First, the Companies state that they examined the attributes of six proposed RNG projects currently in the Companies' interconnection queue, and estimated the net benefits of such RNG interconnections on a per-Dth basis. Second, the Companies

estimated the number of Dth of RNG needed to achieve one basis point of value each year – the Companies set this volume of RNG as the minimum achievement level.

Q. How do the Companies propose to set the midpoint and maximum achievement levels?

A. On page 102 of the FOH Panel’s direct testimony, the Companies state that they based their proposed midpoint and maximum achievement levels on an evaluation of historical trends in RNG connections dating back to the 1980s, with forecasts based on those trends to determine what the RNG market would do itself without utility intervention. The Companies then estimated a trendline for accelerated RNG deployment, with the midpoint set at a threefold improvement in RNG interconnection volumes and the maximum set at fivefold improvement in such volumes. Ultimately, as shown on page 101 of the FOH Panel’s direct testimony, the resulting midpoint and maximum targets for each Company are equal to 2.5 times and five times the minimum target, respectively.

Q. How many basis points of financial incentive do

the Companies propose related to the Supply Diversification metric?

A. The Companies propose four basis points, 10 basis points, and 20 basis points at the minimum, midpoint, and maximum achievement levels, respectively for each Company and each year from 2020 through 2023.

Q. Does the Panel agree with the Companies' proposed Supply Diversification metric?

A. No, we do not.

Q. Why doesn't the Panel agree with the Supply Diversification metric?

A. First, the Companies' rationale for the Supply Diversification EAM is not supported by either the Order Adopting Regulatory Policy Framework and Implementation Plan (the Track One Order), issued February 26, 2015 in Case 14-M-0101 or the Track Two Order. Second, an additional incentive for the Companies to pursue increased RNG interconnections is unnecessary at this time. Third, the Companies' proposed targets and associated financial incentives are unreasonable.

Q. Explain why the Supply Diversification EAM is

inconsistent with the Track One and Track Two Orders.

A. On pages 102 and 103 of the FOH Panel's direct testimony, the Companies claim that their Supply Diversification metric aligns with the Commission's findings in the Reforming the Energy Vision (REV) proceeding, and asserts that such metric incentivizes the Companies to create a more diverse energy supply. The Companies' assertion that its Supply Diversification is consistent with the Commission's findings in the REV proceeding, as communicated in the Track One Order and Track Two Order, is inaccurate. While the Track One Order does identify fuel and resource diversity as one of the six policy objectives of the REV proceeding, see page 4, it does so in the context of diversifying the electric grid away from reliance on natural gas, see page 17. For its part, the Track Two Order, which identified the categories of EAMs that would be considered by the Commission, did not identify diversification of fuel sources within a gas utility's supply portfolio as an EAM-worthy metric. In fact, the only gas-related

metric the Commission did address in the Track Two Order was related to its openness to gas Energy Efficiency-related incentives, see page 82. In addition, as shown in the Companies' FOH Panel Exhibit__(FOH-2), the KEDNY and KEDLI System Efficiency EAM metrics, inclusive of both the Companies' proposed Peak Reduction metric and Supply Diversification metric, are comprised of 98% and 99% avoided CO2 benefits, respectively, making these metrics not System Efficiency metrics, but Carbon Reduction metrics.

Q. Didn't the Commission consider EAM metrics related to Carbon Reduction in the Track Two Order?

A. Yes, but only those specifically related to decarbonization of electric uses such as transportation, through increased electric vehicle penetration, and building heating and cooling uses through increased penetration of heat pumps. The Track Two Order did not include increased penetration of RNG as an EAM metric.

Q. Why is an additional EAM incentive not necessary to spur the Companies to pursue increased RNG

penetration?

A. Given the supply constraints the Companies are currently facing, any increase in non-pipeline gas supply, from increased RNG interconnections for example, will help the Companies manage their gas supply issues. The Companies business model is already inherently aligned with increasing RNG interconnections, therefore, additional customer-funded incentives for the Companies' shareholders are not necessary.

Q. Explain how the Companies' targets and associated financial incentives are unreasonable.

A. As described on pages 101 and 102 of the FOH Panel's direct testimony, the Companies set their proposed minimum target based on the number of Dth of RNG required to achieve one basis point of value, with the midpoint and maximum number of Dths 2.5 times and five times the minimum level, respectively. As a reward for creating one, 2.5, and five basis points of value at the minimum, midpoint, and maximum achievement levels, the Companies propose to be provided four, 10, and 20 basis points of

financial incentives, respectively. Put another way, for each $1 in value created by RNG interconnections, the Companies propose to earn $4 in financial incentives. This level of incentivization is clearly unreasonable.

Energy Efficiency EAMs

Q. Please describe the Companies' proposed Energy Efficiency EAMs.

A. The Companies proposed two Energy Efficiency EAMs: the Incremental Energy Efficiency metric and an Affordability metric.

Q. Please describe the Incremental Energy Efficiency metric.

A. The Incremental Energy Efficiency metric measures incremental reductions achieved through each Company's energy efficiency programs in MMBTUs for each year. The Companies proposed to set the minimum achievement level equal to the targets established by the Commission in the December 2018 Energy Efficiency Order. The Companies proposed to set the maximum achievement level based on achieving cost efficiencies equal to each Company's lowest

average historic actual run rate from the last four years, with some adjustments. The Companies proposed to set their respective midpoint achievement levels equal to the average of the minimum and maximum achievement levels.

Q. Did the Companies propose the same set of targets for each year?

A. No. The Companies proposed to set the targets related to this metric for 2020 based on the method described above, but based on the sum of MMBTU savings for both Calendar Year 2019 and 2020. During 2021, 2022, and 2023 the Companies propose to base the minimum, midpoint, and maximum achievement levels only on the MMBTU targets from each respective year.

Q. Have the Companies' updated their proposed Incremental Energy Efficiency EAM after filing their initial testimony?

A. Yes, the Companies proposed an update to the Incremental Energy Efficiency targets. In their initial testimony, the Companies based the targets on net MMBTU savings. The FOH Panel corrected these targets to be based on gross savings in its Corrections and Update Testimony.

Q. What level of financial awards do the Companies propose related to the Incremental Energy Efficiency EAM?

A. For 2020, the Companies each propose to earn zero basis points at the minimum achievement level, 20 basis points at the midpoint achievement level, and 40 basis points at the maximum achievement level. Annually between 2021 and 2022, the Companies each propose to earn zero basis points at the minimum achievement level, 10 basis points at the midpoint achievement level, and 20 basis points at the maximum achievement level.

Q. Does the Panel agree with the Companies' Incremental Energy Efficiency metric proposals?

A. No, we do not. The Companies' proposed Incremental Energy Efficiency metric is not responsive to the Commission's directives in either the December 2018 Energy Efficiency Order or the March 2018 Energy Efficiency Order. In addition, the Companies' proposed Incremental Energy Efficiency EAM metric is not responsive to the Commission's March 2018 Energy Efficiency Order requiring that programmatic energy

efficiency EAMs be based on Verified Gross Savings. Instead of the unresponsive Incremental Energy Efficiency metric proposed by the Companies, we recommend that the Commission adopt a Share the Savings EAM metric, which we will describe in detail later in this testimony. The incentives provided by an Incremental Energy Efficiency metric would be wholly duplicative of the Share the Savings EAM.

Q. Explain how the Incremental Energy Efficiency metric is duplicative of the incentives provided by the Share the Savings metric.

A. There are three pathways toward earning a financial award under the Share the Savings metric: (1) reducing total energy efficiency portfolio expenditures; (2) increasing the first year MMBTU savings achieved; and (3) increasing the average effective useful lifetime (EUL) of the Companies' portfolios by focusing on longer-lived measures. Pursuing any or all of these three pathways can result in a significant reduction to total cost per lifetime MMBTU. To achieve the Annual MMBTU EAM, however, the Companies would have to increase the first year

MMBTU savings achieved by the portfolio, which the Companies will already be incentivized to do through the Share the Savings EAM.

Q. Please explain your recommended Share the Savings metric.

A. We recommend a three-step process to determine the financial incentives under the Share the Savings EAM. First, the Companies would calculate the actual dollar per lifetime MMBTU program run-rate it achieves each year. To do this, each Company would divide the cost of its total energy efficiency portfolio by the product of the portfolio's first-year MMBTU savings and the EUL of the portfolio. Second, each Company would then subtract the annual lifetime run rate from a baseline annual lifetime run rate. The baseline run rate would be calculated by dividing the Commission-authorized portfolio budget by the product of the Commission-authorized portfolio savings target and the Company's portfolio EUL. Third, each Company would multiply the difference in dollars per lifetime MMBTU calculated in the second step by the actual annual lifetime MMBTU achieved by

that Company's portfolio, multiplied by that Company's share of savings percentage, 30 percent. We provide an illustrative example of this process in Exhibit___(SEAMP-3).

Q. How did the Panel develop its proposed 30 percent Company-share of savings?

A. As shown in Exhibit__(SEAMP-3) we benchmarked the recent Commission-approved electric and gas energy efficiency EAM targets for other utilities. Based on this exercise, we determined that a roughly 50% improvement in energy efficiency program run rates were needed to achieve the maximum EAM award. In each of the utilities benchmarked, the Commission has awarded a total 10 basis points to programmatic gas energy efficiency EAMs. As shown in Exhibit__(SEAMP-3), we then calculated the resulting Company-share of savings percentage for approximately 10 basis points of incentives based on a 50 percent reduction in energy efficiency portfolio costs per lifetime MMBTU. Based on these calculations we determined that a 30 percent Company-share of the savings is appropriate.

Q. Does the Share the Savings EAM run the risk of unintended consequences?

A. Yes. Therefore, we recommend instituting "guardrails" on the Share the Savings EAM.

Q. What "guardrails" does the Panel recommend?

A. We recommend two guardrails. First, we recommend that the Companies be required to meet the minimum lifetime MMBTU target of its energy efficiency portfolio before they can have the opportunity to earn the Share the Savings EAM. Second, we recommend that the Companies use a consistent EUL for each energy efficiency measure for both setting the benchmark cost per lifetime MMBTU and the achieved cost per lifetime MMBTU. These EULs should reflect the EULs stated in the New York Standard Approach for Estimating Energy Savings from Energy Efficiency Programs – Residential, Multi-Family, and Commercial/Industrial Measures (commonly known as the Technical Resource Manual or TRM), Version 7, effective January 1, 2020, submitted on April 15, 2019 in Matter 15-01319. These requirements will ensure that KEDNY and KEDLI achieve at least the minimum lifetime MMBTU

savings from its energy efficiency portfolio each year prior to earning this EAM. The requirements also ensure that the Companies will neither benefit nor suffer due to a change in the measure of EULs as a result of changes to the Technical Resource Manual during a given year.

Q. Did the Companies address the March 2018 Energy Efficiency Order's requirements regarding verified gross savings in its Share the Savings EAM?

A. No, the Companies did not address verified gross savings in their Energy Efficiency EAM proposals. In response to IRs DPS-270 and DPS-271, the Companies stated that they will propose plans and a schedule for developing verified gross savings following the issuance of Staff guidance regarding verified gross savings.

Q. Did the Commission require Staff to develop guidance regarding verified gross savings?

A. Yes. The Commission required Staff, consultation with utilities, New York State Energy Research and Development Authority (NYSERDA) and other interested stakeholders, to

issue Gross Savings Verification Guidance that included the minimum level of gross savings that must be verified for the purpose of awarding EAMs, the methods and approaches that would be considered acceptable to perform the analysis, and any other considerations that would provide clarity and structure to the consistent approach to the verified gross savings policy.

Q. Has Staff issued the required guidance regarding verified gross savings?

A. Yes. Staff issued its verified gross savings guidance document on August 23, 2019 in Case 15-M-0252.

Q. Do you have a recommendation regarding verified gross savings as it relates to your recommended Share the Savings EAM?

A. Yes, we recommend that the Companies apply the realization rate from the most recent third-party EM&V activities to the gross savings reported against the EAM target. We also recommend that any program that has not been subjected to EM&V activities resulting in a realization rate during the two years prior to the rate period should be prioritized, and

activities that result in realization rates for those programs should be included in the schedule of EM&V activities to be conducted within the upcoming three years beginning during the Rate Year. In the event that the Commission approves other programmatic energy efficiency-related EAMs in this proceeding, our recommendations on this matter should apply to those metrics as well.

Q. What is the schedule for awarding EAMs based on verified gross savings?

A. As the Commission noted in its March 2018 Energy Efficiency Order, the verified gross savings approach may result, to some extent, in a lag between programmatic activity and the awarding of incentive payments. While we are supportive of KEDLI and KEDNY's proposal to receive EAMs annually, we recommend a lag period to allow for the verified gross savings process to occur. To the extent the Companies are able to adjust their savings with realization rates to calculate verified gross savings by June 1 of the year following annual programmatic activity, we recommend that the Companies forecast their

verified gross savings achievement at that time and begin to collect their EAM incentive payment based on that forecast in September. For example, if the Companies run six programs in 2020, and two have a verified gross realization rate by June 1, 2021, the Companies could begin to collect any EAM incentive the Companies earned in September 2021. Each Company should reconcile that forecast and adjust its collections as necessary at the time it forecasts its next EAM incentive payment. Each Company should conduct an annual reconciliation of the previous year’s forecasted EAM incentives when setting the incentive payment collection levels to be effective in September of the year following program activity.

Q. Are there ways in which KEDLI and KEDNY could limit the risk of a downward adjustment to incentives previously recognized?

A. Yes. If the Companies have concerns regarding the recognition of EAM revenues, they could recognize a portion of the forecasted incentive level until such time as a full reconciliation can occur based on completion of the verified

gross savings process, thereby limiting the risk of future downward adjustments.

Q. Please describe the Affordability metric.

A. The Companies propose to implement the Affordability metric to provide an incentive for the Companies to help reduce low income customers' bills. As proposed by the Companies, on pages 106 and 107 of the FOH Panel direct testimony, the Affordability metric would measure the percentage reductions in annual low-income customer average monthly bills compared to the previous year's weather-normalized low income customer annual average monthly bill.

Q. Did the Companies propose specific targets related to this metric?

A. No. Instead of defined minimum, midpoint, and maximum achievement levels, and associated basis point awards, the Companies propose to earn one basis point of financial award for every 0.2 percent year-over-year reduction in the average low-income customer bill achieved. To put it another way, the Companies would earn 10 basis points of financial award for achieving a two percent reduction in average low income customer

bills. The Companies propose not to set a maximum cap on the amount of financial awards available for this metric.

Q. Does the Panel agree with the Companies' proposed Affordability metric?

A. No, we do not support this metric for three reasons. First, the Affordability metric, as proposed by the Companies, is not consistent with the EAMs provided for in the Track Two Order. Second, the targets proposed by the Companies are set inappropriately low, resulting in financial rewards for what would likely be the result of random chance and not specific Company actions. Third, the financial awards proposed by the Companies are far greater than the value of the outcomes achieved.

Q. Explain how the Affordability metric is not consistent with the Track Two Order.

A. The Commission considered an Affordability metric as part of its Track Two Order and determined that the outcomes resulting from such a metric are best addressed by other means, see pages 92-93.

Q. Explain how the Companies' proposed targets

proposed are inappropriately low.

A. The amount of variation in recent historical annual monthly average low income customer bills is significant. As shown in Exhibit___(SEAMP-4), the average annual monthly low income customer bill between 2014 and 2018 was $104.66 for KEDNY and $107.35 for KEDLI, with standard deviations of $15.01 and $15.50, or 14.34 percent and 14.44 percent, respectively. A two percent decrease in year-over-year annual average monthly low income customer bills, which would provide the Companies with a 10 basis point financial reward, represents just 0.14 standard deviations.

Q. What are standard deviations?

A. A standard deviation is a statistic that describes how much variation there is in a set of data. The greater the amount of variation in a data set, the greater the standard deviation. The standard deviation of a data set can also tell us how likely a particular outcome is to be the result of random, naturally occurring, variation in a given data set. The greater number of standard deviations a particular

outcome is away from the average of the data set, the greater likelihood that that outcome was not the result of random variation and was instead influenced by some outside factor, for example, utility efforts to reduce low income customer bills.

Q. At 0.14 standard deviations, how likely is it that reductions in low income customer bills are the result of increased utility efforts and not random chance?

A. At 0.14 standard deviations, we can only be about 10 percent certain that the variation from the average is a result of Company intervention and not simple random variation.

Q. Are there examples from other utilities' EAMs where the standard deviation or other similar descriptive statistic have been the basis for setting EAM targets?

A. Yes. The standard errors of regression, a descriptive statistic similar to standard deviations – essentially the standard deviations of a set of data from a linear regression – has been used as the basis for the Electric Energy Intensity metrics at Niagara Mohawk, Central

Hudson, and Orange and Rockland.

Q. What level of statistical certainty were used in those proceedings?

A. In each of those proceedings, the Commission approved targets which were developed based on minimum, midpoint, and maximum achievement levels set equal to 0.25, 1.0, and 1.75 standard errors, respectively.

Q. If targets were developed for the Affordability metric in this proceeding using similar levels, what would those targets be?

A. The targets based on 0.25, 1.0, and 1.75 standard deviations for KEDNY would be 3.59 percent, 14.34 percent, and 25.01 percent at the minimum, midpoint, and maximum achievement levels, respectively. For KEDLI, the targets would be 3.61 percent, 14.44 percent, and 25.27 percent at the minimum, midpoint, and maximum achievement levels, respectively.

Q. How do the achievement levels anticipated by KEDNY and KEDLI compare to the targets you just described?

A. The maximum amount anticipated by the Companies, 2.0 percent, is only roughly half of the minimum

achievement level developed using this method, and about one-twelfth of the maximum achievement level we described.

Q. Explain how the financial awards proposed by the Companies outweigh the value of the outcomes achieved.

A. As shown in Exhibit___(SEAMP-4), a reduction of two percent of average monthly bills would result in savings of approximately $104,000 to $150,000 per year for KEDNY low income customers in aggregate and approximately $20,000 to $28,000 per year for KEDLI low income customers in aggregate. Based on the Companies' proposed 10 basis point maximum and Staff's value of basis points, KEDNY would earn $3.35 million in financial incentives, and KEDLI would earn $2.11 million in financial incentives. Put another way, for each $1 the Companies save low income customers, KEDNY would be rewarded with $22 to $32, and KEDLI would be rewarded with $75 to $106.

Q. Would low income customers retain the full benefit of achievement under this metric if it were approved?

A. No. The revenues not collected from low income customers would be included in the Revenue Decoupling Mechanism and be recovered, at least in part, from these very same low income customers. In addition, these very same customers would also be responsible for a portion of the Companies' EAM awards related to this metric. The combination of statistically insignificant targets, shareholder financial incentives tenfold greater than the value of the outcomes created, and the diminished impact of those outcomes leads us to strongly recommend that the Commission not approve the Companies' Affordability metrics.

Carbon Reduction EAM

Q. Please describe the Companies' proposed Carbon Reduction EAM.

A. The Carbon Reduction EAM comprises two metrics: CO2 Reduced from Full-Service Customers, and CO2 Reduced from Transportation Customers. The Carbon Reduction EAM focuses on the adoption rate of three specific utility programmatic offerings: (1) customers converting from oil to

geothermal heating by taking heating service from utility-owned geothermal systems; (2) sales of RNG via the Green Gas Tariff; and (3) the use of compressed natural gas (CNG) and RNG in natural gas vehicles to offset use of diesel fuel.

Q. Please summarize the CO2 Reduced from Full-Service Customers metric (Full Service metric).

A. The Full Service metric is based on the sum of carbon reductions from customer uptake of utility-owned geothermal systems, and carbon reductions from sales of RNG through the Green Gas Tariff.

Q. How do the Companies propose to measure carbon savings from customer uptake of utility-owned geothermal systems?

A. The Companies propose to multiply the number of active utility-owned geothermal systems by the amount gas usage the customers using said geothermal systems would otherwise use to determine the amount of natural gas offset by customers using utility-owned geothermal systems instead. Next, the Companies propose to multiply the amount of natural gas offset by

geothermal customers by the lifecycle carbon emissions related to burning natural gas to determine the amount of carbon savings resulting from geothermal customers offsetting natural gas usage.

Q. How do the Companies propose to measure carbon reductions resulting from increased use of RNG?

A. The Companies propose a complicated three-part calculation, however, the method proposed by the Companies is mathematically equivalent to a fairly simple equation. Carbon dioxide savings from RNG would be calculated by multiplying the amount of RNG sold, in Dth, by the difference in CO2 emissions rates on a metric ton of CO2 emitted per Dth basis between pipeline natural gas and RNG.

Q. How many basis points do the Companies propose to earn at maximum achievement levels for the Full Service metric?

A. KEDNY proposes a maximum of one basis point for 2020, and a maximum of three basis points for each year from 2021 through 2023. KEDLI proposes a maximum of two basis points for 2020, four basis points for 2021, five basis points

for 2022, and six basis points for 2023.

Q. Please summarize the Companies' proposed CO2 Reduced from Transportation Customers metric (Transportation metric).

A. The Transportation metric would reward the Companies for encouraging CNG vehicle usage. The Companies propose to base this metric on the sales of natural gas to CNG fueling stations, multiplied by the difference in carbon emissions between CNG-fired vehicles and diesel-fired vehicles. The Companies also propose to incorporate carbon reductions related to increased use of RNG at these CNG fueling stations, providing additional carbon savings by comparing the carbon emissions related to RNG to the emissions related to pipeline gas.

Q. How many basis points of financial incentive do the Companies propose to earn at maximum achievement levels for the Transportation metric?

A. The Companies each propose to earn a maximum of five basis points each year from 2020 through 2023.

Q. Does the Panel agree with the Full Service and

Transportation Carbon Reduction EAM metrics?

A. No. There are three reasons why we do not support the Company's Carbon Reduction EAM metrics. First, the metrics proposed by the Companies are not aligned with the requirements of the Track Two Order. Second, the metrics' outcomes do not spur changes to the Companies' traditional business model, but instead reinforce incentives inherent in the Companies' existing business model. Third, the financial incentives the Companies propose are significantly greater than the societal value of the outcomes of the metrics.

Q. Explain how the Companies' Carbon Reduction metrics are not aligned with the Track Two Order.

A. As previously discussed, the Track Two Order specified that greenhouse gas reduction metrics should focus on decarbonization of end-uses, specifically related to heat pump and electric vehicle adoption. Further, the Companies proposals for these metrics are wholly programmatic, depending on customer use of customer-funded utility programs, whereas the

Track Two Order clearly expressed a preference for outcome-based EAM metrics.

Q. Explain how the Companies' proposed Carbon Reduction metric would not spur a change in the business model, but instead reinforce the existing business model.

A. As explained above while discussing the Companies' proposed Supply Diversification EAM, The Companies business model is already inherently aligned with increasing RNG interconnections. Similarly, through increased sales and revenues, the Companies already have a business model incentive to increase use of their systems through increased penetration and use of CNG vehicles. Neither of these outcomes require a change in business model for which the Companies need incentives to pursue. Therefore, additional customer-funded incentives for the Companies' shareholders are not necessary.

Q. Explain how the financial awards proposed by the Companies are greater than the societal value of the outcomes the Carbon Reduction metrics achieve.

A. The current Social Cost of Carbon (SCC), or

rather the societal value of reducing carbon emissions, is approximately $52 per metric ton. Based on the Companies' proposed Carbon Reduction metric targets and financial awards, the Companies would earn more in financial awards than the value of reducing carbon emissions by the proposed amounts. For example, KEDNY proposes to earn one basis point, approximately $335,537 during 2020 as a reward for reducing 3,295 metric tons under the Full Service metric in 2020, or $101.83 in shareholder incentives per metric ton of carbon reduced. This figure is the least egregious of the four total incentives, with $117.86 per metric ton, $290.16 per metric ton, and $182.65 per metric ton of incentives related to KEDLI's Full Service metric, KEDNY's Transportation metric, and KEDLI's Transportation metric, respectively. Put another way, the Companies propose to earn between $1.95 to $5.56 in shareholder incentives for every $1 in societal benefits achieved.

Q. Are there other more pragmatic reasons why the Companies' Carbon Reduction EAM metrics should

not be eliminated?

A. Yes. While the Companies' proposals would reward the Companies for customer adoption of utility-owned heat pumps, the Staff Efficiency and Sustainability Panel recommends eliminating the Companies' proposed utility-owned geothermal heat pump programs.

PLATFORM SERVICE REVENUES

Q. What are Platform Service Revenues (PSRs)?

A. The Track Two Order defines PSRs as a new form of utility revenues associated with the operation or facilitation of distribution-level markets. The Track Two Order also notes that PSRs are expected to take the place of EAMs in the long term as the utilities transition from their traditional role as distribution utilities to their new role as operators of the distributed system platform.

Q. Do KEDNY and KEDLI propose any PSRs in this rate filing?

A. Yes. The Companies propose two PSRs on page 116 of the FOH Panel's direct testimony, one related to their Energy Efficiency E-Commerce Platform,

and another related to their Utility Energy Services Contracts (UESC) program. For each of these PSRs, the Companies propose to retain a portion of the fees collected from program vendors and participants and pass the remainder back to customers.

Q. Please describe the Companies' proposed PSR related to the E-Commerce Platform.

A. On page 117 of the FOH Panel's direct testimony, the Companies propose to retain 20 percent of the fees they collect from vendors participating in their respective E-Commerce Platforms. The Companies propose to defer the remaining 80 percent of fees from the E-Commerce Platform for future credit to customers.

Q. Did the Companies explain why they proposed to retain a 20 percent share?

A. Yes. The Companies explain on page 117 of the FOH Panel's direct testimony that they developed their proposed E-Commerce Platform EAM to be consistent with a similar PSR approved by the Commission as part of the Rate Order issued on March 15, 2018 in Cases 17-E-0238 and 17-G-0239 regarding Niagara Mohawk.

Q. Does the Panel agree with the Companies' proposed E-Commerce Platform PSR?

A. No. The 20 percent share PSR approved by the Commission as part of Cases 17-E-0238 and 17-G-0239 for Niagara Mohawk references electric businesses only.

Q. Did the Track Two Order establish guidelines for consideration of PSRs?

A. Yes, the Track Two Order establishes five criteria for consideration of PSRs: (1) whether the service facilitates the growth of markets; (2) whether there is already a third-party market for the service that adequately serves all sectors of the market; (3) whether utility economies of scale and/or existing utility expertise are likely to result in a cost-effective stimulation of the market; (4) whether utility provision of the service is likely to prevent other providers from entering the market; and (5) the extent to which a utility has proposed placing shareholder funds at risk.

Q. Did the Companies' demonstrate that the criteria established in the Track Two Order were satisfied?

A. We do not believe the Companies provide substantive/reasonable distribution-level market information to justify consideration of the PSR in either their direct testimony or in response to IR DPS-363.

Q. Please describe the Companies' proposed PSR related to the UESC Program.

A. On page 117 of the FOH Panel's direct testimony, the Companies propose to retain one-third of the fees collected from UESC participants. The Companies propose to defer the remaining two-thirds of such fees for future credit to customers.

Q. Did the Companies explain why they proposed to retain a one-third share of the UESC Program fees?

A. Yes. On pages 117 and 118 of the FOH Panel's direct testimony, the Companies explain that the higher Company-share of USEC Program fees is reflective of additional marketing and managerial responsibilities of the UESC program compared to the E-Commerce Platform, and the additional risk that such activities may not result in a contract to participate in the USEC

program. The Companies argue that a higher Company-share of revenues is warranted because customer participation in the UESC is not guaranteed regardless of the Companies' marketing and managerial efforts.

Q. Does the Panel agree with the Companies' proposed PSR related to the UESC program?

A. Partially. We agree that the Companies should be provided a PSR to encourage them to maximize participation and benefits of the UESC program. However, the Companies' proposal to retain one-third of the fees paid by UESC program participants is unreasonably high.

Q. Why do you disagree with the Companies' proposed retention of one-third of UESC-related fees?

A. The Track Two Order established 20 percent as a reasonable utility-share of revenues for PSRs built upon programs without any shareholder funds at risk, and stated that higher sharing percentages might be reasonable depending on the amount of shareholder funds at risk.

Q. Are any shareholder funds at risk related to the UESC program?

A. No. Regardless of whether or not the Companies

are able to convince customers to participate in the UESC program, the Companies are made whole for the costs of the program. No shareholder funds are at risk, as acknowledged by the Companies in response to IR DPS-363, and therefore the Companies should not receive a PSR greater than 20 percent of revenues from the UESC program.

Q. Does the Panel recommend an alternate structure for a PSR related to the UESC program?

A. Yes. Instead of the Companies' proposed PSR based on a one-third share of all UESC program fees, we recommend that the Companies be able to retain a PSR equal to 20 percent of the net revenues collected from UESC program participants.

Q. Explain what you mean by net revenues.

A. The costs of the UESC program are recovered through base rates. Under our recommendation, the Companies would apply 100 percent of the fees collected from UESC program participants toward offsetting the costs of the UESC program until such costs are fully offset. Once the costs of the UESC program are offset, the

Companies would be eligible to retain 20 percent of any further UESC program fees, with the remainder deferred for customer benefit. An example of our recommended PSR is provided in Exhibit___(SEAMP-5).

Q. Are there benefits to adopting your recommended UESC-related PSR instead of the Companies' proposal?

A. Yes. As shown in Attachment 2 to the Companies' response to IR DPS-362, under the Companies' proposed PSR, the Companies would receive approximately double the earnings that accrue to customers. Further, as shown in Exhibit___(SEAMP-5), unlike the Companies' proposal, our recommended PSR only rewards success in the UESC program when it produces fees above and beyond its costs. Under this paradigm, the Companies will have a significant incentive for the UESC program to pay for itself in a given year, and will thereafter have the opportunity to retain a portion of those fees once the program is profitable.

KEDLI EMPOWER REPLACEMENT INCENTIVE

Q. Please describe KEDLI's proposal to continue its EmPower Replacement Program incentive mechanism.

A. The KEDLI EmPower Replacement Program, also called the Heat Program, and associated incentive mechanism were approved as part of the 2016 Rate Order issued by the Commission on December 16, 2016 in Cases 16-G-0058 and 16-G-0059. As part of the EmPower Replacement Program, KEDLI provides low income customers with four tiers of energy efficiency services. Under KEDLI's existing EmPower Replacement Program incentive mechanism, the Company earns financial rewards for each household participating in Tier 3 and Tier 4 of the program, based on a percentage of the program budget. A description of the participation Tiers under the KEDLI EmPower Replacement Program is available in the direct testimony of the Staff Efficiency and Sustainability Panel.

Q. Please describe the current incentive mechanism for participation in Tiers 3 and 4 of the EmPower Replacement Program.

A. Under the current mechanism, the Company earns a

reward of 0.007 percent of the program budget for each household participating in Tier 3, and a reward of 0.014 percent of the program budget for each household participating in Tier 4. Historically, KEDLI has earned approximately $122.88 per household participating in Tier 3, and $245.76 per household participating in Tier 4. Since Tier 4 participants also participate in Tier 3, KEDLI earns a total of $386.64 per household participating in both Tier 3 and Tier 4.

Q. Does the Panel agree with KEDLI’s proposal to continue its EmPower Replacement Program Incentive Mechanism?

A. Partially. We agree that KEDLI should continue to receive an incentive to maximize participation in its EmPower Replacement Program, however, we recommend redesigning the incentive mechanism.

Q. Why should the EmPower Replacement Program Incentive Mechanism be redesigned?

A. Currently KEDLI’s incentive per household will increase or decrease depending solely on the size of the program budget. As shown in

Appendix 7, Schedule 10, page 2, of the Joint Proposal adopted in the 2016 Rate Order, the incentives per participating household were set at $122.88 for Tier 3, and $245.76 for Tier 4. As shown in the FOH Panel's Exhibit___(FOH-12), KEDLI proposes to earn per-household incentives of $148.75 for Tier 3 and $297.50 for Tier 4 based solely on its request for a higher EmPower Replacement Program budget. The current mechanism provides KEDLI with an incentive not only to increase participation in the EmPower Replacement Program but also to increase the EmPower Replacement Program budget. This mechanism provides KEDLI with a positive incentive for continually spending and requesting ever more program funds from customers.

Q. What does the Panel recommend?

A. Instead of providing the incentive on the basis of a percentage of program budget, we recommend setting the incentive as a flat dollar amount per household, regardless of the program budget. In addition, we recommend reducing the incentive related to participation in Tier 4 of the

program.

Q. Why should the Tier 4 incentive decrease?

A. As shown in Exhibit___(SEAMP-6), the incentives earned by KEDLI related to Tier 4 have significantly outpaced the incentives related to Tier 3. The KEDLI EmPower Replacement Program is designed to provide an incentive for KEDLI to promote energy efficiency to low income customers. The EmPower Replacement Program was not designed to encourage the Company to pay for subsidized improvements to customers' premises, however, the majority of incentives earned by the Company are a result of Tier 4 participation, which includes subsidization for non-energy home infrastructure, instead of Tier 3. Further, as shown in Exhibit___(SEAMP-6), the level of energy savings related to Tier 4 measures is only a fraction of the savings achieved under Tier 3.

Q. Should the Tier 4 incentive be eliminated entirely?

A. No. Although there is significant overlap between Tier 3 and Tier 4 – all Tier 4 participants, by definition, also participate in

Tier 3 – there is still good reason to encourage KEDLI to make the most of its Tier 4 funding, provided such incentive is set at a reasonable level.

Q. What incentive levels does the Panel recommend?

A. We recommend setting the Tier 3 incentive equal to $125.00 per household, and the Tier 4 incentive equal to $62.50 per household. Again, as participants in Tier 4 are also participants in Tier 3, under our proposal KEDLI would earn $187.50 for each household participating in both Tiers 3 and 4.

Q. How did the Panel develop its recommended per-household incentives?

A. We set our Tier 3 incentive to closely match the current $122.88 per household incentive, using a round number instead of the current figure. We set our Tier 4 incentive equal to half of the Tier 3 incentive level. Under this incentive regime, the Companies will continue to have a higher marginal incentive to enroll customers in both Tiers 3 and 4, however, the Tier 4 incentive will not outpace the incentive related to the main purpose of the EmPower Replacement

Program itself.

NON-PIPES ALTERNATIVES INCENTIVE MECHANISM

Q. Please provide a summary of the Companies' proposals regarding Non-Pipes Alternatives (NPAs).

A. The FOH Panel briefly discusses its NPA-related proposals on page 118 of its direct testimony, with significant additional detail contained in Exhibit___(FOH-11). In Exhibit___(FOH-11), the Companies propose: (1) a definition of NPAs; (2) screening criteria for traditional infrastructure projects to be considered for development as an NPA; (3) eligibility requirements for resources to participate in an NPA project; (4) a shareholder incentive mechanism; (5) NPA project cost and incentive cost recovery mechanisms; and (6) associated reporting requirements.

Q. How do the Companies define an NPA project?

A. The Companies define an NPA project as a portfolio of resources used to delay, reduce, or eliminate the need for traditional capital investments to reinforce or expand gas supply

and distribution infrastructure, and which reduces net utility plant in service. The Companies note that while their proposed incentive mechanism would be limited to projects that reduce utility net plant in service, they would not be precluded from pursuing opportunities to develop solutions for other types of projects that would not reduce utility net plant in service, but would produce societal benefits.

Q. What types of projects would be included under the Companies' proposed incentive mechanism?

A. The Companies' proposed incentive mechanism would include any project that displaces KEDNY or KEDLI capital expenditures, and therefore net plant in service. For example, the mechanism would apply to a project that would eliminate the need to replace gas distribution main in a particular area.

Q. What types of projects would not be included under the Companies' proposed incentive mechanism?

A. The incentive mechanism would not cover projects that reduce the need for interstate pipelines,

since reductions to interstate pipeline capital costs do not result in a reduction to distribution utility net plant in service. To be clear, the Companies would still have the opportunity to pursue such projects, however, the explicit cost recovery and incentive mechanisms proposed for NPAs in these proceedings would not apply to those projects. Instead, the Companies would have to separately request authorization and cost recovery for such projects.

Q. Please define “NPA screening criteria.”

A. NPA screening criteria are a set of requirements which obligate the Companies to attempt to develop an NPA project when all of the criteria are met. When considering a capital project, the Companies must compare the project against the screening criteria. If the project meets all of the screening criteria, the Companies must attempt to develop an NPA project to implement in lieu of the planned traditional capital investment. The Companies may optionally attempt to develop NPA projects for those traditional capital investments which do

not meet all of the criteria, however, they would not be obligated to do so.

Q. What screening criteria do the Companies propose?

A. The Companies propose to separate projects into two categories, large and small. For small projects, the Companies propose that they be required to attempt to develop an NPA portfolio for traditional projects with capital costs more than $1 million and less than $3 million, which would be designed to be completed within one fiscal year. For large projects, the Companies propose to that they be required to pursue NPAs for projects with capital costs greater than $3 million, which would be designed to be completed over more than one fiscal year. In the event that these screening criteria need to be adjusted, the Companies propose to make such adjustments through a separate filing.

Q. Explain the Companies' proposed eligibility requirements for resources participating in NPA projects.

A. The Companies propose that a resource must be a commercially-proven technology that provides

load relief coincident with the system need, and upon which the Companies can rely on to delay or replace the need for the applicable traditional capital project.

Q. Provide an overview of the Companies' proposed NPA Incentive Mechanism.

A. The Companies propose a shareholder incentive equal to 30 percent of identified societal net benefits, plus or minus 50 percent of portfolio cost savings or overruns, respectively. The Companies propose to cap the incentive at no greater than 50 percent of the societal net benefits, and propose an incentive floor of $0. The Companies propose slightly different mechanisms for large and small NPA projects. The Companies propose to use a streamlined process for small projects based on the calculation of the incentives on a per-Dth basis to recognize the more rapid speed of development for these smaller projects. The Companies propose a more rigorous process for large projects, which would recognize the longer lead time available for projects of this magnitude. For both small and large projects, the Companies

propose adjustment mechanisms to modify the available incentive if an updated needs assessment determines that more or less load relief is required to achieve deferral or replacement of the traditional infrastructure, or if the NPA project will not be able to defer the infrastructure as intended.

Q. Provide an overview of the Companies' proposals regarding recovery of NPA project and incentive costs.

A. The Companies cost recovery proposals are related to three areas: (1) modifications to the Companies' Capital Investment Reconciliation Mechanism; (2) allocation of NPA-related costs to service classifications (SCs); (3) amortization and recovery of NPA-related costs in the Delivery Rate Adjustment surcharge; and (4) roll-in of unrecovered amortized NPA costs into base rates during the Companies' next rate proceedings.

Q. What modifications do the Companies propose to their Capital Investment Reconciliation Mechanism related to NPA projects?

A. To the extent that an NPA project results in the

Companies displacing a capital project reflected in the Average Gas Plant in Service Balances (balance), the Companies propose to reduce the balance to exclude the forecast net plant associated with that capital project. The Companies propose to apply the carrying charges associated with the displaced project as a credit against recovery of NPA-related costs. In the event that those carrying charges are greater than the cost of the NPA project, the Companies propose to defer the difference for the benefit of customers.

Q. How do the Companies propose to allocate NPA costs to the various SCs?

A. The Companies propose to allocate NPA costs to service classifications based on the type of traditional solution the NPA project would defer or eliminate, roughly on the basis of a “beneficiaries pay” principle. Specifically, the Companies propose to allocate peak day need-related NPAs to SCs based on each SC’s contribution to Peak Sendout, and propose to allocate daily demand need-related NPAs to SCs based on each SC’s contribution to Total Gas

Deliveries.

Q. What do the Companies propose regarding amortization and recovery of these costs?

A. The Companies propose to amortize NPA project costs over 10 years. The Companies propose to begin recovery of these costs through the DRA surcharge as they are incurred. For earned NPA-related incentives, the Companies propose to amortize such incentives, once earned, over the remaining deferral period of the traditional infrastructure. The Companies would begin recovery of such incentives through the DRA surcharge once earned. The Companies propose to include any unamortized costs plus carrying charges in base rates the next time gas base rates are reset instead of continuing to recover NPA project and incentive costs through the DRA surcharge.

Q. Provide an overview of the Companies' proposed reporting requirements for NPA projects.

A. The Companies propose to file a detailed Implementation Plan and Benefit Cost Analysis (BCA) result with the Secretary to the Commission for each NPA project they pursue when

they have reasonable certainty as to the costs of the NPA project. The Companies propose to update the Implementation Plan annually, and may file updates more frequently, as needed. The Companies propose to file updated Implementation Plans outside the typical annual filing: (1) if the amount of load relief required increases or decreases; (2) if the deferral period for the traditional infrastructure project lengthens or shortens; or, (3) when the Companies file to earn incentives under their proposed NPA Incentive Mechanism. The Companies also propose to file an update to the BCA results when the amount of load relief or deferral period are modified.

Q. Did the Companies propose any other reporting requirements?

A. Yes. In addition to the annual updates to each NPA project’s Implementation Plan, the Companies propose to summarize all of their active NPA projects annually by January 31 of each year, beginning the first year that the Companies have any active NPA projects. The Companies also propose to file quarterly reports with the

Secretary to the Commission tracking the progress of each NPA project 60 days after the close of each quarter.

Q. Where would the Companies file these reports?

A. All of the reports described above would be filed with the Secretary to the Commission in these proceedings.

Q. What is the Panel’s general philosophy with regard to NPAs?

A. We recommend that the Companies be positioned to begin responding to system needs with non-traditional solutions as quickly as possible. Therefore, we recommend that the Commission approve the necessary NPA development processes and cost recovery mechanisms as part of these proceedings. Given the rapid pace of policy developments in the gas business, however, we do not recommend establishing any specific shareholder incentive mechanism at this time, especially since the Companies have yet to identify any potential projects for which they may pursue NPAs. NPA projects, especially a utility’s first such project, may take a year or more to develop - even after the utility has

decided to pursue a particular NPA. Any NPA shareholder incentive mechanism approved as part of these cases runs the risk of potentially being out of step with the then-current Commission policy by the time the Companies have developed an actual NPA project proposal.

Q. Does the Panel have any specific recommendations regarding the Companies' NPA-related proposals?

A. Yes. We recommend that the Commission approve the Companies' NPA-related proposals with the exception of the Companies' proposed NPA Incentive Mechanism. Instead of approving the Companies' proposed NPA Incentive Mechanism, the Commission should allow the Companies to file a petition for an appropriate shareholder incentive mechanism concomitantly with the Implementation Plan for the Companies' first NPA project.

Q. Are there benefits to approving NPA development criteria and cost recovery mechanisms in these proceedings, but delaying approval of a shareholder incentive mechanism?

A. Yes. Approving the NPA development criteria and cost recovery mechanisms in these proceedings

will provide the Companies with regulatory certainty with regard to their ability to recover NPA costs. This will eliminate barriers so that the Companies can begin seeking NPA opportunities as soon as possible. While delaying approval of a specific shareholder incentive mechanism might not accelerate the Companies' timeline for pursuing these projects, it will ensure that the shareholder incentive mechanism ultimately approved is consistent with the direction of the Commission's policies regarding the natural gas business in place at that time.

Q. Does this conclude your testimony at this time?

A. Yes.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a/ National Grid

Case Cases 19-G-0309 & 19-G-0310

January 2020

Prepared Supplemental Testimony
of:

Staff Earnings Adjustment
Mechanisms Panel

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Utility Engineering Specialist 3

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Q. Members of the Staff Earnings Adjustment Mechanisms Panel (Panel), please state your names.

A. Our names are Robert Cully, Kathryn Mammen, Laurie Kokkinides, and Davide Maioriello.

Q. Has the Panel previously sponsored testimony in these proceedings?

A. Yes. We offered pre-filed direct testimony, submitted August 30, 2019.

Q. What is the purpose of the Panel’s supplemental testimony?

A. The purpose of our testimony is to update the exhibits provided as part of our pre-filed direct testimony, specifically Exhibit___(SEAMP-3), based on the Commission’s recent Order Authorizing Utility Energy Efficiency and Building Electrification Portfolios Through 2025, issued on January 16, 2020 in Case 18-M-0084 (January 2020 Energy Efficiency Order).

Q. Is the Panel sponsoring Exhibits as part of this supplemental testimony?

A. Yes, we are sponsoring one exhibit with our testimony.

Q. Please describe the exhibit.

A. Exhibit___(SEAMP-7) provides an update to Exhibit___(SEAMP-3), which we provided with our pre-filed direct testimony.

UPDATE TO EXHIBIT___(SEAMP-3)

Q. Please describe the changes the Panel recommends to update Exhibit___(SEAMP-3).

A. As shown in Exhibit___(SEAMP-7), we updated Exhibit___(SEAMP-3) to reflect the Companies' budgets and Million British Thermal Units (MMBTU) targets approved by the Commission in the January 2020 Energy Efficiency Order.

Q. Why is it necessary to update Exhibit___(SEAMP-3)?

A. Exhibit___(SEAMP-3) was submitted as part of our pre-filed direct testimony to illustrate how our recommended Share the Savings Energy Efficiency EAM metric would work, and what potential financial award would be available to the Companies based on their performance under that metric. The budgets and targets shown in Exhibit___(SEAMP-3) are now obsolete, based on the Commission's January 2020 Energy Efficiency Order. Accordingly, it is reasonable to provide

an update to Exhibit___(SEAMP-3) to reflect those new budgets and targets, and to provide an estimate of the potential financial awards available under the Share the Savings metric.

Q. Are the budgets and MMBTU targets approved in the January 2020 Energy Efficiency Order described elsewhere in Department of Public Service Staff's supplemental testimony?

A. Yes, the budgets and MMBTU targets required under the January 2020 Energy Efficiency Order are fully described in the supplemental testimony of the Staff Efficiency and Sustainability Panel. At a very high level, the Commission increased the Companies' Energy Efficiency portfolio budgets and targets for the period from 2021 through 2025, which results in an increase to the Rate Year budgets and targets shown in Exhibit___(SEAMP-7) compared to the assumed budgets and targets provided in Exhibit___(SEAMP-3).

Q. Are there any noteworthy changes shown in Exhibit___(SEAMP-7) compared to Exhibit___(SEAMP-3)?

A. Yes. As shown in Exhibit___(SEAMP-7), since

both the energy efficiency budgets and targets have increased as a result of the January 2020 Energy Efficiency Order, the total financial award available to KEDNY and KEDLI has also increased, further strengthening the Companies' incentive to maximize the effectiveness of their energy efficiency program offerings.

Q. Does this conclude your testimony at this time?

A. Yes.

MR. GOODRICH: Okay. And I believe we do not need to do anything with the exhibits at this time?

A.L.J. COSTELLO: That's -- that's correct. If -- if anyone has any objections, they can raise objections. But we're going to deal with the exhibits at the end of the hearing. We'll -- we'll admit them at the end of the hearing.

MR. GOODRICH: Thank you, Your Honors. And Your Honors, this -- the -- this panel is now available for cross examination.

A.L.J. COSTELLO: Thank you. Okay. We have PULP.

MR. BERKLEY: Thank you, Your Honors.

CROSS EXAMINATION

BY MR. BERKLEY:

Q. And good morning, panel. Let me get a little more closely to that since I'm not as loud as Rob. My name is Richard Berkley. I am representing the Public Utility Law Project of New York. Let me start by asking a -- a basic question, panel. What are E.A.M.s?

A. (Cully) So Earnings Adjustment Mechanisms, E.A.M.s are defined in the -- the REV

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Track 2 order, that's the, I believe it's May 26th, 2016 order by the commission which implemented positive only earning -- additional earnings incentives tied to public policy goals. They are -- typically have a -- a defined metric and number of basis points and are to be set in individual utility rate cases.

Q. Thank you. If you turn to pages 3 and 4 of your direct testimony, starting on line 23 panel stated, "If the utility meets the E.A.M. targets, benefits are achieved, and utility's shareholders earn financial rewards." Is that correct?

A. That's correct.

Q. How do the E.A.M.s benefit ratepayers?

A. So the E.A.M. targets and metrics are tied to benefits which the -- the -- rather they are tied to outcomes which result in net benefits to society. Things like reduced energy costs, reduced capacity costs, both on the generation system and the transmission system, reduced carbon dioxide emissions, et cetera.

Q. If I can paraphrase what you said

you're saying that they have a value to all of New York State and society generally. Do they have any quantifiable value for the ratepayers in this specific case or a case in which E.A.M.s are being proposed?

A. Yes.

Q. Are E.A.M.s worth the ratepayer moneys paid?

A. Yes.

Q. Does staff ever disagree with the E.A.M.s proposed by company?

A. Always.

Q. Thank you. On what basis does staff disagree?

A. I believe that we outlined a number of bases; sometimes multiple for each metric. Generally speaking, staff and the companies initially disagree on how the metric is formed. For example, what is measured. Staff and the company frequently disagree on the target levels, the minimum midpoint and maximum levels at which defined basis point allowances are provided. And staff and the company often disagree on the number of basis points assigned to each metric and that each of those minimum,

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midpoint and maximum levels.

Q. Thank you. Would it be fair to say that staff, at least upon occasion, disagrees with the company based on whether the E.A.M.s are worth it or not?

A. Yes.

Q. Thank you. What's the -- I'm sorry, let me withdraw that.

Couldn't the Public Service Commission just order utilities to do all these things that the E.A.M.s stand for for free?

A. Yes. However, the Commission and the Track 2 order provided E.A.M.s as a way to get the utilities to drive after these outcomes in the most effective way possible.

Q. Thank you. I'm -- I'm familiar with the Track 1 and Track 2 orders from the REV proceeding 14-M-0101. But if the utility were to be ordered by the commission to undertake an activity that is reflected by an E.A.M., wouldn't they have to drive after it with all their energy and whatever anyway? To say it in another way, what happens to the company if it disobeys orders by the commission?

A. I'm not a lawyer. I can't speak

to that.

Q. Thank you. Turning to the page 14 in your direct testimony and I am referring -- I'm sorry, starting on page 13 on line 20 but then running over to line 3 on page 14. Staff recommended the rejection by the commission of the company's proposed E.A.M.s based on their basis points. Could you explain why staff lowered the amount of basis points available for the E.A.M.s?

A. So that's a fairly multi-faceted question. One of the reasons why we recommended a lower basis point total is because we did not disagree with the very presence of some of the E.A.M. metrics requested by the Company. And therefore, eliminated the number of basis points associated with those metrics. Other metrics, we agreed with in principle, but provided recommendations tweaking the metric somewhat or -- or changing it such that we could agree with it and recommended a different number of basis points. And it should also be noted that the roughly 78 or 77, depending upon the company, number of basis points requested by KEDLI and KEDNY at the time were approximately 7 to 8 times as high as the commission had ever provided a number

of basis points for any other gas utility.

Q. Thank you. And just to clarify, when the company achieves an E.A.M. target and it receives a, for want of a better phrase, bonus or additional moneys in the amount of basis points, is that added on to the company's rate requests and R.O.E.?

A. Yes.

Q. Thank you.

A. Maybe a -- a clarification to that. It is moneys in addition to the typical revenue model of the utility.

Q. Thank you. And what is the typical revenue model?

A. The companies earn a return on rate base.

Q. And is that called cost of service ratemaking?

A. That's my understanding.

Q. What's the relationship between E.A.M.s and cost of service ratemaking?

A. The commission defined that in the Track 2 order. It stated that the E.A.M.s were to be positive-only incentives. In particular, the

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commission did in fact consider negative incentives as part of the Track 2 order but determined not to do so because of potential risks to the company's financial requirements. The -- particularly the -- I believe, it's called the risk -- essentially the -- the level of risk that utility has, they need to earn a return commensurate with that risk. The commission, the Track 2 order, essentially expressed concern that it would not want to increase the R.O.E. because of the additional risk of E.A.M.s and so initially set the E.A.M.s as positive-only.

Q. Thank you. Is it possible that by achieving all of its E.A.M.s, that the company could receive moneys that are not allocated to any pieces of its infrastructure or any of its operation that are used and useful for ratepayers?

MR. GOODRICH: Objection, vague.

A.L.J. LEARY: Did you say vague?

MR. GOODRICH: Yes. Can you please re -- rephrase the question?

MR. BERKLEY: Sure. Yeah, just -- just --.

A.L.J. LEARY: Do you want to -- do -- ?

A.L.J. COSTELLO: Yeah. Try to rephrase it.

A.L.J. LEARY: Do you want to rephrase?

MR. BERKLEY: I'll rephrase it.

BY MR. BERKLEY: (Cont'g.)

Q. Are you familiar with the term called used and useful, with regard to utilities?

A. (Cully) Yes.

Q. Do you know what it means?

A. It's fairly self-explanatory. A piece of equipment, generally speaking, can only be included in rate base and thus begin to earn a return on that equipment or rather on the cost of that equipment once it is both used in utility service and useful to customers.

Q. Thank you. Are E.A.M.s connected to any of the company's physical infrastructure used and useful for ratepayers?

A. Yes.

Q. Could you explain?

A. So, many of the E.A.M.s that we describe in our testimony are directly related to better use of the utility system and drive things

like avoided transmission and distribution system benefits which in the medium- and long-term are designed to reduce the cost of service to ratepayers.

Q. Thank you. Turning to pages 41 and 42 in your direct and let me start on lines 4 through 7. Wherein, the panel describes the affordability metric and you state, "The company's proposed to implement the affordability metric to provide an incentive for the companies to help reduce low-income customer bills." Skipping to page 42, lines 4 and 5, the question is asked, does the panel agree with the company's proposed affordability metric to which the answer is, no, then a whole bunch of other testimony. So, my question is other then an argument on cost, does staff have other concerns about the affordability metric?

A. Can you explain what you mean other than the concerns on cost?

Q. Yes. And forgive me, I could have -- I could've read this. So, starting on lines 7 through 16. First, the affordability metric is not consistent with E.A.M.s as provided for in the Track 2 order in REV. Second, the targets proposed by the companies are set inappropriately low resulting in

awards -- I'm sorry, rewards for what would likely be the result of random chance and not specific company actions.

Third, the financial awards proposed are far greater than the value of the outcomes achieved. Does staff have any other concerns other than those three that are set forth in the testimony?

A. Those are the three concerns set forth in our testimony. We have no additional concerns at this time.

Q. Thank you. Would lowering the average bill for low-income customers, could that be brought about by simply terminating customers with high bills, and if so, would that trigger the E.A.M.?

A. It's a little difficult to answer because you're asking me to speculate on an E.A.M. that we do not support.

Q. Fair enough. Thank you.

A. I will however say that under the company's affordability metric, the company did not define what actions it would take to lower those customer bills. So theoretically, they could lower the average by eliminating the upper portion of customer bills that are raising that average.

Q. Thank you. What is the -- I am sorry, forgive me. If the company were to receive all of the E.A.M.s it proposed in this case, what would be the increase to their revenue?

A. We did not perform that math, but it should be relatively simple depending upon what level of dollars per basis point you used. We did perform math showing what our basis point total would be. I believe that is in S.E.A.M.P.-2. I'm not sure which Hearing -- Hearing Exhibit number that is.

THE REPORTER: That's 417.

A.L.J. LEARY: It's Hearing Exhibit 417, I believe. Does that -- does that make sense to everybody?

MR. BERKLEY: Yes.

A.L.J. LEARY: Just check it for the record, Mr. Goodrich.

MR. GOODRICH: S.E.A.M.P.-2 is -- is 417, yes.

A.L.J. LEARY: Thank you.

MR. BERKLEY: Thank you.

BY MR. BERKLEY: (Cont'g.)

Q. When staff analyzed the company's proposed E.A.M.s and when it provided its own

estimates of what the E.A.M. should be, that is in terms of potential revenue, did staff give any reference to the ability of customers to afford the E.A.M.s added on top of the company's revenue requirement?

A. I'm sorry. Can you restate the question please?

Q. Yes. And let me -- let me try and restate it a little more simply.

Did considerations of affordability for ratepayers factor into staff's analysis when it was analyzing the Company's proposed E.A.M.s?

A. Yes, in the sense that we ensure that the number of basis points available to the companies were within a reasonable range previously supported by the commission in other cases, supported by the REV Track 2 order, and we always strive to ensure that the E.A.M.s result in outcomes that are on balance much more beneficial to customers and society as a whole than to the company itself.

In -- in other words, the -- we believe the benefits of the E.A.M.s as we have recommended them, the lion’s share of those benefits accrue to customers and not to the company.

Q. Thank you. And one last question on this topic. When staff does that sort of analysis, does it actually examine what the bill impacts might be upon low- and moderate-income customers in the company service territory?

A. So we are not rate-design engineers. We do not do bill impacts. However, we do ensure that the societal net benefits again, accrue predominantly to customers and not to the company.

Q. Thank you.

MR. BERKLEY: No further questions Your Honors.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: Okay. We have A.A.R.P.

MR. RIGBERG: Thank you Your Honor.

A.L.J. LEARY: You found a parking spot?

MR. RIGBERG: On Grand Street.

A.L.J. LEARY: Oh.

MR. RIGBERG: I guess it's lobby day and the visitor lot was full. Yeah, thank you.

BY MR. RIGBERG:

Q. I -- Mr. Berkley covered most of the questions I was going to ask. One -- one question I do have though is, if you turn to page --

THE REPORTER: I'm sorry. What is your name?

MR. RIGBERG: Saul -- I'm sorry, Saul Rigberg with A.A.R.P.

THE REPORTER: Thank you.

BY MR. RIGBERG: (Cont'g.)

Q. In -- in the first few pages of your testimony, you discussed the commission's orders and -- other -- and the energy -- setting out E.A.M.s policies. And you referenced on -- on page 6 starting on the bottom of page 6, several proceedings regard -- that are -- you pointed out on page 7 are -- are electric proceedings. And I'm wondering in -- in designing -- in staff's design of the E.A.M.s in this case, did you look at any gas E.A.M.s that other -- with other utilities in the state?

A. (Cully) Yes.

Q. And -- and which ones were they?

A. Gas E.A.M.s have been part of the 2017 Niagara Mohawk case that's, I believe, 17-E-0238. The Central Hudson 2017 case 17-E-0459, the

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Orange and Rockland case 18-E-0067, and at the time of this testimony, we had also just finished testifying at Con -- Con Edison that was case 19-E-0065. There's also associated gas numbers with each of those cases, but I always remember them by the electric number.

Q. Okay. Yeah. Thank you. Now in -- in designing the E.A.M.s for -- for this -- these companies, did you look at outcomes from any of those gas cases? You know, did you look at actual results and -- and -- and factor those results into your design of the E.A.M.s in this case?

A. Yes.

Q. Okay. Can you explain in -- in which way you did that?

A. So you can see in our exhibit, just trying to remember which exhibit it was from our pre-filed testimony. If you look at Exhibit S.E.A.M.P.-3, page 4 of 4. Again, I'm not sure which pre -- which hearing exhibit that relates to.

A.L.J. LEARY: That is Hearing Exhibit 8 -- 1 -- I'm sorry.

A.L.J. COSTELLO: 418.

A.L.J. LEARY: 418. 418.

THE WITNESS: Thank you, Judge. You can see that we examined the number of basis points and required improvement rates for the energy efficiency E.A.M. metrics that are currently enforced at O&R, Central Hudson and Niagara Mohawk. We also generally looked at things like the number of basis points available for each of those utilities that the commission has already approved.

BY MR. RIGBERG: (Cont'g.)

Q. Thank you. Now, when you use that term improvement rates, what -- what does that mean?

A. Specifically, on Exhibit S.E.A.M.P.-3, we examined the difference between the maximum target for the energy efficiency programmatic E.A.M. at O&R, Central Hudson and Niagara Mohawk compared to the minimum target for that same metric at each of those utilities and determined a percentage of difference between the maximum target and the minimum target. We then used that percentage difference that the commission has already approved for targets in those cases to inform what our maximum number of basis points might be under the energy efficiency share of the savings metric in this case.

Q. And did you set the tar -- your tar -- the targets in this case higher based on the outcomes in the other cases, or – lower, or the same?

A. (Cully) I would say that our considerations and target setting in this case are informed by the other cases. We always strive to provide targets that are reasonable for each utility. That neither provide an incentive for doing nothing, nor are wholly unachievable.

However, the answer is more complicated in this case. Because in the case of KEDLI and KEDNY in the present proceeding, we are recommending a share-the-savings type metric for the energy-efficiency programs instead of the incremental dekatherm of energy efficiency achieved metric in the three prior cases.

The difference is essentially that the other metrics have defined targets and basis points available with each of those targets. Whereas, as we described in our pre-filed testimony, direct testimony rather, the share-the-savings metric is based on the difference in dollar per lifetime M.M.B.T.U. run rates of the energy efficiency program.

So there --it's more of a continuous function as opposed to a function with defined metrics and a linear extrapolation between -- I'm sorry, three defined points in linear extrapolation between those points.

Q. And is it correct that you proposed the same, share-the-savings metric in the -- in the Con Ed case?

A. Yes. And we proposed the same metric in the NYSEG and RG&E rate cases as well.

Q. Okay. Thank you.

A. I should also mention that the share-the-savings metric was specifically called for by the commission in the December 2018 energy efficiency order, which we referenced in our pre-filed direct testimony.

Q. Thank you. And do you remember, where the -- the three utilities that you looked at, how successful were they in achieving the -- the minimum level in the metrics?

A. I do not have that information with me at this time. However, it is publicly available on our D.M.M. system.

Q. Okay. Thank you.

MR. RIGBERG: That's all the questions I have, Your Honors.

A.L.J. COSTELLO: Okay. Thank you. Any re-direct?

MR. GOODRICH: May we take a couple of moments?

A.L.J. COSTELLO: Sure.

MR. GOODRICH: Thanks.

A.L.J. COSTELLO: Okay. We'll go off the record.

(Off the record 9:42 a.m.)

THE REPORTER: On the record.

MR. GOODRICH: Your Honor, staff has no re-direct.

A.L.J. COSTELLO: Okay. Thank you. So then we -- the next -- well, I'd like to thank the panel for their testimony, and you're excused.

MR. GOODRICH: They -- they will be remaining because they are --.

A.L.J. COSTELLO: They are the same -- the next panel.

A.L.J. LEARY: They're the next panel.

A.L.J. COSTELLO: They are? Okay.

A.L.J. LEARY: Yeah.

A.L.J. COSTELLO: Okay.

MR. GOODRICH: We have a couple more than that, Your Honor.

A.L.J. LEARY: The witnesses who are already sworn in continue to be under oath. We need not swear you in again. The two additional witnesses for this panel, would you stand and state your name and business address for the record?

MS. PARK: My name is Jennifer Park and my business address is 3 Empire State Plaza, Albany, New York.

MR. HITT: Joseph Hitt, 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: Could you raise your right hand, please? Do you swear or affirm that the testimony you'll provide is the truth?

MR. HITT: Yes.

MS. PARK: Yes.

WITNESS; JENNIFER PARK; Sworn

WITNESS; JOSEPH HITT; Sworn

A.L.J. COSTELLO: Okay. Thank you. You may be seated.

A.L.J. LEARY: I want to make sure we're back on the record. Are we back on? Okay,

good. Should have asked that earlier. Mr. Goodrich.

MR. GOODRICH: Thank you, Your Honors.

DIRECT EXAMINATION

BY MR. GOODRICH:

Q. This is Brandon Goodrich. So panel, as I did with the previous panel, I'm going to ask you a few questions to -- to get this started. So Staff Efficiency and Sustainability Panel, before you is a document entitled Prepared Testimony of Staff Efficiency and Sustainability Panel, consisting of a cover page and 60 pages of questions and answers dated August 2019.

Was that testimony prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Also before you is a document entitled Prepared Rebuttal Testimony of Staff Efficiency and Sustainability Panel consisting of a cover page and 17 pages of questions and answers dated September 2019. Was that testimony prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Finally, also before you is a document entitled Prepared Supplemental Testimony of

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Staff Efficiency and Sustainability Panel, consisting of a cover page and 6 pages of questions and answers dated January 2020. Was that testimony prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Do you have any changes or corrections to make to any of your three prepared testimonies?

A. (Panel) No.

Q. Do you affirm that the information contained in the panel's prepared testimony is true and correct to the best of your knowledge?

A. (Panel) Yes.

Q. And if you were asked the same questions today under oath, would you answer them the same way?

A. (Panel) Yes.

Q. So also before you are exhibits identified at the time of filing as S.E.S.P.-1 and S.E.S.P.-2. These have now been marked for identification purposes as Exhibits 422 and 423, respectively. Were those two exhibits prepared by you or under your direct supervision?

A. (Panel) Yes.

Q. Do you affirm that the information contained in the two exhibits just identified is true and correct to the best of your knowledge?

A. (Cully) Yes.

A. (Park) Yes.

MR. GOODRICH: Your Honors, I ask that the prepared initial -- initial rebuttal and supplemental testimonies of the Staff Efficiency and Sustainability Panel be entered into the record as if given orally.

A.L.J. COSTELLO: Okay. That is -- a motion is granted, and the testimony will be entered in the record as if given orally here today. At this point, the court reporter should insert the files that will be sent to them under these titles.

It will be D.P.S. Staff Efficiency and Sustainability Panel, direct testimony.

D.P.S. Staff Efficiency and Sustainability Panel rebuttal testimony and D.P.S. Staff Efficiency and Sustainability Panel supplemental testimony. #orally

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Efficiency and
Sustainability Panel

Robert Cully
Utility Engineering Specialist 3

Joseph Hitt
Utility Analyst 2

Laurie Kokkinides
Utility Analyst 1

Kathryn Mammen
Utility Analyst 3

Davide Maioriello
Utility Engineering Specialist 3

Jennifer Park
Assistant Engineer

Office of Markets and Innovation

Office of Electric, Gas and Water
State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Members of the Staff Efficiency and Sustainability Panel (Panel), please state your names, employer and business address.

A. Our names are Robert Cully, Joseph Hitt, Laurie Kokkinides, Kathryn Mammen, Davide Maioriello, and Jennifer Park. We are employed by the New York State Department of Public Service (Department) and our business address is Three Empire State Plaza, Albany, New York 12223.

Q. Mr. Cully, what is your position at the Department?

A. I am currently employed as a Utility Engineering Specialist 3 assigned to the Energy Efficiency and Demand Response Section of the Office of Markets and Innovation (OMI).

Q. Mr. Cully, describe your educational background.

A. I graduated from Clarkson University with a Bachelor of Science degree in Environmental Engineering in May 2010.

Q. Mr. Cully, describe your professional experience and your position at the Department.

A. I have been employed by the Department since August 2012. Since joining the Department, my responsibilities have included examining utility

rates and fees, rate design, benefit cost analyses, demand response programs, electric standby service rates, demand management programs, including Non-Wires Alternatives, and Earning Adjustment Mechanisms (EAMs).

Q. Mr. Cully, have you previously provided testimony before the Public Service Commission (Commission)?

A. Yes. I have provided testimony in numerous rate proceedings. Most recently, I provided testimony in Cases 13-E-0030, 13-G-0031, and 13-S-0032, 16-E-0060, 19-E-0065, and 19-G-0066, Consolidated Edison Company of New York, Inc. (Con Edison); Cases 14-E-0493, 18-E-0067, and 18-G-0068, Orange and Rockland Utilities, Inc. (Orange and Rockland); Case 17-E-0459 and 17-G-0460, Central Hudson Gas & Electric Corporation (Central Hudson); Cases 17-E-0238 and 17-G-0239, Niagara Mohawk Power Corporation d/b/a National Grid (Niagara Mohawk); and Cases 15-E-0283 and 15-E-0285, New York State Electric & Gas Corporation and Rochester Gas and Electric Corporation.

Q. Mr. Hitt, what is your position at the

Department?

A. I am employed as a Utility Analyst II assigned to the Clean Energy section of the Office of Markets and Innovation.

Q. Please provide a summary of your educational background and professional experience.

A. I graduated from the State University of New York Institute of Technology at Utica/Rome in 1998 with a Bachelor of Science degree in Industrial Engineering Technology, where I also received a Master of Science degree in Advanced Technology in 2000. Additionally, I received a Master of Business Administration degree from the University at Albany in 2006. Prior to and while pursuing my education, I held several technical positions of increasing responsibilities. My technical skills and education led to Operations Management positions in the transportation and energy industries. I joined the Department in July 2012. My responsibilities at the Department include reviewing and monitoring utility energy efficiency programs, the Clean Energy Fund, and a monitoring and auditing role of the energy

efficiency programs Technical Resource Manual.

Q. Have you previously provided testimony in proceedings before the Commission?

A. Yes, I provided testimony in the following rate cases: Case 16-W-0259, New York American Water Company, Inc.; and Cases 16-E-0061 and 16-G-0061, Con Edison.

Q. Ms. Kokkinides, what is your position in the Department?

A. I am employed as a Utility Analyst 1 assigned to the Energy Efficiency section of OMI.

Q. Please describe your educational background.

A. I received a Bachelor of Arts degree in Environmental Studies and in Geography from Binghamton University in 1994 and a Master of Arts degree in Geography from Binghamton University in 1996.

Q. Ms. Kokkinides describe your professional experience and responsibilities with the Department.

A. I have been employed by the Department since 2018. Prior to joining the Department, I held positions in private industry and at the New York State Energy Research and Development

Authority that involved site research and investigations, development and evaluation of energy efficiency programs and development of policy implementation metrics and impacts. Currently, my relevant responsibilities include reviewing and monitoring utility energy efficiency programs, evaluation, monitoring and verification (EM&V) policy review and Clean Energy Guidance.

Q. Have you previously testified before the Public Service Commission (Commission)?

A. No. I have not testified before the Commission.

Q. Ms. Mammen, what is your position at the Department?

A. I am a Utility Analyst 3 in the Clean Energy Section of OMI.

Q. Please describe your educational background.

A I received a Bachelor of Arts Degree in sociology from Fordham University in 2006. I also received a Master of Public Administration from Rockefeller College of Public Affairs & Policy at the University at Albany, State University of New York in 2010.

Q. Please describe your professional experience and

responsibilities at the Department.

A. I joined the Department in 2009, first as an intern for the Secretary to the Commission, and later as a Utility Analyst in the Office of Energy Efficiency and the Environment where I was responsible for overseeing the implementation and evaluation of energy efficiency programs offered under the Energy Efficiency Portfolio Standard. Currently, my responsibilities include reviewing and monitoring utility energy efficiency programs and developing policy recommendations for Commission consideration related to energy efficiency and other clean energy activities.

Q. Have you previously provided testimony before the Commission?

A Yes. Most recently, I testified in Cases 19-E-0065 and 19-G-0066, Con Edison. I have also testified in Cases 15-G-0058 and 15-G-0059, KeySpan Gas East Corporation d/b/a National Grid (KEDLI) and The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) on energy efficiency issues; and in Case 16-W-0130, Suez Water New York Inc., on the issue of the water

conservation program.

Q. Mr. Maioriello and Ms. Park, do you discuss your credentials in another testimony submitted in these proceedings?

A. Yes. We discuss our credentials in the direct testimony of the Staff Gas Supply Panel.

Q. Panel, what is the purpose of your testimony in these proceedings?

A. We will address KEDNY's and KEDLI's (jointly, the Companies) proposals and our recommendations regarding: (1) gas energy efficiency portfolios, including increased budgets and savings targets for the Company-administered gas portfolios, cost recovery mechanisms necessary to fund such portfolios, the KEDLI HEAT Program, and the Companies' proposed additional full-time employees (FTEs) to manage the Companies' expanded clean energy activities; (2) the Utility Energy Services Contract Program; (3)the Fuel Switching Calculator; (4) Demand Response; (5) the Expanded Geothermal Demonstration Projects; (6) the Clean Conversion Program; (7) the Low- and Moderate-Income Gas Conversion Programs; (8) renewable natural gas

interconnections; (9) the Power-to-Gas Demonstration Project; and (10) the Hydrogen Blending Study.

Q. In this testimony, will the Panel refer to, or otherwise rely upon, any information obtained during the discovery phase of these proceedings?

A. Yes. We will refer to, and have relied upon, several responses to Department Staff Information Requests (IRs), all of which are included in Exhibit__(SESP-1). We will refer to these IRs by the designation given when issued, e.g., DPS-123.

Q. Is the Panel sponsoring any other exhibits?

A. Yes. The Panel is sponsoring one additional exhibits in our testimony.

Q. Please describe these exhibits.

A. Exhibit__(SESP-2) shows our recommended use of unspent energy efficiency funds.

Q. What time period does the Panel’s testimony address?

A. Although the Company’s testimony includes proposals for the Rate Year, the 12-month period ending March 31, 2021; Data Year 1, the 12-month period ending March 31, 2022; Data Year 2, the

12-month period ending March 31, 2023; and Data year 3, the 12-month period ending March 31, 2024, our testimony focuses on the Rate Year.

ENERGY EFFICIENCY

Background

Q. Provide a brief background of utility-administered energy efficiency programs beginning with Energy Efficiency Portfolio Standard (EEPS).

A. On June 23, 2008, the Commission issued the Order Establishing Energy Efficiency Portfolio Standard and Approving Programs in Case 07-M-0548. Within the Commission's EEPS framework, New York State's investor-owned electric and gas utilities and the New York State Energy Research and Development Authority (NYSERDA) implemented residential, multifamily, and commercial and industrial energy efficiency programs. The Commission approved each EEPS program, and the associated annual budgets and savings targets, through a series of orders issued in Cases 07-M-0548, 09-G-0363, and other related cases. The second and final phase of the Companies' EEPS

programs ended on December 31, 2015. Funding for the EEPS programs was collected through a surcharge mechanism, the System Benefit Charge.

Q. What actions did the Commission take with respect to energy efficiency programs in preparation for the conclusion of EEPS?

A. First, the Commission authorized a new regime for electric efficiency programs in its February 26, 2015 Order Adopting Regulatory Policy Framework and Implementation Plan issued in Case 14-M-0101 (Track One Order). The Track One Order established portfolio-level budgets and savings targets for electric efficiency programs for 2016 and required all affected utilities to file a Budget and Metric Plan and a companion Energy Efficiency Transition Implementation Plan (ETIP) by July 15, 2015, reflecting the authorized 2016 budgets and targets and proposing budgets and targets for 2017 and 2018.

Q. What followed the Track One Order?

A. Following the Track One Order, on June 19, 2015, the Commission issued the Order Authorizing Utility-Administered Gas Energy Efficiency Portfolios for Implementation Beginning January

1, 2016 in Cases 07-M-0548 and 15-M-0252 (June 2015 Gas ETIP Order). The June 2015 Gas ETIP Order authorized 2016 budgets and targets for gas energy efficiency programs and required utilities to incorporate the gas programs into the July 15, 2015 Budget and Metric Plan and ETIP filings.

Q. Through what mechanism were the ETIP programs to be funded?

A. The June 2015 Gas ETIP Order required utilities to file tariff revisions establishing an Energy Efficiency Tracker (EE Tracker) surcharge mechanism to recover the costs associated with the ETIP energy efficiency programs. In its December 18, 2015 Order Approving Tariff Amendments, the Commission approved the Companies' EE Tracker surcharge mechanisms effective January 1, 2016.

Q. When were the energy efficiency portfolios and implementation plans resolved by the Commission?

A. The Commission acted on the utility Budget and Metrics filings to establish portfolio-level electric and gas budgets and savings targets in its January 22, 2016 Order Authorizing Utility-

Administered Energy Efficiency Portfolio Budgets and Targets for 2016-2018 (January 2016 ETIP Order) in Case 15-M-0252.

Q. How did the Commission's ETIP framework change the way in which the Companies administered efficiency programs?

A. In January 2016, the Companies began implementing gas energy efficiency programs under the Commission's new framework. This new framework provided KEDNY and KEDLI more flexibility in designing and implementing their respective electric and gas efficiency programs to meet portfolio-level Dekatherm (Dth) savings targets within Commission-authorized portfolio-level budgets, as opposed to specific program authorizations.

Q. Has the Commission authorized budgets and targets beyond the 2016-2018 period?

A. Yes, the Commission authorized 2019 and 2020 gas budgets and targets in its March 15, 2018 Order Authorizing Utility-Administered Energy Efficiency Portfolio Budgets and Targets for 2019-2020, issued in Case 15-M-0252 (March 2018 ETIP Order).

Q. Did the March 2018 ETIP Order include other directives associated with the implementation of ETIP programs?

A. Yes. In the March 2018 ETIP Order, the Commission: (1) adjusted each utility's targets from net to gross savings and emphasized the importance of realized savings; (2) changed the units of gas savings targets and reported savings from Dth to Million British Thermal Units (MMBtu); (3) required recovery of costs for utility-administered energy efficiency programs to be shifted into base delivery rates; (4) directed any utility EAM based on energy efficiency programmatic performance be calculated and awarded based on verified gross savings; and, (5) required each utility to evolve its ETIP into a System Energy Efficiency Plan (SEEP).

Q. Please explain the Commission's directive to shift to gross savings targets.

A. Previous Commission policy required that both energy efficiency targets, and savings reported against those targets, be expressed on a net basis, accounting for the effects of free-

ridership and spillover by applying a default net-to-gross ratio of 0.90. In recognition of specific barriers that make it difficult to meaningfully calculate accurate net-to-gross ratios, the Commission determined that, going forward, goals should be established, and savings reported, on a gross basis. The March 2018 ETIP Order defined gross savings as the change in energy consumption and/or demand that results directly from program-related actions taken by participants in a program, regardless of why they participated.

Q. Pursuant to the March 2018 ETIP Order, when were utilities required to shift cost recovery for utility-administered energy efficiency programs into base rates?

A. The Commission directed each utility to move the energy efficiency costs collected through the EE Tracker surcharge mechanism into base delivery rates during the first rate proceeding for that utility following the March 2018 ETIP Order.

Q. Did the Commission provide guidance regarding the reconciliation of energy efficiency costs once those costs are collected through base

rates?

A. Yes. The Commission required a downward-only reconciliation to be conducted at the end of a rate plan, and annually thereafter during any stay-out period, with any unspent funds to be deferred, with appropriate carrying charges, for future ratepayer benefit. Utilities, according to the Commission, are allowed to carry funds from year to year during a rate plan.

Q. Please explain the Commission’s direction regarding verified gross savings.

A. At a high-level, verified gross savings include adjustments associated with evaluation, measurement and verification (EM&V) findings that verify key factors used to calculate savings. The Commission’s requirement for verified gross savings results from its increased focus on actual realized savings, which are critical to expanded energy efficiency and the attainment of the State’s greenhouse gas emission reduction objectives. The direct testimony of the Staff EAM Panel includes additional details and recommendations regarding verified gross savings.

Q. What is the SEEP?

A. The SEEP must describe the entirety of the utility's expanded reliance on, and use of, cost-effective energy efficiency to support its distribution system and customer needs.

Q. Did any Commission actions outside of the Utility ETIP or rate case proceedings occur that affected the Company's energy efficiency portfolios?

A. Yes. The Commission issued its Order Adopting Accelerated Energy Efficiency Targets (December 2018 Efficiency Order) on December 13, 2018, in Case 18-M-0084 (Utility Energy Efficiency Proceeding). In that Order, the Commission increased the Companies' 2019 and 2020 gas targets and budgets and established a process through which the Commission will establish targets and budgets for the 2021-2025 period.

Q. Please describe the Commission's actions taken in the December 2018 Efficiency Order and how they affected KEDNY and KEDLI's energy efficiency portfolios.

A. In that Order, the Commission increased the Companies' 2019 and 2020 gas targets and budgets

and required KEDNY, KEDLI, and the other large investor-owned utilities, in consultation with NYSERDA, to file proposed energy efficiency budgets and targets for the 2021-2025 period on or before March 31, 2019.

Q. What preceded the December 2018 Efficiency Order?

A. In April 2018, Department Staff and NYSERDA issued a report entitled New Efficiency: New York (NE:NY Report). Among other things, the NE:NY Report highlighted the importance of energy efficiency in achieving the State’s goal of a 40% statewide reduction of greenhouse gas emissions from 1990 levels by 2030, and established a 2025 statewide energy efficiency target of 185 trillion Btu (TBtu) of energy usage reductions at the customer level, which required utilities to achieve an additional 31 TBtu reduction above and beyond the then-current energy efficiency program target levels. The NE:NY Report also identified a number of strategies to reach the statewide 185 TBtu savings goal by 2025.

Q. What other relevant actions did the Commission

take in the December 2018 Efficiency Order?

A. In the December 2018 Efficiency Order, the Commission determined the Statewide 185 TBtu target was reasonable and adopted the incremental 31 TBtu as a fuel neutral target for the State's utilities to achieve through their electric and gas efficiency portfolios by 2025. The Commission also established sub-targets of at least 5 TBtu from heat pump adoption. The Commission also established a cap of $1.6 billion to achieve the 31 TBtu; required 20% of incremental spending to fund low- and moderate-income customer programs; and established an expectation for cost reduction per unit of energy saved.

Q. Have the Companies made the filing as required by the December 2018 Efficiency Order?

A. Yes, the Companies' proposals were included in the NY Utilities Report Regarding Energy Efficiency Budgets and Targets, Collaboration, Heat Pump Technology and Low- and Moderate-Income Customers (April Utility Proposal). This report was filed on April 2, 2019, in the Utility Energy Efficiency Proceeding. KEDNY-

and KEDLI-specific proposals were set forth in Chapter XII, the National Grid Chapter. Subsequent to the April Utility Proposal filing, the Utilities have jointly filed revisions to that proposal with the most recent dated May 10, 2019 (May Utility Proposal), which includes the Companies' proposals in a revised Chapter XII and a new appendix, Appendix A (Companies' May Proposal). Our testimony is based upon the contents of the May Utility Proposal, the Companies' May Proposal, and the Companies' Corrections and Updates filing in these rate proceedings filed on July 3, 2019.

Q. Does the Companies' May Proposal align with the Companies' initial testimony provided by the Future of Heat Panel?

A. No. Although the targets provided in both sources matched, there is a discrepancy between the calendar year budgets for 2019 and 2020. However, in the Corrections and Updates Testimony of the Future of Heat Panel, the Companies' corrected the budget amounts the Companies propose to move into base rates. The Companies stated that the corrections to the

budgets were the result of a change in the revenue requirement, however we view these corrections to the budgets as a means of aligning with the December 2018 Efficiency Order and the Companies' May Proposal, not as a function of any changes to the revenue requirement.

Budgets and Targets

Q. What budgets do the Companies propose in their Corrections and Updates filings?

A. On page 7 of the Corrections and Updates testimony of the Future of Heat Panel, the Companies state that they would use the authorized budget for calendar year 2020 and the 2021-2025 presumptive targets and budgets set forth in the December EE Order as the proposed budgets and targets for the Rate Year and Data Years until the Commission directs otherwise. The Future of Heat Panel proposed a Rate Year budget of $18.636 million for KEDNY and $9.478 million for KEDLI.

Q. What are the Companies' targets for calendar years 2019, 2020, 2021, 2022 and 2023?

A. The budgets provided by the Future of Heat Panel on page 22 of its direct testimony, were for the Rate Year, however, the targets provided reflect the following calendar years: 2019-2020, which the Companies combined; 2021; 2022; and 2023. For KEDNY the calendar year 2019-2020 gas savings target is 811,814 MMBtu and for KEDLI it is 452,387 MMBtu. The 2021; 2022; and 2023 MMBtu targets for KEDNY are 510,740; 574,740; and 704,740, respectively. For KEDLI the MMBtu targets for 2021; 2022; and 2023 are 268,821; 301,821; and 343,821, respectively.

Q. Do you agree with the budgets and targets proposed by KEDLI and KEDNY in the Companies' May Proposal and the Future of Heat Panel's Corrections and Updates testimony?

A. We recommend that the Companies' proposed targets and budgets be used as placeholders until the Commission acts on the May Utility Proposal in the Utility Energy Efficiency Proceeding. Following Commission action on that proposal, the final targets and budgets authorized for KEDLI and KEDNY in that proceeding should be adopted in these rate

proceedings. If Commission action in the Utility Energy Efficiency Proceeding does not occur before the Commission sets rates in these proceedings, we recommend establishing a deferral mechanism. This mechanism should reconcile the difference between the revenue requirement impacts approved in these rate proceedings with those ultimately approved in the Utility Energy Efficiency Proceeding. Discussion of the energy efficiency targets as they relate to the Companies' proposed Earnings Adjustment Mechanisms can be found in the Staff EAM Panel's testimony.

Cost Recovery Mechanism

Q. How are the Companies' energy efficiency programs currently funded?

A. The Companies currently fund energy efficiency programs through the EE Tracker, a component of the System Benefit Charge. The EE Tracker is designed to collect the authorized funding for the Companies' current ETIP programs.

Q. How do the Companies propose to treat energy efficiency costs in these rate proceedings?

A. On page 24 of the Future of Heat Panel's direct testimony, the Companies propose to collect energy efficiency funds through base rates.

Q. Does the Panel agree with the Companies' proposal to shift energy efficiency costs out of the EE Tracker and into base rates?

A. Yes. The Companies' proposal to move energy efficiency costs into base rates is consistent with previous Commission direction provided in the March 2018 ETIP Order, which required each utility, during its next rate proceeding, to recover the costs for the portfolio of utility-administered energy efficiency programs through base delivery rates. In addition, shifting the energy efficiency funding out of the EE Tracker and into base rates is consistent with the approach adopted by the Commission in the most recent rate proceedings for Niagara Mohawk, in Cases 17-E-0238 and 17-G-0239; Central Hudson, in Cases 17-E-0459 and 17-G-0460; and Orange and Rockland, in Cases 18-E-0067 and 18-G-0068.

Q. Do the Companies propose any additional changes to the treatment of energy efficiency costs?

A. Yes. On page 24 of the Future of Heat Panel's

direct testimony, the Companies propose several additional changes related to the treatment of evaluation, measurement and verification (EM&V) funds associated with energy efficiency programs.

Q. Explain the Companies proposed changes to the collection of EM&V funds?

A. The Companies propose to: (1) move EM&V costs out of the energy efficiency portfolio budget, (2) maintain the portfolio-level energy efficiency budget, and (3) collect the EM&V funds separately through base rates. The Rate Year EM&V costs that would be shifted out of the energy efficiency budget for KEDNY and KEDLI are $924,353 and $473,898, respectively.

Q. What justifications do the Companies provide for the EM&V funding changes?

A. As explained on pages 24-25 of the Future of Heat Panel’s direct testimony, the Companies’ justification for these changes is that it will enable them to fund additional customer energy efficiency programs. The Companies also state that these costs are either not directly related to delivery of the energy efficiency portfolio

contained in the ETIPs or are scoped to integrate initiatives beyond just energy efficiency.

Q. Does the Panel agree with the Companies' proposals related to the EM&V funding?

A. No. The Companies did not provide enough detail in the need or use of the additional budget. The Companies state that the additional monies will enable them to add additional energy efficiency programs and initiatives that would not increase the ETIP budgets. In the Companies' response to IRs DPS-383 and DPS-384, the Companies state that the latest ETIP, filed February 19, 2019, represents their current implementation plan for their respective portfolios of energy efficiency projects, including those that would be funded with the replaced EM&V costs. The Companies' ETIPs do not identify any additional customer programs. In addition, the Commission is currently considering the Companies' energy efficiency budgets inclusive of EM&V in the Utility Energy Efficiency Proceeding. Therefore, we recommend maintaining EM&V within the Companies' energy

efficiency budgets.

Q. Does the Panel recommend a reconciliation of energy efficiency costs?

A. Yes. We recommend a reconciliation that aligns with the Commission's direction in the March 2018 ETIP Order. Specifically, we recommend a downward-only reconciliation of the energy efficiency expenses at the end of the Rate Year. In the event of a multi-year rate plan, we recommend that the Companies be allowed to carry unspent funds from year to year during the term of the rate plan. At the end of the rate plan, and annually thereafter, any unspent funds should be deferred, with appropriate carrying charges, for future customer benefit.

Q. Do the Companies propose to collect the entirety of the energy efficiency budgets through base rates?

A. No. In the Companies' response to IR DPS-268, the Companies propose to use unspent EEPS funds to offset a portion of the energy efficiency budgets and then collect the remainder of the budgets through base rates.

Q. Please explain what is meant by "unspent EEPS

funds."

A. Unspent EEPS funds are those funds that the Companies collected from customers to pay for energy efficiency programs offered during the previous EEPS period but that the Companies have neither spent nor committed to spend.

Q. How do the Companies propose to use unspent EEPS funds to offset the energy efficiency portfolio budget?

A. In the Companies' response to IR DPS-268, KEDLI proposes to allocate approximately $4.869 million of unspent EEPS funds to offset the proposed gas portfolio budgets in calendar years 2019 and 2020. Also in response to IR DPS-268, KEDNY proposes to allocate approximately $20.405 million of unspent EEPS funds to offset the proposed gas portfolio budgets in calendar years 2019 and 2020.

Q. Does the Panel recommend using any other source of funds to offset the gas energy efficiency budgets?

A. Yes. In addition to using unspent EEPS funds, we recommend using any interest that has accrued on the EEPS funds, unspent ETIP funds, and

interest accrued on the NYSERDA Clean Energy Fund (CEF) Bill-As-You-Go (BAYG) collections to offset the ETIP energy efficiency budgets.

Q. Please explain what is meant by “unspent ETIP funds.”

A. Unspent ETIP funds are those funds that the Companies collected from customers to pay for energy efficiency programs offered during the previous ETIP period but that the Companies have not spent as of the beginning of the Rate Year.

Q. How does interest accrue on NYSERDA CEF collections?

A. The NYSERDA CEF collections are collected by the Companies through the SBC surcharge. NYSERDA relies on a quarterly BAYG approach to obtain the CEF funding from the utilities. Under this methodology, KEDLI and KEDNY collect their authorized share of NYSERDA’s CEF budget on an ongoing basis, but only transfer their proportional share of NYSERDA’s actual expenditures to NYSERDA at the end of each quarter. This approach results in the Companies accruing a balance of CEF funds. Further, in accordance with the Commission’s Order

Authorizing the Clean Energy Fund Framework (CEF Order), issued on January 21, 2016 in Case 14-M-0094, the Companies are required to accrue carrying charges on this balance using the Other Customer Provided Capital Rate.

Q. What does the Panel recommend regarding the use of unspent EEPS and ETIP funds and EEPS and BAYG interest to offset the energy efficiency collections?

A. We recommend that the Companies use the combined total of the EEPS and ETIP unspent funds along with the EEPS and NYSERDA BAYG interest to offset the energy efficiency budgets over a five-year period. Exhibit__(SESP-2) provides a summary of the unspent EEPS and ETIP funds and EEPS and BAYG accrued interest and our recommended use of those funds.

Q. Does the change in the cost recovery method affect the manner in which the Companies implement the programs, the reporting requirements, or the implementation plan filing requirements currently associated with ETIP programs?

A. No. We recommend that the Companies continue to

implement the programs in accordance with the rules and requirements set forth in Commission orders and Staff guidance. Pursuant to the March 2018 ETIP Order the Companies are required to begin integrating energy efficiency planning into its forecasted system plans and evolve its ETIP into a SEEP, which will describe the entirety of the utilities expanded reliance on, and use of, cost effective energy efficiency to support its distribution system and customer needs. The method of cost recovery, whether through surcharges or base rates, should have no impact on the continued transparency of the utility-administered energy efficiency programs.

KEDLI HEAT Program

Q. Please provide the background of the KEDLI HEAT Program.

A. The Commission’s January 21, 2016 Order in the Clean Energy Fund proceeding, Case 14-M-0094, required that NYSERDA operate its programs in a fuel-neutral manner and, to facilitate this fuel-neutral approach, established a policy that eliminates collections from non-electric

customers. As KEDLI customers receive electric service from the Long Island Power Authority, which does not contribute to NYSERDA programs, KEDLI customers are not eligible for NYSERDA programs, including NYSERDA's EmPower Program for low-income customers. In order to avoid a gap in energy efficiency services to low-income customers in KEDLI's territory, the Commission required NYSERDA and KEDLI to pursue alternatives for 2017 and thereafter. KEDLI developed the HEAT program as its EmPower replacement program and has been implementing it since 2017.

Q. Please describe KEDLI's HEAT program.

A. The HEAT program is an energy efficiency program designed to serve low-income customers. The program consists of four tiers: Tier 1 includes customer outreach, education, and engagement to educate customers on the benefits of energy efficiency and encourage participation in Tier 2 measure offerings; Tier 2 consists of a home energy audit and direct installation of certain energy efficiency measures; Tier 3 entails the installation of additional energy efficiency

measures identified in the Tier 2 home energy audit; and Tier 4 includes the installation of custom energy efficiency measures that may require extensive work on the home, including the preparation of space needed to accommodate energy efficiency equipment.

Q. Does KEDLI propose any changes to the HEAT program?

A. Yes. On page 29 of the direct testimony of the Future of Heat Panel, KEDLI proposes to increase the annual budget from $1.9 million to $2.5 million. KEDLI also proposes to expand the program to include eligible moderate income customers whose residences do not have central air conditioning.

Q. Does the Panel support KEDLI’s proposed expansions of the HEAT Program to moderate-income customers?

A. Yes, we support increasing the budget to $2.5 million and expanding the program to eligible moderate income customers.

Q. Is the HEAT program a part of KEDLI’s energy efficiency portfolio?

A. No. Currently, KEDLI implements this program as

a standalone program that operates outside of the energy efficiency portfolio.

Q. Does the Panel recommend any changes to the way in which the HEAT program is implemented?

A. Yes. We recommend that KEDLI incorporate the HEAT program into its energy efficiency portfolio. At its core, the HEAT program is an energy efficiency program and KEDLI should operate this program in accordance with the rules and requirements associated with the current energy efficiency portfolio, including any statewide low-to-moderate income (LMI) customer framework developed within the Utility Energy Efficiency Proceeding. We further recommend the $2.5 million budget for the HEAT program be additional to the energy efficiency budget authorized by the Commission in Case 18-M-0084.

Q. Does KEDLI propose an incentive mechanism associated with the HEAT program?

A. Yes, KEDLI proposes the EmPower Replacement Program incentive mechanism, which would award KEDLI for customer participation in Tiers 3 and 4 of the HEAT program. The Staff EAM Panel

provides an overview and Staff's recommendations related to the KEDLI's proposed EmPower Replacement Incentive Mechanism.

Energy Efficiency Labor

Q. Are the Companies proposing any incremental full time equivalent employees (FTEs) associated with the energy efficiency programs?

A. Yes. On page 28 of the direct testimony of the Future of Heat Panel, the Companies propose an additional seven FTEs: four FTEs focused on product development and delivery to customers, one FTE to support sales activity for commercial and industrial products and two FTEs for technical sales support.

Q. Does the Panel support KEDNY and KEDLI's request for additional FTEs?

A. Yes. The Companies are significantly ramping up energy efficiency-related initiatives. Specifically, the KEDNY energy efficiency target increases by 32% in 2019 and 55% in 2020 and for KEDLI the energy efficiency target increases by 26% in 2019 and 45% in 2020. Therefore, we agree that it is reasonable for the Companies to

add seven additional FTEs to help support the enhanced energy efficiency portfolios.

UTILITY ENERGY SERVICE CONTRACT PROGRAM

Q. Please describe the Utility Energy Service Contract (UESC) Program.

A. As described on pages 34 and 35 of the direct testimony of the Future of Heat Panel, the Companies have operated the UESC Program since 2011 and it is an energy services contracting program whereby the Companies and customers enter limited-source contracts for energy management services. The customer pays for the improvements delivered under the UESC Program. The Companies have been offering the UESC Program through the Federal Energy Management Program that targets government facilities. The UESC Program has since evolved to include large commercial customers as well.

Q. Are the Companies proposing changes to the UESC Program?

A. Yes. As described on pages 35 and 36 of the direct testimony of the Future of Heat Panel, the Companies propose changes including: (1)

incentivizing up to 50 percent of the total cost of a feasibility study that assesses energy savings opportunities, calculates the costs of those savings recommendations, and provides implementation assistance at a customer's facility; (2) adopting a more proactive approach to attract participants; (3) increased involvement in the development of project scopes for participating customers; and (4) implementing a Platform Service Revenue fee structure.

Q. Does the Panel support the proposed changes?

A. Overall, we support the changes. However, this Panel will not address the most significant change of the program, the Platform Service Revenue fee structure. The Platform Service Revenue fee structure is discussed in the testimony of the Staff EAM Panel.

Q. Does the Panel have any additional recommendations regarding the UESC program?

A. Yes, we recommend that the Companies be required to develop specific energy efficiency targets and budgets for the UESC Program as part of the ETIP/SEEP and include the program as a part of

the Companies' overall energy efficiency portfolio.

Q. Please explain why the Panel recommends developing energy efficiency targets and budgets for the UESC Program?

A. The UESC program incorporates energy efficiency measures and results in energy savings and should therefore be transitioned into the Companies' energy efficiency portfolios to ensure consistent and comprehensive treatment of energy efficiency programs and their resultant savings. We recommend that the UESC program adhere to all implementation and reporting requirements associated with the Companies' ETIP/SEEP Programs.

Q. How does the Panel recommend the Companies fund the program?

A. We recommend that the Companies fund the UESC Program with the authorized energy efficiency budgets. Historically, the Companies have not spent the annual energy efficiency budgets and can accommodate the UESC program budget without additional funding beyond that of the energy efficiency portfolio budgets.

Q. Are the Companies proposing additional FTEs for the UESC program?

A. Yes. The Companies proposed two additional FTEs: a marketing manager and a project engineer.

Q. Does the Panel agree with the additional FTEs?

A. No. We do not support additional FTEs for the UESC program as the Companies have not sufficiently justified the need for the additional FTEs they are requesting. As we recommend that the Companies shift the UESC program into the energy efficiency portfolio, we recommend that the Companies rely on the existing and incremental FTEs associated with the energy efficiency portfolio to implement the UESC program.

FUEL SWITCHING CALCULATOR

Q. Please describe the fuel-switching calculator as proposed by KEDNY and KEDLI.

A. On page 38 of the direct testimony of the Future of Heat Panel, the Companies describe a proposed fuel-switching calculator as a web-based tool that would inform customers about the energy

savings and financial implications associated with switching from the customer's existing fuel source to an alternative fuel source, including natural gas, geothermal heat pumps, and renewable natural gas. The calculator would provide customers with a side-by-side comparison, including the estimated annual cost, payback period, available incentives, tax credits, conversion costs, and annual fuel savings.

Q. Does the Panel support the fuel switching calculator?

A. Yes. The fuel-switching calculator enables customers to evaluate different heating options to make more informed energy decisions when choosing or replacing the energy equipment in their homes.

Q. Does the Panel have any recommended changes associated with the fuel-switching calculator?

A. Yes, we have two recommendations. First, we recommend that the Companies' adopt standardized assumptions and calculations where appropriate. Second, we recommend that the calculator include other end-uses of fuel in addition to primary

heat such as domestic hot water, cooking, or air conditioning in order to give customers a complete picture of additional energy efficiency upgrades that can be made.

Q. Does the Panel have any other concerns with the fuel-switching calculator?

A. Yes. The Fuel-Switching Calculator is a tool that could be used more broadly by National Grid's operating utilities to inform their customers and, therefore, the $250,000 cost of the Fuel-Switching Calculator should be shared among other National Grid entities. This would reduce costs to KEDNY and KEDLI. To appropriately share the cost with other National Grid entities, we recommend using the general allocator to assign 8.29 percent of the cost to KEDLI and 13.31 percent of the cost to KEDNY.

DEMAND RESPONSE

Q. Please describe the Companies' Demand Response (DR) programs.

A. On page 42 of the direct testimony of the Future of Heat Panel, the Companies state that they have two DR programs. The first is the

Interruptible and Temperature Control (IT/TC) Service that offers a reduced delivery rate for customers that interrupt their natural gas service when demand peaks. The IT/TC service has two pricing tiers that customers are placed into based on the customer's fuel switching capabilities. The Companies do not propose changes to IT/TC service in this proceeding.

Q. What is the second DR program?

A. The second DR program is currently a Demonstration Project providing market-based payment to commercial firm customers who commit to minimum gas usage reductions during peak events, based on their performance during specified DR events. This project was approved as part of the 2016 Rate Order, issued on December 16, 2016, in which the Commission adopted a Joint Proposal and established rate plans for the Companies in Cases 16-G-0058 and 16-G-0059. Our testimony in these proceedings will focus on the Companies' proposals related to this program, and we will refer to this program as the Performance-based Gas DR Program.

Q. How do customers currently participate in the

demonstration-scale program?

A. Currently, customers enroll load relief associated with direct utility control of specific equipment within customers' premises, with an associated bid price. The Companies use a competitive selection process based on the load relief and price proposed in the bids received from customers. The Companies develop a load relief portfolio seeking to maximize load relief and minimize costs. Thereafter, during a load relief event, KEDNY and KEDLI directly shut off enrolled customer equipment for the duration of the Event. A more complete description of the current Gas Demand Response Demonstration Project is available as part of the Companies' implementation plan filings made on July 27, 2017 in Cases 16-G-0058 and 16-G-0059.

Q. Do KEDNY and KEDLI propose changes to the Performance-based Gas DR Program?

A. Yes. On pages 44 and 45 of the direct testimony of the Future of Heat Panel, the Companies propose scaling the project up from demonstration size and including the costs of the Performance-based Gas DR Program in base

rates. Additionally, the Companies propose making three operational changes which will impact participant responsiveness to peak events.

Q. What is the first operational change that the Companies propose?

A. The Companies propose certain new requirements for customer sizes and proposed load relief amounts. Specifically, the Companies to require that firm customers meet a minimum annual demand value, that participants provide a minimum reduction value to be eligible to participate in the program, and that the amount of load relief pledged by participants must be at least 10 percent of either the customer’s gas usage during a peak period or the sum of such customer’s nameplate gas usage requirements. Specifically, KEDNY proposes that firm customers must consume at least 6,000 Dth per year, and must pledge a minimum load relief amount of 2 Dth/hour during events. Similarly, KEDLI proposes that firm customers must consume at least 4,000 Dth per year, and must pledge a minimum load relief amount of 2 Dth/hour during

events.

Q. What is the second operational change that the Companies propose?

A. The Companies propose to require that, in order to receive an incentive payment, participants must reduce their gas demand for the entire duration of the 3-hour peak event for each event called.

Q. What is the third operational change that the Companies propose?

A. The Companies propose to impose a penalty when participants do not provide their enrolled load relief, and may seek or revoke customers' eligibility to participate in the program if the participating customers fail to respond to peak events.

Q. Does the Panel support these changes?

A. Yes. These changes would more closely align the Companies' Performance-based Gas DR Program with other Electric DR programs focused on reducing Commercial and Industrial customers' peak demand, and would closely align with a similar performance-based Gas Demand Response pilot currently being operated by Con Edison.

Aligning KEDNY and KEDLI's program rules with other programs targeting the same customer segment should help the Companies attract customers that may already be familiar with the Electric demand response programs, and allow participants with greater flexibility regarding the actions they take to reduce gas demand when called upon since customers will be free to choose how to provide load relief instead of enrolling specific pieces of equipment for utility control. Overall, we generally agree that these changes will likely result in a more successful DR program.

Q. Does the Commission typically require a significant degree of identical requirements for DR programs across utility service territories?

Q. Yes, however, although there are a number of minor differences between the Performance-based Gas DR Program proposed by the Companies and Con Edison's Gas DR Pilot, these differences are reasonable at this time.

Q. Why?

A. Unlike the Electric DR programs, which have been operational across the major electric investor-

owned utilities since the summer of 2015, and since 2009 at Con Edison, gas DR programs designed for firm service customers are in their infancy. The program proposed by KEDLI and KEDNY will be only the second such performance-based gas demand response program in New York State. Therefore, although the Commission has required a significant level of consistency among the Electric DR programs, it makes sense to allow KEDNY and KEDLI to test the effectiveness of several program features against Con Edison's similar program.

Q. What are the capital expenditure (CapEx) costs related to the Performance-based Gas DR Program?

A. As stated by the Companies on pages 47 and 48 of the Future of Heat testimony, for KEDNY, the CapEx costs of the Performance-based Gas Demand Response Program are $236,000 during the 12-month period ending March 31, 2020, with additional CapEx costs of $59,000 per year during the Rate Year, and Data Years 1 and 2. For KEDLI, the CapEx costs are $107,000 during the 12-month period ending March 31, 2020, with additional CapEx costs of $27,000 per year

during the Rate Year, and Data Years 1 and 2.

Q. What are the Operations and Maintenance (O&M) costs related to the Performance-based Gas Demand Response Program?

A. O&M costs related to the program include both labor and non-labor O&M. For non-labor O&M costs at KEDNY, the Companies identify $1.216 million during the Rate Year, and $1,531 million, $1,743 million, and $2.104 million during Data Years 1 through 3, respectively. For non-labor O&M costs at KEDLI, the Companies identify $0.577 million during the Rate Year, and $0.720 million, $0.814 million, and $1.004 million during Data Years 1 through 3, respectively. These forecast non-labor O&M costs would be recovered through base rates. For labor costs, the Companies propose to hire one incremental FTE during both the Rate Year and Data Year 1, with an additional two FTEs each year for Data Year 2 and Data Year 3. The Companies propose to split these FTEs 70/30 between KEDNY and KEDLI.

Q. Does the Panel agree with the Companies' cost forecasts and cost recovery mechanisms?

A. We do not support the Companies' proposal to hire additional FTEs for this program, and we recommend an alternate cost recovery mechanism for the non-labor O&M costs.

Q. Why does the Panel not agree with the Companies' request for additional FTEs?

A. The Companies already have personnel assigned to the Demand Response Demonstration Project, which should be sufficient to run the Performance-based Gas Demand Response program given the simultaneous and significant investments the Companies propose to make for management software and automated systems to help manage this program. Therefore, we recommend eliminating the Companies' requested additional FTEs for this program.

Q. Why does the Panel disagree with the Companies' proposal to include non-labor O&M costs of the Program-based Gas Demand Response program in base rates?

A. Non-labor O&M costs related to most demand response programs are related to incentive costs paid to participants. As these incentive payment costs can vary significantly since they

are based on the number of participants enrolled in the program, the amount of load relief enrolled in the program, the number of events called in a given year, and, of course, participant performance during those events. These factors can lead to significant swings in incentive payment costs from year to year, and can be difficult to forecast. Therefore, we recommend that, instead of recovering these costs in base rates, the Companies recover the non-labor O&M costs as-incurred through the Delivery Service Adjustment surcharge.

Q. Do other utilities recover the costs of their demand response programs in a similar way?

A. Yes. The other investor-owned utilities recover the costs of their respective demand response programs through a non-bypassable delivery surcharge mechanism. Applying this same cost recovery mechanism to KEDNY and KEDLI would bring their programs in line with other electric and gas demand response programs statewide.

SUSTAINABLE HEAT INITIATIVES

Expanded Geothermal Demonstration Pilot Program

Q. Please summarize KEDLI's Geothermal Demonstration Project, which was implemented through the 2016 Rate Order.

A. The KEDLI Geothermal Demonstration Project, described on pages 50-52 of the direct testimony of the Future of Heat Panel, was established to test a ground-source heat pump solution using a shared-loop system and its ability to provide heating and cooling to homes in Glenwood Village, an area not currently served by KEDLI's gas network. This test-and-learn demonstration project provides ten homes with an alternate heating and cooling and energy cost savings through a geothermal loop system that is owned by the Company. The geothermal system is operational in the Glenwood Village community and KEDLI is moving forward with the demonstration project at a site in Medford, where construction will begin in the third quarter of 2019.

Q. Please describe the performance of the Geothermal Demonstration Project to date.

A. KEDLI'S quarterly report for the Geothermal Demonstration Project, covering the period from April 1, 2019 through June 30, 2019, filed July 31, 2019 in Case 16-G-0058, states that the Glenwood Village geothermal system continues to be monitored to ensure that it is functioning properly; and that performance data continues to be collected and analyzed. The Medford geothermal system is at the beginning stages of installation.

Q. Does KEDLI propose changes to the Geothermal Demonstration Project?

A. Yes. On page 52 of the direct testimony of the Future of Heat Panel, the Companies propose to expand the project to other parts of KEDLI's service territory and to KEDNY's as well, prioritizing LMI and commercial and industrial customers. The Companies propose to charge a fixed monthly fee to participating customers for the design and the installation of the geothermal system that will be owned by KEDLI or KEDNY. The Companies state that utility-ownership of the ground loop will increase the adoption rate of this type of technology.

Q. Does the Panel support the expanded Geothermal Demonstration Project?

A. No, not at this time. The existing Geothermal Demonstration Project is just entering the data analysis phase and has not yet been proven to be cost-effective. In addition, KEDNY and KEDLI have not demonstrated that utility ownership of a geothermal system would not adversely affect third-party market participants that may be able to provide additional affordable alternatives. Finally, the Companies have not performed a solicitation, such as for a non-pipeline alternative, seeking third-party developers to demonstrate that there is a need for utility ownership.

Q. Do the Companies request any incremental FTEs associated with this program?

A. Yes, the Companies have requested one FTE to support the expanded geothermal program.

Q. Does the Panel support the proposed incremental FTE?

A. No. We do not support the Companies' proposal for an additional FTE as we do not support expanding the Geothermal Demonstration Project.

Clean Conversion Program

Q. What is the Clean Conversion Program?

A. KEDLI proposes to replace the existing Neighborhood Expansion Program with the Clean Conversion Program. Under the Clean Conversion Program, KEDLI proposes to work with customers that request a new gas connection to educate them about their environmentally friendly energy options. If the customer does not choose an environmentally friendly non-gas option, KEDLI will offer an opportunity to switch from oil to natural gas. Under the Clean Conversion Program, KEDLI will extend gas service to customers under the same density and minimum customer connections requirements established in the Order Directing Investments and Allowing, In Part, Deferral Authority for Costs Associated with Incremental Capital Expenditures and Establishing a Surcharge, issued on December 15, 2014 in Case 14-G-0214.

Q. What is the proposed cost of the Clean Conversion Program?

A. KEDLI proposes $26 million in CapEx in the Rate Year, $23 million in Data Year 1, $19 million in

Data Year 2, and $16 million in Data Year 3, as shown in Exhibit___(GIOP-1), Schedule 1.

Q What are the differences between the Neighborhood Expansion Program and the Clean Conversion Program?

A. As detailed in the response to IR DPS-596, the Clean Conversion Program is primarily a reactive program that builds off of the Neighborhood Expansion Program's customer interest, density, and minimum threshold customer participation requirements that is applied when a customer requires natural gas facilities, whereas the Neighborhood Expansion Program is meant to take a proactive approach into identifying areas of potential large project expansion.

Q. Does the Panel have any issues with the KEDLI's proposal?

A. Yes.

Q. Please explain.

A. As explained by the Staff Policy Panel, Staff's position in these rate proceedings reflects an assumption that the Northeast Energy Supply Enhancement (NESE) project will not be built during the Rate Year. Without the NESE Project,

KEDLI’s ability to connect new customers is limited. Consequently, a program based upon potential new natural gas customers will not be of significant use. We recommend not approving the Clean Conversion Program or its budget.

LMI Conversion Program

Q. Please describe the Companies LMI Conversion Program

A. On page 59 of the direct testimony of the Future of Heat Panel, the Companies propose to expand the low-income conversion program to moderate income customers. The program pairs energy conversions with incentives for high efficiency heating and hot water equipment to reduce a customer’s carbon footprint and energy bills.

Q. Would the Companies pursue the LMI Conversion Program if the NESE project does not get built?

A. In its supplemental filing on June 11, 2019 in which the Companies discussed the implications of NESE not being built on the proposals in these rate proceedings, the Companies indicated that they would not implement the LMI Conversion Program.

Q. What is the Panel's position regarding the LMI Conversion Program?

A. Given that, for the purposes of these rate proceedings, Staff is assuming that NESE will not be built during the Rate Year, we accept the Companies' position to not implement the LMI Conversion Program.

REDUCING CARBON EMISSIONS ON THE GAS NETWORK

Renewable Natural Gas Interconnections

Q. Please describe what KEDNY and KEDLI propose regarding renewable natural gas (RNG) interconnections.

A. As described on page 89 of the Testimony of the Future of Heat Panel and set forth in KEDNY and KEDLI's GIOP testimonies, the Companies propose to facilitate RNG interconnections by installing and owning the interconnection facilities of successfully implemented RNG developer projects and also propose to create a new Future of Heat Engineering department that will include additional FTEs to review and process future interconnection applications.

Q. What does the Panel recommend regarding the RNG

interconnection proposal?

A. While we support this proposed program, KEDNY's Rate Year RNG projects are still in the infancy stages of development. Additionally, there is the potential reallocation of FTE's due to our recommended capital program changes associated with the Clean Conversion Program. Accordingly, we recommend denying the additional requested FTEs for both Companies and remove the associated O&M related to the KEDNY projects for the Rate Year. The associated KEDNY Rate Year CapEx should also not be allowed and will be addressed by the Staff Gas Infrastructure and Operations Panel.

Power-to-Gas Demonstration Project

Q. Please describe what KEDLI proposes for the power-to-gas demonstration project.

A. KEDLI, as described on page 83 of the Testimony of the Future of Heat Panel, proposes to develop a project that converts excess renewable energy to hydrogen and then converts the hydrogen to pipeline-quality natural gas to be ultimately injected into the existing gas network and

delivered to customers. The Company proposes to begin scoping, design, and engineering of the project in 2019 with an anticipated full project design completion in October of 2023. The projected cost of the project scoping and design phase includes O&M costs of $0.65 million in the Rate Year, $1.825 million in Data Year 1, $0.725 million in Data Year 2, and $0.325 million in Data Year 3. KEDLI would propose project construction costs and costs in excess of those above in its next rate case.

Q. Does the Panel support this project?

A. No. The current energy storage targets set forth by the Commission in its Order Establishing Energy Storage Goal and Deployment Policy (Storage Order), issued on December 12, 2018 in Case 18-E-0130, provide for the envisioned solutions and goals for excess renewable electricity. We recommend that the Commission deny the project and the associated funding.

Hydrogen Blending Study

Q. Please describe what the Companies propose for the hydrogen blending study.

A. As described starting on page 87 of the Testimony of the Future of Heat Panel, the Companies propose a hydrogen blending study to analyze the impact of blending hydrogen into the natural gas system, to determine the percentage of hydrogen that can be safely blended into the Companies' gas network without negatively impacting the network or customer appliances. This would be a precursor to the development of a hydrogen blending demonstration project. The projected cost of the study would be $0.446 million in the Rate Year split evenly between KEDNY and KEDLI, assuming no additional grant funds are received.

Q. Does Staff support this project?

A. No. As with the Power-to-Gas Project, the current energy storage targets set forth in the Storage Order provide for the envisioned solutions and goals for excess renewable electricity. Therefore, there is no need at this time to study utilizing that excess

renewable electricity to inject hydrogen into the Companies' gas networks. Accordingly, we recommend that the Commission deny this project and the associated funding.

Q. Does this conclude your testimony at this time?

A. Yes.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan East Gas Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

September 2019

Prepared Rebuttal Testimony of:

Staff Efficiency and
Sustainability Panel

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Utility Analyst 2

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Q. Members of the Staff Efficiency and Sustainability Panel (Panel), please state your names.

A. Our names are Robert Cully, Joseph Hitt, Laurie Kokkinides, Kathryn Mammen, Davide Maioriello, and Jennifer Park.

Q. Has the Panel previously sponsored testimony in these proceedings?

A. Yes. We offered pre-filed direct testimony, submitted on August 30, 2019. Our pre-filed direct testimony addressed the Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan East Gas Corporation d/b/a National Grid's (KEDLI), collectively the Companies, proposals related to Energy Efficiency programs; the Utility Energy Services Contract Program; the Fuel Switching Calculator; Demand Response; the Expanded Geothermal Demonstration Projects; the Clean Conversion Program; the Low-and Moderate-Income Conversion Program; renewable natural gas interconnections; the Power-to-Gas Demonstration Project; and the Hydrogen Blending Study.

Q. What is the purpose of the Panel's rebuttal

testimony?

A. The purpose of our rebuttal testimony is to address: (1) the direct testimony of the Environmental Defense Fund (EDF) witness Ferit Ucar with regard to the requirements of the recently passed Climate Leadership and Community Protection Act (CLCPA); (2) the direct testimonies of Estates NY Real Estate Services LLC (Estates) witness Jared Rodriguez, and intervenor Bob Wyman with regard to their proposals to transition the Companies from Natural Gas businesses to “thermal utilities” which would sell heat and cooling services supplied by utility-owned geothermal loop fields; and (3) the City of New York (City) Policy Panel’s (City Policy Panel) proposal to undertake a district heating study and pilot.

Q. Is the Panel sponsoring Exhibits as part of this rebuttal testimony?

A. No.

Climate Leadership and Community Protection Act

Q. Are you familiar with the CLCPA?

A. Yes. It is a New York State law enacted this

year.

Q. How is the CLCPA relevant to these proceedings?

A. Witness Ucar refers to the CLCPA as the basis for certain of his proposals. While the CLCPA's provisions speak for themselves, we address witness Ucar's proposals and the stated rationales for those proposals.

Q. Provide an overview of witness Ucar's proposals with respect to the CLCPA.

A. On pages five and six of his direct testimony, witness Ucar outlines various provisions of the CLCPA with respect to the Act's modifications to New York State's Environmental Conservation Law. Specifically, witness Ucar states that the CLCPA requires New York State to implement measures to reduce statewide greenhouse gas (GHG) emissions levels by 85 percent of the 1990 levels by the year 2050, with up to an additional 15 percent reduction sourced from carbon offsets. On pages six and seven of his direct testimony, witness Ucar also summarizes the Act's requirements for the Public Service Commission, which he describes as including: (1) ensuring that 70 percent of statewide electric generation be

"generated by renewable energy systems" by the year 2030, with 100 percent of statewide electrical demand met by carbon-free generation resources by the year 2040; (2) establishing programs to procure nine gigawatts (GW) of electricity generation from offshore wind resources by the year 2035, six GW of solar photovoltaic generation by the year 2035, and three GW of energy storage capacity by the year 2030; (3) requiring that a minimum amount of the programs implemented by the Commission benefit low and moderate income customers and disadvantaged communities. In addition, on page seven of his direct testimony, witness Ucar summarizes the CLCPA's requirement that all New York State agencies, including the Public Service Commission, consider whether administrative approvals and decisions are inconsistent with attaining statewide emissions limits established in a new addition to the New York State Environmental Conservation Law.

Q. Does witness Ucar provide an opinion as to what he believes the Commission should to do to consider whether its decisions are consistent

with the CLCPA?

A. Yes. On page seven of his direct testimony, witness Ucar contends that the Commission would need a comprehensive current baseline of GHG emissions in each regulated utility service territory. Further, witness Ucar opines that the Commission would need utilities to provide sufficient information to assess actual progress being made in each utility service territory towards GHG emissions reductions.

Q. Does witness Ucar propose specific actions in this rate proceeding related to the CLCPA?

A. Yes. On pages nine and 10 of his direct testimony, witness Ucar proposes that the Companies develop a comprehensive GHG assessment methodology, and that such methodology should include the development of a utility-specific economy-wide GHG emissions inventory and analysis. Witness Ucar states, on page 24 of his direct testimony, that the Companies should develop their own emissions data repositories in a similar fashion to the methodology required of the New York State Department of Environmental Conservation.

Q. Are there important facets of the CLCPA which witness Ucar did not highlight in his testimony?

A. Yes, there are three main areas that witness Ucar did not highlight which deserve further discussion. First, the CLCPA itself defines a process for developing recommendations for how to meet New York State's climate obligations. Second, while the CLCPA did require that the Commission consider whether its decisions would be consistent with meeting the GHG emissions reductions requirements, it did not specify how the Commission must make such discretionary determinations. Third, the CLCPA requires that the New York State Department of Environmental Conservation (NYSDEC), not individual utilities, develop a statewide emissions inventory.

Q. What process does the CLCPA define for developing recommended actions to meet New York State's climate requirements?

A. The CLCPA implements a new section of New York State Environmental Conservation Law, §75, which establishes the New York State Climate Action Council (Climate Action Council), a body of 22 members ranging from the commissioners of

various State agencies, the presidents of various State authorities, the chairperson of the Commission, as well as a number of other appointees designated by the Governor and members of the State Senate and Assembly. The CLCPA requires the Climate Action Council to convene advisory panels and working groups to help it deliberate, and, within two years of the effective date of the Act, prepare and approve a scoping plan outlining recommendations for attaining emissions reductions of 40 percent of 1990 levels by the year 2030 and 85 percent reductions of such levels by 2040, achieving emissions reductions beyond the 85 percent level required by 2040, as well as achieving net-zero emissions from all sectors of the New York State economy. The CLCPA requires that the Climate Action Council's recommendations be incorporated into the next State Energy Plan following the development of such recommendations.

Q. Does the Panel have any specific recommendations related to witness Ucar's proposals?

A. Yes. We have two recommendations.

Q. Please describe your first recommendation.

A. We recommend that the Commission continue to adopt the programs as we recommended in our direct testimony, which are consistent with Commission policy, while the Climate Action Council goes about its work as required by the CLCPA to develop further specific recommendations. That is, the programs which result from this proceeding should be consistent with both Commission policy and the requirements of the CLCPA, but should not pre-judge the recommendations of the Climate Action Council nor make an end-run around the process required by the CLCPA.

Q. Please describe your second recommendation.

A. We recommend rejecting witness Ucar’s proposal for comparing carbon emissions reductions to a baseline emissions inventory in a utility’s service territory in these proceedings. Witness Ucar’s proposal is but one of any number of possible methods the Commission might employ to determine whether its decisions are consistent with the requirements of the CLCPA. Alternatively, the Commission could require the Companies to work closely with the NYSDEC to

ensure that the statewide GHG emissions report and models developed by that agency are sufficiently granular to be useful for utility-specific purposes without requiring significant duplication of efforts. A third option would be to establish a consistent methodology for use across utility service territories. The Commission should decide for itself how to determine whether its actions are inconsistent with the carbon emissions reduction goals set forth in the CLCPA. Thus, such decision is beyond the scope of these rate proceedings, which affect only two utilities and address their rates only for the Rate Year.

Q. Could there be a significant overlap or duplication of effort if each utility in New York State were required to develop and maintain their own service territory-specific GHG inventories?

A. Yes. A KEDNY customer taking Electric service from Con Edison, for example, could be paying for a KEDNY specific inventory, a Con Edison-specific inventory, a New York City-specific inventory, and the statewide inventory.

Gas Utility Transition to Thermal Utility

Q. Provide an overview of Estates witness Rodriguez's direct testimony regarding transitioning the Companies from gas utilities to thermal utilities.

A. On pages nine and 10 of his direct testimony, witness Rodriguez states that the Companies may be able to develop alternative revenue sources through an affiliated thermal company which would sell thermal energy sourced from geothermal and water-sourced loop fields instead of gas. On page 11 of his direct testimony, witness Rodriguez encourages the Commission to investigate allowing regulated monopolies to establish thermal company affiliates. Witness Rodriguez further proposes, on page 15 of his direct testimony, that the Commission provide the Companies with access to additional customers by allowing the Companies to develop new business models such as shared loop geothermal projects and development of a thermal company affiliate. On pages 9 and 11 of his direct testimony, witness Rodriguez states, without elaboration, that there may be legal

hurdles which currently restrict gas utilities from engaging in a line of business selling thermal energy.

Q. What is a thermal company?

A. On page 10 of his direct testimony, witness Rodriguez defines a thermal company as “a company that builds, maintains and manages shared infrastructure … that provides a heating and cooling medium for water source heat pumps or similar technologies.” Witness Rodriguez states that such a thermal company’s distribution facilities would likely be horizontal water pipe networks, heat exchangers, pumps and pumping stations, and physical connections to customers.

Q. Provide an overview of intervenor Wyman’s direct testimony regarding transitioning the Companies from gas utilities to thermal utilities.

A. On page 13 of his direct testimony, intervenor Wyman states that the Companies should invest in non-gas assets, and should “pump heat and not gas.” Intervenor Wyman states that shared district geothermal heating systems have the potential to displace the use of gas-fired space

and water heating requirements. On pages 13 and 14 of his direct testimony, intervenor Wyman alleges that the installation of such district geothermal systems by utilities "fits within existing and anticipated law and policy."

Q. Are there currently legal or statutory issues which preclude gas utilities from transitioning to thermal companies as proposed by witnesses Rodriguez and Wyman?

A. Yes. Staff Counsel has advised us that there are legal issues which prohibit gas corporations from transitioning to thermal companies. Staff Counsel has advised us that they will address such legal and statutory issues in briefs as part of these proceedings.

Q. What does the Panel recommend with respect to witness Rodriguez's and intervenor Wyman's proposals?

A. We recommend that the Commission reject such proposals.

District Heating Study

Q. Provide a summary of the City Policy Panel's proposals regarding its proposed District

Heating Study.

A. On pages 26 through 28 of its direct testimony, the City Policy Panel proposes that the Companies conduct a four-phase feasibility study to examine, pilot, and report on the potential for utility-owned district geothermal systems to replace the need for leak-prone pipe (LPP) replacement capital projects. Phase one of the study would require that the Companies hire a consultant to examine the feasibility of deploying district energy systems, identifying and targeting areas of high LPP concentrations. Phase two of the study would consist of selecting ten to twenty high-potential or unique use-case sites for further detailed assessment and design, ultimately ranking such sites based on potential benefits, costs, and feasibility. Phase three would consist of a pilot program to build and test district energy systems at several of the sites identified and examined in phases one and two. Phase four of the study would assess the scalability of the pilot locations, examine the costs and benefits of expanding the pilot, and would identify required

changes to the pilot structure and regulatory construct needed to increase adoption of district energy systems.

Q. Did the City Policy Panel provide a forecast of District Heating Study costs, or propose a way to pay for the study?

A. While the City Policy Panel did not provide a forecast of the costs related to the study, it did propose that the Commission consider changes to the regulatory model to allow the avoided costs of LPP replacements to be used to offset the capital costs of district energy systems.

Q. Does the Panel agree with the City Policy Panel’s proposed District Heat Study?

A. Partially. While we agree that displacing LPP replacement capital projects is attractive as a potential area where gas capital expenditures could be avoided using geothermal heat pumps or other technologies, we disagree with the City’s implication that utility ownership of shared loop fields should be the default solution.

Q. Why should utility ownership of shared loop fields not be the default option?

A. First, as noted earlier, Staff Counsel has

advised us that there may be legal issues with utility ownership of non-gas assets. Second, setting aside those legal issues for the sake of argument, we see utility ownership of geothermal loop fields and/or heat pumps as a last resort in the event that the private market is unwilling or unable to provide a cost-effective solution. Whether market conditions warrant utility ownership of geothermal systems should be tested under the same criteria as utility ownership of any other type of distributed energy resource (DER).

Q. Has the Commission adopted a set of criteria for utility ownership of DER?

A. Yes, the Commission's February 26, 2015 Order Adopting Regulatory Policy Framework and Implementation Plan in Case 14-M-0101 (Track One Order) provides such criteria. On page 68 of the Track One Order, the Commission establishes that utilities will not be allowed to own DER assets as a general rule, unless markets have had an opportunity to provide service and have failed to do so in a cost-effective manner. On page 70 of the Track One Order, the Commission

lays out four exceptions to the general rule under which utility ownership would be allowed where: (1) procurement of DER has been solicited to meet a system need, and the utility has demonstrated that competitive alternatives proposed by non-utility parties are clearly inadequate or more costly than a traditional utility infrastructure alternative; (2) a project consists of electric energy storage integrated into electric distribution grid infrastructure; (3) a project will enable low and moderate income (LMI) customers to benefit from DER where markets are not likely to satisfy the need; and (4) a project is part of a Demonstration Project.

Q. Has the City Policy Panel demonstrated a market failure, or that its proposed District Energy Study and pilot are exempt from the general rule prohibiting utility ownership of DER?

A. No, it has not.

Q. Should the District Energy Study be considered a Demonstration Project?

A. No. The Track One Order provided Demonstration Project authorization and budgets for Electric

utilities only. As a procedural matter, utility Demonstration Projects require Department of Public Service Staff approval to move forward – clearly, by our testimony on this matter, approval by Staff of a utility-owned district energy system is by no means guaranteed or even likely. Further, the Companies are already operating one utility-owned shared geothermal loop field as a Demonstration Project approved by the Commission in the Companies' last rate proceeding – a second such demonstration project is not needed. Furthermore, our rationale for not supporting the Companies' proposals to expand that Demonstration Project as expressed in our direct testimony apply equally to the City's proposal.

Q. Does this conclude your testimony at this time?

A. Yes.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid &
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

January 2020

Prepared Supplemental Testimony of:

Staff Efficiency and Sustainability Panel

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Q. Members of the Staff Efficiency and Sustainability Panel (Panel), please state your names.

A. Our names are Robert Cully, Joseph Hitt, Laurie Kokkinides, Kathryn Mammen, Davide Maioriello, and Jennifer Park.

Q. Has the Panel previously sponsored testimony in these proceedings?

A. Yes. We offered pre-filed direct testimony, submitted August 30, 2019.

Q. What is the purpose of the Panel’s supplemental testimony?

A. The purpose of our testimony is to update the recommended budgets and targets for the KEDLI and KEDNY energy efficiency portfolios based on the Commission’s recent Order Authorizing Utility Energy Efficiency and Building Electrification Portfolios Through 2025, issued on January 16, 2020 in Case 18-M-0084 (January 2020 Energy Efficiency Order). We also address the balances of unexpended Energy Efficiency Portfolio Standard (EEPS) funds, EEPS interest, Energy Efficiency Transition Implementation Plan (ETIP) funds, ETIP interest, and New York State

Energy Research and Development Authority (NYSERDA) Bill-As-You-Go Interest.

Q. Is the Panel sponsoring Exhibits as part of this supplemental testimony?

A. No.

Update to Energy Efficiency

Q. Since the Panel filed its initial testimony, has anything occurred that would change the Panel's recommendations?

A. Yes. On January 16, 2020, the Commission issued the January 2020 Energy Efficiency Order, which formally adopted budgets and targets for the KEDLI and KEDNY energy efficiency portfolios. In addition, the January 2020 Energy Efficiency Order established an additional source of funding to offset the energy efficiency budgets.

Q. How does the January 2020 Energy Efficiency Order change the recommendations regarding the budgets and targets contained in the Panel's initial testimony?

A. Since the Rate Year encompasses the period from April 1, 2020 through March 31, 2021, it reflects three quarters of the Companies' 2020 budgets and targets, and one quarter of the

Companies' 2021 budgets and targets. The Companies' 2020 budgets and targets were previously authorized in the December 2018 Efficiency Order. In our initial testimony, we recommended using the 2021 budgets and targets proposed by the Companies in their May Utility Proposal as placeholders until the Commission acted upon that proposal, which it did in the January 2020 Energy Efficiency Order. The January 2020 Energy Efficiency Order authorized company-specific budgets and targets for calendar years 2021-2025. Therefore, we now recommend that the Rate Year budgets and targets reflect three quarters of the 2020 values adopted in the December 2018 Efficiency Order and one quarter of the 2021 values adopted in the January 2020 Energy Efficiency Order. Specifically, we recommend Rate Year budgets of $10,710,572 for KEDLI and $18,557,009 for KEDNY and Rate Year targets of 290,245 MMBtu for KEDLI and 457,309 MMBtu for KEDNY.

Q. How does the January 2020 Energy Efficiency Order change the recommendations regarding the use of unspent funds contained in the Panel's

initial testimony?

A. The January 2020 Energy Efficiency Order does not change the recommendations regarding the use of unspent funds in the Rate Year from our initial testimony. Although the January 2020 Energy Efficiency Order provides KEDLI with an additional $6.73 million and KEDNY with an additional $10 million in unspent NYSERDA EEPS gas funds, we recommend those additional funds be used to offset energy efficiency funds in future years rather than in the Rate Year.

Q. Do you have any comment regarding the appropriate balances of deferred unexpended EEPS, EEPS interest, ETIP, ETIP interest and NYSERDA Bill-As-You-Go Interest?

A. Yes. In our pre-filed direct testimony, we provided the balances of those accounts through June 30, 2019, based on information provided to us by the Companies. As a significant amount of time has passed, those balances should be updated before the Commission issues its final rate order in these proceedings.

Q. Do you recommend any updates to the balances at this time?

A. No.
Q. Why not?
A. While we have sought updated balance information from the Companies, at this time we have not received all of the necessary information and cannot verify it sufficiently to rely on it for a recommendation.
Q. Please explain.
A. We asked the Companies information request (IR) DPS-1082, in which we sought updated unspent fund balances for EEPS, EEPS interest, ETIP, ETIP interest, and NYSERDA Bill-As-You-Go Interest. The Companies provided a response to that IR on January 23, 2020. Due to the timing of the receipt of the IR response, we have not had time to verify the accuracy of the updated balances provided by the Companies. In addition the information provided in response to IR DPS-1082 caused us to ask a follow up IR, DPS-1085 on January 27, 2020, to have the Companies explain discrepancies between their responses to IRs DPS-1082, DPS-268 and DPS-893.
Q. Given that you are still waiting for information regarding the unspent funds balances, what do

you recommend at this time?

A. At this time, we recommend maintaining the unspent fund balances reflected in our initial testimony and maintaining the amortization amounts as reflected in Staff's initial Exhibit__SRRP-1, Exhibit__SRRP-2 and Exhibit__SESP-2. We are hopeful that we will have more clarity on this issue at the time of the hearing and will, if possible, provide further information at that time.

Q. Does this conclude the Panel's supplemental testimony at this time?

A. Yes.

MR. GOODRICH: Your Honors, the witnesses are now available for cross-examination.

A.L.J. COSTELLO: Okay. The company?

MR. GAVILONDO: Thank you, Your Honor. Carlos Gavilondo for the companies.

CROSS EXAMINATION

BY MR. GAVILONDO:

Q. Good morning, panel. My cross examination is first going to focus on the -- the company's proposed fuel switching calculator. And I have some interrogatory requests that the panel prepared in response to the question from the company and counsel is going to circulate these to the parties or to the panel, I should say.

MR. GOODRICH: This is the two you just gave me?

MR. GAVILONDO: Yes.

A.L.J. LEARY: Have these already been introduced --?

MR. GAVILONDO: They have not yet been marked.

A.L.J. LEARY: These are new?

MR. GAVILONDO: These are new.

A.L.J. LEARY: Okay. So --.

A.L.J. COSTELLO: Okay. What we'll do is mark the first one, which is request number N.G.-55 as Exhibit 615. And the second document, which is request number N.G.-56 as Exhibit 616. And just please describe those documents for the record.

MR. GAVILONDO: Thank you, Your Honor.

BY MR. GAVILONDO: (Cont'g.)

Q. Yes, panel. Exhibit pre-marked for identification 615 is a 1 page document, entitled N.G.-55. This is an interrogatory request dated November 12th, 2019, propounded by the companies on the Staff Efficiency and Sustainability Panel responded to on November 22nd, 2019.

Is this a true representation of the panel's response to this interrogatory request?

A. (Cully) Yes.

Q. Thank you. The second document is -- in identified, I'm sorry. Pre-marked identification as Exhibit 616. It's entitled N.G.-56 dated November 12th, 2019, a response date of November 22nd, 2019. And I asked the panel if this is a true representation of the panel's response to this interrogatory request?

A. Yes.

Q. Thank you. With respect to the fuel switching calculator that the company has proposed, is it staff's position that the allocation of costs associated with the fuel-switching calculator should be based on the company's general allocator, which would allocate 8.29% to KEDLI and 13.31% of the cost to KEDNY?

A. Can you point us to where on the testimony those specific percentages are listed?

Q. Yes. On line -- I'm sorry, on page 40 of your direct testimony, lines 5 through sev -- 17. And if you go to the bottom, you can see on lines 15 through 17 the panel's recommendation.

A. Yes, that's an accurate representation of our recommendation.

A.L.J. LEARY: I just want to clear up the record. I thought you said 2.9%. Did you say 8.29%?

MR. GAVILONDO: 8.29%.

A.L.J. LEARY: Okay. Thanks.

MR. GAVILONDO: Yes, Your Honor.

A.L.J. LEARY: And 13.31% --

MR. GAVILONDO: That is correct, Your Honor.

A.L.J. LEARY: -- as I see it in the testimony. Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. Thank you, Mr. Cully. Now, is it staff's position that the -- that companies that do not benefit from a service provided by the service company should be allocated a portion of the cost of that service?

A. Say that again one more time, please.

Q. Certainly. Is it staff's position that costs of a service company service should be allocated to a company that does not benefit from those services?

A. It's my recollection that we did not submit testimony to that effect. However, we do believe that the other service companies of National Grid could benefit from this web-based tool because it is not specific to a particular service company nor does it necessarily need to be specific to a particular fuel type.

Q. Thank you. Is it staff's position that companies that -- the company should pay for a service from the service company before the

service company provides that service to the companies?

A. Can -- can you help me disentangle the companies and the service companies a little bit, please?

Q. National Grid structure has a central service company that provides services to a multitude of operating companies. And in doing so, it allocates costs to the companies that are benefiting from those services. And my question to you -- to the panel is whether or not it is staff's position that companies -- operating companies should pay for a service --

A. Yeah, KEDNY and KEDLI?

Q. KEDNY and KEDLI are operating companies, yes. Should pay for a service from the service company that the central service company before the central service company provides those services to the operating company.

A.L.J. LEARY: Can I just ask counsel for purposes of this record? Rather than saying or calling a service company, a service company, why don't you get it down to the central services company versus KED -- so if you could use those names, I

think it would help the panel understand a little bit more rather than the theoretical should a company pay for a service before the service is provided or I mean you can --

MR. GAVILONDO: Your Honor --.

A.L.J. LEARY: -- ask both questions I think, but I -- I -- I see some confusion developing here. So --.

MR. GAVILONDO: Your Honor, in -- in this instance, some of the companies that would be allocated costs are not KEDNY and KEDLI. So if I were to refer to the operating companies as a receiver of costs from the service company. I am not exclusively referring to KEDNY and KEDLI. I can try to clarify the question --

A.L.J. LEARY: And then --.

MR. GAVILONDO: -- and that might help the panel and -- and -- and the bench. So let --.

A.L.J. LEARY: Yeah. Get -- get, there's a foundational question that I think might be helpful and then you get to what I think you were just talking about.

MR. GAVILONDO: Okay.

A.L.J. LEARY: So I'm looking for

help.

BY MR. GAVILONDO: (Cont'g.)

Q. Can I just ask the panel, would you like me to repeat the question?

A. I think I have the thrust of it essentially should KEDNY and KEDLI pay for a service before that service is provided by National Grid.

Q. I think if you answer that question, that is fine, yes.

A. As I understand it, rather as we, the panel understand it, the service is being provided within the rate year, thus they're rate-year costs. So yes, KEDNY and KEDLI should pay for the costs of that service as it's being provided during the rate year.

Q. Very good. Thank you. I'd like to turn the panel's attention to its testimony relative to the power to gas demonstration project and that appears beginning at page 57 of this panel's direct testimony. Please let me know when you're there.

A. We're there.

Q. Thank you. As the panel's aware, the companies proposed a power to gas demonstration

19-G-0309/19-G-0310 - 2-11-20 - Brooklyn Union Gas project. And beginning at line 12 through 21 of page 58 of the panel's direct testimony, you indicate that the panel does not support the demonstration project. Could you describe the basis for the panel's opposition to the demonstration project?

A. Our basis for the -- for our position is laid out on lines 13 through 19. Essentially the power to gas project as the companies laid forth in the -- in their testimony as we understood it, essentially stores power in the form of gas. We do not believe that that meets the criteria for storage as laid out in the storage order. That is the December 12th, 2018 order, in case 18-E-0130.

A.L.J. LEARY: I just want to clarify for the record that you were rev -- referring to lines 13 through 19 on page 58 of the testimony.

MR. GAVILONDO: That is correct, Your Honor. And actually line 12 has the question, so to put it in context lines 12 through 21.

A.L.J. LEARY: So perfect to say the lines, but make sure you say the page as well.

THE WITNESS: (Cully) We'll do. Thank you.

MR. GAVILONDO: Your Honor, if I may just one moment.

BY MR. GAVILONDO: (Cont'g.)

Q. Thank you, panel. So the basis for the panel's opposition is that you've just referenced is that it's not identified as an option or a potential solution in the commission's December 13, 2018 order on energy storage. Is that -- is that a fair characterization of your testimony?

A.L.J. COSTELLO: I believe it's a December 12th order according to the testimony.

MR. GAVILONDO: Yes, Your Honor. According to the testimony, it says December 12th. When I look at the face page of the order itself, it says issued and effective on December 13th.

A.L.J. COSTELLO: Thank you.

MR. GOODRICH: Judge, I would object. I mean, it's asked and -- he asked -- it's asked and answered. He asked the -- the panel what's their position, they stated their position, he's saying so is this your position again.

A.L.J. LEARY: I think we allow it, but I want to make sure that counsel understands that the word criteria does not meet the criteria was, I

believe, the testimony. So your characterization --

MR. GOODRICH: The criteria.

A.L.J. LEARY: -- was somewhat different than what I wrote down, at least in what the record probably reflects. If I'm good at hearing what the panel has stated. So I would just ask that you sort of stay pretty tight to --

MR. GAVILONDO: I'll withdraw the question, Your Honor.

A.L.J. LEARY: -- what their testimony is.

MR. GAVILONDO: Thank you.

A.L.J. LEARY: Okay.

MR. GAVILONDO: I'll withdraw the question.

A.L.J. LEARY: Or you can rephrase. That's up to you.

MR. GAVILONDO: Yeah.

THE WITNESS: (Cully) I would just like to add that our testimony does have a typo in it. It should be December 13th, not December 12th.

BY MR. GAVILONDO: (Cont'g.)

Q. Okay. Thank you.

A. Of -- of 2018. That's on page

58, line 16, at the very end. The storage order was in fact issued on December 13th, 2018.

MR. GOODRICH: I thought, sorry. This might be that I heard wrong. But I thought that Carlos had said the order was on the 14th.

MR. GAVILONDO: No, the December 13th, I'm sorry.

A.L.J. LEARY: No, the 13th.

MR. GOODRICH: I just heard wrong. Okay. I'm sorry, I'm sorry. I thought we were adding a third date.

A.L.J. LEARY: I'm -- I'm just going to exercise some executive authority here and say no correction is required to that error. No correction need be required -- is required. I think everybody in the room can go and look at the order and I've changed my version of the testimony. So don't worry about the date.

BY MR. GAVILONDO: (Cont'g.)

Q. So panel referencing the December 13th, 2018 order. I'm going to read from page 10 of that order, in case number 18-M-0084 -- I'm sorry. No, it is not. I apologize.

A.L.J. LEARY: 18-E-0130?

MR. GAVILONDO: Thank you, Your Honor.

BY MR. GAVILONDO: (Cont'g.)

Q. Okay, page 10 under the discussion, roman numeral 1, energy storage deployment goals. Achieving the state's ambitious system, clean energy, and CO2 reduction goals will require contributions from a variety of -- of resources and energy storage will be a critical resource enabling New York's clean energy future.

Did I read that accurately?

A. That's what the order says.

Q. Thank you.

A.L.J. LEARY: Does -- does the panel have the order in front of it? Thank you.

MR. GAVILONDO: Thank you.

A.L.J. LEARY: The answer was yes?

MR. GAVILONDO: Yes.

A.L.J. LEARY: Thank you.

THE WITNESS: Yes. That's what the order says on that page.

BY MR. GAVILONDO: (Cont'g.)

Q. Thank you.

A. Specifically. The order, however, later says and lays out a deployment roadmap

to meet those goals, which the company's power to gas program is not part of.

Q. Thank you. So in the panel's opinion, should the state foreclose potentially beneficial clean storage alternatives that may not be specifically identified in the storage order?

A.L.J. LEARY: Do you mean to say should the department or are you asking this panel about a state policy and whether that should be foreclosed? Is what -- what -- where are you going with this?

MR. GAVILONDO: I'll rephrase the question.

A.L.J. LEARY: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. In the panel's opinion, should the Department of Public Service or the State Public Service Commission foreclose potentially beneficial clean storage alternatives that may not be specifically identified in the energy storage order?

A. No. However, the definition of clean is, I think, up for debate with respect to the power to gas program as the company stated in its response to one of the I.R.s that was entered into

the record yesterday in the company's own panel's cross examination.

I can't remember the number of that I.R. response directly off the top of my head. I believe it was 850, D.P.S. 850. The company noted that the efficiency of the power to gas project is approximately 48%. And that is with using the very most beneficial assumptions around that.

For example, that 48% is the efficiency of using solely the hydrolysis of water using electricity, from H2O into hydrogen H2 and O2, oxygen. And then reacting that in a bioreactor, using carbon dioxide as a source to -- into the hydrogen stream to make a CH4 methane.

That is the overall efficiency, as I understand it, of the hydrolysis and a bioreactor portion of that step. However, you also have to consider the efficiency of the source fuel used to create the electricity. And whatever the elect -- end use of the -- the gas after it has, in fact, been made in the power to gas project as well.

For example, the company, in my opinion, did not show that it would only use renewable generation to fire the hydrolysis reaction

in its rebuttal -- I'm sorry, in its testimony. If the company were to use, say, bulk electric power, that would itself be -- chances are, fired from natural gas.

You have to multiply that 0.48 efficiency factor by another 0.6 to 0.8, depending upon the efficiency of the natural gas burners, that generate that electricity. And if in fact that methane on the backend of this reaction is then used, to be burned in a gas turbine to make then power, in fact storing electricity instead of methane. You have to then multiply that again by 0.6 to 0.8.

In short, the overall efficiency in our view is anywhere between 48% and 30% depending upon the various assumptions and makeup of how exactly you go about doing that power to gas project and what it's ultimately used for. The storage order, on the other hand, requires a minimum efficiency of 80%, I believe, for these storage resources that are incentivized under the various programs that the com -- that the commission required in that order.

Q. Thank you. Turning your attention to the hydrogen blending study proposed by

the company and the panel's testimony relative to that proposal, which begins on page 59 of the panel's direct testimony. Thank you.

Staff also opposed the company's hydrogen blending study proposal, is that correct?

A. (Cully) That is correct.

Q. And the basis for staff's opposition to the hydrogen blending study is set forth beginning at lines 18 of -- I'm sorry, page 59 line 18, beginning there. Is that correct?

A. Page 59 line 18?

Q. That's correct. That's the question.

A. That's correct, yes.

Q. All right. Thank you.

MR. GAVILONDO: I have no further questions. Thank you.

A.L.J. COSTELLO: Okay. The next party up is PULP.

MR. BERKLEY: Thank you, Your Honors.

BY MR. BERKLEY:

Q. And still good morning, panel. On pages 17 and 18 of your direct testimony, which is entitled, Prepared Testimony of Staff Efficiency and

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Sustainability Panel. Starting on line 8 on page 17 and going through line 15 on page 18, you discuss the new efficiency New York report and also the December 2018 efficiency order by the commission.

Starting on line 8 of page 18, you then talk about some of the things that were in the order and if you don't mind my skipping down to line 11, it required 20% of incremental spending to fund low and moderate income customer programs. Is that correct?

A. (Mammer) Yes.

Q. And if I don't give you enough time to turn the page, I'm drinking coffee, so bear with me. I'll try. Are there any projections or studies of how this spending on energy efficiency would lower, moderate and low income bills?

A. (Cully) So first off, the commission order speaks for itself. Second, we did not perform such an analysis and third, the utilities are required to run their energy efficiency portfolios in a cost-beneficial manner. Therefore, we expect the energy-efficiency programs as a whole to be beneficial to customers.

Q. Thank you. When you say the

commission's order speaks for itself in the 2018 efficiency order, were there any requirements to do projections or studies of how the spending on energy efficiency would lower, low- and moderate-income bills?

A. Could you point me to where it says that in the order?

Q. I'll withdraw it. I don't have it with me.

A. I have a copy if you'd like.

Q. Sure. And just a -- in order to be somewhat more efficient with time, let me continue asking questions. We'll come back to this one when we've found something in the order. Is there an allocation formula for directing the spending of energy efficiency between low- and moderate-income households?

A. Can you define allocation formula please?

Q. A formula or a plan that would allot amounts of money as in a ratio between low- and moderate-income households.

A. I am not aware -- we are not aware of any such allocation factor between low- and

moderate-income households.

Q. Thank you. Has staff quantified whether or not this 20% spending on low-and moderate-income customer programs would affect the commission's and state's goal of reaching a 6% energy burden for low-income households? And that's the goal established in case 14-M-0565.

A. Can you repeat the question, please?

Q. Yeah. Let me go back a couple of pages. Has staff quantified whether the 20% of incremental spending that you describe on lines 11 through 13 of page 18 of your testimony, has staff quantified if that spending would affect -- reaching the commissions and state's goal of reaching the 6% energy burden for low-income households?

A. So our testimony in this case is based on the commission order. We did not develop the 20% of incremental-funding requirement set forth in the commission order, nor did we provide such an analysis as part of this proceeding. However, we do expect that the increased budgets to the L.M.I. program required in that order will help toward the 6% energy burden set forth in the affordability

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Q. Thank you. I'm -- thank you. I'm now turning to pages 31 and 32 of your direct testimony. On lines 14 -- beginning on line 14 on page 31 and going through line 6 on page 32. You described something called the HEAP program for KEDLI. Does staff have any quantitative results from the low-income dwellings served so far in KEDLI's HEAP program and by so far, I mean since 2017?

A. So to answer the question a bit pedantically, this panel does not, however, the Staff Earning Adjustment Mechanism Panel in its Exhibit S.E.A.M.P. 6 lays out the KEDLI EmPower replacement program, which is the same as the HEAP program, historical data from 2017 through 2019, the year to date at that time which is itself based on an interrogatory D.P.S.-975, attachment 1, which is included in our Exhibit S.E.A.M.P.-1.

Q. Thank you. Would this panel, and I'm not asking for the members of the panel who repeated from the previous panel necessarily, but would this panel be able to answer the question whether or not staff believes that the program was effective?

MR. GOODRICH: Objection. Just -- effective is a very vague term. Could counsel --?

A.L.J. LEARY: What do you mean by effective, Mr. Berkley?

MR. BERKLEY: I'm willing to rephrase but by effective I meant whether it resulted in the lowering of bills for people enrolled in the program.

A.L.J. LEARY: I think that addresses your objection, Mr. Goodrich.

MR. GOODRICH: Yes.

A.L.J. LEARY: Panel?

THE WITNESS: (Cully) We do not have data to support that conclusion one way or another. The data that we requested is the -- the actual program data as far as number of installs and number of Million British Thermal Units, M.M.B.T.U. of energy savings resulting from those installations.

BY MR. BERKLEY: (Cont'g.)

Q. Thank you. On lines 19 through 21 of -- still on page 32. Staff says we support, yes, we support increasing the budget to 2.5 million and expanding the program to eligible- and moderate-income customers. What is the definition of an eligible- and moderate-income customer?

A. We adopted the definition of moderate income as proposed by the company, which we understand to be based on a certain percentage of the area median income.

Q. Thank you. Is staff aware of the requirement in the low-income order that the company enroll more individuals in its low-income program on Long Island?

A. We are generally not aware of that.

Q. Thank you. I'll move on then. I'm turning to page 33 starting on line 19, the panel says, does KETLI propose an incentive mechanism associated with HEAP? And then starting on line 21 panel says, yes, KEDLI proposes the EmPower replacement program incentive mechanism, which would award KEDLI for customer participation in tiers three and four of the HEAP program. Is this E.A.M. based solely on participation? I'm sorry, on participation rate or numbers?

A. So I would not describe the KEDLI EmPower replacement program incentive mechanism as an E.A.M. specifically. However, it is based on the number of households participating in tiers two --

sorry, three and four of the KEDLI EmPower replacement program.

Q. Thank you. If there were no positive results, would KEDNY still collect -- I'm sorry, KEDLI still collect this E.I.M., Earnings Incentive Mechanism?

A. Could you define positive results, please?

Q. If there was no lowering of the bills of the individuals' households in the program.

A. So while the KEDLI EmPower replacement mechanism -- or EmPower replacement program incentive mechanism is solely based on the number of households participating in tiers three and four of the program, those households will necessarily benefit from the energy efficiency measures installed in their premises likely lowering their bill.

Q. Thank you. I'm moving to page 34 lines 9 through 15. There is a discussion by the panel of a proposal by the company to add some full-time equivalents in the energy efficiency programs that it operates. My question is do any of the F.T.E.s proposed by the company or currently in the

program do public outreach that would alert the public to the existence of the energy efficiency programs?

A. So while we do not have the specific information related to how many F.T.E.s or partial F.T.E.s are related to specifically outreach and education for the energy efficiency program. Outreach and education is a -- a major part, integral part of energy-efficiency programs and utility programs in general. And it would be surprising to us that there is no outreach and education related to energy efficiency as part of the labor costs.

Q. Thank you. Moving to page 56. There is a discussion of the L.M.I. conversion program that starts partway down page 55. And I'm going to just read lines 3 through 7 on page 56, which is a response to the question, what is panel's position regarding the L.M.I. conversion program?

The response is, given that for the purpose of these rate proceedings, staff is assuming that NESE, the NESE pipeline will not be built during the rate year. We accept the company's position to not implement the L.M.I. conversion program.

My question is: Would the L.M.I.

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conversion program help low- and moderate-income households even without the completion of the NESE project and by help I mean lower bills?

A. Can you point us to where in our testimony we made such an assertion?

Q. The -- forgive me. The assertion that you would accept the company's position?

A. No, that the L.M.I. conversion program would decrease customer bills.

Q. I don't believe anywhere in your testimony there is a -- is such an assertion, which is why I'm asking the question. Would the L.M.I. conversion help lower bills for low- and moderate-income households even without the NESE Project?

A. We have no data to support such a conclusion one way or another.

Q. Thank you. I have one last question. On page 57, concerning the power to gas demonstration project, lines 19 through line 1 of page 58. It says, KEDLI as described on page 83 of the testimony of the Future of Heat Panel proposes to develop a project that converts excess renewable energy to hydrogen and then converts the hydrogen to pipeline quality natural gas to be ultimately

injected into the existing gas network and delivered to customer.

And my question echoes one that was asked yesterday, which is, was staff able to identify any existing excess renewable energy in the company service area that would be proposed to be used in this project?

A. Staff is unaware of any such excess renewable energy.

Q. Thank you. I just want to return, not asking another question. I want to return to the question I'd asked earlier about whether or not the commission had plans to do a study.

I think I'm going to withdraw that question. Although the commission does request a great deal of follow-up from staff, the utilities and NYSERDA and other parties. I think I'll just put this in my brief. So thank you. Thank you, Your Honors.

A.L.J. LEARY: Okay.

A.L.J. COSTELLO: Thank you. We're going to take a short break and we'll take like a 15-minute break. Come back at 11 o'clock and we'll

finish up with this panel. Thank you.

A.L.J. LEARY: Who is up?

A.L.J. COSTELLO: The City of New York.

BY MR. FUNG: (Cont'g.)

Q. Thank you, Your Honors. Good morning, panel. Justin Fung for the City of New York. Could we go to the panel's rebuttal testimony, starting at page 12. Just let me know when you get there.

A. (Cully) We're there.

Q. So starting on page 12 of your rebuttal, the panel addressed a proposal from the city to conduct a district heating study on the feasibility of using geothermal systems to replace leak-prone pipe replacement capital projects. Is that correct?

A. That's correct.

Q. And skipping over to Page 14 of the rebuttal under the Q, does the panel agree with the city policy panels proposed district heating study. The panel answered partially.

Is it a fair characterization of the panel's testimony that the panel's objections to the

city's proposal centered around the concept of utility ownership of geothermal assets or one of the -- one of the objections to it. Is that correct?

A. I think that's a fair characterization. On lines 19 through 21 of page 14, we state that we disagree with the city's implication that utility ownership of shared loop fields should be the default solution.

Q. And later on in the rebuttal, I believe it is on page 16 going into 17. The panel also disagreed with characterized -- characterizing such a study as a demonstration project. Is that correct?

A. That is correct.

Q. Would it be fair to say that, in the panel's opinion, there is merit in at least examining the feasibility of using district geothermal systems to replace leak-prone pipe replacement capital projects. Is that fair to say?

A. So as usual the answer to that question is a relatively complicated one. The company has already done a district geothermal heating study through its existing utility-owned, district-energy system that it describes in its

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testimony, the -- the heat pump demonstration project approved by the commission in the prior case.

We don't necessarily support utility ownership of such a shared loop field, though we would love to see a third party own such a system. Further, if the company finds an area where it can avoid replacing leak-prone pipe, through the use of geothermal facilities, we believe that it will already be able to do so through the use of non-pipe alternatives.

And finally, the January 2020 energy-efficiency order and the December 2018 energy-efficiency order laid the responsibility for general heat pump panel penetration at the feet of NYSERDA and the electric utilities and we would like to see that continued.

Q. Thank you. Is the panel aware of the results of the company's prior district heating study that you just stated that it performed?

A. As we understand it, the final results of such study are currently being worked on by the company. The company does routinely report on the progress of its demonstration projects, the geothermal demand response, et cetera. And I believe

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quarterly filings are publicly available on D.M.M.

Q. Do you know which proceeding that's in?

A. The prior KEDLI & KEDNY rate proceeding.

MR. GOODRICH: For the record that's 16-G-0058 and 16-G-0059.

MR. FUNG: Yes.

BY MR. FUNG: (Cont'g.)

Q. And I believe, Mr. Cully, you testified earlier as part of the E.A.M. panel that you participated in the Con Edison rate cases 19-E-0065 and 19-G-0066. Did any other members of the panel participate in the Con Ed rate case or are they familiar generally with that rate case?

A. (Mammen) Yes.

Q. And --

A. (Cully) Let me just --

Q. Sorry. Go ahead.

A. -- clarify that a little. Many of the members of this panel, but not all, were assigned to the Con Edison rate proceeding as well.

Q. And for those panel members that did work on the Con Ed rate case, are you generally

familiar with the joint proposal that was filed in those rate cases on October 16th, 2019?

A. I think it's fair to characterize that the panel as a whole is generally familiar with the highlights of the joint proposal. As far as I know, no one on the panel here has a copy of the joint proposal in front of them, but on a very high level basis, we are familiar with it.

Q. Okay. I -- I don't have any specific questions regarding exact details about the joint proposal, but I just wanted to follow-up with whether the panels are generally aware that as part of that joint proposal there were provisions for a district heating study similar to what the city has proposed in this rate case.

MR. GOODRICH: Objection. If Mr. Fung wants to introduce some -- some awareness of that he can bring in the J.P. and point to where in the J.P. that it's stated instead of a characterization and asking the panel to -- to agree with the characterization.

MR. FUNG: Your Honors, I'm just asking if they're generally aware if they respond with no, we can move on.

A.L.J. LEARY: I agree. Does anybody on the panel know about that portion of the J.P. that involves a district heating study that was approved in the J.P. and I guess, as proposed by both Consolidated Edison and D.P.S. staff, if not other parties signing on to the J.P.

Anybody familiar with that provision?

THE WITNESS: (Cully) Yes, Your Honor.

BY MR. FUNG: (Cont'g.)

Q. And are the panel members generally aware that the commission adopted that joint proposal in an order dated January 16th, 2020 and a high low?

A. Yeah. So just going back as -- as I understand it as -- well, rather as the panel understands it, the proposal set forth by the city as -- again as -- as we understand it, in this proceeding is relatively consistent with the results of that negotiated joint proposal for Con Edison, and we do understand that the commission approved such joint proposal in a recent commission order.

Q. Thank you.

MR. FUNG: Nothing further, Your Honors.

A.L.J. LEARY: I just want to clarify something. I believe the panel's testimony --.

THE WITNESS: (Cully) I'm sorry, Judge, I'm ready to hear now.

A.L.J. LEARY: Is there something further that the panel wants to offer in its testimony as a result of the colloquy that's been going on or is that not what you were up to?

You're -- you're -- you're ready for me to ask your question. Okay. Good. I just want to make sure I wasn't interrupting something that --

THE WITNESS: (Cully) No, Your Honor.

A.L.J. LEARY: Okay. You indicated that in response to the city's question about the company's existing heating study that there were no final results. They were still being worked on by the company but there were "routine reporting efforts as well as quarterly filings" and this is in the right -- prior rate case 16-G-0058 and 0059.

So, I don't want to be forced to go look at the quarterly filings. What's -- what -- what are you seeing in the quarterly filings in terms of the results of the existing heating study that the company has in place now.

THE WITNESS: (Cully) Your Honor, we have about five general observations from our understanding of the quarterly reports as they come in.

So first off, the quarterly reports typically track more of the -- the pedestrian elements of this -- things like tracking the quarterly cash outflows. Typically the -- I call them program results are summarized on an annual basis generally.

What we have seen to date is that for individual participants, the equipment efficiency has been lower than what we would typically expect for such systems. However, customer satisfaction with the project -- I'm sorry, participant satisfaction with the project is good.

I think there needs to be a distinction drawn between participant and customer. Because these participants are not gas customers, never have been gas customers and probably never will be gas customers.

So the participant satisfaction with the project is good. However, with the caveat that these customers are not paying for the full cost of

the system in the payments that they are making to KEDLI.

It should be noted that the project, as our understanding goes, has been proceeding under budget, however, this program does not address leak-prone pipe or any sort of gas infrastructure replacement since it is targeting customers that are outside the utilities current gas footprint.

A.L.J. LEARY: Thank you. Is that it? I mean, that was plenty.

THE WITNESS: (Cully) Yes, yes, yes that --.

A.L.J. LEARY: Anything else.

THE WITNESS: No further observations at this time.

A.L.J. LEARY: Okay.

MR. FUNG: Your Honor --

A.L.J. LEARY: Thank you.

MR. FUNG: -- can I jump back in real quick? Just to follow up on that.

A.L.J. LEARY: I have one other question. Again, it's a clarification for the record. I believe in response to a question about the effectiveness of the empower replacement program.

Actually, I'm sorry. I may have written this, this is PULP's cross examination about the low and moderate income. You were referring to S.E.A.M.P.-1 and -6, which are hearing Exhibit 614 and whatever, maybe 620. Does that sound right Judge Castello?

A.L.J. COSTELLO: No, 416, 421.

A.L.J. LEARY: 421. And I believe counsel for PULP asked if the program was effective. He then defined effective at your request to mean results and that the program was resulting in lowering bills.

Your response was no data to support that conclusion either way, but then you indicated that you were asking for data from the company to support the program's effectiveness. I just wanted to follow up on the data request. Is that -- did I misunderstand that?

THE WITNESS: (Cully) I think so, Your Honor. We had asked in our data requests D.P.S., I think it was 390. I'm sorry, totally wrong on that. D.P.S.-975, which is included as one of the I.R. responses in our S.E.A.M.P.-1 Exhibit.

We had asked for the number of pieces

of energy efficiency equipment installed, and the number of M.M.B.T.U. savings achieved through such installations for both tier 3 and tier 4. I understood Mr. Berkeley's question to be about overall low-income customers savings as a result of the program as a whole, not as individual customers participating.

We would expect to see that when an individual customer participates in this program and has a piece of energy-efficiency equipment installed in their residence that they would see a concomitant bill decrease.

A.L.J. LEARY: Okay. So thank you for that clarification. Mr. Jung.

MR. FUNG: Fung, sorry. F-U-N-G in case the --

A.L.J. LEARY: Sorry.

MR. FUNG: -- court reporter needs it. Just --

A.L.J. LEARY: Fung.

BY MR. FUNG: (Cont'g.)

Q. Just going back to the panel's testimony. You stated that the -- the district study that or the district project that you're referring

to previously is one that does not address leak-prone pipe and is geared more towards non-gas customers. Is that correct?

A. (Cully) That's correct.

Q. And so would the panels -- are the panels opposing a study that specifically analyzes whether it is feasible to use geothermal systems as a leak-prone pipe replacement mechanism?

A. So we believe that the ability of geothermal systems when appropriately clustered to avoid leak-prone pipe is non-controversial. We don't need a study to tell us that a load relief can relieve load.

However, to the extent that the company finds an area, where leak-prone pipe can be or the need to invest in a leak-prone pipe replacement program can be avoided through the use of load relief, like geothermal.

We believe the companies already have the ability to go after those through their non-pipe alternative screening criteria and cost recovery mechanisms that were testified to by both the company and staff in this case.

It should be noted, however, that Con

Edison does not have such criteria established already. Those are being part of the J.P. developed as part of a collaborative following on to the end of that proceeding.

MR. FUNG: Thank you. Nothing further, Your Honors.

A.L.J. COSTELLO: Okay. A.A.R.P.

MR. RIGBERG: Thank you, Your Honor. Saul Rigberg with A.A.R.P.

CROSS EXAMINATION

BY MR. RIGBERG:

Q. Good morning, panel. I have a few question starting on page 26 of your direct testimony and going into 27 and 28 regarding unspent EEPS and ETIP moneys. Do you have those pages handy?

A. (Cully) Sorry, 20 -- 26 through 28.

Q. Twenty-six, 27, 28. There is a discussion referencing unspent moneys. And my question is, has the panel determined why the moneys were unspent?

A. We did not include such an analysis as part of our testimony.

Q. Okay. Well, if you could just

turn to page, okay, 37 of your testimony on lines 21 and 22. You say historically the companies have not spent the annual energy-efficiency budgets.

So going -- going back to -- to pages, on page 27, lines 10 and 11. You say that KEDLI has accumulated $4.869 million of unspent EEPS funds and then lines 14 and 15 you say that KEDNY has accumulated $20.405 million of unspent EEPS funds.

A.L.J. LEARY: Could you just define EEPS for the record, please?

MR. RIGBERG: Sorry. Yeah, it's the Energy Efficiency Portfolio Standard.

BY MR. RIGBERG: (Cont'g.)

Q. So I'm wondering, as you decided whether or not to continue spending at these levels, or in the case of KEDLI to increase spending by 600,000, why you didn't look at -- why -- the reasons that the existing money that was recovered from ratepayers was not spent?

BY MR. RIGBERG: (Cont'g.)

A. (Cully) So first off, the commission itself established the budgets for the low income and more general energy-efficiency programs. Our testimony does not develop such budgets but

recommends that such budgets be implemented and also makes recommendations with respect to use of the unspent EEPS money toward those efforts.

Q. Well, I think on page -- let's see if I can find it. But on -- on page 32 on lines -- it was starting line 7 through 18. There is a discussion regarding expanding or adding $600,000.00 to the -- to a budget -- to the budget.

Did you take into account the company's underspending in the past when you -- when you agreed to add $600,000.00 going forward?

MR. GOODRICH: Objection. Actually, Your Honor, A.A.R.P. hasn't established that there is any underspending in the HEAP program. He -- he was talking about underspending in the energy-efficiency program and the HEAP Program is the EmPower replacement program, which as testified to before is separate from the energy efficiency program.

A.L.J. LEARY: Mr. Fung?

MR. FUNG: No, it's --.

A.L.J. LEARY: I'm -- I'm sorry --

MR. RIGBERG: Okay.

A.L.J. LEARY: -- Mr. Rigberg.

MR. RIGBERG: Okay. That I think Mr.

Goodrich --

A.L.J. LEARY: Is it lunch time?

MR. RIGBERG: -- raises a good point.

BY MR. RIGBERG: (Cont'g.)

Q. So does the panel know if KEDLI has fully spent the allocated 1.9 million in the HEAP program?

A. (Cully) We are unaware of any underspending in that program.

Q. Are you saying that they fully spent their allocated amount?

A. It is my belief that they spent their amount.

Q. Okay. Thank you. All right. So going back to EEPS. Has the panel -- is the panel aware of any staff follow-up with the companies about why they've underspent and why -- why they're -- why they have ratepayer money, but have not spent it on the required -- the program?

A. So it's our understanding that the companies historically have generally hit their savings targets related to those programs under budget and on a going forward basis, the commission orders have been pretty clear about what the

commission expects, as far as use of unspent funds toward future program costs.

Q. If -- if the targets can be hit with a lower amount of money, what -- what's the logic in continuing to bill ratepayers at -- at the higher amount?

A. So our goal in these rate cases is to implement the commission-approved budgets and targets. However, it should be noted that the targets going forward for these companies as recently set by the commission reflect a much higher amount of savings per dollar spent.

So we believe the commission orders essentially reflect the past performance of the programs in achieving those savings at the lower rate and incorporates that into the target -- budgets and target -- yes, budgets and targets moving forward.

Q. Okay. Good. Could you make sure I understand this. You -- you said the company is -- the commission has said that targets can be achieved spending less money. Is that what you said as part of the -- the commission order?

A. No, the commission sets the budgets and targets.

Q. Right. But you said -- I thought you said something about the commission is now expecting companies to achieve savings more cost effectively spending less money per -- per quantity of savings.

A. That is our understanding of what the recent commission orders are based on.

Q. Okay. Did the commission also increase the target amount of savings as -- as goals?

A. Yes.

Q. Yeah. Okay.

A. It should also be noted that the commission also implemented the share the savings metric, E.A.M. metric which further puts financial pressure on the utilities to implement the energy-efficiency programs as cost effectively as possible.

Furthermore, the energy-efficiency programs are a downward-only reconciliation. So any money not spent in a particular period will be held over for future customer benefit.

Q. That's going forward, but -- but not in this case?

A. That's how they have the deferrals in the first place.

Q. Okay. But -- but in this case you're -- you're recommending that the unspent monies not be returned to customers, but to be used for more energy-efficiency programs?

A. The energy-efficiency deferrals are used as an offset for the present and future energy-efficiency program costs. So they are in fact essentially being used as a rate moderator specifically for the energy-efficiency program costs.

Q. Okay.

A. So for example, if the energy-efficiency program -- and I'm just making up numbers here is $5 million and there's a $1 million dollar deferral. The total incremental that the company will have to recover is $4 million.

So the customers are in fact receiving the benefit of that offset in this case.

Q. Okay. And -- and you're saying that's -- that's common in -- in all -- with all utilities regarding the EEPS program.

A. I didn't say that, but that is true.

Q. Okay. But -- but if that's the -- if that's the default approach, why -- why is it

even -- why do you recommend that that'd be done in this case? To -- to use the unspent money to offset that gas propo -- you know, gas portfolio budgets in calendar years 2019 and '20?

A. The commission required such action and we believe that the commission order should be followed.

Q. Okay. All right. Could you remind me what ETIP stands for?

A. Efficiency Transition Implementation Plan.

Q. Okay. Thank you. Now, on the bottom of page 27 lines --lines 22 through 24, you mentioned unspent ETIP funds. Are -- are those funds different from the EEPS funds that -- that you mentioned above?

A. Yes, but functionally, no.

Q. Okay.

A. Let me explain that a little bit more. The original energy efficiency budgets were called EEPS, E-E-P-S.

Q. Okay.

A. In 2015, the commission in, I believe, it was the REV Track 1 order changed the

acronym to ETIP and there's also another acronym as a result of the December 2018 energy efficiency order SEEP, S-E-E-P.

I don't remember off the top of my head what that one stands for. System Energy Efficiency Portfolio, SEEP. However, they are all essentially different names of the vintage of energy efficiency program that the utilities have run throughout the years.

A. Okay. What is -- is -- you said C.E.E.P?

A. S-E-E-P.

Q. S-E-E-P, okay. Then -- so is -- is when you -- when you refer to unspent ETIP funds, is that the same money that you were talking about earlier for the unspent EEPS money funds of like $4.869 million for KEDLI and $20.405 million for KEDNY or is it different money?

A. It's different money accrued over a different set of years.

Q. Okay. And do you know how much money when you refer to unspent ETIP funds? Do you know how -- what the dollar quantities are?

A. The total amounts at the time of

our initial testimony are shown in Exhibit S.E.S.P.-2. I don't know what hearing exhibit that is, 423. One of the questions that we had asked the company panel yesterday is for an update of such totals given that these balances reflect the reality as of the time of testimony and these balances have been accruing interest and whatnot since then.

So these will be updated with the results of the company provided I.R. response when it provides such response.

A.L.J. LEARY: And is that my -- is my understanding correct that that response is due 10 days from the 6th, last week?

MR. GOODRICH: No, Your Honor. The response was due on the 6th that was 10 days from when it was asked. It's -- I think that's the one. So they asked for additional time and they've said that they're going to provide it on the 13th which is tomorrow and that -- no, Thursday.

A.L.J. LEARY: Thursday.

MR. GOODRICH: Thursday. And then staff would like to review it to see if we have issues with it. And you know, before we actually asked you to -- to admit it, but --.

A.L.J. LEARY: Logistically, how is this going to work out? Does this panel come back and talk about it or --?

MR. GOODRICH: So my hope is that we won't have any issues with the I.R. -- the I.R. response and we would be able to get it in with an affidavit if -- if necessary, so you don't have to bring witnesses from either of the company's panel or ours back.

But if -- if not, then we'd have to see what -- what the issue is and -- and how we would deal with it.

A.L.J. LEARY: So is the 13th written in stone or can you get it out tonight?

MR. GAVILONDO: I don't know, Your Honor, but I could certainly ask the question on a break.

A.L.J. LEARY: Okay. I had this same concern yesterday and that's why extensions on the eve of the hearing don't work because there are these loose issues that are not addressed in the record potentially, I'm trying to prevent that.

So see what you can do. My request respectfully to the company, and I'm sorry to

interrupt here. I just -- I -- there are six people here that might have to come back and that's the concern I have, and I don't want staff to -- or any party to be faced with the proposition of inconveniencing a panel because the courtesy of an extension was provided to the company.

There is some critical information, this is the second time this information has come up in two days. So it needs to be in the room and provided to all the parties, not by Thursday, but in hand now.

So I'm just asking you to do the best you can so that we don't have to bring people back and have these extraneous issues addressed in the record. Hopefully, we will not have to anyway. So is there any other outstanding discovery that is due?

A.L.J. COSTELLO: Let me just -- let me just ask a question going to that. If I can just refer the panel to the supplemental testimony starting on page 4. I just want to clarify whether what is discussed on page 4 starting at lines -- line 12 and following over to page 6, line 10.

I just want to make sure we're talking about the same information there as what we've just

been discussing.

THE WITNESS: (Mammen) Yes.

A.L.J. COSTELLO: Okay. Thank you.

A.L.J. LEARY: So just for the record, this is D.P.S. 18 -- 1082?

THE WITNESS: The -- the update that we're expecting is D.P.S. 1085.

A.L.J. LEARY: 85.

THE WITNESS: The -- the original --

A.L.J. LEARY: I see it.

THE WITNESS: -- response was 1082.

A.L.J. LEARY: I see.

THE WITNESS: And we asked to follow up.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Okay. Thank you for that clarification. Do the best you can on getting that into the room A.S.A.P. I'm sorry, where were we? Did you have something Mr. --?

MR. GAVILONDO: No, I was just going to agree, Your Honor. Thank you.

A.L.J. LEARY: Okay. Thanks.

MR. RIGBERG: Thank -- thank you, Your Honors.

BY MR. RIGBERG: (Cont'g.)

Q. Now, I have a question about the -- the clean -- the clean energy fund bill-as-you-go collections. Are there -- are -- is there money accumulated in that category as well?

A. (Cully) Yes.

Q. And -- and how much money is -- is -- is -- is there that is sitting unused?

A. At the time of our direct testimony, the bill-as-you-go interest through June 30th, 2019 for KEDLI was $187,000.00 through June 30th, 2019, the bill-as-you-go balance for a KEDNY is $506,000.00 -- sorry, $506,311.00 and those will also be updated in response or as a result of D.P.S. 1085.

Q. Now, is that interest -- what does that interest based on or where does that interest come from?

A. The NYSERDA Clean Energy Fund bill-as-you-go collections and the balances related to that are described in our direct testimony, beginning at line 11 of page 28 through page 29 line 5.

Q. Okay. Thank you. But is the interest -- is there -- is there unspent money that

the interest is associated with? I mean, how does the interest develop if -- if the money is spent as its collected?

A. Beginning on page 28 line 16. Under this methodology, KEDLI and KEDNY collect their authorized share of NYSERDA C.E.F. budget on an ongoing basis, but only transfer their proportional share of NYSERDA's actual expenditures to NYSERDA at the end of each quarter.

This approach results in the company's accruing a balance of C.E.F. funds. Further, in accordance with the company's -- the commission's order authorizing the Clean Energy Fund framework or C.E.F. order issued on January 21st, 2016, in case 14-M-0094 the companies are required to accrue carrying charges on this balance using the other customer provided capital rate.

Q. Okay. Thank you. Can you please turn to page 55 of your direct testimony?

A. We're on it.

Q. Okay. Thank you. I have a question about the -- the current conversion program, which is -- as I understand it is focused on low-income customers?

A. We can speak to what the companies proposed in this case, but not the current version.

Q. Okay. So -- so the panel would not know how many low-income customers have participated in the current program?

A. We do not know that information.

MR. RIGBERG: That's all the questions I have, Your Honor.

A.L.J. COSTELLO: Thank you.

MR. O'BRIEN: Your Honor, just for the record, I -- I would note that response to 1085 was sent this morning around 9:53.

A.L.J. LEARY: Great. Can you get it to us, Judge Costello and I?

MR. GOODRICH: With all due respect, Your Honors, as is normal with discovery, staff would like to review the response and see if it actually wants to add it to the record.

MR. O'BRIEN: Correct. I wasn't implying anything, I'm just answering Your Honor’s question about when it would be sent.

A.L.J. LEARY: I'm not making any ruling on its admissibility. I -- it -- it seems to

me that it fills in a couple of blanks for me in terms of the testimony. If you really don't want us to see it, I need to know why?

MR. GOODRICH: I -- I don't mind it being shared. My concern though is that this is the third or fourth in the line of I.R.s on this. There have been issues with it. Staff and the companies have -- have spoken to try to resolve those issues, but it is not inconceivable that there may still be issues with the response.

And therefore this response in and of itself could only serve to further confuse the record or it could clarify the record, but we would like to see what's there before we -- we -- we bring it in -- we asked to bring it in into the record.

A.L.J. LEARY: Got it. We are -- we are in a hearing now, disputes about discovery are over. If there is a continuing dispute between D.P.S. staff and the company, I -- I would urge you to get an intermediary, not Judge Costello and I, but someone to resolve that --

MR. GOODRICH: There is no dispute.

A.L.J. LEARY: -- if you think that there is a dispute.

MR. O'BRIEN: Yeah, Judge we --

MR. GOODRICH: There is no dispute. This is just --.

A.L.J. COSTELLO: There is a dispute?

A.L.J. LEARY: I'm sorry. I misheard that then.

MR. O'BRIEN: Yeah.

A.L.J. LEARY: I thought that --

MR. GOODRICH: We just said that we --

A.L.J. LEARY: -- there was a potential for a dispute because this is one of several discovery requests on which there had been disputes.

MR. GOODRICH: There -- not disputes in the sense of discovery disputes.

A.L.J. LEARY: Got it.

MR. GOODRICH: Just confusion as to -- we asked a question, the information provided perhaps was not exactly as we have thought we were going to see or raised additional questions and so on and so forth.

A.L.J. LEARY: Okay. So this isn't -- we don't want to answer that or we're not --

MR. MOLONEY: It's -- it's issues

associated with understanding the questions --

A.L.J. LEARY: Got it.

MR. MOLONEY: -- and the answers.

A.L.J. LEARY: Okay. Then my misunderstanding, great. I hope all goes well and there is not a dispute. Well, I would like to know, and I think Judge Costello would like to know is if there is still a problem associated with this particular I.R.

And then we can help by giving you some mediator to deal with it and get everybody on the same page and get it wrapped up before the hearing is over is all I'm suggesting. Our record needs to have the information that you want to have in it, that's --.

MR. GOODRICH: At this time, I don't foresee that being necessary. Should --

A.L.J. LEARY: Okay.

MR. GOODRICH: -- should the -- should something that requires that arise. We would obviously come to Your Honors expeditiously. I think this is just a factor of the response being provided after the majority of these folks were on the stand.

A.L.J. LEARY: Exactly -- exactly, my

point.

A.L.J. COSTELLO: Do you have any sense of when -- when you may have your staff -- people may be reviewing the I.R.s or be able to get back to us or one way or another, whether it's going to come in as an exhibit or not?

MR. GOODRICH: I am assuming that our folks are going to look at it this afternoon when they're -- when they're --.

A.L.J. COSTELLO: I'm not asking for a promise. I'm just trying to get a sense.

MR. GOODRICH: Yeah, I -- I assume they're going to look at it this afternoon. I -- I have in the past underestimated the effort that it requires to look at things, so I don't want to -- want to --

A.L.J. COSTELLO: Again, I just wanted to get a sense of --

MR. GOODRICH: -- on that.

A.L.J. COSTELLO: -- timing. Thank you.

A.L.J. LEARY: So we still have --

A.L.J. COSTELLO: The Environmental Defense Fund.

A.L.J. LEARY: E.D.F. but --

MR. GOODRICH: Your Honors.

A.L.J. LEARY: -- the -- with -- with the schedule -- here is my concern. Can we go off the record for a second, please?

(Off the record 12:04 p.m.)

THE REPORTER: On the record.

A.L.J. COSTELLO: Environmental Defense Fund.

MS. MURPHEY: Thank you. This is Erin Murphy on behalf of Environmental Defense Fund. I wanted to start briefly with a housekeeping matter before I dive into questioning, which is just to clarify that in the Efficiency and Sustainability panel's rebuttal testimony, there are several references to the testimony of E.D.F. witness Ferit Ucar.

And E.D.F. has filed testimony by James Fein who essentially adopted the testimony of Ferit Ucar due to personnel changes at E.D.F. And James Fein's testimony is substantively identical to Ferit Ucar's testimony.

So I'm not going to be asking the -- the panel to look at any pages in the Fein testimony,

but I just wanted to -- to clarify that.

CROSS EXAMINATION

BY MS. MURPHEY:

Q. So my questions will be primarily focused on the efficiency and sustainability panel's rebuttal testimony. Your testimony indicates that you are familiar with the Climate Leadership and Community Protection Act, which I will refer to as the C.L.C.E.P.A. enacted in July 2019 and that took effect on January 1st, 2020. Is that correct?

A. (Cully) Yes.

Q. If you could please turn to page 6 and reference line 8 of the rebuttal testimony. Your testimony states, second, while the C.L.C.P.A. did require that the commission consider whether its decisions would be consistent with meeting the G.H.G., greenhouse gas emissions reductions requirements, it did not specify how the commission must make such discretionary determinations. Is that correct?

A. I'm sorry. Which -- which page and line was it again?

Q. I'm on page 6 --

A. I see it now.

Q. -- line 8, yeah.

A. Yes. That's what it says.

Q. So your testimony indicates that you are aware that under the C.L.C.P.A., the Public Service Commission must consider whether its decisions would be consistent with meeting the greenhouse gas emissions reductions requirements. Correct?

A. Yes.

Q. Do you agree that the commission's order resolving this rate case, or these rate cases will be such a decision subject to the C.L.C.P.A.?

MR. GOODRICH: Objection. Calls for a legal conclusion.

MS. MURPHEY: I would point out that this panel's testimony discusses the C.L.C.P.A. at some length and I'm merely asking a question that's based on their testimony.

A.L.J. COSTELLO: They can answer to the extent, you know, based on their understanding.

MR. GOODRICH: Sure.

A.L.J. COSTELLO: So that's overruled.

BY MS. MURPHEY: (Cont'g.)

Q. Would you like me to restate the question?

A. Yes, please. Thank you.

Q. Do you agree that the commission's order resolving these rate cases will be such a decision subject to the C.L.C.P.A.?

A. I agree that the commission is making a decision and the C.L.C.P.A. has certain requirements of New York State governing bodies with respect to decisions under the C.L.C.P.A. As far as what the commission must do in order to make that decision, I think that's very open ended at this point.

And the thrust of our rebuttal testimony was essentially that the E.D.F. witness recommended one such method that the commission might go about making a decision on those grounds but that it is certainly not the only way the commission could make such a decision and that is -- the commission is free to make whatever determinations it need to -- it needs to, to make such decision for this case.

Q. Thank you. Please refer to the panel's rebuttal testimony at page 8, line 7. I'll give you a minute.

A. I'm there.

Q. The panel's rebuttal testimony states that is, the programs which result from this proceeding should be consistent with both commission policy and the requirements of the C.L.C.P.A., but should not prejudge the recommendations of the Climate Action Council nor make an end run around the process required by the C.L.C.P.A. Is that correct?

A. That's a correct reading of our testimony, yes.

Q. So now -- sorry to make you jump pages. Please refer to page 9 line 7 of the rebuttal testimony. Your testimony states, the commission should decide for itself how to determine whether its actions are inconsistent with the carbon emissions reductions goals set forth in the C.L.C.P.A.

Thus, such a decision is beyond the scope of these rate proceedings which affect only two utilities and address their rates only for the rate year. Is that correct?

A. That's what it says, yes.

Q. Thank you. All right. So are you suggesting here, in fact, that the commission's resolution of this rate case does not have to

consider whether the decision is consistent with the C.L.C.P.A.?

A. That is not what I said.

Q. Could you clarify?

A. The commission will have to make a determination of whether its decisions are consistent with the C.L.C.P.A. We are stating that the commission can make such determination on its own and does not need to rely on this rate case to make those decisions.

Q. Sure. But -- I mean, to be clear, the testimony states that such decision of how to determine whether the commission's actions are consistent or inconsistent with the goals of the C.L.C.P.A. is "beyond the scope of these rate proceedings". Is that correct?

A. That's what it says.

Q. Thank you. And when you say the commission should decide for itself how to determine its own consistency with the C.L.C.P.A., are you suggesting that the parties should not address this issue in the course of this rate case?

A. Rate cases are generally to determine the revenue requirement of one or more

utilities for a given span of a year or more. Generally speaking, we see matters of important state policy being much more -- better suited to statewide generic proceedings as opposed to one off individual utility rate cases.

When you handle issues of such important and individual utility rate cases, you wind up with a hodgepodge of policy decisions, some of which may or may not be very different from one another as a result of the individual outcomes of those rate cases.

Therefore, we believe in general that the high level policy determinations should be made on a statewide generic basis as opposed to an individual utility rate case. So in the sense that this is an individual utility rate case, we believe that the better part or the better venue for such decision making should be a separate different proceeding statewide in front of the commission.

Q. Thank you for that response. I want to circle back a little bit because I -- I hear what you're saying, but I also want to point out that what we've covered a couple minutes ago is that your testimony states that the programs which result from

this proceeding should be consistent with both commission policy and the requirements of the C.L.C.P.A.

So is there a tension between those two statements?

A. No.

Q. Can you clarify?

A. The commission can decide for itself what is or is not consistent in its order on these -- on these cases. To the extent that it is a statewide policy, we believe that the way that those decisions should be made should be an individual utility -- no, I'm sorry.

For statewide policy those decisions should be made as part of a statewide generic proceeding with the results of which implemented in individual utility rate cases. A fantastic example of this very concept is the energy efficiency orders that have come out over the last year in early 2020 with the January 2020 energy efficiency order and in late 2018 with the December 2018 energy efficiency order.

In both of those cases, the commission, through a statewide proceeding,

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We see this again and again and again in things like the storage order of December 2018. The ongoing energy efficient -- sorry, electric vehicle proceeding where the Department of Public Service Staff has issued a white paper that is being considered for com -- party comments -- sorry, public comments and later commission determination, et cetera. All of which will be implemented in utility rate cases.

Utility rate cases are not the place for wide sweeping changes to New York State public policy.

Q. So does this preference for -- excuse me, does this preference for statewide proceedings allow an individual rate case decision to not comply with the C.L.C.P.A.?

A. No.

Q. Can you -- can you explain that answer?

A. The commission will have to comply with the C.L.C.P.A. in any decision that it makes.

Q. Thank you. And circling back a bit, when you say that the commission should decide for itself how to determine its can -- determine its consistency with the C.L.C.P.A.? Are you suggesting that the Department of Public Service staff should not consider this issue in these -- these rate cases?

A. These rate cases result in a set of testimony and either a joint proposal or recommended decision to the commission of what we recommend the commission should do.

In this case, we're saying, commission you should decide for yourself as opposed to coming up with a -- a recommendation as part of this relatively small slice of New York State Utility Industry Regulation.

Q. So let's return to page 9 and at line 11 of your rebuttal testimony, which I read previously. I'm going to read a subset of it. That piece of your testimony states, (Reading) Thus, such decision is beyond the scope of these rate proceedings which affect only two utilities and

address their rates only for the rate year.

So in that statement, when you say that this case addresses only two utilities rates, are you suggesting that the number of utilities addressed has an impact on the applicability of the C.L.C.P.A.?

A. No.

Q. Okay. One moment. So you've talked about this a little bit, but I'd like to clarify because the testimony states that this proceeding only addresses the company's rates. But isn't it correct that in deciding a rate case, the commission addresses a series of terms and conditions of service that include multiple programs like programs for low-income communities, incentives to improve import -- performance, as well as, safety and environmental initiatives such as leak-prone pipe replacement and any other number of -- of topics?

A. Yes, but only for two utilities or for the number of utilities that are currently in for rates in any given proceeding. For example, this case is two utilities, however, other cases have one utility.

Q. And when you say that this

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proceeding only addresses the company's rates for a rate year, isn't it correct that what is decided in this rate case is anticipated to be in place until the companies file and the commission decides another rate case?

A. Yes.

Q. And the company could wait weeks or even months to file a new rate case. Is that correct? Or years?

A. The utility has discretion as to when it files a rate case, is my understanding.

MS. MURPHEY: Thank you. That concludes my cross examination.

A.L.J. LEARY: Lunch?

A.L.J. COSTELLO: Redirect. Do you want to take a short break and then redirect, or do you want --

MR. GOODRICH: Yes, please.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Do you want lunch instead and then redirect or --?

MR. GOODRICH: I think I would like to discuss with the panel because depending on how much redirect we have, if it's not much, I think they

would appreciate being able to leave and not come back after lunch.

A.L.J. LEARY: Yeah. Okay. Let's make this quick. It's I think 12:20 according to my watch and can you come back by 12:30?

MR. GOODRICH: Sure.

A.L.J. LEARY: And make that decision then. Thanks.

A.L.J. COSTELLO: Okay. Off the record.

A.L.J. LEARY: Off the record.

(Off the record 12:23 p.m.)

A.L.J. COSTELLO: Okay. We want to thank the panel for its testimony, and you're excused. Thank you.

THE WITNESS: (Cully) Thank you.

A.L.J. LEARY: Thank you. Off the record -- a couple of questions --

(Off the record 12:33 p.m.)

(On the record 1:23 p.m.)

THE REPORTER: On the record.

A.L.J. COSTELLO: And just because it looks like we've lost a few people. I just want to ask for appearances again and start with the company.

MR. GAVILONDO: Carlos Gavilondo for KEDNY and KEDLI and with me at counsel table are Patrick O'Brien, Kenneth Maloney with the law firm of Cullen and Dykeman and Phillip Decicco.

A.L.J. COSTELLO: And D.P.S. staff.

MR. FORST: Nicholas Forst for D.P.S. staff and with me is Brandon Goodrich.

A.L.J. COSTELLO: Okay. And we have the City of New York.

MR. FUNG: Justin Fung for the City of New York.

A.L.J. LEARY: And it looks like E.D.F.

MS. MURPHEY: Erin Murphey for Environmental Defense Fund and I'm joined by Martin Rothfelder with Rothfelder Stern on behalf of E.D.F.

A.L.J. COSTELLO: Thank you. Is there anyone else making an appearance? No. Okay, thank you.

THE REPORTER: Can we go off the record?

A.L.J. COSTELLO: Sure. We'll go off the record on the record.

(Off the record 1:24 p.m.)

A.L.J. COSTELLO: On the record.

A.L.J. LEARY: On the record.

A.L.J. COSTELLO: And again I just ask you to state your name and business address for the record.

MR. MANGO: You got it. Alex Mango, 40 Sylvan Road, Waltham, Massachusetts.

A.L.J. LEARY: I don't think your microphone is on.

MR. MANGO: Okay. We'll try it again. Alex Mango, 40 Sylvan Road, Waltham, Massachusetts.

A.L.J. LEARY: Thanks.

A.L.J. COSTELLO: Thank you.

MR. MARTIN: Good afternoon. Jeff Martin, 300 Erie Boulevard, Syracuse, New York.

MR. DEMAURO: Dan Demauro, 300 Erie Boulevard, Syracuse, New York.

MR. OLIVE: Steve Olive, 40 Sylvan Road, Waltham, Massachusetts.

MR. RAVIPATY: Mukund Ravipaty, 40 Sylvan Road, Waltham, Massachusetts.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: Spell your last name.

MR. RAVIPATY: R-A-V for Victor, I-P-

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A-T-Y, Ravipaty.

A.L.J. LEARY: Thank you.

A.L.J. COSTELLO: I'll ask you to stand and please raise your right hand. Do you swear or affirm that the testimony you will provide is the truth?

MR. MARTIN: I do.

MR. DEMAURO: I do.

MR. OLIVE: I do.

MR. RAVIPATY: I do.

WITNESS; JEFF MARTIN; Sworn

WITNESS; DANIEL DEMAURO; Sworn

WITNESS; STEVE OLIVE; Sworn

WITNESS; MUKUND RAVIPATY; Sworn

A.L.J. LEARY: Okay. Thank you. You may be seated.

THE REPORTER: Can I just have the spelling of Alex's last name, please?

THE WITNESS: (Mango) Sure. Mango, just like the fruit.

THE REPORTER: Okay. Perfect.

THE WITNESS: Yeah.

THE REPORTER: It's easy.

A.L.J. LEARY: So we have another

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witness here, two, three, four, five, okay.

A.L.J. COSTELLO: Okay.

MR. GAVILONDO: Thank you, Your Honors. I would just like to call the bench's attention to the addition of Mr. Mango to the panel. He was not a sponsor or I'm sure I should say he was not listed on the pre-filed testimony submitted on behalf of this panel.

However, he has supported the work of the panel throughout the process and as we go through the testimony, I believe he will be prepared to adopt the testimony.

A.L.J. COSTELLO: Okay.

DIRECT EXAMINATION

BY MR. GAVILONDO:

Q. Good afternoon, panel.

A. (Panel) Good afternoon.

Q. Do you have before you a document entitled the direct testimony of the Information Technology Panel dated April 2019 consisting of 55 pages, a cover sheet and a table of contents?

A. (Panel) We do.

Q. Okay. And -- and just for clarity, when you do answer a question, please do so

orally so the court reporter can report it, so no shaking of heads up or down, thank you.

Do you also have before you a document entitled correction -- corrections and updates Testimony of the Information Technology Panel dated July 3, 2019 consisting of 10 pages, a cover sheet and a table of contents?

A. (Panel) Yes.

Q. Thank you. And do you also have before you a document entitled rebuttal testimony of the Information Technology Panel dated September 18th, 2019 consisting of 28 pages, a cover sheet and a table of contents?

A. (Panel) Yes.

Q. And were these documents that you just identified prepared by you or under your direction and supervision?

A. (Panel) Yes.

Q. Thank you. If I were to ask you the questions contained in those documents today, would your answers be the same?

A. (Panel) Yes.

Q. And do you adopt those questions and answers as your sworn testimony in these

19-G-0309/19-G-0310 - 2-11-20 - Brooklyn Union Gas proceedings?

A. (Panel) Yes.

Q. Thank you.

MR. GAVILONDO: Your Honors, I'd like to ask that the direct, corrections and updates, and rebuttal testimony of the company's Information Technology Panel be transcribed into the record as if orally given.

A.L.J. COSTELLO: That is granted and at this point the court reporter will insert the following files. KEDNY-KEDLI Information Technology Panel direct testimony. KEDNY-KEDLI Information Technology Panel corrections and updates testimony. And KEDNY-KEDLI Information Technology Panel rebuttal testimony. #orally

Before the Public Service Commission

KeySpan Gas East Corporation d/b/a National Grid and

The Brooklyn Union Gas Company d/b/a National Grid NY

Direct Testimony

of

Information Technology Panel

Dated: April 2019

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I. Introduction and Qualifications

Q. Please introduce the members of the Information Technology ("IT") Panel.

A. The IT Panel consists of Stephen Olive, Daniel J. DeMauro, Mukund Ravipaty, and Jeffrey Martin.

Q. Mr. Olive, please state your full name and business address.

A. My name is Stephen Olive. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid USA Service Company, Inc. ("National Grid Service Company"), a subsidiary of National Grid USA ("National Grid"), as Senior Vice President and US Chief Information Officer ("CIO"). My responsibilities include leading and continuously improving the performance of the IT organization in the US, overseeing internal IT workforce development, and partnering with National Grid's electric and gas distribution businesses to develop, build, and implement new technologies and IT strategies to support business initiatives and customer needs.

Q. Please describe your educational background and professional experience.

A. I graduated from Northeastern University in Boston, Massachusetts in 1987 with a Bachelor of Science in Electrical Engineering Technology. In 1996, I received a Master of Business Administration degree with an IT concentration from Northeastern University. From 2002 to 2011, I had various roles at Raytheon Company including Vice President and CIO, Information Technology - Integrated Defense Systems. At Raytheon, I provided leadership in the design and execution of the IT strategy and ensured that IT investments contributed to Raytheon's business goals and objectives. From 2012 to 2016, I was with Royal Philips, an organization spanning 100 countries with over 117,000 employees and annual revenue over $33 billion dollars. I held various senior positions at Royal Philips, including Global CIO, and led the IT organization on transforming the business by designing end-to-end processes and deploying IT solutions. From 2016 to 2018, I was the Global CIO for Owens & Minor, a leading healthcare industry solutions provider. At Owens & Minor, I helped develop a transformational strategy and deployment roadmap focused on maximizing productivity and cost effectiveness across several integrated supply chain services. In 2018, I joined National Grid as Senior Vice President and US CIO.

Q. Mr. DeMauro, please state your name and business address.

A. My name is Daniel J. DeMauro. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company, as Director, IT Regulatory. My responsibilities include oversight of IT activities in regulatory proceedings for all of National Grid's New York electric and gas distribution operations, including Niagara Mohawk Power Corporation d/b/a National Grid ("Niagara Mohawk"), KeySpan Gas East Corporation d/b/a National Grid ("KEDLI"), and The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") (KEDNY and KEDLI hereafter referred to collectively as "the Companies").

Q. Please describe your educational background and business experience.

A. I graduated from LeMoyne College in Syracuse, New York in 1983 with a Bachelor of Science in Accounting. In 1986, I joined Niagara Mohawk as an Accountant. In 1991, I became Plant Accounting Manager after holding several other positions in the financial accounting department for Niagara Mohawk. In 2006, I became Director, Internal Audit US, for

National Grid Service Company. From 2007 through 2015, I held several positions in the accounting department, including Director – Finance Integration, Director – Balance Sheet Integrity, and Accounting Program Director. In these roles, I was directly involved with several National Grid accounting processes, including oversight of the Fixed Asset Accounting, Cash Accounting, Account Reconciliations, and Revenue Accounting work teams. I was promoted to my current position in January 2016.

Q. Have you previously testified before the Commission?

A. Yes. I have previously testified before the Commission on behalf of Niagara Mohawk in Cases 17-E-0238 and 17-G-0239 (the "2017 NMPC Rate Case").

Q. Mr. Ravipaty, please state your full name and business address.

A. My name is Mukund Ravipaty. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. By whom are you employed and in what capacity?

A. I am employed by National Grid Service Company as Director, Security Services, Design, and Architecture. My responsibilities include overseeing the development of cyber-security strategy and architecture to

ensure that National Grid's cyber and security protections are developed to keep pace with the evolving threats and capabilities of hostile individuals, groups, and nations, thereby enabling and supporting resilient business operations and customer service.

Q. Please describe your educational background and professional experience.

A. In 1998, I received a Bachelor of Science degree in Computer Science from Andhra University in India. In 2018, I received a Master of Business Administration degree at the Massachusetts Institute of Technology Sloan School of Management. I have over 20 years of professional experience focused exclusively on cyber-security. I joined National Grid in 2011. Prior to joining National Grid, I worked across multiple information security domains in the outsourcing, local government, banking, utility, telecom, new media, and internet sectors.

Q. Have you previously testified before the Commission?

A. Yes. I testified before the Commission on behalf of Niagara Mohawk in the 2017 NMPC Rate Case.

Q. Mr. Martin, please state your full name and business address.

A. My name is Jeffrey P. Martin. My business address is 300 Erie Boulevard West, Syracuse, New York 13202.

Q. By whom are you employed and in what capacity?

A. I am employed by the National Grid Service Company as Director, Billing and Revenue Strategy. My current responsibilities include developing customer strategy and designing the future state of National Grid's customer systems.

Q. Please describe your educational background and professional experience.

A. I hold a Bachelor of Science degree in Information Systems Management from the State University College at Buffalo. In 1994, I joined Niagara Mohawk's Information Systems department and transitioned to National Grid following its 2002 acquisition of Niagara Mohawk. Prior to assuming my present responsibilities, I was the Director of Billing Operations for all National Grid operating companies and oversaw billing for the Long Island Power Authority ("LIPA") during the term of National Grid's management services contract with LIPA. I have also managed National Grid's Retail Choice, Meter Data Services, and Accounts Processing departments. I have worked on and managed various aspects

of several major customer system conversion projects, including system conversions for Niagara Mohawk in 1999, New England Electric in 2008, Narragansett Electric (Rhode Island electric and gas) in 2012, and KEDLI in 2013.

Q. Have you previously testified before the Commission?

A. Yes. I have testified before the Commission in several proceedings.

Q. Does the Panel sponsor any exhibits as part of its testimony in this proceeding?

A. Yes. The Panel sponsors the following exhibits that were prepared or compiled under our supervision and direction:

(i) Exhibit __ (ITP-1) presents a graphical depiction of the IT management organization;

(ii) Exhibit __ (ITP-2) presents the allocation codes used to determine the forecast annual rent expense to KEDLI and KEDNY for IT projects in the Rate Year (the twelve months ending March 31, 2021), Data Year 1 (the twelve months ending March 31, 2022), Data Year 2 (the twelve months ending March 31, 2023), and Data Year 3 (the twelve months ending March 31, 2024) (Data Years 1, 2, and 3 are collectively the "Data Years");

(iii) Exhibit __ (ITP-3) presents the operating expenses incurred by KEDLI and KEDNY, respectively, for IT during the twelve months ended December 31, 2018 ("Historic Test Year");

(iv) Exhibit __ (ITP-4) presents in-flight and new IT capital projects and investments planned for the Rate Year and Data Years;

(v) Exhibit __ (ITP-5) summarizes the IT Technology Modernization projects in the Rate Year and Data Years;

(vi) Exhibit __ (ITP-6) presents a list of National Grid's cyber security IT projects in the Rate Year and Data Years;

(vii) Exhibit __ (ITP-7) provides details of incremental operating expenses associated with in-flight and planned IT projects in the Rate Year and Data Years, shown as incremental, project-specific operating expenses and "run-the-business" costs; and

(viii) Exhibit __ (ITP-8) presents the IT operating expenses in the Historic Test Year, as well as projected IT operating expenses for the Rate Year and Data Years.

II. Purpose of Testimony

Q. What is the purpose of the IT Panel's testimony?

A. The IT Panel's testimony describes National Grid's IT function, specifically as it relates to KEDNY and KEDLI. The testimony will

explain the services provided by IT and how IT provides these services. The IT Panel will describe the major IT investments and initiatives during the Rate Year and Data Years, including investments in technology modernization, cyber and physical security, a customer information system replacement, and other investment priorities. Finally, the IT Panel will discuss the operating costs that National Grid and KEDNY/KEDLI will incur in the Rate Year and Data Years for IT services.

III. Structure of and Services Provided by National Grid IT

Q. Please describe the role of the IT function at National Grid.

A. IT services range from critical gas transmission/distribution support systems to standard office desktop applications. These services underpin the safe, reliable, and secure physical and commercial operation of KEDNY and KEDLI's gas distribution businesses. IT also provides a suite of software applications that serve the needs of customers and allow for the effective management and operation of KEDNY and KEDLI. National Grid operates these applications on hardware systems that are centrally located as well as distributed to employees in the form of personal computers and other devices. IT also provides KEDNY and KEDLI with asset and work management systems that provide the administrative support required to manage gas service.

The utilization of information technology is an important tool for utilities to improve the efficiency of energy delivery networks, and will be a central enabler for all parts of network operation. Reducing energy usage, balancing energy demands, and providing more reliable energy delivery can only be achieved by advancements in technology. All this change comes with a new regulatory landscape, and changes in National Grid's risk portfolio.

To prepare for current and future needs, the IT organization is transitioning from being primarily an order taker and service delivery organization to a key partner with the business in developing strategies to address the most pressing business needs. This has resulted in National Grid reskilling its IT workforce to become more customer focused and changing our engagement model with suppliers.

Q. Please summarize the structure of National Grid's IT organization.

A. The IT organization is structured to ensure accountability for achieving its goals. The organization includes four CIO positions (US CIO, UK CIO, Group Functions CIO, and Ventures CIO) who report to the Group Chief Information and Digital Officer. The US CIO is fully responsible for US IT operations including developing the current and future capabilities and

skills of the teams that serve National Grid's US businesses. The US CIO is a member of the executive management team overseeing US business operations and is the face of IT for National Grid's US business, as well as the conduit for presenting business requirements to the overall IT organization.

Reporting to the US CIO are five US IT Business Partners (*i.e.*, Electric, Gas, Transmission and Capital Delivery, Customer, and Finance and Business Services). The Business Partners sit on the management teams of their respective business segments and are responsible for the relationship with the business stakeholders. They also are responsible for long-term planning, development of annual budgets, and ensuring IT programs and projects deliver the promised business benefits and enable the businesses to deliver their commitments to customers.

The Business Partners work with National Grid's IT Delivery Center leaders to ensure that resources with appropriate skill sets are available to meet the forecast demands of National Grid's businesses. Delivery Center leaders are accountable for providing services at agreed levels of cost, time, and quality by developing and implementing common ways of

working across the US. The IT organization contains the following Delivery Centers: Strategy and Innovation; Architecture; Solution Delivery; Infrastructure and Operations; Digital Risk and Security; Commercial and Vendor Management; and Data Management. A simplified graphic depiction of National Grid's IT organization is provided in Exhibit __ (ITP-1).

Q. What services does the IT function provide to the Companies?

A. National Grid's IT function provides, maintains, and manages the computer hardware, software, security (cyber and physical) telecommunications, and other related infrastructure, systems, and services for National Grid's business operations across all National Grid service territories.

Q. Please explain National Grid's approach to IT service delivery.

A. As described earlier, National Grid has a robust IT organizational structure designed to effectively manage the IT needs of the entire business. To deliver on the service needs of the business, National Grid's IT organization utilizes several external partners, each of which performs a specific IT function. These functions include:

- Application development and maintenance: This includes a full range of application services, such as developing new applications and day-to-day support of existing applications.
- Internet, collaboration, and e-mail: This function manages email, web conferencing, instant messaging and collaboration tools, such as SharePoint, operated on vendor-owned and hosted infrastructure.
- Networks and communications: This function manages a single network service that consolidates National Grid's Local Area Network ("LAN"), Wide Area Network ("WAN"), telephony, and video and audio conferencing.
- Data center and client services/enterprise services: This function provides data center services (*e.g.*, servers, data storage); manages hardware, software, and storage located in data centers; back-up capability and disaster recovery services; and client services, such as the provision and support of end user devices (*e.g.,* laptops), and deployment of the operating systems and applications that run on those devices.
- Managed print: This function manages support services for a refreshed and standardized fleet of print devices, enabling increased security for printing, copying, faxing, and scanning.

These core IT services enable the effective operation of the entire organization.

Q. Please describe how the Company develops the IT investment plan.

A. To determine which projects to include in the IT investment plan, IT engages in an iterative process that involves collaboration with representatives of the various jurisdictions (*e.g.*, New York, Massachusetts, Rhode Island, and FERC) and business units (*e.g.,* Gas, Electric, Transmission and Capital Delivery) to identify the highest priority projects. IT collaborates with the jurisdictions and business units to assess infrastructure requirements, sequencing, resource availability and capability, size and impact of investment requests, and the overall benefits of the investments. First, IT considers in-flight multi-year investments and identifies the amount of work remaining and the delivery timeline for those projects. Second, IT adds new investments needed to comply with legal or regulatory requirements/mandates or other critical corporate objectives. Third, IT adds projects to address IT infrastructure - primarily investments to upgrade/replace/add functionality to maintain reliability in systems and establish foundational capabilities required for future operational needs of the businesses. Finally, IT works through the Business Partners to prioritize project requests of the business. After this

process, IT distributes a draft investment plan to the business units, jurisdictional representatives, and other key stakeholders for feedback on the plan. A final investment plan review occurs each fiscal year ("FY") with the Jurisdictional Presidents and business unit leadership. Once the Jurisdictional Presidents and business unit executives review and approve the investment plan, the consolidated financial information from the plan, which is organized by jurisdiction and operating company, is incorporated with the IT operation and maintenance ("O&M") requirements into the IT budget. The IT budget is then reviewed and approved by the US CIO, Global CIO, and Corporate Finance, and incorporated into a consolidated National Grid budget.

IV. IT Cost Allocation and Historic Test Year Costs

Q. How are the costs of IT projects and investments charged to KEDNY and KEDLI?

A. IT capital projects and investments that are shared investments across operating companies are implemented and owned by National Grid Service Company. The costs of these investments are allocated to KEDNY and KEDLI (and the other operating companies that utilize and/or benefit from the investments) in the form of rent expense. Exhibit __ (ITP-2) provides the allocation factors utilized to determine the forecast

annual rent expense for the IT projects planned for the Rate Year and Data Years.

Q. How are the costs of IT capital projects recovered?

A. The costs of IT capital projects are amortized over a period of years. The amortization period is determined by the average service lives of the assets. In general, many of the IT-related assets on the books of the National Grid Service Company are either software intangibles or portable general plant equipment that is being depreciated/amortized and recovered over a period equal to their planned usefulness or expected lives. The actual physical replacement of assets is determined by business decisions to maintain, refresh, replace, and/or decommission assets, and that evaluation is conducted based on variables such as operational risk, cost to be incurred, and the availability of replacement options. Recovery of hardware and equipment and amortization of software intangible assets will approximate the useful life of the new system, taking into consideration technological and functional obsolescence.

Q. What were KEDNY's and KEDLI's IT costs for the Historic Test Year?

A. KEDNY's and KEDLI's IT costs for the Historic Test Year were $45.3 million and $28.9 million, respectively, and consisted of $27.8 million and $18.7 million of operating expenses, as set forth on Exhibit __ (ITP-3), and $17.5 million and $10.2 million of service company rents respectively, as set forth on Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 1.

Q. What are the projected total rent expenses for IT projects and investments to KEDNY and KEDLI in the Rate Year and Data Years?

A. The rent expense in the Rate Year and Data Years is shown in KEDNY and KEDLI's Revenue Requirement Panels' Exhibit __ (RRP-11), Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 2, 3, 5, 6, 8, 9, 11, and 12. As shown in Exhibit ___ (RRP-3), Schedule 9, Page 6, KEDNY and KEDLI are forecasting IT rent expense in the Rate Year of $34.6_million and $22.0 million, respectively. This level of rent expense is driven by new IT capital projects planned to be implemented in the Rate Year and Data Years. Tables 1 and 2, below, summarize the IT rent expense to KEDNY and KEDLI, respectively, for the Rate Year and Data Years, along with the investment areas contributing to the incremental rent expense.

Table 1. KEDNY IT Rent Expense: FY21-FY24

IT KEDNY SC Rents Revenue Requirement				
	RY21	DY22	DY23	DY24
Existing Projects	$ 14,414,762	$ 13,302,817	$ 10,478,537	$ 6,330,145
Technology Modernization	$ 3,527,402	$ 5,894,552	$ 6,088,924	$ 7,270,845
Cyber Security	$ 802,449	$ 1,023,620	$ 1,375,803	$ 1,836,355
Compliance & Mandates	$ 1,388,400	$ 2,930,678	$ 2,502,262	$ 2,952,053
GBE	$ 11,726,441	$ 18,379,685	$ 17,636,185	$ 16,892,685
CIS	$ -	$ -	$ 5,064,783	$ 4,866,501
Other Initiatives	$ 2,736,139	$ 2,706,279	$ 5,483,480	$ 6,678,920
Grand Total	**$ 34,595,594**	**$ 44,237,631**	**$ 48,629,975**	**$ 46,827,503**

Table 2. KEDLI IT Rent Expense: FY21-FY24

IT KEDLI SC Rents Revenue Requirement				
	RY21	DY22	DY23	DY24
Existing Projects	$ 8,646,310	$ 7,980,386	$ 6,246,031	$ 3,813,610
Technology Modernization	$ 2,053,355	$ 2,898,625	$ 3,656,431	$ 4,397,739
Cyber Security	$ 498,734	$ 618,519	$ 855,799	$ 1,142,591
Compliance & Mandates	$ 843,910	$ 1,211,475	$ 1,526,302	$ 1,807,881
GBE	$ 8,585,954	$ 8,336,551	$ 7,991,201	$ 7,645,851
CIS	$ -	$ -	$ 2,350,836	$ 2,258,803
Other Initiatives	$ 1,403,929	$ 2,060,358	$ 3,164,342	$ 3,954,816
Grand Total	**$ 22,032,193**	**$ 23,105,913**	**$ 25,790,941**	**$ 25,021,291**

V. Planned IT Capital Projects and Initiatives

Q. How are IT capital investments presented in these rate filings?

A. This Panel discusses the need for and benefits of IT investments that enable the Companies to provide safe, reliable, and efficient service to customers. The Panel also specifically addresses some major programs and projects covered by the proposals in these filings, including the Technology Modernization program, Cyber Security program, and SAP S/4 HANA project. In addition, the Panel provides support for the

Customer Information System ("CIS") replacement project and Gas Business Enablement ("GBE") program. Primary support for the CIS and GBE programs are provided in the testimony of the Shared Services Panel and Company Witness Christopher J. Connolly, respectively. Exhibit __ (ITP-4) lists the IT capital investment projects included in the Rate Year and Data Years. The exhibit breaks out the projects by program (*e.g.,* cyber security, CIS, Customer Transformation, Corporate IT, Gas Operations, *et cetera*). Although other witness panels discuss the business need for and benefits of some of the significant IT investment programs, the costs of all IT capital investment projects included in the Rate Year and Data Years are supported by this Panel and are included in Exhibit __ (ITP-4). Exhibit __ (RRP-3), Schedule 9, Page 7, to KEDNY and KEDLI's Revenue Requirements Panels' testimony further breaks out the IT investments reflected in these rate filings.

Q. What are National Grid's main IT investment priorities that will affect KEDNY and KEDLI during the period covered by this filing?

A. In addition to the baseline information technology needed to simply run the business, there also are several significant IT programs and initiatives already underway or planned for the period covered by this filing that will benefit KEDNY and KEDLI. These include: (i) Technology

Modernization; (ii) Cyber and Physical Security; (iii) CIS Replacement; (iv) SAP S/4 HANA; (v) GBE program; and (vi) and Customer Transformation. National Grid also is implementing IT projects in support of Gas Operations, Complex Capital Delivery, Compliance and Mandates, Corporate IT, and other work areas that also will benefit KEDNY and KEDLI customers.

A. Technology Modernization

Q. Please describe the Technology Modernization Program.

A. The Technology Modernization Program is an investment program aimed at updating National Grid's IT infrastructure and applications to enable continued reliable and efficient operation. The key investment areas include: the infrastructure and network that are used to compute, store, and transfer data and the business applications that support National Grid's Gas and Electric Operations. National Grid's long-term legacy systems have served it well in the past. However, the IT infrastructure and applications estate has become aged and increasingly prone to failure, resulting in costly disruptions to business operations. Further, because of the fast pace of technology change, many of the applications and infrastructure components within the IT estate are no longer supported by the vendors, such that some software applications are currently supported

by application maintenance vendors on a "best endeavors" basis only, and some replacement hardware is no longer available from the manufacturer and must be sourced from secondary markets. There are risks with operating critical applications and infrastructure that are aged and no longer supported including reduced reliability, resiliency, and maintainability; operational inefficiencies due to manual workarounds; data security issues; decreased service levels and vendor support; and potential system failure. Additionally, National Grid's current IT systems and platforms are heavily customized, lack the flexibility to accommodate new business needs, and will not support near term change and regulatory initiatives.

Investment to replace and upgrade applications and infrastructure components will improve the ability to respond to evolving business and market demands and improve operation of the energy system in the future so that it runs more efficiently, reliably, and safely. Specifically, the IT Technology Modernization Program will deliver the following benefits:

- With respect to applications, the program will:
  - upgrade applications at end of life to current and supported versions
  - retire at-risk applications
  - consolidate core systems to simplify the application landscape

- develop architecture that allows reuse and easy application "plug-in"
- reduce cyber risk through upgrades to operating systems and/or applications
- implement new tools to manage applications and automate business processing
- ensure applications are upgraded in accordance with application roadmaps

- With respect to infrastructure, the program will:
  - remediate high risk, failure-prone assets to ensure business continuity
  - improve network capabilities to enable cloud and mobile adoption, improve cyber-security, and ease maintainability
  - modernize the end user environment to provide new and flexible collaboration capabilities (internally and externally)
  - adopt strategic data centers to mitigate risk and enable robust, scalable infrastructure solutions

Q. What projects are included in the Technology Modernization Program?

A. The Technology Modernization program includes several substantial, multi-year projects planned for the Rate Year and Data Years. Among the more significant projects within the Technology Modernization program are the Application Rationalization project, the Infrastructure Remediation and Lifecycle Refresh project, and the Network Security Infrastructure program.

Q. Please describe the Application Rationalization initiative that is part of the Technology Modernization program.

A. National Grid has a complex application landscape, with over 500 applications. The Company is seeking to rationalize the portfolio through consolidation and decommissioning. Many of these applications have reached, or are reaching, end-of-life and are in dire need of upgrade or replacement. This is impacting the organization by creating needs for intense manual processing (*e.g.*, manual bill creation, US field force time entry). To maintain existing efficiencies, it will be important to consolidate National Grid's application portfolio.

Q. What is the Infrastructure Remediation and Lifecycle Refresh project?

A. The Infrastructure Remediation and Lifecycle Refresh project will deliver refresh and updates to the Data Center infrastructure to ensure that services continue to be appropriately secure and maintain the reliability required by the business. It is estimated that approximately 70 percent of the infrastructure is no longer supported and storage capability is nearing capacity. This investment is needed to maintain the operating environment at a supportable level, and represents a positive step change in reducing infrastructure nearing the end of its useful life.

Q. What is the Network Security Infrastructure program?

A. For network services to remain secure and reliable, it is important that the services are maintained with vendor support and new capabilities are added as technology advances. In addition, the changing threat landscape and expanding use of cloud-based services requires a change from premise-based security appliances towards more agile cloud and virtual security platforms. The Network Security Infrastructure program will deliver capability improvements to network security infrastructure to ensure continued network security and reliability.

Descriptions of all the individual capital projects and programs included in the Technology Modernization Program for the Rate Year and Data Years are provided in Exhibit __ (ITP-5).

Q. What capital costs are associated with the IT Technology Modernization Program?

A. Details by year of the specific project and program capital costs in the Technology Modernization program are provided in Exhibit __ (ITP-4). The capital costs of the Technology Modernization program include

$121.277 million through FY20, $53.342 million in the Rate Year, $53.922 million in Data Year 1, $56.721 million in Data Year 2, and $55.699 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $3.527 million and $2.053 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years.

B. Cyber Security and Physical Security

Q. How are KEDNY and KEDLI addressing cyber security threats?

A. In 2010, National Grid established the Digital Risk and Security organization within the IT organization to protect National Grid's energy networks, IT systems, and confidential company and customer information from cyber security threats. Digital Risk and Security is responsible for identifying and managing cyber risk, development and delivery of secure solutions, identifying, evaluating, and addressing known and emerging threats against National Grid's environments, and implementing a program of cyber security initiatives to proactively identify and protect National Grid from emerging threats.

The Digital Risk and Security organization established a Cyber Security Program outlining a multi-year investment strategy, continuously evaluated based on emerging threats, to enhance National Grid's cyber capabilities. The aim of the Cyber Security Program is to address cyber security threats to the Enterprise, Critical Network Infrastructure ("CNI"), and Operational Technology network systems and environments. The Cyber Security Program focuses on cyber capabilities and the development of safeguards to address key threat areas, including:

- Unauthorized Access / Insider Attack
- System Availability / Malfunction
- Malware / Virus Attack
- Advanced Persistent Threat / External Attack
- Data Leakage / Loss / Privacy
- Regulatory Non-Compliance

Q. Does National Grid work with others on cyber security issues?

A. Yes. National Grid leverages industry and governmental agencies to stay informed of emerging threats. National Grid participates in government information sharing programs that support threat identification and risk reduction. National Grid collaborates with government agencies such as

the Federal Bureau of Investigation, Electricity Information Sharing and Analysis Center, and the US Department of Homeland Security, through the National Cybersecurity and Communications Integration Center. These agencies provide situational awareness on security threats, including alerts, near-real time cyber threat indicator sharing, remediation, incident response, and other security specific resources.

Q. What is National Grid's plan to address cyber security in the Rate Year and Data Years?

A. National Grid's approach to cyber security continues to evolve to keep pace with changing threats across the entire enterprise. Digital Risk and Security's strategy aims to enable the strategic goals and objectives of the business, deliver the appropriate safeguards to protect the organization's most critical assets, ensure a robust cyber security workforce and first line of defense, and establish risk management and governance frameworks to appropriately identify, manage, mitigate, and communicate cyber risk and ensure proportionate measures are delivered to address the most pertinent areas of risk.

Q. What steps does National Grid take to review its cyber security program to ensure it is meeting its objective?

A. National Grid continuously reviews and updates the program to enhance the ability to protect enterprise and CNI systems and operations as technology changes and new threats emerge. National Grid will continue to advance its cyber security capabilities to address threats while also making investments focused on improving the cyber security posture of operational technology systems.

National Grid has adopted the National Institute of Standards and Technology ("NIST") Cyber Security Framework ("CSF") as the methodology to manage cyber risk and associated mitigation plans and improve its cyber posture. The NIST CSF provides a structured approach to identifying and managing risk, providing a comprehensive range of security controls and measures that when implemented will decrease the likelihood of threats of unauthorized access, compromise, disruption, and destruction to core systems and infrastructure. The Cyber Security program will ensure continued deployment of capabilities to enterprise, CNI, and operational technology systems, in alignment with the NIST CSF, to predict and detect attacks, respond effectively to incidents through continued investment in defense-in-depth architectures, and controls to prevent unauthorized attacks and compromise of critical environments.

Q. Please describe the major work streams National Grid is implementing to address cyber security issues.

A. The Cyber Security Program has been categorized into work streams that highlight the primary focus of the initiatives. These are:

Enhance the Foundation Cyber Security Program – Improve foundational cyber security capabilities, security services, and controls to be better postured to respond to cyber threats as threats emerge in alignment with industry leading best practices.

Threat Resistant Networks Cyber Security Program – Implement greater network segmentation, continuing to build upon defense-in-depth to improve redundancy and resiliency of networks, and localize failures and vulnerabilities from affecting the wider enterprise networks.

Robust Identity and Controls Cyber Security Program - Improve and simplify identity management across the organization, to safeguard cloud and application access, prevent insider threats, and deliver actionable identity intelligence of user access and activity.

Secure Endpoints Cyber Security Program - Strengthen endpoints against attack, increase visibility of anomalous endpoint activity, and ensure only known devices are authorized to access the network.

Security Operations and Monitoring Cyber Security Program – Improved monitoring and discovery across data centers, networks, and endpoints. Improved visibility of network activity across the organization and strengthened capability to detect anomalous and malicious activity.

Culture and Awareness Cyber Security Program - Improve cyber security culture and awareness across the enterprise, strengthening the organizations first-line of defense through appropriate training, as well as ensure the organization can recruit, retain, and grow best in-class talent.

Q. What projects are included in the IT Cyber Security Program?

A. The capital projects included in the IT Cyber Security Program for the Rate Year and Data Years are listed in Exhibit __ (ITP-6).

Q. What are the projected costs of the cyber security projects National Grid has planned for the Rate Year and Data Years?

A. The cyber security projects are forecast to require incremental capital investment of $27.841 million through FY20, $10.232 million in the Rate Year, $16.504 million in Data Year 1, $17.729 million in Data Year 2, and $17.329 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $0.802 million and $0.499 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years.

Q. What is the purpose of National Grid's physical security investments?

A. National Grid operates electric and gas energy infrastructure that could be subject to physical attack. Physical security threats include the risk that unauthorized individuals will attempt to enter the Companies' facilities with the intent of vandalizing property, equipment or facilities, theft of tangible or intellectual property, or damage of critical equipment. National Grid's physical security measures are intended to mitigate the risk of such threats by deploying enhanced security systems across the Companies' networks and facilities to detect and deter physical attacks.

Q. Please describe the physical security counter measures in place for KEDNY and KEDLI.

A. There has been a heightened understanding of evolving physical security risks over the past several years. To address this, National Grid significantly increased its deployment of physical security assets, such as facility access card readers, security cameras, and alarm systems, to meet regulatory requirements, enhance customer and employee safety, and safeguard assets. In 2013, National Grid created a centralized organization to address physical security issues, the Physical Security Control Center ("Security Control Center"). The Security Control Center operates around the clock to monitor physical security incidents and provide immediate operations and emergency response for all National Grid facilities, including those of KEDNY and KEDLI. The staff of the Security Control Center disseminates security information and notifications, monitor cameras, and receives and responds to alarms. National Grid has, and is continuing to increase, the number of sites monitored by the Security Control Center.

Q. What are the projected costs of the physical security projects National Grid has planned for the Rate Year and Data Years?

A. The physical security projects are forecast to require incremental capital investment of $2.643 million through FY20, $1.724 million in the Rate Year, $2.125 million in Data Year 1, $2.215 million in Data Year 2, and $2.235 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $0.165 million and $0.106 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years.

C. Customer Information System ("CIS") Replacement Project

Q. Please describe the planned implementation of a new customer information system for KEDNY and KEDLI.

A. As described in the testimony of the Shared Services Panel, National Grid is planning to implement a new CIS to replace its legacy systems. KEDNY currently is supported by the Customer Related Information System ("CRIS"), which is 30 years old. KEDLI is supported by the Customer Service System ("CSS") (the same system that supports Niagara

Mohawk), which is more than 20 years old. These systems are increasingly challenged in meeting the heightened expectations of customers, and the increasing needs of an ever-advancing technology and regulatory billing environment. The Shared Services Panel describes the need to replace these aged systems with a modern, fully functional customer system designed to meet customer expectations and address marketplace and regulatory objectives.

Q. Has the Company prepared a business case that supports the need for the planned CIS replacement?

A. Yes. The Company has prepared a detailed CIS Business Case, which is included in Exhibit __ (SSP-5) to the testimony of the Shared Services Panel.

Q. What is National Grid's projected capital spend for the CIS replacement project?

A. The CIS replacement project is forecast to require incremental capital investment of $46.235 million through FY20, $62.675 million in the Rate Year, $58.666 million in Data Year 1, $41.201 million in Data Year 2, and $39.946 million in Data Year 3. The CIS replacement project is not expected to go into service for KEDNY or KEDLI until Data Year 2

(FY23). Therefore, there would be no Rate Year rent expense associated with this project. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 9 shows the forecast rent expense for the project beginning in Data Year 2 of $5.065 million and $2.351 million to KEDNY and KEDLI, respectively.

D. SAP S/4 HANA Project

Q. What is the SAP S/4 HANA project?

A. National Grid uses SAP ERP as its back-office platform supporting various essential functions, including Finance, Payroll, Human Resources, and Supply Chain. S/4 HANA refers to SAP's Business Suite 4, which is SAP's next generation platform.

Q. How does the S/4 HANA product compare to National Grid's existing SAP platform?

A. National Grid implemented its current version of SAP (SAP ECC 6.0) near the end of calendar year 2012. That implementation was very complex and involved replacing disparate legacy systems that had been in place over many years for multiple companies. Although the current version of SAP is providing accurate and robust information and services

to meet the organization's needs, as discussed below, it will no longer be supported by SAP.

The new S/4 HANA platform is built on a modern, in-memory database and offers personalized user experiences through SAP Fiori. SAP S/4HANA provides a unified financial and management platform to centralize process execution, planning, and reporting based on the same (single source of the truth) data. This solution is equipped with a simple and intuitive user experience, and the solution offers one common, real-time view of financial and operational data to help ensure enterprise-wide consistency and reduce reconciliation time and errors, thus minimizing and/or eliminating the need for human touch-points to enable financial process success.

Q. Why is the S/4 HANA Project proposed now?

A. SAP will no longer support the version of the SAP platform currently deployed at National Grid after 2025. Allowing a critical system such as the organization's back-office financial engine to go out of support carries a high degree of risk. This includes the inability to implement any critical security or compliance patches issued by SAP to address any vulnerabilities discovered, software incompatibility with future products

and services offered by SAP, inability to scale in an increasing digital world to take advantage of leading practices, and overall higher operating costs, decreased performance, and reduced overall reliability. Therefore, National Grid is initiating the effort now to determine the best path forward to transition onto the next generation SAP platform, S/4 HANA.

Q. The 2012 initial SAP implementation did not go smoothly. What confidence does National Grid have that the move to S/4 HANA will go well?

A. The 2012 implementation indeed was challenging. However, National Grid has learned much from that implementation to inform the move to S/4 HANA. As an initial matter, the scope of the 2012 implementation and the S/4 HANA project are vastly different. The 2012 implementation involved replacing disparate, aged systems with a brand-new system. Here, the transition is from an earlier generation of SAP to the latest version of the application. National Grid also is planning for ample upfront review and exploration to determine the optimum solution to be implemented. In addition, the IT organizational structure described previously, along with the scaled agile delivery model, will provide for improved oversight and accountability as well as greater management and control of solution deployment risk. Lastly, as discussed in more detail

below, National Grid is bringing together all change programs such as CIS and S/4 HANA into a central transformation team to better implement and coordinate these programs.

Q. What is the anticipated schedule for developing and implementing the S/4 HANA project?

A. As mentioned above, support for National Grid's current SAP application terminates at the end of 2025. National Grid plans to perform discovery phase work in FY20. Such efforts would consider future process inventories and mapping those future processes to software solutions to limit and rationalize customizations; evaluating onsite vs. cloud hosting options; future technology and infrastructure; organizational and human resources; and sequencing the scope (processes/systems) into potential deployment scenarios.

Following the discovery phase, the project would move into the design phase in FY21. The design phase would refine the accuracy of the operations and capital cost developed during the discovery phase, as well as the timeline of the project. This would include developing design principles and a detailed Target Operating Model; creating an overall release, sprint and test plan; developing a detailed business case, with

benefit profiles, governance and planning; establishing a quality gate process; creating a Program Management Office; and designing an appropriate change management plan.

The results of the design phase would be used in the realize/build and test phase on the project. The build and test phase is anticipated to be structured on the scaled agile model, which will incorporate multiple incremental releases (minimum viable products) to reduce risk and manage the process. However, details of how this would be structured necessarily must await the results on the preceding project phases.

Q. What costs are included for the S/4 HANA project during the period covered by this rate filing?

A. National Grid forecasts $34.650 million of operating expense in the Rate Year for the design phase of the project that it will be undertaking in FY21. This amount is reflected in Exhibits __ (ITP-7) and __ (ITP-8) and contributes to the incremental IT operating expenses allocated to KEDNY and KEDLI in the Rate Year. There are no capital costs included in the Companies' filings for the S/4 HANA project.

Q. Are these the only costs the Company expects to incur during the period?

A. No. However, until the design phase is matured, National Grid will not have a clear estimate of potential project costs. Rather than present a project cost estimate based on insufficient information, the Companies propose to (i) provide the Commission with an update of the costs as part of their corrections and updates filing and (ii) defer for future recovery the differences between the costs recovered in rates and the Companies' actual costs.

E. Gas Business Enablement ("GBE")

Q. Please describe National Grid's GBE Program.

A. The GBE Program is a comprehensive framework of new technology solutions and business process changes necessary to strengthen and improve the performance of National Grid's US gas business. The US gas business faces several challenges, including the need to replace aged computer systems, drive continuous improvement in gas safety performance, deliver an expanding and increasingly complex capital investment program, and meet evolving customer expectations. The GBE Program was developed through a collaboration among the US gas business, IT, and various other functions within National Grid. IT's role

within GBE is to support the delivery of improvements across systems and technology to strengthen operational and safety performance and build a modern platform that supports customer demands. The GBE Program is described in detail in the testimony of Company Witness Christopher J. Connolly.

Q. What costs are associated with the GBE Program in the Rate Year and Data Years?

A. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year rent expense forecast for the GBE Program is $11.726 million and $8.586 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the program to KEDNY and KEDLI in the Data Years.

F. Customer Transformation

Q. Please describe the Customer Transformation Program.

A. The Customer Transformation program (also referred to as Customer Experience Transformation ("CXT")) is a comprehensive program to change how National Grid interacts, serves, and communicates with customers. This program will replace out-of-support platforms to mitigate

existing risk to National Grid's self-service billing, payment, and communications portals. It will set the foundation for the processes and technology changes needed to drive step improvements to the customer experience. The program will focus on re-engineering customers' digital interactions to create a universal and seamless customer experience through multiple service options, including web, mobile, text, email, and future emerging channels.

Q. What capital costs are associated with the Customer Transformation projects in the Rate Year and Data Years?

A. Details by year of the specific project and program capital costs in the Customer Transformation program are provided in Exhibit __ (ITP-4). The aggregate annual capital costs of the Customer Transformation program include $19.460 million through FY20, $6.143 million in the Rate Year, $3.400 million in Data Year 1, $3.000 million in Data Year 2, and $3.000 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $0.662 million and $0.281 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9,

Workpapers 6, 9, and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years.

G. Other IT Capital Work Streams

Q. Please describe the other work streams the IT function will be supporting with capital projects in the Rate Year and Data Years.

A. In addition to the work streams we have previously described, the IT function will be implementing capital projects in support of National Grid's Gas Operations area, Complex Capital Delivery, Compliance and Mandates, and Corporate IT.

Q. Please describe the IT projects associated with Gas Operations.

A. The IT projects in the plan to support Gas Operations are primarily aimed at supporting the new consolidated gas control rooms and upgrades to the hardware and operating systems that are considered to be end of life. Further, the current version of the SCADA application will require an upgrade due to its incompatibility with the new operating systems.

Q. What are the projected capital costs for Gas Operations projects in the Rate Year and Data Years?

A. Details by year of the specific project and program capital costs in the Gas Operations program are provided in Exhibit __ (ITP-4). The aggregate annual capital costs of the Gas Operations program include $19.373 million through FY20, $2.982 million in the Rate Year, $6.500 million in Data Year 1, $5.000 million in Data Year 2, and $5.500 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $1.267 million and $0.622 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years

Q. Please describe National Grid's IT projects in support of the Complex Capital Delivery work stream.

A. National Grid recently created a Capital Delivery organization that is tasked with systematically improving the ability to develop and execute its capital plan more efficiently for complex projects and eventually for the entire capital program. As part of this effort, National Grid is working to identify efficiencies and opportunities for better ways of working that may

reduce costs. Essential to this effort is the updating and upgrading of IT systems that support the capital delivery process.

Q. What are the projected capital costs for the IT Complex Capital Delivery work stream in the Rate Year and Data Years?

A. Details by year of the specific project and program capital costs in the Complex Capital Delivery work stream are provided in Exhibit __ (ITP-4). The aggregate annual capital costs of the Complex Capital Delivery program include $5.711 million through FY20, $4.937 million in the Rate Year, $5.200 million in Data Year 1, $7.013 million in Data Year 2, and $1.240 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $0.182 million and $0.113 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years

Q. Please describe the Compliance and Mandates work stream.

A. This program also includes various support capabilities that are driven by new regulations, safety and reliability requirements and include Gas

Control Simulator and Integration of standard operating procedures with work management. The work stream also includes equipment such as remotely operated flood sensors and fixed communication systems to monitor and control shut-off valves that will be installed in FEMA flood-zones to mitigate the risk of flood damage to the gas facilities.

Q. What are the projected capital costs for the Compliance and Mandates work stream in the Rate Year and Data Years?

A. Details by year of the specific project and program capital costs in the Compliance and Mandates work stream are provided in Exhibit __ (ITP-4). The aggregate annual capital costs of the Compliance and Mandates projects include $25.062 million through FY20, $22.155 million in the Rate Year, $21.659 million in Data Year 1, $21.200 million in Data Year 2, and $21.200 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $1.388 million and $0.844 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years

Q. Please describe the Corporate IT work stream.

A. The size and nature of National Grid's operations, as well as the comprehensive regulatory environment in which it operates, demand that it has comprehensive and sophisticated systems in place to manage and support corporate activities, including human resources, assurance, legal, tax, treasury, audit, claims, and other functions. Investments in this area support that goal. The Future of Finance program is an example of a significant program of work in the Corporate IT work stream. The Future of Finance program will enhance, upgrade, and replace applications within the Finance function (including Tax, Treasury, Audit, Claims, and Assurance) to address such areas as digitization, robotic process automation, workflow tooling, analytics, mobility reporting, and controls. In addition, this program will support strategic assessments and implementations/upgrades of tax software (Sabrix/Vertex), PowerPlan fixed asset system, and Governance Risk and Compliance.

Q. What are the costs for the Corporate IT projects for the Rate Year and Data Years?

A. Details by year of the specific project and program capital costs in the Corporate IT work stream are provided in Exhibit __ (ITP-4). The aggregate annual capital costs of the Corporate IT projects include

$18.668 million through FY20, $25.646 million in the Rate Year, $14.619 million in Data Year 1, $15.011 million in Data Year 2, and $11.600 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $0.208 million and $0.130 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9 and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years.

Q. What other IT capital projects are planned for the Rate Year and Data Years that affect KEDNY and KEDLI?

A. The list of all IT capital projects planned for the Rate Year and Data Years, inclusive of all the projects the Panel previously described, is provided in Exhibit __ (ITP-4).

Q. What are the total projected capital costs of the IT projects shown in Exhibit __ (ITP-4) for the Rate Year and Data Years, and the corresponding Rate Year impacts on KEDNY and KEDLI?

A. The aggregate annual capital costs of the IT projects include $294.901 million through FY20, $194.288 million in the Rate Year, $185.595

million in Data Year 1, $171.090 million in Data Year 2, and $160.561 million in Data Year 3. As shown in Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpaper 3, the Rate Year incremental rent expense forecast for these investments is $8.430 million and $4.785 million for KEDNY and KEDLI, respectively. Exhibit __ (RRP-11), the Workpapers to Exhibit __ (RRP-3), Schedule 9, Workpapers 6, 9, and 12 also show the forecast rent expense for the projects to KEDNY and KEDLI in the Data Years.

VI. Delivering the IT Capital Plan

Q. National Grid is planning to develop and implement a significant number of major IT projects over the next few years. How will National Grid ensure its ability to deliver these projects in a timely and cost-effective manner?

A. National Grid appreciates that today's utilities rely increasingly on information technology, and the pace of change is unprecedented in both the utility industry and in the IT landscape in general. That is why National Grid has made a focused effort to create a robust, well-organized IT organization that can meet the challenges facing it. The structure of the IT organization provides the visibility and controls necessary across the organization to effectively coordinate and manage multiple resources and

projects. IT delivery centers are organized such that resources can be appropriately rationalized to meet prioritized business needs. Further, National Grid's supplier relationships enhance resource availability and flexibility.

Q. Please expand upon the steps being taken to ensure the timely and cost effective delivery of IT projects.

A. The steps being undertaken to ensure timely and cost effective delivery of the IT investments are as follows:

- IT has modified its organizational structure so that it is tightly aligned to the National Grid business units such that the IT Business Partners are accountable for the delivery of programs to time and budget.
- At the same time, the Company has insourced and will continue to insource numerous key technology roles including Project Managers and Enterprise Architects. IT is bringing these roles back in-house to enhance project delivery capability and modernize the solution delivery process. The Architects will provide simplification and standardization across the application portfolio. The Project Managers that are being brought on-board are skilled in modern delivery frameworks which will allow projects to be delivered in releases. resulting in faster time to market and an increase in productivity and

quality.

- In addition, the recent re-negotiation of the Application Development and Application Maintenance contract resulted in an increase to the number of IT solution vendors from two to four; thereby increasing National Grid's overall ability to scale up and respond to the increase in business demand.
- National Grid is bringing together all large change programs such as CIS and SAP S/4 HANA into a central transformation team. This consolidation will help ensure that these transformative investments are delivered into the business on time and in a manner that maximizes value to customers. The creation of a central team will ensure that the multiple change programs are implemented in a coordinated and efficient manner across the organization.

For major projects, such as the ones the Panel describes above, National Grid also establishes individual project control boards. These project control boards manage project scope and work to assure the availability of resources needed to deliver the respective project.

VII. IT Operating Costs in the Rate Year and Data Years

Q. What comprises operating costs for the IT function?

A. In simple terms, IT operating costs consist of: (i) initial project expenses and post-implementation costs incurred with new projects; and (ii) costs of running the projects once they are placed in service. The first category of costs includes O&M incurred to support investment planning during the startup stage and post application development costs that may not be capitalized under applicable accounting rules. The second category of costs are those costs of running a project once it is in service ("run-the-business" or "RTB" costs). These RTB costs include: software licensing and maintenance agreements, hardware maintenance agreements, third party contracts, consultant/contractor costs, and internal labor.

Q. What are the Companies' projected IT incremental operating costs for the Rate Year and the Data Years?

A. KEDNY's and KEDLI's projected incremental operating costs for the Rate Year and Data Years (excluding the costs of the GBE Program, which are presented in the testimony of Company Witness Christopher P. Connolly) can be found in Exhibit __ (ITP-8). These incremental costs consist primarily of incremental IT project operating expenses and RTB costs for new capital projects (details of which appear in Exhibit __ (ITP-

7), incremental operating expenses associated with the CIS Replacement program, design phase costs for the S/4 HANA project, and inflation from the Historic Test Year. The projected amounts of these costs for National Grid appear on page 1 of Exhibit __ (ITP-8).

These costs are then allocated to KEDNY and KEDLI using the allocators identified in Exhibit __ (ITP-8) to arrive at the following incremental IT operating expense amounts:

KEDNY:

- Rate Year: $18.1 million
- Data Year 1: $16.5 million
- Data Year 2: $9.7 million
- Data Year 3: $3.7 million

KEDLI:

- Rate Year: $9.1 million
- Data Year 1: $7.6 million
- Data Year 2: $4.4 million
- Data Year 3: $1.7 million

These incremental operating costs already reflect substantial operating cost savings National Grid is planning to realize in connection with its Accelerate Program, which is described in KEDNY's and KEDLI's Revenue Requirements Panels' testimonies. IT-related Accelerate Program initiatives include the re-negotiation of the Verizon Network and Application Development and Maintenance contracts, and the transition to a new SAP hosting provider. These savings are shown in Exhibit __

(RRP-3), Schedule 34, Workpaper 1, and reflected in the Infrastructure and Operations line on page 1 of Exhibit __ (ITP-8). Absent these savings, the projected incremental IT operating costs would be higher.

VIII. Reporting

Q. What steps do the Companies propose to implement to keep the Commission advised of their progress in implementing the IT investment plan?

A. As part of the 2017 NMPC Rate Case, Department of Public Service Staff proposed that Niagara Mohawk implement periodic reporting of IT project development and investment activity in a manner comparable to the comprehensive periodic reporting already performed for electric and gas capital investment activities. Niagara Mohawk's implementation of those reporting measures have not only served to better apprise Staff of the status of the company's IT investment projects, it has also brought greater internal attention to IT project status and monitoring. Given the beneficial effects such reporting has achieved for Niagara Mohawk, the Companies propose to establish IT-related reporting requirements like those adopted by the Commission in the 2017 NMPC Rate Case. Specifically, the Companies propose that:

(i) at the beginning of each Rate Year, the Companies will file

with the Secretary their IT capital investment plan for that Rate Year, including a narrative explanation of the plan and an identification of all proposed IT projects and their costs;

(ii) they will file IT-related quarterly variance reports within 60 days of the end of each Rate Year quarter that will contain the same information as is being provided by Niagara Mohawk; and

(iii) on a semi-annual basis, the Companies will meet with Department of Public Service Staff to discuss the IT investment plan and the progress being made on fulfilling the goals of the plan.

Throughout this process, the Companies will work in good faith with Staff to identify any additional information that would be useful to include in the quarterly reports.

Q. Does this conclude your testimony?

A. Yes.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of the

Information Technology Panel

Dated: July 3, 2019

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I. Introduction

Q. Please introduce the members of the Information Technology ("IT") Panel.

A. The Panel consists of Stephen Olive, Daniel J. DeMauro, Mukund Ravipaty, and Jeffrey Martin.

Q. Is this the same Panel that previously submitted prepared direct testimony in these proceedings?

A. Yes. This Panel provided direct testimony as part of the Companies' April 30, 2019 filings. The defined terms in the Panel's direct testimony have the same meanings in this corrections and updates testimony.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The Panel's corrections and updates testimony explains and supports certain corrections and/or updates to the Companies' proposals and projected costs of various IT projects included in the Companies' April 30, 2019 base rate filings in these proceedings.

Q. Does the Panel sponsor any exhibits as part of its Corrections and Updates testimony?

A. Yes. The Panel sponsors the following exhibits, all of which were prepared by or under the supervision and direction of one or more members of the Panel:

Exhibit ___ (ITP-4CU) IT Capital Projects;

Exhibit ___ (ITP-7CU) IT Project OPEX Costs; and

Exhibit ___ (ITP-8CU) IT Total and Incremental Operating Expenses.

Q. Please identify the corrections and/or updates presented by the Panel.

A. The Companies have identified the following corrections and updates related to National Grid's IT projects and costs:

i. Correction for the inadvertent omission of an allowance for funds used during construction ("AFUDC") associated with the replacement of the customer information system ("CIS");

ii. Update to the planned in-service dates for CIS;

iii. Update to the capital and operating expense budgets for mandated IT investments anticipated to occur during the Rate Years and Data Years;

iv. Update to reflect revised allocation of costs from project INVP 5130;

v. Update to the in-service date for the IT/OT program, which was identified in the Companies' response to Information Request No. DPS-577;

vi. Update to the costs projected for the SAP S/4 HANA project; and

vii. Correction to include inadvertently omitted run-the-business ("RTB") operating expenses associated with certain projects that went, or will go, into service prior to the end of the Rate Year.

II. Corrections & Updates

CIS Project

Q. Please describe the corrections and updates associated with the proposed CIS project.

A. There are two changes related to the CIS project: (i) inclusion of inadvertently omitted AFUDC; and (ii) an update to the in-service dates initially proposed.

Q. Please describe the correction to include AFUDC in the determination of the cost of the CIS project.

A. Large, longer-term capital projects, such as the planned CIS project, require utilities to finance the cost of development prior to the asset going in-service and becoming used and useful. Utilities in New York are compensated for their costs to finance such projects through the accrual of AFUDC for recovery in rates. In their initial filings in these proceedings, the Companies inadvertently failed to include any AFUDC amount associated with the CIS project in the model used to determine the revenue requirements for KEDNY and KEDLI. The Companies are correcting that error by including the appropriate AFUDC amount in their corrected and updated revenue requirements.

Q. What is the amount of AFUDC included for the CIS project?

A. The total forecast AFUDC amount attributable to the CIS project is approximately $41 million. Of that amount, $26.7 million is associated with projects that go into service during the Rate Years and Data Years, and it is therefore appropriate to include KEDNY and KEDLI's allocable share of this amount in the determination of the Companies' revenue requirements in these proceedings.

Q. Please describe the update to the planned in-service date for the CIS project.

A. As described in the testimony of the Shared Services Panel, National Grid has moved the CIS project start date from September 2019 to April 2020. This change results in a corresponding change of the in-service date for the CIS project from April 1, 2022 to October 1, 2022 for both KEDNY and KEDLI. See Exhibit __ (ITP-4CU), page 1 of 5, which reflects the new planned in-service dates.

Q. What are the revenue requirement impacts from the CIS project AFUDC correction and in-service schedule update?

A. There is no change in the Rate Year or Data Year 1 revenue requirements for KEDNY or KEDLI from the inclusion of the CIS project AFUDC and the update to the in-service date. In Data Year 2, these changes result in a decrease in rent expense to KEDNY and KEDLI of $1.316 million and $0.611 million, respectively. In Data Year 3, these changes result in an increase in rent expense

to KEDNY and KEDLI of $4.098 million and $1.902 million, respectively. Exhibit __ (RRP-11CU), the Workpapers to Exhibit __ (RRP-3CU), Schedule 9, Workpapers 3, 6, 9, and 12 show the updated forecast rent expense associated with the CIS project for KEDNY and KEDLI.

Changes To The Mandated Projects Budget

Q. Please describe the updates to the capital and operation expense budgets for mandated IT investments anticipated to occur during the Rate Year and Data Years.

A. Each year, National Grid is required to invest significant amounts in IT projects needed to implement requirements imposed by governmental authorities. Such authorities include state and federal utility regulators, state or federal lawmakers, taxing authorities, and similar bodies. Based on the experience of being subject to annual emergent mandatory IT obligations, National Grid proactively budgets for such expenditures and includes an amount in its annual budget for anticipated mandatory projects, the details of which are generally not known at the start of the budget year. In their April 30, 2019 filings, the Companies included $20 million per year of capital investment for mandated IT projects across National Grid's US operations for the Rate Year and each Data Year as shown in Exhibit __ (ITP-4), page 1 of 5, and annual operating expenses of between $1.5 million and $2.5 million for such projects as shown in Exhibit __ (ITP-7), page 1 of 2. Subsequent

to the April 30, 2019 filing, National Grid reexamined the inputs and historical information used to estimate the annual mandatory projects capital and operating expense budgets and determined that the capital investment amount should be reduced to $14 million per year for each of the Rate Year and the Data Years, and the operating expense budget should be increased to $6 million per year. These changes are reflected in Exhibit __ (ITP-4CU), page 1 of 5, and Exhibit __ (ITP-7CU), page 1 of 2.

Project INVP 5130

Q. Please describe the update to reflect revised allocation of costs from project INVP 5130 – CIAC Process Systems Enhancements.

A. In their April 30, 2019 filings, the Companies included a project INVP 5130 in Exhibit __ (ITP-4), page 2 of 5, with a total project capital investment of $1 million, and corresponding annual operating expenses shown on Exhibit __ (ITP-7), page 1 of 2. The Companies anticipated that a portion of the costs associated with this project would be allocated to KEDNY and KEDLI. Subsequent to the April 30 filings, the Companies determined that KEDNY and KEDLI would not benefit from this particular project. Accordingly, the project has been removed from the IT capital projects list for KEDNY and KEDLI and its associated costs are being removed from the Companies' revenue requirements.

IT/OT Discovery Project

Q. Please describe the update concerning the status of the IT/OT project.

A. The IT/OT Discovery Project is a cybersecurity project designed to enable National Grid to better protect high value operational assets. Exhibit __ (ITP-4) mistakenly included a projected in-service date of July 2019 for this project; whereas the proper date is February 2021. The date has been updated in Exhibit__ (ITP-4CU)

SAP S/4 HANA

Q. Please describe updates projected for the SAP S/4 HANA project.

A. In the April 30, 2019 filings, the Companies forecast $34.650 million of operating expense in the Rate Year for the design phase of the SAP S/4 HANA project that National Grid will be undertaking in FY21. Although there were no capital costs in the initial filing for this project, the Panel noted (at page 40) that until the design phase had matured, National Grid would not have a clear estimate of potential project costs. The Panel further indicated that it would provide the Commission an update of project costs as part of this corrections and updates filing. Although the design phase has not begun, as a result of work completed during the "Discovery Roadmap" phase of the project, which began after the April 30, 2019 rate filings, National Grid now has a more detailed estimate of

future costs. This estimate will be further refined during the "Design phase" once the project is started.

Q. What work was completed in the Discovery Roadmap phase that resulted in the revised estimate?

A. The project team developed a draft project scope by consulting with various National Grid process areas affected by the project. As part of that review, the team also looked at the integration assumptions, change assumptions, control impacts, capability gaps, improvement opportunities, and implementation considerations for each of the major process areas. This information allowed National Grid to develop an initial proposed solution that will be further refined during the design phase. The Discovery Roadmap phase also produced a brief overview and timeline for the pre-implementation phase of the project as well as potential sequencing options designed to maximize the proposed investment and avoid rework with other existing or future initiatives.

Q. What are the projected costs for the SAP S/4 Hana project?

A. Based on the investigation performed in the Discovery Roadmap phase described above, National Grid estimates capital expenditures of $373.5 million and operating expenditures of $93.5 million over a 5-year period CY20 – CY24. KEDNY and KEDLI's allocated share of these costs is estimated to be $80.6

million in capital expenditures and $20.3 million of operating expenditures. These estimates do not include AFUDC or RTB costs expected to be incurred in the Rate Years and Data Years. Future updates after the Design phase will provide these amounts.

Updated RTB Costs

Q. Please describe the correction to address omitted RTB costs for projects that went or will go into service prior to the end of the Rate Year.

A. RTB costs are the costs of operating and maintaining an IT project once it is in service. Such costs include costs associated with software licensing and maintenance agreements, hardware maintenance agreements, and application support contracts, as well as consultant/contractor costs, and internal labor.

National Grid continually puts new IT projects into service and incurs RTB costs associated with such projects. In developing its operating expense forecast for the April 30, 2019 filing, the Companies failed to identify the RTB costs for the projects that went in-service, or are planned to go in-service, in the time period between the Historic Test Year and the Rate Year. Additionally, the Companies inadvertently omitted RTB costs in Data Years 1, 2, and 3 for certain projects that were properly reflected in the forecast for the Rate Year. The updated RTB costs are reflected on Exhibits __ (ITP-7CU) and (ITP-8CU) and total $28.5 million in

the Rate Year, an increase of $1.7 million and $1.0 million for KEDNY and KEDLI, respectively, above the level included in the Companies' April 30 filings.

Q. Does this conclude the Panel's corrections and updates testimony?

A. Yes, it does.

Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY AND KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of the

Information Technology Panel

Case 19-G-0309
Case 19-G-0310

September 18, 2019

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I. Introduction

Q. Please introduce the members of the Information Technology ("IT") Panel.

A. The Panel consists of Stephen Olive, Daniel J. DeMauro, Mukund Ravipaty, and Jeffrey Martin.

Q. Is this the same Panel that previously submitted direct and corrections and updates testimony in these proceedings?

A. Yes. Capitalized terms defined in the Panel's direct and corrections and updates testimony have the same meanings here.

Q. What is the purpose of the Panel's testimony?

A. The purpose of the Panel's rebuttal testimony is to respond to certain recommendations presented in the testimonies of the following Department of Public Service Staff ("Staff") witness panels: Staff Policy Panel ("SPP"); Staff Consumer Services Panel ("SCSP"); and Staff Information Technology Panel ("SITP").

Q. Does the Panel sponsor any exhibits as part of its rebuttal testimony?

A. Yes. The Panel sponsors the following exhibits, all of which were prepared by or under the supervision and direction of one or more members of the Panel:

- Exhibit ___ (ITP-1R) – Linear Regression Analysis, FY2015-2019.
- Exhibit ___ (ITP-2R) – Service Company Rents Adjustments Reconciliation.

Q. Please summarize the Companies' response to Staff's testimony.

A. The Companies filed rates for the Rate Year commencing April 1, 2020, and submitted additional information to enable the Public Service Commission (the "Commission") to set rates for the subsequent three years (*i.e.*, April 1, 2021 through March 31, 2024, which are collectively referred to hereinafter as the "Data Years" and individually as a "Data Year"). Staff conducted a review of the Companies' IT plans; however, the SITP limited its testimony and recommendations for the Rate Year only. As we explain below, in some instances, the Companies agree with Staff's recommendations. In other areas, however, the Companies disagree with Staff's proposed adjustments. Where the Companies accept an adjustment or agree with an issue raised by Staff, such acceptance or agreement is affirmatively noted. Where the Companies

disagree with Staff's position, or where more information is needed to clarify the record, we indicate that in our rebuttal testimony.

II. Staff's IT Recommendations

Q. What recommendations does Staff make regarding the Companies' IT plans?

A. Staff's recommended adjustments and changes to the Companies' IT plans are summarized at pages 4-5 of the SITP's testimony, which proposes the following:

(i) an adjustment to remove certain projects from the IT capital forecast;

(ii) an adjustment to remove $57.4 million from the GBE Program budget;

(iii) a "forecasting" adjustment to the Companies' IT capital budget forecast;

(iv) an adjustment to the Companies' return on IT capital investments;

(v) a downward-only Service Company Rent IT and GBE Program Net Utility Plant and Depreciation Expense Reconciliation Mechanism; and

(vi) rejection of the Companies' proposal to defer the difference between the projected and actual costs of its SAP S/4 HANA project.

In addition to the foregoing adjustments and recommendations, Staff also proposes a number of changes relating to the process for sanctioning IT projects and performance metrics related thereto, as well as a recommendation for enhanced reporting of and visibility on the Companies' IT efforts.

III. Removal of Projects from IT Capital Forecast

Q. Please describe the SITP's proposal to remove certain projects from the Companies' IT capital forecast.

A. The SITP proposes to remove capital expenditures and operating costs from the Rate Year associated with the Community Distributed Generation ("CDG") Low Income Discount program and the Customer Information System ("CIS") project.

The Companies agree that the costs of the CDG Low Income Discount program should be removed. That program is an electric-only program and was included in the Companies' costs inadvertently. The Companies acknowledged this error in their response to Information Request ("IR") DPS-882 (Exhibit __ (SITP-1, p. 1427)). Although the Companies agree that the costs of the CDG Low Income Discount program should be removed, however, the amount of the adjustment the SITP made for this item is

incorrect. The correct amount of the adjustment is discussed later in the Panel's testimony.

Q. What is the Companies' position regarding the SITP's adjustment to remove CIS costs from the Rate Year?

A. The Companies strongly disagree with the SITP's proposed adjustment. The SITP proposes removing $15.411 million of operating expense associated with the CIS project in the Rate Year. The SITP further proposes to reduce the Service Company's Rate Year capital forecast by $70.2 million to reflect the elimination of CIS-related capital expenditures.

As described by the Companies' Shared Services Panel in its direct testimony, National Grid is planning to implement a new CIS to replace legacy systems that are currently 30 years old for KEDNY and over 20 years old for KEDLI. A new CIS will create business agility to enable the Companies to bring new customer services and billing system options to market quickly and deliver on National Grid's goals of an effortless customer experience with increased choice and flexibility. To the extent that the Companies do not obtain funding for CIS, those benefits will be delayed and the Companies will be exposed to increased risks from having to operate and maintain outdated legacy systems.

While the Companies acknowledge that there is a degree of uncertainty surrounding the final costs of the CIS project and that the Companies continue to look for ways to finalize the project as efficiently as possible, there is no doubt that the Companies need to incur at least the $15.4 million of operating expenses and $70.2 million of capital expenditures forecast for the Rate Year to proceed with the project. Moreover, there is no doubt that, regardless of whether there may be adjustments to the final CIS solution, National Grid will need to expend far more than the CIS-related capital and O&M expenditures that are forecast for the Rate Year to complete the project. Under these circumstances, it is reasonable for the Commission to authorize the CIS-related expenditures that would enable the Companies to proceed with the first phase of the CIS project.

The failure to sufficiently fund the CIS project will only result in further delay in the implementation of a modern solution needed to address the needs of utility customers today.

Q. What is the Companies' position regarding the proposal by the SCSP that the Companies continue to advance the CIS project and file a separate petition at a later date for approval?

A. The Companies disagree with the SCSP's proposal. The Commission will be establishing rates for the Companies for the period beginning April 1, 2020. The Companies are projecting that they will incur substantial costs during the Rate Year and beyond to implement the CIS project. The Companies acknowledge that they continue to refine cost estimates and schedules, but the best information available today supports the conclusion that there will be material expenditures during the Rate Year to progress this project for the benefit of customers. The SCSP's proposal of a separate petition outside of the ongoing rate case process would introduce an element of uncertainty into the Companies' financial planning that is both unreasonable and unnecessary. Given that the best information available at this time indicates the Companies will be making significant expenditures during the Rate Year on the CIS project, and that the Companies filed the necessary cost and schedule information to support recovery of CIS project costs as part of this rate case, the SCSP's proposal that the CIS project costs be handled outside the context of this rate case should be rejected.

IV. $57.4 Million Adjustment to GBE Program Budget

Q. Please address the SITP's proposal to remove $57.4 million from the Companies' GBE Program forecast budget.

A. The SITP proposes an adjustment to reduce the investment forecast for the GBE Program by $57.4 million. As discussed in the Rebuttal Testimony of the Companies' Witness Connolly, the Companies disagree with this proposed adjustment.

V. "Forecasting" Adjustment

Q. Please address the SITP's proposed "forecasting" adjustment.

A. The SITP proposes a "forecasting" adjustment that, combined with Staff's other recommended adjustments to the Companies' IT plans, would reduce the Rate Year IT capital expenditure amount to $129.4 million from the $243.8 million Rate Year forecast proposed by the Companies. The SITP's testimony describes using a linear regression analysis of the Companies' actual IT capital spending in FY 2015, FY 2016, FY 2018, and FY 2019 to arrive at its recommended $129.4 million budget level. The SITP omits the actual results of FY 2017 from its analysis based on its finding that the level of IT capital spending in that year was an "outlier."

The SITP then calculates an additional adjustment factor by: (i) reducing the Rate Year IT capital forecast by $70.2 million related to the CIS capital adjustment discussed above; and (ii) comparing the post-CIS adjusted forecast ($173.6 million) to the $129.4 million result of their linear regression analysis. The result is a 25.5 percent adjustment factor presented by Staff (1 – 129.4/173.6 = 25.5%).

Q. How would the SITP's recommendation affect the Companies' ability to carry out its IT capital plan in the Rate Year?

A. Basically, the SITP's proposed capital spending level would provide funding only for the Companies' baseline technology, Technology Modernization, and Cyber and Physical Security projects, but would provide no capital for the proposed CIS or SAP S/4 HANA projects. The Rate Year capital expenditures forecast for those programs, $70.2 million for CIS and $48.6 million for SAP S/4 HANA, are effectively eliminated from the Companies' IT capital plan by the SITP's adjustments. These major projects are vital to the effective future operations of the Companies and failure to provide any capital funding for them is unreasonable.

Q. Do the Companies agree with the $129.4 million adjusted IT capital level and the forecasting adjustment proposed by the SITP?

A. No. The Companies disagree with the SITP's recommended IT capital level of $129.4 million and the proposed forecasting adjustment for several reasons.

First, the need for incremental IT investment in the utility industry continues to accelerate. The Companies' proposed IT capital investments for 2020 and beyond were not based on their IT budgets in 2015 or 2016. Rather, the budgets were based largely on a bottom-up determination of individual projects that the Companies plan to invest in before the end of the Rate Year, including the CIS and SAP S/4 HANA projects. That analysis is presented on a project-by-project basis on Exhibit __ (ITP-4CU). Establishing IT budgets based primarily on a backward-looking analysis would leave the business insufficiently funded to meet the future needs of customers.

In reviewing the Companies' IT capital budget, the SITP appears to have become overly focused on the Companies' overall annual capital spending as compared to budgets in past years and lost sight of the need for all of the IT projects on which the Companies are proposing to expend capital in the Rate Year. As we explained in our direct testimony, the need for the capital projects listed on Exhibit __ (ITP-4CU) is driven by the fact that many of the critical systems employed by the Companies are nearing the end of their useful lives and must be replaced and upgraded in the near future to meet the

demands of the business. Moreover, many other new systems are required by the business to address operational, financial, regulatory, and other needs. The Companies will not be able to meet the demands of the business or their customers and regulators if their IT capital plan is significantly curtailed. The SITP's forecast adjustment would impose an arbitrary 25.5 percent curtailment on the Companies' non-CIS IT capital programs without any consideration of the impact on individual IT projects or the ability to meet current and future demands for information technology services.

Q. Is there evidence in this case that demonstrates that the need for incremental IT investment will continue in the future?

A. Yes. In testimony in this case, the New York City Policy Panel (at 33-34) recommends further modifications to the Companies' systems to include the ability to collect data on energy efficiency and demand response programs, incentives provided to customers, and data associated with renewable gas projects. Leaving aside the merits of these recommendations, they are nonetheless illustrative of the ongoing and frequently changing demands that are made upon the Companies' IT systems. Such changing demands will continue to require the Companies to invest in their IT systems.

Q. Is National Grid the only utility that has seen a need for increased investment in IT systems in recent years?

A. No. Consolidated Edison Company of New York, Inc.'s ("ConEd") most recent rate filing with the Commission reflected $989 million in requested capital and O&M expenditures for IT projects over a three-year period. Like National Grid, ConEd is proposing a significant level of incremental IT capital spending to meet accelerating business demands. ConEd's proposed IT spending level provides a relevant point of comparison that demonstrates that National Grid's proposal to spend $243 million on IT projects in the Rate Year is reasonably driven by the demands placed upon utilities in today's marketplace.

Q. Does the Panel have any specific concerns with the particular regression analysis performed by the SITP to forecast IT capital expenditures in the Rate Year?

A. Yes. The linear regression analysis used by the SITP provides no useful information concerning what the appropriate level of IT investment should be in the Rate Year. Linear regression is used to determine whether a linear relationship exists in a data set such that the relationship can be used to predict future results. The data set is comprised of points that measure the relationship between at least two variables: an independent variable, which is

expected to be the causative variable, and a dependent variable, which is supposed to be the effect. There is no basis for concluding that a linear relationship exists or should exist between fiscal years and IT capital investment levels. Indeed, the Companies' proposed IT capital investment levels fluctuate up and down year over year depending on the nature of projected capital investments planned for a given year. Therefore, a linear regression analysis is simply not an appropriate tool to use to forecast future investment levels or to develop appropriate levels of IT investment for ratemaking purposes.

Further, for a regression analysis of past performance to be used to predict future results, there should be an expectation that future conditions will be similar to the past conditions used for the analysis. Here, however, the regression analysis covers a period that does not include spending such as proposed for the CIS or SAP S/4 HANA projects, and therefore is not representative of the Company's future investment expectations.

Q. Does the Panel have any concerns with the data used by the SITP in obtaining its regression result?

A. The weaknesses of the SITP's regression analysis are demonstrated by a review of the data itself. While the SITP proposed to analyze the levels of IT

spending in the period 2015 - 2019, the SITP simply ignored the data for one of the five fiscal years, FY 2017, arbitrarily dismissing it as an "outlier." At the same time, however, as the SITP acknowledges (at 32-33), a significant portion of the 2017 IT capital spend characterized as an "outlier" was the result of advancing planned investments on projects in the Companies' Technology Improvement Project from "2018 and beyond." In other words, a significant portion of the FY 2017 actual spend had the effect of suppressing actual spending in FY 2018 and beyond; a result that Staff's regression equation makes no adjustment for in 2018 and 2019.

Q. What result would the SITP's regression analysis have produced if FY 2017 data had been included?

A. The results are shown on Exhibit __ (ITP-1R). Most significantly, the R-squared value of the equation would have dropped from 0.932 in the SITP's results to 0.526; a result the SITP itself would have rejected. Given this result, it is clear that Staff's regression analysis provides no meaningful information concerning what the Companies' actual IT budget should be in the Rate Year or what the actual level of such spending is likely to be.

Q. Do the Companies disagree with Staff's observations that the Companies have underspent their projected IT capital budgets?

A. No. The Companies acknowledge that, over the past five fiscal years, actual total IT capital spend has not met budget. In addition, the Companies do not disagree with the SITP that in FY 2015 - FY 2019, the Companies underspent their annual budget by an average of approximately $8.8 million, or eight percent. However, an eight percent average of budget underspending over the past five fiscal years does not support the SITP's proposed downward IT capital investment adjustment of nearly 47 percent or even the 25.5 percent reduction derived from the SITP's forecasting adjustment.

Q. Have the Companies taken steps to ensure that they will be able to complete the projects as budgeted for the Rate Year?

A. Yes. The SITP unreasonably ignores the efforts that National Grid has undertaken to ensure the timely and cost-effective delivery of the forecast Rate Year capital program. Specifically, as described at pages 50-51 of the Panel's direct testimony, the Companies have insourced numerous key technology roles to enhance project delivery, capability, and modernize the solution delivery process. Moreover, the Companies have increased the number of IT solution vendors from two to four, thereby increasing National Grid's ability to scale up and respond to increases in business demands. The Companies have taken reasonable steps to address past resource deficiencies

that may have prevented them from completing IT capital projects in a timely manner.

Q. Do the Companies agree with the SITP's calculation to adjust the Service Company rent expense to reflect the SITP's recommendations described above?

A. No. Even assuming the SITP described adjustment were adopted, there are several errors in SITP's calculations of how those adjustments would be reflected as shown in Exhibit __(ITP-2R). These errors include: a double count of adjustment amounts, the application of a rate of return ("ROR") for certain adjustments that does not agree with the SITP's proposed ROR, clerical errors, and most notably, the misapplication of the proposed 25.5 percent forecasting adjustment as a direct 25.5 percent reduction to Service Company rent expense.

Q. Do the identified errors have a significant impact on the SITP's proposed adjustment?

A. Yes, even accepting for the sake of argument all of the SITP's recommended reductions, the SITP's total proposed Service Company rent expense reductions of $10.32 million for KEDNY and $7.35 million for KEDLI would be accurately calculated as reductions of only approximately $3.73 million

and $2.91 million for KEDNY and KEDLI, respectively. The SITP has over-adjusted Service Company rent expense by $6.6 million for KEDNY and $4.4 million for KEDLI.

Q. Please describe the errors in the SITP's proposed adjustment to remove $57.4 million from the Companies' GBE Program forecast budget.

A. The SITP's adjustment to remove $57.446 million of forecast GBE Program spending was calculated for both KEDNY and KEDLI using an ROR of 7.515 percent rather than the SITP's proposed ROR of 7.584 percent for KEDNY and 7.543 percent for KEDLI. In addition, there were clerical errors in the SITP's calculation of the rent adjustment specific to the GIS investment and the Maximo investment for KEDLI. These errors resulted in an excess adjustment of approximately $0.155 million for KEDNY and $0.086 million for KEDLI.

The SITP also prorated the $57.446 million adjustment among the individual GBE Program investments based on the capital balances at Fiscal Year 2020. It would have been more appropriate to prorate the removal of the forecast spending based on the proposed FY 2021 incremental spending in which these forecast amounts are included. Although this results in a skewed adjustment

to the individual investment balances, it has no net effect on the combined GBE Program calculated rent expense adjustment.

Q. Please describe the errors in the SITP's proposed adjustment to remove the CDG Low Income Discount program.

A. The full amount of the revenue requirement included in rent expense was $83,406 and $50,405 for KEDNY and KEDLI, respectively. The SITP's proposed adjustment did not remove the full revenue requirement associated with the program resulting in an under adjustment of $21,775 and $13,158 for KEDNY and KEDLI, respectively.

Q. Please describe the errors in the SITP's proposed 25.5 percent forecasting adjustment.

A. The SITP incorrectly applied the proposed forecasting adjustment of 25.5 percent directly to the Service Company rent expense rather than to the forecast Rate Year capital spending associated with the Service Company IT projects. This results in overstating the adjustment to reduce rent expense by approximately $4.446 million for KEDNY and $2.908 million for KEDLI. In response to IR NG-05, the SITP indicated that "Staff did not apply a 25.5% reduction to Service Company Rent expense. Rather, the SITP applied a 25.5% reduction only to the forecasted capital projects portion of service

Company Rent expense." However, this is incorrect. As demonstrated by the SITP's calculation found on page 5 of Exhibit__(SITP-2), Staff applied a 25.5 percent reduction directly to the Service Company rent expense remaining after adjustments to remove the forecast GBE Program contingency capital spending and the CDG Low Income Discount program were taken from the Companies' corrections and updates filed Service Company rent expense of $20.124 million and $13.318 million for KEDNY and KEDLI, respectively. It should also be noted that the SITP's calculation of the Net Service Company rent expense of $17.951 million for KEDNY and $11.731 million for KEDLI on page 5 of Exhibit__(SITP-2) contained a formula link error that resulted in KEDLI's CDG Low Income Discount program adjustment being applied to KEDNY's net Service Company rent expense calculation and vice versa for KEDLI's net Service Company rent expense calculation. The correction of these errors is appropriately reflected in the Companies' corrected calculation presented in Exhibit__(ITP-2R).

To illustrate the magnitude of the SITP's overstatement of the forecasting adjustment, the rent expense reductions proposed by the SITP of $4.572 million for KEDNY and $2.988 million for KEDLI could not be achieved with a complete elimination of the entire forecast rate year IT capital program spending, including all of the forecast Rate Year spending for the GBE

Program. All else equal, removing 100 percent of the forecast capital spending for the Rate Year from the Service Company rent model would only result in a reduction to the Companies' filed Service Company rent expense of $2.902 million for KEDNY and $2.005 million for KEDLI, amounts that are significantly less than the amounts presented by the SITP on Exhibit __ (SITP-2) to reflect the 25.5 percent forecasting adjustment.

Q. Please describe the errors noted in the SITP's proposed ROR adjustment.

A. The SITP's proposed ROR adjustments of $2.443 million for KEDNY and $1.902 million for KEDLI are significantly larger than what would be expected given that small changes in the ROR do not typically yield significant adjustments to Service Company rent expense. The re-calculated ROR adjustments should be $0.435 million for KEDNY and $0.441 million for KEDLI. It appears from the calculation of the SITP's ROR adjustment on page 6 of Exhibit__(SITP-2) that the large discrepancy results primarily from the SITP incorrectly including the adjustments for the removal of GBE Program capital expenditures and the CDG Low Income Discount program in the SITP Forecast Expense balance used to measure the ROR adjustment. This overstates the ROR adjustment and has the effect of double counting these adjustments in the SITP's total Service Company rent expense reductions. To illustrate the magnitude of the SITP's incorrect proposed ROR

adjustments, the service company rent model used to calculate the Companies' filed Service Company rent expense would require a ROR of approximately 3.36 percent for KEDNY and 4.37 percent for KEDLI to achieve the SITP's proposed ROR adjustments (shown in the SITP Forecast Expense balance on page 2 of Exhibit __ (SITP-2)). Such RORs are clearly far below what the SITP has recommended.

Q. The SITP recommends the adoption of a downward only Service Company Rents IT and GBE program Net Utility Plant and Depreciation Expense Reconciliation Mechanism. Do you agree with this recommendation?

A. No. Although a tracker may be appropriate for a multi-year rate plan, such as the plan adopted by the Commission for NMPC, the rationale for such a mechanism is less persuasive for a litigated one-year rate case. Accordingly, the Companies do not support a downward-only tracker for the IT program.

VI. Adjustment to the Companies' Return on IT Capital Investments

Q. Please address the SITP's proposed adjustment to the return applicable to IT capital investments.

A. The Companies disagree with the SITP's proposal to apply the Staff's proposed cost of capital and capital structure to the IT investments of the

Service Company. The matter is addressed in the Rebuttal Testimony of the Companies' Witness Jonathan Cohen.

VII. SAP S/4 HANA

Q. The SITP recommends (at 59-60) that (i) the Companies' rates should reflect the Companies' portion of Service Company's projected operating expenditures of $30.6 million for the SAP S/4 HANA project, (ii) the Companies should be required to manage the capital costs for the project within the context of the SITP's IT capital budget of $129.4 million, and (iii) the Companies' request for deferral of costs in excess of the amount of costs reflected in rates should be rejected. Do you agree?

A. The Companies appreciate the SITP's recognition of the importance of the SAP S/4 HANA project and willingness to include the operating costs of the project in rates. Moreover, the Panel already explained why it disagrees with the SITP's recommendation concerning the amount of the IT capital budget to be reflected in rates.

In addition, given the importance and size of the SAP S/4 HANA project, the Companies request that the Commission grant the Companies authority to defer costs in excess of the current cost projections. As the SITP recognizes, the Companies' cost projection for this project are very preliminary and

therefore there is a possibility that the actual costs of the project will materially differ from the current projection. As an alternative to a blanket deferral authorization, at a minimum, the Companies request the Commission to explicitly determine that the Companies would have the right to seek deferral of costs of the SAP S/4 HANA project in excess of the level reflected in rates by filing a separate petition. Specifically, the Companies request the Commission to confirm that such a deferral petition would not be rejected because there is an allowance for the costs of the SAP S/4 HANA project reflected in rates.

VIII. Proposed Sanction Process Changes and Performance Indicators

Q. Please address the SITP's recommendation that the Companies adopt certain IT-related process improvements.

A. The SITP proposes that (i) IT process improvements implemented for NMPC in Case 17-E-0233 *et al*. be applied to the Companies, and (ii) that Service Company should make improvements to its investment request summary ("IRS") and sanction papers. The Companies generally agree with these recommendations as set forth on pages 67-72 of the SITP testimony.

Q. What is the Companies' position regarding the SITP's recommendation to implement specific performance benchmarks related to sanction papers?

A. The Companies oppose the Staff's recommendation regarding developing performance indicators relating to sanction papers. The metrics Staff proposes would require substantial resources and time to develop and track. Further, although performance benchmarks may be appropriate as part of a negotiated resolution of a multi-year rate plan, there is less of a basis for them in a one-year litigated case. Therefore, unless this matter were to proceed to negotiation that resulted in a multi-year resolution, the Companies oppose establishing performance indicators relating to sanction papers.

Q. The SITP further recommends (at 42) that KEDNY and KEDLI's cyber security program be included in the quarterly IT project update meetings that are held with Staff, and that the Companies notify appropriate Staff as cybersecurity projects get sanctioned or partially sanctioned. Do you agree with their recommendations?

A. Yes.

Q. Please address the SITP's recommendation that the Companies implement enhanced reporting requirements and meet with Staff periodically regarding the IT program.

A. The Companies support periodic reporting on IT project status to Staff. The Companies believe greater transparency on IT project selection, prioritization, timeline, and budget status will promote increased collaboration between the Companies and Staff on the selection and delivery of IT projects and facilitate Staff's review of IT projects as part of rate proceedings for the benefit of New York customers. Indeed, NMPC, KEDNY, and KEDLI provide quarterly reporting and meet with Staff on a regular basis regarding their respective electric and gas capital investments; and similar to the reporting processes for electric and gas assets, NMPC has agreed to provide regular status reports on IT projects as part of the Joint Proposal adopted by the Commission in Case 17-E-0238 *et al*. These regular status reports include:

(1) an IT Capital Investment Plan for each rate year, including a narrative explaining the overall IT investment plan and identifying the proposed IT projects and their estimated costs;

(2) quarterly variance reports on the status and budgets of major projects, details and explanations of budget and timeline variances, reports on budget exceptions, and identification of allocations to NMPC; and

(3) a year-end report that includes, in addition to quarterly reporting, sanction papers for any projects exceeding $1 million that were authorized during the previous Rate Year, and reporting on performance on any IT projects selected for a cost incentive with Staff.

In addition to these regular status reports, NMPC has agreed to meet with Staff on a semi-annual basis to:

- discuss budget and actual spending to date, status of in-flight projects, proposed projects for the next six months;
- review quarterly filings;
- develop mutually agreeable sanction paper enhancements needed to justify IT investments and allow for adequate review; and
- discuss whether different or additional information would be useful to include in the quarterly reports.

The Companies would agree to include KEDNY and KEDLI in these requirements and processes.

Q. The SPP (at 23) recommends that the Companies' adopt the following process improvements:

(1) proposed budgets included in a rate filing should be supported by sanction papers;

(2) once the sanction papers are developed, National Grid's Chief Information Officer ("CIO") should attest to the costs and benefits of the programs that make up the approved budget;

(3) after the projects are closed, National Grid's CIO should attest to and report on the actual costs and benefits achieved through the IT investment plan, and

(4) these improvements should be incorporated into the IT reporting requirements.

Do you agree?

A. Not entirely. The Companies oppose the recommendations for attestation of costs and benefits by National Grid's CIO. From the Companies' perspective, it is not clear what the benefit of such attestations would be or what they might be used for. As we noted previously, the Companies are willing to provide their IT Capital Investment plans and reports on the performance of those plans. This information should be sufficient to permit Staff to make a determination concerning the reasonableness of the costs and benefits of the Companies' IT programs.

With respect to the SPP's proposal that future proposed rate case budgets be supported by projects only with completed sanction papers, the Companies disagree. The dynamic nature of an operating utility results in the maturity of

IT projects within investment plans varying with time. Although the Companies strive to ensure that the information supporting the proposed IT budget reflects the most current and developed information available at the time, it may not be possible or practical to establish going-forward budgets based solely on projects that have completed the National Grid sanction process. This is especially the case where the Companies are attempting to present multi-year spending plans and budgets. Although the Companies are open to collaborating with Staff on opportunities to improve the process for developing IT plan budgets, we do not accept this SPP recommendation.

IX. Conclusion

Q. Does this conclude the Panel's rebuttal testimony?

A. Yes, it does.

MR. GAVILONDO: Thank you, Your Honor.

A.L.J. COSTELLO: Can I just before -- before we begin, I just want to -- I don't know if anyone is going to have any questions about confidential information. Just in case, if your answers -- even if the -- if the person isn't questioning you about it and you need, but you need to respond in your answer relying on confidential information, you have to let us know before you respond because we'll have to -- we may have to clear the room. Okay. You understand?

MR. GAVILONDO: Thank you, Your Honor.

THE WITNESS: (Panel) Yes.

MR. GAVILONDO: Your Honor, may I take a moment, please?

A.L.J. COSTELLO: Certainly.

MR. GAVILONDO: Thank you, Your Honor.

BY MR. GAVILONDO: (Cont'g.)

Q. Now, panel, I turn your attention to your direct testimony and associated with that, did the panel sponsor eight exhibits identified as I.T.P.-1 through I.T.P.-8, which for purposes of the proceeding have been pre-marked for identification as Exhibits 119 through 126?

A. (Panel) Yes.

Q. And for your corrections and updates testimony, did the panel -- panel sponsor three exhibits identified as I.T.P.-4C.U., I.T.P.-7C.U. and I.T.P.-8C.U. respectively, which have been pre-marked for identification as Exhibits 280 through 282?

A. (Panel) Yes.

Q. And with respect to rebuttal testimony, did the panel sponsor two exhibits identified as I.T.P.-1R. and I.T.P.-2R. which have been pre-marked for identification as Exhibits 341 and 342?

A. (Panel) Yes.

Q. Okay. And were all the exhibits that the panel just identified prepared by you or under your direction and supervision?

A. (Panel) Yes.

Q. Thank you.

MR. GAVILONDO: Your Honors, the panel has some additional direct testimony today. We've spoken to staff counsel as well and -- and if it's okay with staff counsel, we'd like to proceed with the additional direct -- and okay with the bench,

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we'd like to proceed with the additional direct today.

A.L.J. LEARY: Okay. Explain why you do not have this included in either the direct -- the corrections/updates or the rebuttal testimony. What -- what happened that it wasn't included prior to this point and you're going to bring it in to the hearing?

MR. GAVILONDO: It's actually an accepting of a position that staff has identified in their testimony and when we submitted our rebuttal testimony, we were not -- that was in September and we now find ourselves in February.

And given the passage of time and the concerns of staff, expressed with respect to the delay on investments and the fact that we're looking at a one-year case, we believe that adopting staff's proposal with respect to tracking of capital investments from the -- I'm sorry, for the information technology investments made by the company would be appropriate and rather than have to go through the process of -- of the witnesses testifying to that on cross-examination.

We'd prefer to do it in -- in direct

if, again, if the staff --

A.L.J. LEARY: Hold on a second.

MR. GAVILONDO: -- counsel. Yes.

A.L.J. LEARY: So you referred to issues raised in staff's testimony. Be more specific about the testimony in which this new direct testimony is -- is being responsive to. What testimony are you referring to that staff submitted that required this?

MR. GAVILONDO: In staff's testimony filed -- let me see.

A.L.J. LEARY: Direct.

MR. GAVILONDO: Direct testimony -- Staff filed a direct and supplemental testimony in this case. Testimony dated August 2019, no day, but August 2019 staff in -- in a few different locations and I can identify them if I have a minute, but --

A.L.J. LEARY: No, it's okay.

MR. GOODRICH: In a few different locations proposed that as a protective measure for customers that a one way downward tracker or reconciling mechanism associated with service company costs for investment technology investments should be implemented and -- and with that, I guess --.

A.L.J. LEARY: You don't need to explain that.

MR. GAVILONDO: Okay.

A.L.J. LEARY: But you had the opportunity to respond to direct, but you didn't do it. And what was the reason for that? Was it that you did not agree with it at the time, but now you do?

MR. GAVILONDO: That is correct, Your Honor.

A.L.J. LEARY: Okay.

MR. GAVILONDO: And the passage of time is -- is -- yes.

A.L.J. LEARY: Got it. So I just want to hear from D.P.S. staff about their position on this proposal that we're bringing in new direct testimony rather than be a cross. Is that acceptable to D.P.S. staff?

MR. FORST: Staff has no objections to the direct testimony on this issue.

A.L.J. LEARY: Okay. And anyone else appearing, do you have any objections to additional direct testimony being submitted by the company?

MR. FUNG: No objection from the City.

MS. MURPHEY: E.D.F. has no objection.

A.L.J. LEARY: Anyone else still in the room that we need to hear from? Okay. You can proceed then. We will allow you to supplement your -- the company's testimony.

MR. GAVILONDO: Thank you, Judge Leary.

BY MR. GAVILONDO: (Cont'g.)

Q. Panel, do the companies have any additional direct testimony today?

A. (Demauro) Yes. In light of staff's concerns -- concerns raised by staff about delays in the implementation of I.T. projects, the companies are willing to agree to a downward reconciliation mechanism with regard to service company rent expense. In our rebuttal testimony, we previously had opposed.

A.L.J. LEARY: Can you give me a reference in the rebuttal testimony to what the witness is citing where there was --

MR. GAVILONDO: Yes, Your Honor.

A.L.J. LEARY: -- a dispute about the downward reconciliation that's now an agreement.

MR. GAVILONDO: I'm -- I'm asking the -- I'm directing the Information Technology Panel, but

I'm looking at my paper copy if that's okay.

A.L.J. LEARY: Yeah, yeah.

MR. GAVILONDO: Okay, okay.

A.L.J. LEARY: Me too.

MR. GAVILONDO: Your Honor, the company's rebuttal testimony dated September 18th, 2019 page 21 of 28 beginning at line 8 refers to the recommendation of the staff I.T. panel that the company's adopt a downward only service company rents I.T. and G.B.E. program, net utility plant and depreciation expense reconciliation mechanism.

And the company's answer in its rebuttal of September 2019 begins at page -- I'm sorry. Yes. Page 21 line 12 and it indicates that our answer is no, we do not agree. Although a tracker may be appropriate for a multi-year rate plan such as the plan adopted by the commission for N.M.P.C. which is Niagara Mohawk Power Corporation.

The rationale for such a mechanism is less persuasive for a let -- litigated one year rate case. Accordingly, the companies do not support a downward only tracker for the I.T. program.

A.L.J. LEARY: Thank you. Very helpful.

MR. GAVILONDO: At this time, I have no further questions for the panel, and I tender the panel for cross examination.

A.L.J. COSTELLO: Just identify yourself. Just identify yourself for the record when you're --.

MR. FORST: Your Honor, I was under -- of the understanding PULP had cross-examination first based on our schedule.

A.L.J. COSTELLO: PULP has waived cross-examination of the I.T. panels.

MR. FORST: Understood. Thank you so much. My name is Nicholas Forst for the Department of Public Service. I have here a response to information request, D.P.S 1036 and I'll pass that around.

A.L.J. LEARY: And do you have the hearing exhibit number, Mr. Forst, to which that information resp -- request is related?

MR. FORST: Your Honors, this is a new exhibit.

A.L.J. LEARY: New exhibit, thank you. So we're up to 616?

A.L.J. COSTELLO: This will be 61 --

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this will be marked for identification as Exhibit 617.

MR. FORST: And if you could just take a couple of minutes to review that. Thank you.

A.L.J. COSTELLO: And just identify the document for the record, please.

MR. FORST: So this is information request, D.P.S. 1036 it was a request from staff to the National Grid Shared Services Panel. It was issued November 7th, 2019 and it was responded to on November 18th, 2019 and it's two pages.

A.L.J. COSTELLO: Thank you.

MR. FORST: All set? Is the panel ready?

THE WITNESS: (Mr. Demauro) Yes.

MR. FORST: Okay.

CROSS EXAMINATION

BY MR. FORST:

Q. Was this response prepared by you or under your supervision?

A. (Panel) Yes.

Q. Okay. And would you say that this is a true and accurate copy of the company's response to information request, D.P.S. 1036?

A. (Panel) Yes.

Q. If you could refer down to the bottom of the first page, the second to last sentence of the response states, The teams continue to work on the coordinated plan for submission of a sanction paper to National Grid Senior Executive Steering Committee or S.E.S.C. during the first quarter of calendar 2020. Is that correct?

A. (Panel) Correct, yes.

Q. And the sanctioning paper as referred to in the response refers to the C.I.S. or customer information system and S/4 HANA I.T. projects. Is that correct?

A. (Panel) Yes. Correct.

Q. Can you explain who the Senior Executive Steering Committee is?

A. (Martin) So the Senior --

A.L.J. LEARY: Speak in to the microphone, please.

THE WITNESS: (Martin) The Senior Executive Steering Committee is made up of key leadership members of the organization. The C.F.O. is the chair for the S.E.S.C. And then there are other members, voting members that sit on the

BY MR. FORST: (Cont'g.)

Q. And you said key leadership. Can you define key leadership from where, what company, et cetera?

A. So they're all National Grid, they're leadership members representing the different functional groups. So you have representation from the Gas Business Unit, the Electric Business Unit, the I.T. Business Unit. So the different business units are represented on the committee.

Q. So in the response it states that the team is planning on submitting a sanction paper to the S.E.S.C. during the first quarter of calendar 2020. Has that submission occurred?

A. Yes, it has.

A. (Demauro) Yes.

Q. On what date did that submission of occur?

A. (Martin) The --

A. (Mango) January 27th.

A. (Martin) 27th.

A.L.J. LEARY: Anyone responding must speak into a microphone. So share the wealth over

there of the microphones.

THE WITNESS: (Mango) Yes, I believe the submission was on the 18th, the actual meeting with the committee was on January 27th.

BY MR. FORST: (Cont'g.)

Q. And did the S.E.S.C. take any action at that meeting on January 27th?

A. (Mango) Yes, the S.E.S.C. approved both papers and recommended that the papers move forward to the group exec for presentation.

Q. And can you elaborate on what that process entails?

A. So that the -- which process?

Q. The process to which you just referred, you said, presented to, I believe, the exec.

A. Yes, yes. So we presented to the S.E.S.C. What we presented in both papers is the overview of both programs stating the articulation of how those programs will be delivered. The costs for those programs, the timelines, the breakdown of programs based on the anticipated phases for implementation, that's what we presented to the S.E.S.C. for approval.

Q. And as part of that submission, there's a -- an anticipated start date associated with the projects. Is that correct?

A. Correct.

Q. And what is the anticipated start date?

A. So the anticipated start dates of the programs would be the first quarter of fiscal '21, which would be April of 2020.

Q. Just bear with me for a second. So I have, here is a -- sorry, Brandon is going to help me. I have a response to information request D.P.S. 699. We're just going to pass that out and it is fairly large, so.

MR. FORST: For Your Honors, this is a new exhibit as well.

A.L.J. COSTELLO: Okay. And that will be marked for identification as Exhibit 618.

A.L.J. LEARY: Don't forget to identify for the record, the document --

MR. FORST: Most definitely.

A.L.J. LEARY: -- date, et cetera.

MR. FORST: I was just letting Brandon get started. However, this response -- this is an

19-G-0309/19-G-0310 - 2-11-20 - Brooklyn Union Gas information request response to information request, D.P.S. 699, it was issued on June 26, 2019 and a response was submitted on July 8th, 2019. It contains eight attachments and is 64 pages.

A.L.J. LEARY: I think you can proceed.

MR. FORST: Thank you, Judges. I was just waiting for the panel to acknowledge that they're --

A.L.J. LEARY: I see.

MR. FORST: -- they're ready. Is the panel ready?

THE WITNESS: (Panel) Yes.

MR. FORST: Okay. Excellent.

BY MR. FORST: (Cont'g.)

Q. The response to D.P.S. 699, was this prepare -- was this response prepared by you or under your supervision?

A. (Panel) Yes.

Q. And you'd say that this is a true and accurate copy of the company -- the company's response to the information request D.P.S. 699?

A. (Panel) Yes.

Q. Okay. I have no direct questions

-- or no cross questions on this document. If I could now turn the panel's attention to page 12 of its direct testimony?

On page 12, the panel discusses the basics of linear regression analysis. Is that correct?

A. (Demauro) Could you repeat the question, please?

Q. Pardon me. If I could direct the panel's attention to page 12 of the staff panel's testimony regarding the basics of linear regression analysis.

A. (Demauro) I don't know that we have staff's testimony in front of us.

Q. If you could then please refer to page 12 of your rebuttal testimony actually, thank you.

A. I'm there.

Q. If you could turn your attention to specifically line 18. On line 18 you just -- you -- the panel describes that linear regression is used to determine whether a linear relationship exists in a data set such that the relationship can be used to predict future results. Is that correct?

A. Yes.

Q. Would the panel stand by the premise that effective prediction of future results is reliant upon the similar -- similarity to past performance?

A. No.

Q. Could you please elaborate why you would no longer stand by that?

A. You're talking about the -- you're talking about an I -- I.T. project function and I think as a basis of our rebuttal testimony, we spoke in terms of looking at past performance as far as, you know, its capital expenditures within a given year, three and four years into the past and -- and utilizing that to establish the capital expenditures that will be needed for future projects.

It doesn't take into consideration, you know, emerging demand and changing business circumstances. So looking backwards with a linear regression analysis really doesn't take in -- into consideration the changing circumstances of a I.T. investment priorities.

Q. So I would just ask again. Would the panel stand by the premise as far as linear

regression analysis is concerned as a general matter that linear regression looks at past performance in order to predict future expenditures?

A. I guess from a purely statistical perspective, I would -- I would grant you that.

Q. Okay. Earlier we discussed the customer information, systems upgrade project in the S/4HANA project. Would the panel agree that those projects are not routine projects?

A. You know the -- the C.I.S. replacement project and the S/4HANA project are projects that, you know, in the normal course of business every software product that the company utilizes eventually needs to be replaced.

So in that regard having to make a cyclical investment to replace, you know, I.T. software that is either going to be potentially unsupported in the future or have other operational risk and issues with it is something that happens in the normal course of business.

The size and complexity of the projects are unique in their nature, but the fact that they eventually need to be replaced is not.

Q. So the panel would agree that

these projects are atypical in terms of how often a project of their scope, size and cost are necessary or undertaken?

A. I will concede that.

A. (Mango) We're agreeing, yes.

A. (Martin) Yes, yes.

Q. And approximately how -- pardon me. Is it not the case that the systems being upgraded or replaced have not been upgraded or replaced in approximately thirty to forty years?

A. (Martin) Yeah, the C.I.S. system is in its 31st year of service for KEDNY. KEDLI has been moved over to our legacy Niagara Mohawk System which has been in service for now 21 years, so yes, extended periods.

Q. So the panel would therefore agree that these projects don't occur regularly, say for instance, yearly or every five years?

A. (Martin) Correct.

A. (Mango) Yes.

Q. If we could please turn your attention to page 13, line 13 in the panel's rebuttal testimony? It states here on line -- page 13, line 13, here however the regression analysis covers a

period that does not include spending such as proposed for the C.I.S. or SAP S/4HANA projects and therefore is not representative of the company's future investment expectations. Is that correct?

Just to be clear for the record, that's lines 13 through 16 on page 13 of the company's rebuttal testimony for the Information Technology panel.

A. (Mango) Correct.

A. (Martin) Yes, I would say that --

A. (Mango) Yes.

A. (Martin) -- that's an accurate reading of it.

Q. So the company has undertaken a joint project like the C.I.S. and S/4HANA project in -- in the past?

A. (Martin) Not a joint project, correct.

Q. And as you stated earlier, the C.I.S. system has not been upgraded or enhanced in the last 30 to 40 years, correct?

A. It's not accurate to say it hasn't been enhanced, you know, the systems are regularly enhanced. It has not been replaced in that

many years, correct.

Q. Okay. So you would say that the expenditures related to C.I.S. and S/4HANA are atypical of previous capital I.T. investment? At least in the last five years?

A. (Mango) Last five years, yes.

A. (Martin) Yes, I would say in the last five years. Yeah.

Q. Regarding the staff -- the panel's analysis of staff's linear regression analysis, on page 14 of its rebuttal testimony, the panel states that it disagrees with staff's exclusion of fiscal year 2017 spending from its analysis. Is that correct?

A. (Olive) That's correct.

Q. And would the panel agree that the fiscal year 2017 spending was the result of technology -- of its technology improvement plan?

A. The fourth quarter of that period represented a period in which the technology modernization program had just been approved by our senior management team and an evaluation was taking place as to what projects that had been, you know, queued up and were ready for implementation could be

pulled forward from fiscal year '18 planning. And as such, we, knowing that we had operational risk and technical debt that had to be addressed as an immediate priority. We pulled those investments forward into fiscal year '17 that were planned for fiscal '18 and beyond. And that's representative of the fact that in fiscal year '18, we had a budget amount and our actuals were significantly under that amount in fiscal '18 and you can see where the overage in '17 is directly connected to the underages against budget in fiscal '18.

A.L.J. LEARY: So just for the record, that would be 2017 and 2018 you are referring to when you say '17 and '18?

THE WITNESS: I'm -- I'm sorry, Your Honor. Yes, we -- we refer to our fiscal years which run from April 1st through March 31st as our fiscal year '17 and '18 even though they cross calendar year boundaries. F.Y. '17 would have run from April 1st of '16 and ended on March 31st of '17, F.Y. 18 --

A.L.J. LEARY: 2017?

THE WITNESS: 2017.

A.L.J. LEARY: Thank you.

BY MR. FORST: (Cont'g.)

Q. Do the companies intend to accelerate their I.T. spending similar to how they did in fiscal year 2017 in the near future?

A. All of our investment decisions are made on evaluations of critical priorities and we have a portfolio calibration board that evaluates on a monthly basis spending priorities for potential emerging walk-ins and walk-outs and we evaluate our -- our investment plan priorities monthly in -- in that regard.

MR. FORST: I have no further questions for the panel.

A.L.J. COSTELLO: Do you have any redirect?

MR. GAVILONDO: Your Honor, if I might take just five minutes.

A.L.J. COSTELLO: Sure.

MR. GAVILONDO: Thank you.

A.L.J. COSTELLO: Go off the record.

A.L.J. LEARY: Wait, before you do that. I just want to make sure I understand the testimony.

A.L.J. COSTELLO: Back on the record.

A.L.J. LEARY: Sorry.

THE REPORTER: I never went off.

A.L.J. COSTELLO: Okay.

A.L.J. LEARY: So the question was will I.T. spending accelerate, I think you are referring to the next few years that -- is that correct?

MR. FORST: Yes, Your Honors. We were asking whether the similar -- whether the I.T. spending acceleration as it occurred in fiscal year 2017 was likely to reoccur or be repeated in some fashion in the next few years.

A.L.J. LEARY: And the response as I understand it from the panel and this is what I need a little bit more information on is that your priorities are established and I believe it was priorities for not just I.T. investment or technology investment, but for other kinds of investment are set sort of on a monthly basis. Is that --

THE WITNESS: (Olive) No. No, Your Honor, that wasn't the intent of the response.

A.L.J. LEARY: Okay.

THE WITNESS: The intent of the response was that the -- the nature and type of projects are evaluated monthly as far as our spending

plans and looking at our actuals on a fiscal year basis, you know, the trend is obviously such that just a few years ago we were spending in the neighborhood of 70 to 80 million dollars a year on I.T., new forecasted I.T. projects.

This year alone through January, we've already spent a $132 million and we will absolutely meet our budget target of a 151 million by the end of this fiscal year which is two months from now. You know, the I.T. organization has reorganized, upskilled, and brought on additional vendor partners to address our ability and capacity to take on, you know, increasing demands of technological improvements across the enterprise to deal with things like, you know, system modernization and other foundational investments and enabling future functionality in -- in customer enablement portals and an engagement with our customers as well as, you know, making sure that aging underlying infrastructure upon which our systems are -- are operational and safe and -- and also there has been an upward trend of our cyber security investments that are required to be made.

So technology is an enabler of

National Grid's business as it pertains to KEDLI and KEDNY is definitely on an upward climb. So to answer that question more fully, yes, you know, our intentions are to continue to increase our investments to address cyber security, technology modernization, you know, gas business enablement and -- and other elements of our -- our overall I.T. technological program to -- to deal with, you know, what's needed to effectively run the business on a safe and reliable basis.

A.L.J. LEARY: I'm -- I'm -- I'm sort of staying with this acceleration that happened in 2017, but I'm taking -- I haven't reached back to look at the previous rate case. I'm taking based on this testimony that this is kind of a -- a new thing that accelerated in 2017 that the company is continuing to pursue. Is that -- is that what --?

THE WITNESS: Your Honor, to fully understand that line of questioning, you have to look back at what transpired in calendar year 2016. The investment plan would have been approved for the following fiscal year at or around April 1st of 2016.

In November of that year, the company pulled together an investment proposal for a multi-

year investment portfolio called technology modernization. The C.I.O. at the time recognized that the company had sweated its underlying infrastructure and asset base for a number of years, and it worked in customer's favor because we were able to maintain our systems and -- and derive as much value out of that underlying infrastructure as we could.

But in 2016 it became apparent that replacement parts and vendor support of software were going to be harder to come by and increase the operational risks to the business. So in November of 2016, our executive committee approved the technology modernization program, which was a multi-year investment portfolio to deal with aging technical infrastructure and unsupported software applications.

Between November of '16 and March of '17, the company began to do preliminary planning and design of projects that could be considered low-hanging fruit and shovel-ready type projects to implement and rather than wait until the following fiscal year in March of 2017, there were enough of those projects that were capable of being addressed and -- and the risk was such that we felt it would

have been imprudent to wait that we pull the trigger on those investments.

In -- in March of 2017 a lot of equipment purchases, and a lot of software projects were launched in March of '17 and -- and those projects if not incurred in March of '17 would have obviously been incurred in -- in fiscal year '18. The -- the company's objection was to the fact that all of the capital expenditures in the year 2017 were excluded from this linear regression analysis which had a punitive effect on the statistical results and -- and brought down the amount of capital expenditures in -- in the forecasting adjustment made by staff.

A.L.J. LEARY: Thank you. I'm going to give -- based upon what just occurred, I would like to give D.P.S. staff another opportunity to pursue anything if you want to pursue it.

MS. FORST: No, Your Honor, we don't need to pursue any further questions.

A.L.J. LEARY: Thanks, redirect.

MR. GAVILONDO: Two minutes, Your Honor.

A.L.J. LEARY: Right, right.

MR. GAVILONDO: Thank you.

A.L.J. LEARY: Sure.

A.L.J. COSTELLO: Go off the record. Redirect.

THE REPORTER: Off the record.

(Off the record 2:04 p.m.)

MR. GAVILONDO: Your Honors, the company has no redirect of the panel. I would like to ask some questions of Mr. Mango to establish his -- his credentials for being on the panel today, if I may? Thank you.

DIRECT EXAMINATION

BY MR. GAVILONDO: (Cont'g.)

Q. Mr. Mango, you are testifying today as part of the Information Technology Panel. Could you please state what your title is with National Grid?

A. Yes. So my title is vice president of finance excellence.

Q. Okay. And what role do you play with respect to the SAP S/4HANA project which is described in the I.T. panel's testimony?

A. So my role is project sponsor. So I am the sponsor of the S/4HANA team that has been

doing the discovery work that was presented in the testimony.

Q. When you say discovery work, discovery has a particular meaning with respect to hearings and -- and rate-case proceedings. What do you mean by discovery work?

A. My apologies. I should have recognized that. Our discovery work is the establishment and implementation of the overall program, so discovery -- by discovery I mean the project planning, preparation, implementation, the phasing of the program, it's directly related to the implementation of S/4HANA National Grid.

Q. Okay. Thank you. And now I have lost my train of thought, I apologize.

MR. GAVILONDO: Pardon me, Your Honors.

A.L.J. LEARY: The reason that -- I think I'm going to help you out.

MR. GAVILONDO: Okay.

A.L.J. LEARY: I hope I am, but I think the last piece is the reason or rationale that this witness was added to the direct rebuttal and corrections and updates testimony.

MR. GAVILONDO: Yes, thank you, Your Honor. That -- that is where I wanted to go.

BY MR. GAVILONDO: (Cont'g.)

Q. So Mr. Mango, with respect to your expertise with the company, why is -- what is your understanding of why you were added to the panel to testify today?

A. So I believe I was added as a result of being the program sponsor for S/4HANA and its implementation, all of the work done to date over the past year-and-a-half as well as my role as -- in finance excellence which includes looking at all finance processes, the redesign and improvement.

Q. Okay. Thank you. And earlier today when the panel first took the stand, I presented to you the testimonies and asked whether or not you adopt the various testimonies, the direct rebuttal, corrections and updates testimony submitted by the Information Technology Panel in this proceeding.

Could you advise the -- the tribunal whether or not you adopt that testimony as your own today?

A. Yes. So I adopt the testimony as

my own.

Q. Thank you.

MR. GAVILONDO: No further questions, Your Honor.

A.L.J. LEARY: Thank you. Thank you to the panel. No other questions or cross-examination or anything else? Thank you very much to the panel from the company. You're excused.

THE WITNESS: (Mango) Thank you.

THE REPORTER: Do you want to go off the record while they --

A.L.J. COSTELLO: Yes, we'll go off the record.

A.L.J. LEARY: Yeah.

(Off the record 2:12 p.m.)

THE REPORTER: On the record.

A.L.J. LEARY: Mr. Goodrich --

MR. GOODRICH: One second.

A.L.J. COSTELLO: And I would ask each of you starting with the gentleman who just sat down to just state your name and business address for the record.

MR. FISHER: Brian Fisher, 3 Empire State Plaza, Albany, New York.

MR. HAUGEN: Keith Haugen, 3 Empire State Plaza, Albany, New York.

MS. PARK: Jennifer Park, 3 Empire State Plaza, Albany, New York.

MR. QUACKENBUSH: Luke Quackenbush, 3 Empire State Plaza, Albany, New York.

MS. KRUGER: Chelsea Kruger, 3 Empire State Plaza, Albany, New York.

A.L.J. COSTELLO: Okay. And can I ask you to please stand and raise your right hand? Do you swear or affirm that the testimony you will provide is the truth?

MR. FISHER: Yes.

Mr. HAUGEN: Yes.

MS. PARK: Yes.

MR. QUACKENBUSH: Yes.

MS. KRUGER: Yes.

Brian Fisher; Sworn

Keith Haugen; Sworn

Jennifer Park; Sworn

Luke Quackenbush; Sworn

Chelsea Kruger; Sworn

A.L.J. COSTELLO: Okay, thank you. You may be seated. Okay.

DIRECT EXAMINATION

BY MR. FORST: (Cont'g.)

Q. Good afternoon, panel. Panel members, before you is a document entitled prepared testimony of Staff Information Technology Panel consisting of a cover page and 73 pages of question and answers dated August 2019 and three exhibits submitted with your testimony labelled SITP-1 through SITP-3 including both redacted and confidential versions of SITP-1. Is that correct?

A. (Panel) Yes.

Q. These exhibits have been pre-marked as Exhibits 477, 478, 479 and 480.

Panel members, before you is also a document entitled prepared supplemental testimony of Staff Information Technology Panel consisting of a cover page and three pages of question and answers dated January 20, 2020 and two exhibits labeled SITP-4 and SITP-5. Is that correct?

A. (Quackenbush) Yes.

Q. And these exhibits have been pre-marked as 518 and 519. Was this set of testimony and exhibits prepared by you or under your direct

supervision?

A. (Quackenbush) Yes.

A. (Park) Yes.

Q. Do you have any changes or corrections to make to that testimony?

A. (Quackenbush) No.

A. (Park) Uh-huh.

Q. If you are asked the same questions today under oath, would you answer them the same way?

A. (Quackenbush) Yes.

Q. And do you affirm that the information contained in your testimony and exhibits true to be -- to the best of your knowledge, information and belief?

A. (Quackenbush) Yes.

A. (Park) Yes.

MR. FORST: Your Honors, I'd like to move that the pre-filed direct and supplemental testimony of the Staff Information Technology Panel be entered into the record as if given orally during the hearing today.

A.L.J. COSTELLO: That's granted and at this point in the record, the court reporter

19-G-0309/19-G-0310 - 2-11-20 - Brooklyn Union Gas should insert testimony entitled D.P.S. Staff Information technology -- Technology Panel direct testimony and D.P.S. Staff Information Technology Panel supplemental testimony. #orally

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

August 2019

Prepared Testimony of:

Staff Information Technology Panel

Jennifer Park
Assistant Engineer

Brian Fisher
Utility Engineering Specialist 2
Office of Electric, Gas and Water

Keith Haugen
Utility Analyst 3 (Cyber)
Office of Resiliency and Emergency Preparedness

Luke Quackenbush
Auditor 2 (Public Utilities)
Office of Accounting, Audits, and Finance

Chelsea Kruger
Utility Analyst 2
Office of Consumer Services

State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Please introduce the members of the Staff Information Technology Panel.

A. The panel members are Jennifer Park, Brian Fisher, Keith Haugen, Luke Quackenbush, Chelsea Kruger

Q. Ms. Park, are you the same Jennifer Park testifying as part of Staff’s Gas Supply Panel?

A. Yes, my credentials are provided in that Panel’s testimony.

Q. Mr. Fisher, did you provide your educational and professional experience in other testimony in this proceeding?

A. Yes, my professional and educational experience are discussed in the prepared testimony of the Gas Infrastructure and Operations Panel.

Q. Mr. Haugen, in what capacity are you employed by the Department?

A. I am employed by the Department of Public Service as a Utility Analyst 3 – Cyber Security, assigned to the Utility Security Section within the Office of Electric, Gas and Water.

Q. Please provide a summary of your educational and professional experience.

A. I have a B.S. in Information Technology from

Empire State College. I am also certified as a Certified Information Systems Security Professional (CISSP) and a GIAC Systems and Network Auditor (GSNA). Beyond that, I have attended numerous courses and workshops on cyber security. My previous professional work experience consists of 5 years as a computer programmer for Newkirk Products, where I started as a junior programmer and worked my way up to senior programmer. I also became supervisor of my unit, overseeing the work of 6-7 programmers of varying skill levels. For two years following Newkirk, I developed workflow applications for Higher Education Systems Corporation as an IT Specialist 2.

Q. Please describe your current duties with the Utility Security Section of the Office of Reliability and Emergency Preparedness.

A. I joined the Utility Security Section in 2008. My current responsibilities include conducting cyber security vulnerability assessments of critical facilities and corporate IT systems, which are owned and operated by the energy, gas, telecommunications and water utilities.

Q. Have you previously testified before the Commission?

A. Yes. I testified on behalf of the Utility Security Section in Case 16-E-0060 and 16-G-0061, Consolidated Edison Company of New York, Inc.

Q. Mr. Quackenbush, are you the same Luke Quackenbush testifying as part of the Staff Accounting Panel?

A. Yes, my credentials are provided in that Panel's testimony.

Q. Ms. Kruger, are you the same Chelsea Kruger testifying as part of the Staff Consumer Services Panel?

A. Yes, my credentials are provided in that Panel's testimony.

Summary of Testimony

Q. What is the purpose of the Panel's testimony in this proceeding?

A. Our testimony will summarize The Brooklyn Union Gas Company d/b/a National Grid NY's (KEDNY) and KeySpan Gas East Corporation d/b/a National Grid (KEDLI), together the Companies, request for its

new Information Technology (IT) programs and projects, discuss our review process, including the review of Cyber Security and Gas Business Enablement (GBE) projects, recommend a number of adjustments related to the proposed IT projects, and make recommendations to improve the transparency of the Company's IT sanctioning and reporting processes going forward. Our testimony will recommend an overall level of IT Capital Expenditures for KEDNY and KEDLI for the Rate Year, and a level of rent expense to be reflected in the Companies' revenue requirements. It is our position that the Companies should spend at the levels they deem appropriate to provide safe and adequate service.

Q. What adjustments are you recommending to the Companies proposed IT investments?

A. We recommend the following revenue requirements adjustments: (1) an adjustment to address the removal of IT projects; (2) an adjustment to address the removal of the GBE contingency amount; (3) a forecasting adjustment to KEDNY and KEDLI's IT capital budget forecast; and (4)

an adjustment to the KEDNY and KEDLI return on IT capital investments; and (5) a downward-only Service Company Rent IT and GBE Program Net Utility Plant and Depreciation Expense Reconciliation Mechanism. .

Q. Please define the Historic Test Year, Rate Year, and Data Years in these proceedings.

A. The Historic Test Year is the twelve-month period ending December 31, 2018. The Rate Year is the twelve-month period ending March 31, 2021. The Data Years, One, Two, and Three, are subsequent twelve-month periods ending March 31, 2022, March 31, 2023, and March 31, 2024, respectively.

Q. What are KEDNY’s and KEDLI’s fiscal years (FY)?

A. The Companies’ FY runs from April 1 to March 31. They are identified by the calendar year in which the FY ends, such that FY 2021 begins on April 1, 2020, ends on March 31, 2021, and is identical to the Rate Year.

Q. In your testimony, will you refer to, or otherwise rely on, any information obtained during the discovery phase of this proceeding?

A. Yes. We rely on several responses provided by

the Company to information requests, or IRs. These responses are included in Exhibit___(SITP-1) and will be identified using the reference number originally assigned by the Department. For instance, the Department's first IR was identified as "DPS-001."

Q. Is the Panel sponsoring any other exhibits?

A. Yes, we are sponsoring the following additional exhibits: Exhibit___(SITP-2), which includes a summary of, and calculations related to, the Panel's proposed adjustments, and Exhibit___(SITP-3), which shows the GBE adjustment.

The Companies' IT Proposal

Q. What is the role of Information Technology (IT) at National Grid?

A. According to pages 9 and 10 of the pre-filed direct testimony of the Companies' IT Panel, IT serves a wide range of services, from critical gas transmission / distribution support systems to standard office desktop applications, and is vital for the "safe, reliable, and secure physical and commercial operation of KEDNY and

KEDLI's gas distribution business". IT also provides the necessary software applications to serve the customer needs and to efficiently and effectively manage and operate KEDNY and KEDLI.

Q. What does IT provide to the Companies?

A. On page 12 of the pre-filed direct testimony of the Companies' IT Panel, IT "provides, maintains, and manages the computer hardware, software, security (cyber and physical) telecommunications, and other related infrastructure, systems, and services for National Grid's business operations across all National Grid service territories".

Q. Do the Companies develop their own IT projects and investments?

A. No, as most of the IT projects are used by multiple operating companies and subsidiaries of National Grid, IT projects are designed and accounted for by National Grid USA Service Company (Service Company). The associated project costs are allocated to the appropriate operating companies using the allocation factors showed in Exhibit___(ITP-2). The associated project costs for each operating company is in

the form of rent expense.

Q. Describe the total Service Company Rent expense for IT projects and investments to KEDNY and KEDLI in the Historic Test Year and Rate Year.

A. The Service Company Rent expenses associated with IT projects and investments to KEDNY and KEDLI in the Historic Test Year was $27.1 million and $21.3 million, respectively, as shown in Exhibit___(RRP-3CU), Schedule 9. The projected rent expense associated with total IT costs and investments for KEDNY and KEDLI in the Rate Year is $43.7 million and $32.2 million, respectively, as set forth in Exhibit___(RRP-3CU). The Companies claim that the major driver of the increase in rent expense are the forecasted IT capital projects.

Q. What are the types of IT capital programs that the Service Company proposed and why are these investments necessary?

A. There are several types of IT capital programs that the Service Company proposes in this rate case: Technology Modernization; Cyber and Physical Security; Customer Information System (CIS) Replacement; SAP S/4 HANA; Gas Business

Enablement (GBE) Program; Customer Transformation; and other IT capital work streams.

First, Technology Modernization is a program that is aimed to update “National Grid’s IT infrastructure and applications to enable continued reliable and efficient operation”. Some of the major projects in this program include the Application Rationalization project, the Infrastructure Remediation and Lifecycle Refresh project, and the Network Security Infrastructure program.

Second, the Cyber Security Program is aimed to enhance National Grid’s cyber capabilities and “address cyber security threats to the Enterprise, Critical Network Infrastructure (“CNI”), and Operational Technology network systems and environments.” The Physical Security Program is intended to mitigate physical security threats, such as vandalizing property, equipment or facilities, by distributing enhanced security systems across National Grid’s networks and facilities.

Third, the proposed CIS will replace its aging

legacy system in order to have a “fully functional customer system designed to meet customer expectations and address marketplace and regulatory objectives”. A detailed Business Case for CIS is shown in Exhibit___(SSP-5) in the Company’s Shared Service Panel.

Fourth, the SAP S/4 HANA project will replace the Company’s current SAP ECC 6.0, which is a back-office platform that supports various functions, such as Payroll and Human Resources. The main reason for the timing of this decision is because SAP will no longer support the current version National Grid is using after 2025.

Fifth, the GBE Program is “a comprehensive framework of new technology solution and business process changes necessary to strengthen and improve the performance of National Grid’s US gas business.”

Sixth, the Customer Transformation Program would change how National Grid interacts with customers. The program will help “mitigate existing risks to National Grid’s self-service billing, payment, and communications portals”

while improving customer experience through different service options, i.e., web. Seventh and finally, IT will be implementing projects in the following areas: Gas Operations, to help consolidate gas control rooms and upgrades to hardware and operating systems, including current SCADA system; Complex Capital Delivery, to update the IT systems that support the capital delivery service in order to further improve developing and executing the capital plan; Compliance and Mandates, to account for necessary implementation of support capabilities and equipment which are driven by new regulations and safety and reliability requirements; and Corporate IT, to ensure systems are in place to manage and support corporate activities, including Human Resources, assurance, legal, tax, treasury, audit, claims, and other functions.

Q. Are the IT investments divided into spending categories?

A. Yes. As shown in Tables 1 and 2 on page 18 of the Companies' pre-filed Informational Technology Panel testimony, IT Investments are

broken down into the following six categories: Existing Projects, Technology Modernization, Cyber Security, Compliance & Mandates, Gas Business Enablement or GBE, Consumer Information System or CIS, and other initiatives.

The Development of the IT Investment Plan

Q. Describe the Service Company's proposed IT capital spending plan for the Rate Year and FYs 2022 through 2024, which the Companies refer to in their filings as Data Years One, Two and Three.

A. As shown in the Exhibit___(ITP-4CU), the proposed IT spending levels are $243.8 million, $382.8 million, $180.8 million and $299.2 for the Rate Year, FY 22, FY 23, and FY 24, respectively, for the Service Company.

Q. Describe the Companies' Corporate IT budgeting process.

A. As stated in the Companies' response to IR DPS-432, the Companies state that their Corporate IT budgeting process consists of two components: the IT Capital Investment Planning Process and the O&M Planning Process. The IT Capital

Investment Planning Process starts in May, where the budgets are developed, projects and prioritized, and estimates refined for the upcoming fiscal year, which begins the following April. The O&M Planning Process starts in September, where the associated operating expenses are developed by using historical spending trends and estimating the impact of any new projects. In November, the investment plan and O&M budget are implemented in the IT budget process and submitted to the global and U.S. Chief of Information office for approval. After implementation of the investment, different types of monthly meetings are held to discuss actual versus budget variances, updates on investment spending, and validate or review an investment plan that may need to be changed.

Q. Is the Companies' Corporate IT budgeting process appropriate?

A. Yes.

Q. Describe the typical life cycle analysis of an IT project and how the individual project budgets developed.

A. As shown in the response to IR DPS-1003, the

Company explained the five stages of an IT project life cycle: pre start-up, start-up, requirements and design, development and implementation, and close. The pre start-up phase captures the problems and needs to develop scope, context, and cost estimates, which is shown in the Investment Request Paper (IRS). The IRS papers provides a cost estimate of project to have a variance of plus 200 percent or minus 50 percent. Next, the project goes into the start-up phase, where a project manager is assigned to the project to develop a team, work plan, and a partial sanction paper. The partial sanction narrows down the variances of the projects estimates to plus 25 percent and minus 25 percent. After the partial sanction is approved, the project goes into the Development and Implementation phase, where the team determines the technical requirements needed for the project, designs the solution and develops the full sanction paper. The full sanction further reduces the variances of the project estimates to plus 10 percent to minus 10 percent. The sanction paper is submitted to

either the US Sanctioning Committee for projects exceeding $1 million or the IT Sanctioning Committee for project below $1 million. After the sanction is approved, the team builds the solution and tests to see if it operates as required and designed in the development and implementation phase. The solution is implemented, and the transitions begins with necessary support provided. Finally, a closure paper is developed to ensure a clear, efficient, and cost-effective shutdown to the project activity.

Q. Did you, or members of Staff under your supervision, conduct a review of projects in each budget category?

A. Yes, we sampled projects to analyze the status of the projects proposed projects. In the response to IR DPS-487, the Companies provided IRS, partial sanction, and sanction papers for each project depending, on the stage of the project development.

Q. Describe the project sanctioning process.

A. The sanctioning process identifies the appropriate spending levels, by specific

programs or projects. The process is used to seek and obtain approval to spend money on project development. The sanction request may address the full project cost, or a partial sanction may be submitted to request enough funding to advance to the next stage of the project development.

Q. What types of sanctions does the Companies employ for IT capital programs or projects?

A. There are four types of sanctions that the Companies employ: partial sanction, sanction, re-sanction, and closure paper. A partial sanction paper is generally submitted to advance a project when a sanction paper cannot be submitted due to lack of a complete scope and final cost. Unlike the partial sanction paper, the sanction paper is generally prepared for the full scope and cost of the project and is considered the final approval to undertake the project. A re-sanction paper must be filed within the 60 days of notice that the cost of a project is forecasted to exceed the tolerance approved in the sanction paper. A closure paper is prepared at the completion of a project that

details the financial and objective outcomes of the project.

Q. What information is contained in the sanction papers?

A. The sanction papers typically provide cost and project details, as well as potential alternatives and the ramifications of those alternatives, so that the Companies can make informed decisions regarding capital projects, including the risks and benefits to the Companies and their customers. More specifically, the sanction paper includes a summary of the amount being requested for sanctioning, broken down into capital and operating expenditures by year, and a summary of the project, including what is being proposed, what is being replaced, drivers, background, benefits, and any business or customer issues. As some sanctions can be done for multiple projects, a summary of projects is listed. The prior sanctioning history shows each partial or prior sanction before the current sanction paper, along with the sanctioned amount, the next planned sanction, all key milestones, and

the cost estimation tolerance around the sanction requested amount. Each sanction paper categorizes the project as mandatory, policy-driven, justified net present value or other.

Q. What other information do the sanction papers include.

A. Each sanction paper also defines an asset management, risk score, primary risk score driver, complexity level, and process hazard assessment. The resources to complete the project, whether internal or external, availability of the resources, and any potential operational and procurement impact are also noted. Other key issues and potential impact to climate change are also noted. The project alternatives that were considered are listed, along with potential risks faced in project implementation. Any cost assumptions and cost benefit analysis or net present value analysis performed are listed or marked not applicable. The recovery of the project costs and financial impact to the Service Company are defined. If a fully developed sanction is completed, there will be an estimate of expected implementation

operating costs and ongoing run the business expenses. Finally, a list of operating companies that will benefit from – and pay for – the project is included, which a plan for customer outreach, if applicable.

Q. Do all papers in the sanctioning process include all of the information you described?

A. No. Depending on the status of a project's development, it may be in different stages of sanctioning and only preliminary information is included in the documentation.

Q. What information is contained in the IRS papers?

A. IRS papers show the key personnel involved in developing the project, as well as the project category, primary policy driver, description and background, expected benefits, scope, dependencies, and assumptions. Estimated costs by year are listed, as well as costs by delivery phase. A breakdown of the project prioritization and cost by capital, operating, and expected run the business costs is included, along with a score for investment risk and complexity. An estimate of the resources needed to complete the project, the key, known

milestone dates, and benefitting operating companies are also listed.

Q. Do all IRS papers include the information you described above in the sanction process?

A. No. Depending on the status of each individual project, the amount of detail in each IRS paper will vary.

Q. Why is less information available in the IRS papers?

A. IRS papers may contain less information, or more broadly defined information, than full sanction papers because, as described previously, these are used at the earliest stage of project development.

Q. Explain how the Companies estimate the preliminary project costs for IT projects on the IRS papers.

A. As stated in the Companies' response to IR DPS-880, the IT Business Consultants (BC) and Project Managers (PM) develop the preliminary costs, based on their experience with delivering projects of similar size, scope, and complexity. Costs may also be estimated by using labor hours spent on prior projects of similar size, scope,

and complexity, combined with vendor rate cards and internal labor. When applicable, historical documents such as previous statement of work from vendors that show estimated hardware or network costs and sanctioning/closure papers are used.

IT Investment Recovery

Q. How do the Service Companies' IT capital expenditures impact the KEDNY and KEDLI revenue requirements?

A. IT project costs are incurred at the Service Company level. The costs are then allocated to the individual operating companies that use the IT services. KEDNY and KEDLI thus are allocated its proportionate share of IT projects costs for each solution it utilized that was developed or obtained by the Service Company.

Q. What types of IT costs are allocated to KEDNY and KEDLI?

A. The Companies divide their IT program costs into three categories: capital expenditures, operating expenses, and "run the business," or RTB, expenses.

Q. Please describe the capital expenditures category.

A. Capital expenditures represent the costs to buy or create the project that will be included as an asset at the Service Company. Exhibit___(ITP-4CU), shows the forecast capital expenditures, by project, for the Rate Year, as well as FYs 2022, 2023, and 2024.

Q. How do these capital expenditures translate to KEDNY and KEDLI's revenue requirements?

A. Once the project is closed to plant in service on the Service Company's books, the Service Company begins to charge KEDNY and KEDLI for its portion of the amortization expense of the projects, as well as a return on the unamortized project costs. This process is similar to that used for "traditional" gas plant, whereby the Companies incur depreciation expense and also earns a return on the net book value when the plant is included in rate base.

Q. Do the IT assets move to KEDNY or KEDLI's books after being placed in service?

A. No. These assets remain on the Service Company's books after KEDNY and KEDLI begin

using them. The Service Company recovers both the return on and the return of the IT asset investment through Service Company Rent Expense, which is a component of Operations and Maintenance expense, or O&M. Service Company Rent expense is shown in both of KEDNY and KEDLI's Exhibit____(RRP-3), Schedule 9.

Q. How much Service Company Rent expense do KEDNY and KEDLI forecast during the Rate Year?

A. KEDNY and KEDLI forecast $43.7 million and $32.2 million of Rate Year Service Company Rent expense, respectively. Of these amounts, $9.2 million and $10.3 million are facilities, $14.4 million and $8.6 million are for existing gas projects, and $20.1 million and $13.3 million are for new gas IT projects, for KEDNY and KEDLI, respectively.

Q. Please describe the IT operating expenses.

A. As described on page 52 of the Companies' IT Panel's Direct Testimony, IT operating expenses consist of initial project expenses and post-implementation costs incurred with new projects. These costs are spread throughout multiple components of the revenue requirement.

Q. How are operating expenses incurred in the Historic Test Year reflected in the Rate Year revenue requirement?

A. As shown in Exhibit___(ITP-8CU), the Service Company incurred operating expenses of approximately $30.6 million in the historic test year. The Companies expect the level of operating expenses incurred in the historic test year to continue in the Rate Year. After accounting for inflation and allocations to KEDNY and KEDLI, the various components of the Rate Year revenue requirement include approximately $4.271 million and $2.660 million of these expenses for KEDNY and KEDLI, respectively.

Q. What level of operating expenses associated with new IT projects are forecast to be incurred during the Rate Year?

A. As shown in Exhibit___(ITP-7CU), the Service Company expects to incur an additional $33.375 million of operating expenses in the Rate Year associated with new IT projects, not including the CIS project and SAP S4 HANA. After allocating to KEDNY and KEDLI, these forecast

costs result in incremental Rate Year expenses of $3.386 million and $2.109 million for KEDNY and KEDLI, respectively. These expenses are included in the Other Initiatives expense line in O&M, as shown in Exhibit___(RRP-3CU), Schedule 27. Additionally, the calculation is shown in Exhibit___(SITP-2).

Q. Please describe the IT RTB expenses.

A. As explained on page 52 of the Companies' IT Panel Direct Testimony, RTB expenses are costs of running a project once it is in service, inclusive of licensing and maintenance agreements, third party contracts, consultant and contractor costs, and internal labor. Similar to the upfront operating expenses, RTB expenses are included in many areas of the revenue requirement.

Q. How are the RTB expenses incurred in the historic test year reflected in the Rate Year revenue requirement?

A. As shown in Exhibit___(ITP-8CU), the Service Company incurred approximately $213.3 million of RTB expenses in the historic test year. The Companies expect the level of RTB expenses

incurred in the historic rate year to continue into the Rate Year. After accounting for inflation and allocations to KEDNY and KEDLI, the various components of the Rate Year revenue requirement include approximately $29.773 million and $18.544 million of these expenses for KEDNY and KEDLI, respectively.

Q. What level of RTB expenses associated with new IT projects are forecast to be incurred during the Rate Year?

A. As shown in Exhibit___(ITP-7CU), the Service Company expects to incur an additional $28.543 million of RTB expenses in the Rate Year associated with new IT projects, not including the CIS project and SAP S/4 HANA. After allocation to the Companies, these forecast costs result in incremental Rate Year expenses of $2.896 million and $1.803 million for KEDNY and KEDLI, respectively. These expenses are included in the Other Initiatives expense line in O&M, as shown in Exhibit___(RRP-3CU), Schedule 27. Additionally, the calculation is shown in Exhibit___(SITP-2).

Q. Does the Rate Year revenue requirement reflect

upfront operating expenses for CIS?

A. Yes. The Companies included operating expenses of $10.526 million and $4.885 million for KEDNY and KEDLI, respectively, to implement CIS. These expenses are included in the Other Initiatives expense line in O&M, as shown in Exhibit___(RRP-3CU), Schedule 27. Additionally, the calculation is shown in Exhibit___(SITP-2).

Q. Does the Rate Year revenue requirement reflect RTB expenses for CIS?

A. No. As shown on Exhibit___(ITP-7CU), the Companies do not include any Rate Year RTB expenses related to the CIS project because the in-service date does not fall within the Rate Year.

Q. Please summarize KEDNY and KEDLI's Rate Year revenue requirement as it related to IT projects.

A. KEDNY and KEDLI have included approximately $94.598 million and $62.237 million of IT-related project costs in the revenue requirements for their gas businesses, respectively. This is comprised of gas capital-related costs of $43.746 million and $32.236

million for KEDNY and KEDLI, respectively, which are incurred as Service Company Rent expenses; upfront operating expenses of $18.183 million and $9.654 million, and RTB expenses of $32.669 million and $20.347 million for KEDNY and KEDLI, respectively.

Q. How much of this revenue requirement is incremental to what was included in the historic test year and associated with new IT projects?

A. Of the amounts previously provided, approximately $36.932 million and $22.115 million are incremental for KEDNY and KEDLI, respectively. This is comprised of incremental Service Company Rent expense of $20.124 million and $13.318 million, respectively, and operating and RTB expenses of $16.808 million and $8.797 million for KEDNY and KEDLI, respectively.

Staff's Review Process

Q. Describe the process you used to review the Companies proposed IT investments.

A. We took several steps to review the proposed IT investments. First, we held two technical sessions with the Companies to discuss their IT

budgeting process, sanctioning process, proposed IT investment plan and the overall goals and objectives of the Companies' proposed IT investments. Next, we reviewed the Service Company's historic IT capital spending from Fiscal Year 2015 to Fiscal Year 2019 to gauge its ability to complete IT projects. Then, we reviewed the proposed IT projects and associated expenses, through IRS and sanction papers. Finally, we analyzed the variances between the cost estimates of the IT capital projects and the final estimates shown in the closure papers.

Q. Describe the Panel's method of reviewing the proposed projects.

A. Exhibit___(ITP-4CU) shows the IT projects and their associated capital expenditures for the Rate Year and Data Years, totaling 163 IT projects. We focused on reviewing projects that were either starting or continuing in the Rate Year. We reviewed the project using the IRS and sanction papers that were provided in response to IR DPS-487, and found that most of the projects only had IRS papers.

Q. Is the Panel comfortable with most of the

proposed projects only having IRS papers?

A. No. As discussed earlier, IRS papers are generated in the pre start-up phase and show cost estimates that can have a variance of minus 50 percent to plus 200 percent. In addition, the IRS papers do not provide a well-defined scope, therefore, we are not confident that the Companies' can execute the proposed IT investments on time given where the investments currently stand.

Q. What other information did you learn from your review of the Companies' IRS and sanctioning processes?

A. To get a better understanding of how accurate the preliminary project costs shown in the IRS papers or early sanction papers are compared to the final project costs shown in the closure papers, we asked for the variances between the two costs, shown in the Companies' response to IR DPS-880. We found that the variances from the preliminary and final project costs ranged from minus 100 percent to plus 918 percent. We understand that the cost estimates shown on the IRS could range from minus 50 percent to plus

200 percent; however, the variances between the preliminary and final costs show, in actuality, that the Companies' proposed project costs could significantly change and thus significantly impact the Rate Year budget.

Historic Review

Q. What did you observe when you reviewed the historic IT Capital Spending?

A. We made several observations. First, we reviewed the historic IT Capital Spending and found the actual IT capital spending levels in FYs 2015 through 2019 were $74.3 million, $81.4 million, $121.9 million, $94.6 million, and $115.2 million, respectively. Second, we determined that the Service Company underspent its budgets in FY 2015, 2016, 2018, and 2019, but significantly overspent in FY 2017. Lastly, for completed IT projects, we observed significant variances between the preliminary project costs and final costs that were included in the closure papers.

Q. Please explain the variances you have mentioned above.

A. As shown in Exhibit___(SITP-2), which was developed using the Companies' response to IR DPS-432, there were significant variances between the actual and budgeted spending in each of the last five years, FYs 2015 through 2019. In each of those years, the Companies underspent their annual budget by an average of $8.8 million, or 8 percent. The most significant underspend was in FY 2015, when the Companies underspent their $113.8 million budget by $39.6 million, or 35 percent. In FY 2017, however, the Companies' significantly exceeded their budget, spending $122 million, or 84 percent, over their budget of $66.4 million. However, $62.1 million, or 49 percent, of the FY 2017 overspend was incurred in March 2017, which is the last month of the fiscal year.

Q. Explain the reasoning behind the significant overspend in March 2017, the last month of FY 2017.

A. The Companies' stated in response to IR DPS-937 that $54.5 million of the $62.1 million, or 88 percent, of the spend was due to the projects associated for the Technology Improvement

Program (TIP), which was the starting point for Technology Modernization Program. The TIP brought forward several investments that were planned for FY 2018 and beyond, prioritizing the infrastructure, software, and applications that needed to be replaced first.

Q. Does the Panel think the Companies' overspending of their budget in 2017 is an outlier?

A. Yes. The Companies' overspend in FY 2017 was a unique occurrence due to ramping up TIP in one month and we do not expect a significant accelerated spend of capital expenditures like this to happen again.

Q. What is the Panel's opinion of the Companies' proposed IT capital budgets considering their historic IT spending performance?

A. Despite historical IT budgets being significantly lower than the proposed Rate Year Service Company IT budget of $243.8 million, the Companies have consistently underspent their budget. As such, we have concerns that the Companies can deliver on their Rate Year IT budget. While the Service Company has made improvements stemming from the Commission's rate

order in Cases 17-E-0238 and 17-G-0239, these process improvements have not yet resulted in improved performance.

Cyber Security

Q. What do KEDNY and KEDLI propose with respect to cyber security?

A. KEDNY and KEDLI explain that the Service Company proposes to complete 12 projects in the Rate Year, and eight in the Data Years. KEDNY and KEDLI also propose to complete 17 cyber security projects that are expected to be in service after the Historic Test Year, but prior to the Rate Year. There are an additional seven ongoing annual programs that will incur costs throughout the Rate Year and Data Years.

Q. What will these projects address?

A. These projects will address a wide range of cyber security issues, including:

Ensuring adequate security protection to cyber assets supporting critical reliability functions across KEDNY and KEDLI's networks and systems;

Providing security information and event management capabilities to analyze security

alerts in real time, and to collect, store and report on data for compliance and forensic analysis across critical assets and business functions;

Strengthening identity, access and authorization management capabilities to ensure that only authorized personnel can access critical assets and perform only those functions for which they are authorized; and

System modernization cyber security framework utilizing a “defense-in-depth” approach for ensuring a reliable and secure electricity and gas infrastructure and protecting the confidentiality and integrity of sensitive data. Defense in depth, or “layered security” as it is also known, reduces the chance of a successful attack by incorporating redundant security mechanisms. This would require an attacker to circumvent each mechanism to gain access to a digital asset.

Q. What are the proposed costs of these projects?

A. The cyber security projects are forecast to require incremental capital investment of $27.8 million through FY 20, $10.2 million in the Rate

Year, $16.5 million in Data Year 1, $17.7 million in Data Year 2, and $17.3 million in Data Year 3.

Q. Please identify the Companies' five Rate Year Cyber Security proposals that you found especially notable due to cost and complexity.

A. The five Cyber Security proposals are: High Value Discovery & Implementation, CNI Forensic Packet Capture, Enhanced DLP Gateway & Endpoint, CNI Security Enhancements Phase I, and Insider Threat Protection.

Q. Please describe the High Value Discovery & Implementation program.

A. The High Value Asset Discovery & Implementation program, formerly referred to as the IT/OT Discovery and Implementation Phase, is intended to provide KEDNY and KEDLI a holistic view of its company assets across the corporation, including infrastructure, business systems, operational technology and information technology assets. High Value Assets are primary targets for cyber-attacks. Having the ability to effectively identify, control, and monitor High Value Assets is a key component of ensuring

continued security and reliability.

Q. Please describe the CNI Forensic Packet Capture project and how it will enhance the Companies' ability to protect their critical infrastructure?

A. The end goal for the CNI Forensic Packet Capture project is to provide the Companies the capability to capture network packets for forensic analysis to help identify threats laterally across CNI networks. The current solution in place lies on the Internet perimeter gateway, allowing only for analysis of vertical network traffic.

Q. Please describe the CNI Security Enhancements Phase I project and the cyber security concerns it will address?

A. The CNI Security Enhancements Phase I project will address the lack of visibility of cloud services being used through the company and the data that is hosted on those services that can lead to unauthorized access and disclosure of information. As well as lack of visibility and monitoring on those services, the organization has limited ability to enforce security policies

on those services. This project will address these architectural weaknesses.

Q. Please describe the Enhanced DLP Gateway & Endpoint project and the cyber security benefits it will deliver?

A. DLP stands for Data Loss Prevention. The Enhanced DLP Gateway & Endpoint project will deliver new processes and enhance the identity, detection and response capabilities associated with malware attacks by introduction of a DLP gateway at the perimeter. This gateway will inspect network traffic and keep known malware off the network, monitor for malware, and continuously update malware signatures and internet addresses of servers hosting the malware.

Q. Explain the risks of insider threats and how the Insider Threat Protection program will address them.

A. Insider threats pose a significant risk to an organization, having the intimate knowledge of systems and information that if used maliciously can affect operations and put disclosure of confidential information at risk. A high

percentage of breaches are associated with insiders, usually inadvertently by employees. Insider Threat Protection will provide the Companies with the capabilities of continuous monitoring and behavior anomaly detection. The solution will integrate with systems and applications to collect data and analyze users, security events and physical access to identify abnormal behavior associated with data misuse, fraud, or IT sabotage.

Q. What is your overall assessment of the need for the capital and O&M programs described in the Companies' filing?

A. We believe that these investments are needed to address ongoing cyber threats. The implementation of these cyber security countermeasures is essential for establishing a high level of cyber security monitoring and protection. The field of cyber security is one in which the risks, threat actors/vectors, and technologies involved are constantly changing and increasing in complexity at a breakneck pace. As a component of the nation's critical energy infrastructure, there is every reason to

believe that probes and surveillance of the Companies' network and supporting electronic devices will increase in frequency for the foreseeable future, especially as more information is shared at ever-increasing volumes using next-generation mobile devices and cutting-edge technologies, such as cloud-based storage and data analysis tools. Furthermore, the types of information handled on KEDNY and KEDLI's network will inevitably become more critical to their operations in the future and thus, a more tempting target for cyber criminals and hostile nation-states. This ever-escalating cyber threat to business and critical infrastructure information requires that utilities remain constantly vigilant and seek new and innovative ways to bolster their cyber security posture.

Q. Explain how the cyber security programs are developed.

A. Each project in the cyber security program has its own Investment Proposal Summary, benefits case, cost estimate, list of benefitting companies, RTB breakdown, project plan, and

project milestones. The Information Security (IS) Portfolio Delivery Board reviews and approves each project based on that information. A short form Investment Proposal paper for all projects approved by the IS Portfolio Delivery Board is provided to, and formally noted by, the US Sanctioning Committee. Each project will report financial actual spending, forecasts, and budget variances back to the IS Portfolio Delivery Board. The Board will summarize financial data to the program level and manage within the authorized spending cap.

Q. How well has the process you described worked in recent years?

A. Despite the project development process that I just described, actual spending versus budget for cyber security for the previous five rate years was 23% less than forecasted. The Companies have also demonstrated a history of delaying projects due to shifting priorities. For example, the High Value Asset Discovery & Implementation program was delayed due to a review of cyber security strategy in 2018.

Q. What do you recommend?

A. We recommend that KEDNY and KEDLI's cyber security programs be included in the IT project update meetings to ensure timely, consistent progress of the cyber security projects proposed in these proceedings. The IT project update meetings are held quarterly to provide Staff with a progress report on KEDNY and KEDLI's IT projects. We also recommend that the Companies notify appropriate Department Staff as projects get sanctioned or partially sanctioned.

Gas Business Enablement

Q. Please explain the Gas Business Enablement (GBE) Program.

A. The GBE program is an overhaul of National Grid USA's gas business systems that are at or approaching end of life. The overhaul is needed due to the age of the systems in order to support improved safety performance, new capital investment, and evolving regulatory and customer demands.

Q. When did the GBE program commence?

A. The GBE program was proposed by Niagara Mohawk, thoroughly reviewed by Staff, and authorized,

with customer protections, in the Niagara Mohawk Rate Order. The first costs were incurred in FY 2017 for the development of the business case, assessment of processes and applications, and high-level program design. The FY 2017 costs were $20.1 million, which were not included in the approved rates in Case 17-G-0239 and are not proposed for recovery in these cases. The total cost of the GBE program at the Service Company level, $458.1 million - exclusive of the $20.1 million - was extensively vetted in Cases 17-G-0238 and 17-G-0239, as the source of the authorized portion allocated to Niagara Mohawk through rent expense. In addition, there was also a Service Company level contingency of $61.0 million.

Q. Do the Companies propose changes to the GBE program in the instant cases?

A. Yes. The Service Company is transitioning from a multi-vendor integrator to a single system integrator.

Q. Did the Companies explain why?

A. Yes. According to the response to IR DPS-684, the Service Company selected a multi-integrator

approach because it provided the best risk score during evaluation and the Service Company was able to align its vendor with the perceived strengths of program integration. However, the implementation schedule became a challenge and by November 2018, the Service Company determined that the increased complexity of managing dependencies across multiple vendors out-weighed the perceived benefits for which it originally selected a multi-vendor approach. The Service Company worked to simplify the strategy and went through a rebid and recontract for the remaining program scope in order to mitigate increased cost and delayed implementation duration that would result due to the multi-vendor related complications.

Q. Are there any changes to the forecasted budget?

A. The Service Company has sanctioned $57.4 million of the $61.0 million contingency and reflected this amount in the Companies rate request. The increase is related to several changes.

Q. Please describe the changes.

A. The Service Company is planning increased testing and training costs of $12.196 million

related to technical training, Subject Matter Expert testing resources, external testing, and end-user training support. Changes include costs related to greater than anticipated complexity, despite foresight knowing it would be a very difficult and complex integration of the legacy interfaces, of approximately $21.168 million. The final change is an additional $24.025 million for Allowance for Funds Used During Construction (AFUDC). The GBE program initially had $0 for AFUDC because of the "agile" development method the Service Company was using for the proposed program. This approach means working in short-cycles and prioritize functionality and implementing of minimum viable products for portions of the program as they becomes available. In addition, based on the implementation of the program thus far, the Service Company has adjusted various project implementation timelines to ensure a better, smoother rollout.

Q. Does the Panel have any adjustments to the proposed increase of $57.4 million?

A. Yes. At this time, we recommend reflecting $0

of the $57.4 million request in the Companies' rates. The Service Company went through a rigorous development process to forecast the GBE program using two independent consultants; one to develop the costs and roadmap, and the other to ensure cost accuracy and that the program was aligned with industry standards. The Service Company has since run into some trouble with implementation, yet took proactive steps to keep the overall program on a path towards success by moving from a multiple to single system integrator and modifying the roadmap and rollout. However, we recommend, to shield customers from delays and excessive costs, the Service Company work towards a successful on-time and on-budget implementation that provides the forecast customer benefits without seeking recovery of additional expenses from customers during this or subsequent rate proceedings.

Q. How do you recommend reflecting the $57.4 million adjustment?

A. Based on the Companies' responses to IRs DPS-684 and DPS-725, we recommend adjustments to several GBE programs at the Service Company level.

These programs – Asset Management, Asset Management/GIS, and Work Management (Maximo) – and the adjustments to each are shown on Exhibit__(SITP-3). We recommend splitting the $24.1 million adjustment for AFUDC between FY20 and FY21 based on the Companies' response to IR DPS-725, part 3, which stated that only $2.114 million was in FY21, and $21.968 million prior to FY21. The AFUDC adjustments for the previous amounts, and the $12.2 million for testing and training and $21.2 million adjustment for legacy integration complexity, were all based on the forecast FY21 ending balance of each program. The revenue requirement impact of our recommendation on Service Company Rent expense is discussed later in our testimony.

Q. Are you proposing customer protections?

A. Yes, we recommend the same customer protections recommended by the Staff Gas Business Enablement Panel (SGBEP) in Case 17-G-0239 be imposed in the instant cases.

Q. Please describe those protections.

A. The protections include the general protections for all IT investments, which we will recommend

later in our testimony. These include a downward-only reconciliation of net plant and depreciation expense and quarterly reporting. In addition to these general protections, we also recommend additional protections to ensure that the Companies deliver the GBE program benefits, which they identified to support the program, as scheduled and to the full degree envisioned. The additional protections include an overall cap on the amount that can be recovered from customers for GBE, and instituting benchmarks to ensure that the Service Company delivers the incremental benefits of GBE compared to the backbone alternative. Our recommended protections can be found in the Joint Proposal adopted by the Commission in the rate order issued in Cases 17-E-0238 and 17-G-0239. The protections are discussed on pages 44-48 of that Joint Proposal and in Appendix 10 to that Joint Proposal.

Q. What cap do you recommend imposing on the amount KEDNY and KEDLI can recover from ratepayers for GBE?

A. The Companies' response to IR DPS-970 shows the

breakdown of total GBE costs proposed in these cases, $515.4 million, into total capital expenditures, $381.7 million, and total operating expenses, $133.7 million. However, we recommend reducing the proposed capital expenditures by $57.4 million. The result is a cap on the portion of the GBE Project recovered from KEDNY and KEDLI customers of $101.9 million and $47.2 million in total capital expenditures, respectively, and $45.2 million and $20.9 million in total operating expenses, respectively.

Staff Adjustments

Removal of IT Projects

Q. Are you addressing the capital and operating costs associated with any specific IT projects?

A. Yes. We recommend removing the capital expenditures and operating costs associated with the CIS project and the Community Distributed Generation (CDG) Low Income Bill Discount program.

Q. Explain why the Panel recommends removing the CIS project.

A. The reasons for removing the CIS program are discussed in the Staff Consumer Services Panel's testimony.

Q. Explain why the Panel recommends the removal of the CDG Low Income Bill Discount program.

A. As stated in the Companies' responses to IR DPS-620, the wrong allocation code was included on this program, as this program only applies to Niagara Mohawk. The Companies stated in response to IR DPS-882 that the error was identified shortly prior to the corrections and updates and therefore Exhibit__(ITP-4) and the revenue requirement was not updated to remove this program. The Companies agree that an adjustment to the revenue requirement is warranted.

Q. Please explain the Rate Year costs associated with the CIS project.

A. CIS project costs are found in the line items Service Company Rent Expense and Other Initiatives. Capital costs associated with the forecasted project are included the Service Company Rent Expense capital budget, however because the project is not projected to go into

service until October 2022, there is no expense related to the project included in Service Company Rent Expense during the Rate Year. The operating costs associated with CIS, included in Other Initiatives for the Rate Year, are $10.526 million and $4.885 million for KEDNY and KEDLI, respectively.

Q. Please explain the Rate Year costs associated with the CDG Low Income Bill Discount program.

A. The CDG Low Income Bill Discount program project costs are found in the line items Service Company Rent Expense and Other Initiatives. For Service Company Rent expense, the costs for the Rate Year are $0.061 million and $0.032 million for KEDNY and KEDLI, respectively. The operating expense amounts included for this program in Other Initiatives are immaterial.

Q. Do you recommend making an adjustment for these costs?

A. Yes, we are removing the capital and operating costs associated with CIS. For Service Company Rent Expense, because the project is not in service, there are no impacts to the Rate Year, however it does reduce the capital budget. For

Other Initiatives, we recommend removing the operating costs associated with CIS totaling $10.526 million and $4.885 million for KEDNY and KEDLI, respectively. We also recommend removing Service Company Rent expense costs associated with the CDG Low Income Bill Discount program, which amounts to $0.061 million and $0.032 million for KEDNY and KEDLI, respectively.

Removal of GBE Contingency Amount

Q. What is the Panel's adjustment to reflect removal of GBE sanctioned $57.4 million contingency?

A. We are removing capital expenditures of $57.4 million from Service Company Rent expense.

Q. Explain how the removal of $57.4 million was calculated.

A. As stated above, the $57.4 million was allocated to, and removed from, GBE capital costs through the line items 'Asset Management,' 'Asset Management / GIS,' and 'Work Management (Maximo).' Once allocated, and removed, we calculated the adjustment for the return on and return of GBE in the Forecasted Projects Rent

expense model.

Q. What is the impact on Service Company Rent expense in KEDNY and KEDLI's revenue requirements?

A. This adjustment reduces KEDNY and KEDLI Service Company Rent expense by $2.136 million and $1.526 million, respectively.

Forecasting Adjustment

Q. What is the Panel's forecasting adjustment?

A. Our forecasting adjustment reflects a decrease to Rate Year capital expenditures based on a linear regression of the historic actual spending.

Q. What is linear regression?

A. A linear regression is a statistical technique used to model the relationship for a set of data. The results of a linear regression provide an R-squared value, and two coefficients, one being the slope of the line and the other being the y-intercept. Using these two coefficients and the equation of a line (y=mx+b), it is possible to reasonable predict future trends.

Q. What is the R-squared value and what does it represent?

A. The R-squared value is a statistical term that indicates how well the line fits to the data, which ranges from 0.0 to 1.0, 1.0 being a perfect fit. For any R-squared value below 0.9 it was determined that the historical data did not show a high enough correlation and the regression analyses for that group was not utilized in our forecast.

Q. Describe how the Panel used linear regression analysis to develop its forecasted rate year budget.

A. We employed linear regression over FY 2015, 2016, 2018, and 2019. We did not use FY 2017 because we determined the year to be an outlier, as discussed earlier in our testimony.

Q. Describe the results of the linear regressions.

A. As shown in Exhibit___(SITP-2), our analysis of the historic data showed a good statistical fit, an R-squared of 0.932, which shows that the historical data we used was a good fit. We found the forecasted Service Company IT capital expenditures to be $129.4 million.

Q. Why is a forecasting adjustment appropriate in this case?

A. As discussed previously, we are not supportive of the Companies' proposed Rate Year Service Company IT capital expenditures of $243.8 million based on the Companies' historic spending on IT projects and the fact that the proposed IT investments in the Rate Year are primarily based on IRS documents. We acknowledge the fact that the Companies' actual spending compared to the budgeted amount has improved each year in the last five years, however, we recommend the Service Company IT capital expenditures budget be $129.4 million in the Rate Year.

Q. Please explain how the forecasting adjustment impacts the Companies' proposed Rate Year budget.

A. The proposed Rate Year Service Company IT capital expenditures is $243.8 million. We recommend first removing CIS from the total capital budget in the Rate Year, which would result in an adjusted IT capital budget of $173.6 million, or a $70.2 million downward

adjustment. Our recommended Service Company IT capital expenditures budget for the Rate Year is $129.4 million or a 25.5% reduction to our adjusted IT capital budget. In total, with the CIS and forecasting adjustment, there is a total downward adjustment of $114.4 million for the Companies' proposed IT capital expenditures budget.

Q. What is the impact on Service Company Rent expense in KEDNY and KEDLI's revenue requirements?

A. This adjustment reduces revenue requirement for KEDNY and KEDLI Service Company Rent expense by $4.527 million and $2.988 million, respectively. Staff calculated this adjustment by taking the forecast IT projects expense according to the Companies and making the following adjustments: (1) remove the impact of reducing GBE contingency CapEx, (2) remove the impact of reducing the CDG Low Income Program costs, then (3) applying the 25.5 percent reduction to the remaining forecast IT project Rent expense portion.

Q. What is the impact on Other Initiatives expense

in KEDNY and KEDLI's revenue requirements?

A. This adjustment reduces revenue requirement for KEDNY and KEDLI Other Initiatives by $1.600 million and $0.996 million, respectively.

SAP S/4 HANA

Q. Briefly explain the SAP S/4 HANA project.

A. As explained on pages 35 and 36 of the initial testimony of the Companies' IT Panel, the Companies' currently use the SAP ECC 6.0, implemented in 2012, for their back-office platform supporting various essential functions, including Finance, Payroll, Human Resources, and Supply Chain. The Companies' propose to replace the SAP ECC 6.0 with SAP S/4 HANA because SAP will no longer support the SAP ECC 6.0 by 2025. The Companies claim that without the support, they would have the "inability to implement any critical security or compliance patches issued by SAP to address any vulnerabilities discovered, software incompatibility with future products and services offered by SAP, inability to scale in an increasing digital world to take advantage of leading practices, and overall

higher operating costs, decreased performance, and reduced overall reliability."

Q. What is the timeline for this project?

A. On pages 38 and 39 of the initial testimony of the Companies' IT Panel, the Companies stated that they would start their discovery phase in FY 20 and design phase in FY 21.

Q. What is the current status of this project?

A. The Companies have finished the discovery phase of the project. As shown in response to IR DPS-942, the Companies stated that the discovery phase was a 12-week effort that ran from February 2019 to May 2019 and resulted in a "roadmap" for future implementation of the SAP project. The Companies worked with PricewaterhouseCoopers and SAP consultants to map the Companies' current process inventory to a best practice utility standard process; determine the SAP modules that should be included in the solution; define a draft future process scope and structure to improve and optimize current process based on user feedback and lessons learned; and rationalize level of customizations needed based on experience with

current version of SAP. The Companies then developed estimated capital and operating expenditures. The Companies plan to begin the design phase in January 2020 but this may change as this phase has not yet been approved or sanctioned.

Q. Briefly describe what the Companies propose for project costs associated with the SAP S/4 HANA project.

A. As shown on page 7, Exhibit___(ITP-4CU), and Exhibit___(ITP-8CU) in the Companies' corrections and updates filing, the Companies propose a Service Company capital and operating expenditure of $48.6 million $30.6 million, respectively. The Companies also propose to defer for future recovery the differences between the costs recovered in rates and the Companies' actual costs, as explained on page 40 of the initial testimony of the Companies' IT Panel.

Q. How did the Companies develop the capital and operating expenditures for the SAP S/4 HANA project?

A. According to the Companies' response to IR DPS-

942, the Companies developed the estimated capital and operating expenditures utilizing a PricewaterhouseCoopers model. The model utilizes a detailed bottom up resource plan for labor for all parts of the project and adds the estimated non-labor costs associated with the project.

Q. Does the Panel agree with this project?

A. Yes. We agree with the need for the project and the proposed operating expenditures needed for the design phase of the project. However, due to the design phase not being sanctioned yet and uncertainty as to when the design phase will actually begin, the Companies should adjust the capital expenditures for the project for the Rate Year to work within our forecasted budget of $129.4 million.

Q. Does the Panel agree with the Companies' proposal to defer the differences between the costs recovered in rates and the Companies' actual costs for future recovery from customers?

A. No. We understand that the Companies' proposed costs are preliminary estimates and may significantly vary; however, we believe

customers should not be required to pay if the Companies' overspend the budget they developed in the discovery phase. Instead, as we will describe later in our testimony, we recommend a downward only mechanism for all IT projects.

Service Company Asset Recovery Charge

Q. As discussed earlier, the IT project assets remain on the Service Company's books after KEDNY and KEDLI begin using them. The Service Company recovers the return on the IT asset investment from KEDNY and KEDLI, and the other National Grid operating companies, through Service Company Rent Expense. What rate of return did the Companies request to apply to the return on of IT capital costs in the Rate Year?

A. The Companies proposed to use a pre-tax weighted average cost of capital of 8.69 percent in their original filings, which is based on a Return on Equity, or ROE, of 9.65 percent with a capital structure comprised of 50 percent common equity and 50 percent long-term debt. This is shown on pages 16 and 17 of the direct testimony of the Companies' witness Jonathan Cohen. The

Companies updated this amount during Corrections and Updates, filed on July 3, 2019. In the update filing, the Companies proposes to use an updated pre-tax weighted average cost of capital of 8.52 percent.

Q. Do you agree with Mr. Cohen’s proposal to use the Service Company rate of return, which includes a 50 percent common equity ratio?

A. No. We understand that the Staff Finance Panel is recommending for KEDNY and KEDLI a common equity ratio of 48 percent and a return on equity of 8.20 percent. Accordingly, we recommend that the common equity ratio and cost rates for common equity and long-term debt recommended by the Staff Finance Panel also should be used in the development of revenue requirement for Service Company Rent expense. This would result in a pre-tax weighted average cost of capital of 7.58 percent and 7.54 percent for KEDNY and KEDLI, respectively. We recommend this rate be applied to assets at the Service Company level so as to avoid requiring customers to pay a higher return on these assets solely because the Companies have chosen to have a

Service Company provide the assets.

Q. What is your adjustment for this reduction in the use of the Staff Finance Panel rate of return?

A. This adjustment reduces KEDNY and KEDLI Service Company Rent expense by $3.553 million and $2.805 million, respectively.

Adjustments Summary

Q. Please summarize your revenue requirement adjustments related to IT projects.

A. For KEDNY and KEDLI, respectively, our revenue requirement adjustments decrease Service Company Rent expense by $10.322 million and $7.355 million; and IT Operating and RTB expenses, which are included in Other Initiatives expense, by $12.126 million and $5.881 million.

Downward Only Reconciliation

Q. Is the Panel concerned that the Service Company will under-spend its Rate Year IT budget?

A. Yes. As previously discussed, the Service Company's historical data shows that there have been significant variances between the capital

budget and actual expenditures. As discussed earlier, the Service Company is planning a substantial increase in IT spending. However, the Service Company has not provided enough support to show that it can ramp-up its spending to complete all of the proposed IT investments. For these reasons, there is a significant risk that the Service Company will again fail to execute its spending plan fully and timely, thereby forcing KEDNY's and KEDLI's customers to pay rates based on a level of new plant that is not actually deployed.

Q. Does the Panel's forecasted budgeting adjustment address this concern?

A. Partially. Our forecasted budget adjustment, as well as the adjustment to remove CIS, reduces the allowed Service Company capital IT spending to $129.4 million in the Rate Year. However, despite this reduction to the Companies' request of $243.8 million, it still exceeds the IT capital spend in prior years by a significant amount. FY 18 and 19 had IT capital spends of $94.6 million and $115.2 million, respectively. Although FY 17 reported an IT capital spend of

$122.0 million, we have concerns about the data for that period, as we previously discussed.

Q. What do you recommend to address this concern?

A. Similar to what the Commission approved in the Niagara Mohawk Rate Order, we recommend a Service Company Rents IT and GBE Program Net Utility Plant and Depreciation Expense Reconciliation Mechanism to protect customers from paying delivery rates higher than necessary to support the IT assets actually implemented.

Q. Briefly describe the proposed Service Company Rents IT and GBE program Net Utility Plant and Depreciation Expense Reconciliation Mechanism.

A. Our recommended mechanism would require the Companies to reconcile its respective actual IT and GBE program average net utility plant and depreciation expense revenue requirements to the forecast revenue requirements. The IT and GBE program average net utility plant and depreciation expense revenue requirement would be calculated by applying the Companies' authorized pre-tax weighted average cost of capital in the Rate Year to the IT and GBE program average net utility plant balance and

adding the depreciation expense to the product. At the end of the Rate Year, if the actual IT and GBE program average net utility plant and depreciation expense revenue requirement is less than the target average net utility plant and depreciation expense revenue requirement, the Companies would defer the revenue requirement impact for the benefit of customers.

Q. Should this mechanism be symmetrical?

A. No. The mechanism should be a one-way, downward only true-up. Therefore, if actual Service Company Rent expense exceeds the approved Rate Year allowance, a regulatory liability would not be established for the Companies to recover from customers at a later date. The calculations for the effects of this mechanism for the Rate Year should be filed with the Secretary on or before July 31st after the close of the Rate Year.

Q. Why does the Panel recommend that the mechanism be a one-way, downward-only true-up mechanism?

A. Budgeting and spending are activities wholly within the Companies' control. Improving their performance in these areas also is within the Companies' control. A symmetrical true-up will

not provide an incentive for the Companies to improve their budgeting and spending processes. Customers have no control over the Companies' level and pace of spending, yet they bear the risk that the Companies' historic challenges in spending to projected levels will continue and will be reflected in rates. The true-up mechanism, therefore, should reconcile only underspending to allocate these risks equitably between the Companies and customers.

Process Improvements

Q. Do you have any recommendations for process improvements?

A. Yes. We recommend to: (1) continue the process improvements that were required in the Niagara Mohawk Rate Order; (2) make improvements to the Service Company's IRS and sanction papers; and (3) implement Key Performance Indicators (KPIs) for projects.

Q. Please explain the process improvements required in the Niagara Mohawk Rate Order.

A. The improvements addressed reporting and processes for IT. For reporting requirements,

Niagara Mohawk was required to make quarterly and annual filings. Prior to each of its rate years, Niagara Mohawk is required to file its IT Capital Investment Plan for that upcoming rate year, including the overall IT investment plan for that year, the process for developing the plan, and identifying the proposed IT projects and their estimated costs. Niagara Mohawk is also required to file quarterly variance reports within 60 days of the end of each rate year quarter to address the status of the Service Company IT programs/projects and budgets. The variance reports include (i) a narrative explaining overall IT investment plan approach; (ii) a description of defined program categories that form the spending rationale for IT projects; (ii) detail on budgets and actual sending for each of the program categories, and for the top ten projects by budget within the program categories; (iv) explanations of variances between budgets and actual spending; (v) identification of allocations to Niagara Mohawk; and (vi) a report on budget expectations by program category and top ten projects based

on allocations to Niagara Mohawk. The variance report in the last quarter would also include (i) an explanation of any cost or timeline differences exceeding ten percent for projects completed; and the latest, partial, full, re-sanction or closure paper for any projects exceeding $1 million that were authorized during the previous rate year.

Q. Are there additional requirements?

A. Yes. Niagara Mohawk is required to meet with Staff on a semi-annual basis to (i) discuss proposed projects for the next six months; (ii) provide an update on the status of in-flight projects; (iii) discuss proposed projects for the next six months; and (iv) review quarterly filings. The Commission required that, in the first meeting, Niagara Mohawk work with Staff to develop a mutually agreeable sanction paper enhancements needed to justify the IT investments and allow for adequate review. The enhancements would include minimum cost alternative analyses, as appropriate, based on the type of IT investment (e.g., regular upgrades versus build of new function), and the

consensus enhancements were to be implemented no later than the end of that first rate year. Niagara Mohawk and Staff would also determine if additional or different information would be useful for the quarterly variance reports.

Q. Does the Panel recommend the same reporting and process requirements, which were approved in the Niagara Mohawk Rate Order, for KEDNY and KEDLI?

A. Yes, we recommend the Companies implement the same reporting requirements and meet with Staff on a semi-annual basis to discuss the topics mentioned above. We realize that the reporting and meeting can be combined for all New York operating companies for efficiency purposes. We also recommend additional reporting and process requirements.

Q. Please explain your recommendation for additional reporting and process requirements.

A. We recommend the following additional requirements to be included in the variance reports: charts showing the governance structure for each of the top ten projects; identification of major implementation risks and the plans to manage & mitigate them; and an explanation of

the root cause(s) of said variances & plans to eliminate the root causes(s). For the semi-annual meetings, we recommend that the following also be discussed: updates on the status of any attendant shut-down operations that are supposed to produce savings; review strategic-level risks and issues related to the implementation of in-flight projects; and discuss any salient changes to governance, contracted resources, or staffing.

Q. Should these reporting requirements continue beyond the Rate Year?

A. Yes. We believe it is important for the Commission to monitor the Companies' capital investment plans on an ongoing basis. Informational reports at regular intervals are critical to maintain oversight of the IT investment plans. These recommendations are consistent with existing reporting requirements for the Companies' gas operations and for Niagara Mohawk. They are critical to establishing a foundation for Staff to conduct comprehensive, project-specific examinations of IT projects in future rate proceedings,

comparable to Staff's current examination of gas capital plans.

Q. Please explain your recommendations regarding information provided in the IRS and sanction papers.

A. We recommend that the Companies more fully support its cost estimates and work collaboratively with Staff to show that such estimates are reasonable. We also recommend that the Service Company IT sanction paper or IRS document should state if the solution chosen was the minimum-cost alternative. If the Companies chose a higher cost, or enhanced, program, the sanction paper should present an analysis that compares the benefits and costs associated with the project life cycle. The analysis should also further explain how the results support the decision to pursue the alternative solution.

Q. Please explain your recommendations for the KPIs.

A. We recommend the Companies develop KPIs to track and quantify benefits stated in Sanction papers and compare and report on the actual benefits

versus what was originally proposed.

Q. Does this conclude your testimony at this time?

A. Yes.

BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Case 19-G-0309 & 19-G-0310

January 2020

Prepared Supplemental Testimony of:

Staff Information Technology Panel

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Q. Members of the Department of Public Service Staff (Staff) Information Technology (IT) Panel (Panel), please state your names.

A. The panel members are Jennifer Park, Brian Fisher, Keith Haugen, Luke Quackenbush, Chelsea Kruger.

Q. Did the members of the Panel previously file testimony in this case?

A. Yes. We submitted pre-filed direct testimony on August 30, 2019.

Q. What is the purpose of the Panel's testimony in this proceeding?

A. Our testimony will address the rebuttal testimony of the Information Technology Panel, filed on behalf of the Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY) and KeySpan Gas East Corporation d/b/a National Grid (KEDLI) (collectively, the Companies). We will discuss a correction to our pre-filed direct testimony and modeling errors.

Q. In your testimony, will you refer to, or otherwise rely upon any information obtained during the discovery phase of this proceeding?

A. Yes, our testimony will refer to and otherwise

rely upon an Information Request (IR), our response to the Companies' IR NG-33. This response is contained in Exhibit___(SITP-4).

Q. Are you sponsoring any additional exhibits?

A. Yes, we are sponsoring one exhibit: Exhibit___(SITP-5) which includes a summary of, and corrected calculations related to, the Panel's originally proposed adjustments.

Corrections and Updates

Q. Does the Panel have any corrections to their pre-filed direct testimony?

A. Yes. We would like to correct certain errors in our adjusted calculation of Service Company rent expense. These errors included: a double count of adjustments, the application of a rate of return for certain adjustments different than the rate of return that we otherwise recommended using, clerical errors, and the misapplication of the proposed 25.5 percent forecasting adjustments as a direct 25.5 percent reduction to Service Company rent expense.

Q. Has the Panel quantified the identified errors from what the Panel originally filed?

A. Yes. Our proposed downward adjustments in our

pre-filed direct testimony of $10.32 million for KEDNY and $7.35 million for KEDLI should be, when accurately calculated, downward adjustments of approximately $3.73 million for KEDNY and $2.91 million for KEDLI. The original adjustments over-adjusted Service Company rent expense by $6.6 million for KEDNY and $4.4 million for KEDLI.

Q. Have the Companies requested that Staff acknowledge the errors?

A. Yes, in IR NG-33, the Companies asked the Panel about these errors. In the response to IR NG-33, we agreed with the Companies' rebuttal testimony and the reconciliation of our adjustments just described and presented in the Companies' rebuttal Exhibit___(ITP-2R).

Q. Would the Panel agree that there are no remaining issues regarding the reconciliation of service company Rent Expense provided in Exhibit___(ITP-2R)?

A. Yes. Staff and the Company are in agreement on this issue.

Q. Does this conclude your testimony at this time?

A. Yes.

MR. FORST: Your Honors, I now proffer the panel for cross-examination.

A.L.J. LEARY: I think the company is first --

A.L.J. COSTELLO: Yeah.

A.L.J. LEARY: -- is that right?

A.L.J. COSTELLO: That's it.

A.L.J. LEARY: First and only.

MR. GAVILONDO: Thank you, Your Honor. Carlos Gavilondo on behalf of KEDNY and KEDLI.

CROSS-EXAMINATION

BY MR. GAVILONDO:

Q. Good afternoon, panel. So the Staff I.T. panel has reviewed the company's I.T. proposal for the rate year. Is that correct?

A. (Park) Yes.

Q. Okay. Thank you. And with respect to the overall I.T. proposal, it's my understanding that the panel or actually the staff has proposed that the C.I.S. project be removed and handled through a separate proceeding as opposed to -- within this rate case. Is that correct?

A. (Kruger) That -- the reasoning behind that is discussed in the Staff Consumer

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Q. Okay. Great, thank you. So are there other I.T. projects that the company has proposed for inclusion in the I.T. investment plan that staff recommends should be removed from the I.T. capital budget that's reflected in this case?

A. (Park) No, we don't have any other projects that we were proposing to re -- for the company to remove in their rate --

A.L.J. LEARY: Excuse me.

THE WITNESS: -- budget.

A.L.J. LEARY: Would you excuse me?

THE WITNESS: Okay.

A.L.J. LEARY: You need to speak right into that microphone.

THE WITNESS: Okay. Yes, we did not have any other projects that we did not -- we denied in the rate year.

BY MR. GAVILONDO: (Cont'g.)

Q. Okay. Thank you. And -- and I guess the corollary to that, are there any recommendations that the company should eliminate any particular projects from their investment plans other

than as Ms. Kruger mentioned the C.I.S. program to be addressed in a separate proceeding?

A. (Quackenbush) Could you please repeat the question?

Q. Certainly. Does the panel recommend that the company eliminate any particular I.T. projects from its investment plans reflected in this case?

A. (Kruger) We also discussed removing the C.D.G. low-income bill discount program on page 50 of our testimony.

Q. Okay, thank you.

A.L.J. LEARY: Could you -- I'm just asking the witness. Is there a way that you can define the C.D.G. program on page 50? Would that -- it -- it's community distributed generation. Is that correct?

THE WITNESS: Yes, that's correct.

A.L.J. LEARY: Okay. I just want to remind the panel that acronyms are tough for the record, so please spell out, then you can use an acronym occasionally.

THE WITNESS: Sorry.

A.L.J. LEARY: No worries.

MR. GAVILONDO: Thank you, panel.

BY MR. GAVILONDO: (Cont'g.)

Q. Panel, I direct your attention to page 4 of your direct testimony. In lines 13 through 16 of page 4, the panel states its position that, "The companies should spend at the levels they deem appropriate to provide safe and adequate service." Is that a correct characterization of that testimony?

A. (Quackenbush) Yes.

Q. Thank you. So the panel's recommendation in this case includes a forecasting adjustment. Is that correct?

A. (Park) Yes.

Q. Thank you and that forecasting adjustment would be a downward adjustment of 25.5% to the company's proposed I.T. capital project budget after the top side adjustments that staff made relative to removing the C.I.S. program and the community distributed generation low-income program. Is that correct?

And -- and if it's helpful, I believe the forecasting adjustment is described in your direct testimony beginning at pages -- at page 31. So page 31 through 34 and then again page 53 through

57.

THE WITNESS: (Park) We also removed the G.B.E. contingency amount before applying the 25.5% forecasting adjustment.

A.L.J. LEARY: Excuse me, can I get that cell phone -- that rang? No, I'm kidding. Everybody turn off your cell phones. I said this yesterday, if your phone rings, it's mine. Actually, I'm going to give it to Judge Costello. So I'm sorry for the interruption, panel. Go ahead.

BY MR. GAVILONDO: (Cont'g.)

Q. Thank you, Ms. Park. So just to clarify, the company removed the customer information system C.I.S. program, the C.D.G. low-income program and an amount associated with the G.B.E. project. And -- and Ms. Park, you mentioned an amount -- did you mention an amount, I'm sorry, I was distracted by a ringing telephone?

A. For the G.B.E.?

Q. Yes.

A. Contingency, it's in their testimony. Do you know what?

Q. And just to clarify, I believe I said the company removed, but I believe it was staff

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recommendations removed. Thank you.

MR. FORST: If I may interject a second. It's Nick Forst for the panel. Could the panel please define what G.B.E. stands for?

A.L.J. LEARY: Thank you. I was just about to do that.

THE WITNESS: (Park) G.B.E. stands for Gas Business Enablement.

THE WITNESS: (Quackenbush) And can you just repeat the question?

MR. GAVILONDO: I will try.

BY MR. GAVILONDO: (Cont'g.)

Q. The staff panel proposed a 25.5% downward adjustment, a so-called forecasting adjustment to the remaining I.T. capital investment portfolio or budget after it removed some top side adjustments. And those top side adjustments that I believe the panel removed were the C.I.S. program, the Customer Information System program costs from the rate year, the C.D.G. Community Distributed Generation, low-income program costs and the G.B.E., Gas Business Enablement costs.

And -- and is -- are those the adjustments that staff made before it applied its

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25.5% downward forecasting adjustment?

A. (Quackenbush) Initially no, but we did put in the supplemental testimony, addressing the reconciliation that the company had performed, and I believe it's Exhibit I.T.P.2R., the company's exhibit.

Q. Thank you, Mr. Quackenbush. And I believe the staff did submit some supplemental testimony and supplemental exhibits that address this as well?

A. That's correct.

Q. And I'm not sure the exit -- the exhibit number for this proceeding, but I believe that is Exhibit S.T.I.P.-5 which consists of 6 pages, is that correct?

MR. FORST: Your Honors, for the purposes of the record, the Exhibit S.T.I.P.-5 has been pre marked as 519.

A.L.J. LEARY: Try to remember everybody we need to hear an exhibit number, not just, it's fine if you want to talk about the exhibit number. In fact, it would best to do both. But we really need to hear an exhibit number in this record. Thank you, Mr. Forst.

MS. FORST: You're welcome, Your Honor.

THE WITNESS: And I believe the company asked if that exhibit was 6 pages?

BY MR. GAVILONDO: (Cont'g.)

Q. Yes.

A. Yes, it is.

Q. Okay. And the adjustments that the staff made up here at the top of Page 1 of Exhibit 519?

A. That's correct.

Q. Okay. Thank you. So I'd like to just turn the attention to the derivation of the forecasting adjustment, if we may. So the panel reviewed the company's actual I.T. capital investment spend compared to the budgeted spend for fiscal years and we heard the prior panel describe fiscal years, I'm happy to find those if this panel would like. But for fiscal years 2015 through 2019, is that correct?

A. (Park) Yes, that's correct.

Q. And in the -- I'm sorry, in the staff's analysis of the spending compared to budget over that time, is it true that the company

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A. Yes, that is correct.

Q. And is it also correct that that underspend was on average approximately 8% less than the company's annual budgets for the period F.Y.15 through F.Y.19?

A. Yes.

Q. In driving the panel's recommended forecasting adjustment of 25.5%, the panel ignored the company's actual and budgeted I.T. investment or spend in F.Y.17, correct?

A. So we didn't ignore 2017, we acknowledge that they did spend significant amounts on their technology modernization program. But we believed that it was a one-time spend, significant spend right near the end of the fiscal year and something that would have happened during the rate year. And I know that the -- the company's I.T. panel mentioned that, that -- because of the significant spend in 2017, they're underspend in 2018 and 2019.

But I -- from what I believe the

service company, they -- they create their budget on an annual basis. So I would -- I would have thought that they would have budgeted accordingly.

Q. Thank you. So you testified that you didn't ignore it, but you didn't include it in the derivation that you used to come up with the forecasting adjustment. Is that correct?

A. That is correct.

Q. And of the 5 years that you reviewed, all 4 years that you used for calculating the adjustment are underspends, correct?

A. Yes, that's correct.

Q. And the only year that you didn't use was an overspend, correct?

A. Yes -- yes, we don't include --.

A. (Quackenbush) I believe the question was that the company exceeded its budget in fiscal year 2017?

Q. The question was whether or not the only year that was excluded when calculating the -- the adjustment -- the forecasting adjustment was F.Y.17, which was the only year in which the company overspent of the five years that the staff evaluated question.

A. Yes.

Q. Thank you. I'd like to turn your attention to page 33 of the staff direct testimony. On lines 2 through 6, staff acknowledges that the T.I.P. which is the Technology Improvement Program, brought forward several investments that were planned for F.Y. -- for F.Y.2018 and beyond, prioritizing infrastructure, software, and applications that needed to be replaced first. Is that -- did I correctly read that testimony?

A. Yes.

Q. Thank you. And just to confirm, when the staff calculated its forecasting adjustment, it did not make any adjustments to the fiscal 2018 or fiscal 2019 data after it removed the fiscal 2017 data, is that correct?

A. (Park) Could you repeat the question again, I'm sorry?

Q. I have a very short memory so I'm going to try.

A. Okay.

Q. But when staff made its forecasting adjustment and removed the fiscal 2017 data, did it adjust the fiscal 2018 or fiscal 2019

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data before it calculated its adjust -- its -- I'm sorry, forecasting adjustment?

A. Could you please specify what you mean by adjustment? Because we -- I think I -- like I said before 2018 and 2019 budgets were based on what the service company, what they -- I mean what they set and then we just looked at the actuals.

Q. Certainly. So the -- on page 33, lines 2 through 6, the staff I.T. panel indicated that the 2017 spend was in part the -- the overspend as staff -- staff has identified it. It was in part due to quote the fact -- was due in part to the bringing forward of several investments that were planned for 2018 and beyond. And my question is simply knowing that programs were brought forward into 2017 from 2018 and beyond, did staff make any adjustments in 2018 and beyond to reflect that programs were removed from those years and implemented in 2017.

A. No, we didn't adjust that because we use the company's actual budgets.

Q. Very good. Thank you. I'd like to turn your attention to page 39 of the panel's direct testimony. So beginning at line 11 there,

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there is discussion of the panel's view of cyber security investments. And the panel states, and I'll read from line 14. Well, I'll go ahead and read the question just for context. What is your overall assessment of the need for the capital and O.N.M. described in the company's filing? And again, for context, this is relating to cyber security investments.

And the answer, we believe that these investments are needed to address ongoing cyber threats. The implementation of the cyber security countermeasures is essential for establishing a high level of cyber security monitoring and protection. The field of cyber security is one in which the risks threat vectors and technologies in -- involved are constantly changing and increasing in complexity at a breakneck pace. Was that accurate reading of that testimony?

A. (Haugen) Yes.

Q. Thank you, Mr. Haugen. Is the panel's position that the increasing complexity associated with cyber security is limited to cyber security investments and does not affect any other information technology investments?

A. I strictly looked at the cyber security aspect of this rate case, nothing to do with I.T. investments whatsoever.

Q. Is there anyone on the panel that has specific I.T. background or I.T. experience that can address the question of whether or not I.T. is becoming more complex as time goes on?

MR. FORST: Objection, Your Honors. I think that question is a little vague in terms of what is meant by complex.

A.L.J. LEARY: My own opinion about this is it's axiomatic that I.T. every day gets different. And you may call it more complex. Where are you going with this? What -- what's -- what are you really looking for here?

MR. GAVILONDO: Your Honors, the panel has testified that the information technology spend that they're recommending should be based on a linear year extrapolation of prior spend. And what I'm trying to establish is that, in fact, a linear, the -- the -- the difficulty or the complexity is not necessarily linear.

A.L.J. LEARY: As time goes on is a very linear term to me, and that was part of your

question, wasn't it? And you're now saying, it's not linear?

MR. GAVILONDO: Time is all relative, right, You Honor. No, Your Honor --.

A.L.J. LEARY: Let me ask you to rephrase because I understand, you know, what's -- what the objection is about. So see if you can get there in another way, maybe laying some additional foundation for the panel to work with.

MR. GAVILONDO: Okay.

A.L.J. COSTELLO: Just -- just -- just noting also that in the testimony from the panel, they refer to complexity at a breakneck speed. So maybe clarify what they mean by complexity. And then you can ask the question.

MR. GAVILONDO: Okay. Thank you, Your Honor. Can I take a moment?

A.L.J. LEARY: Yeah.

MR. GAVILONDO: Thank you. Thank you, Your Honors.

BY MR. GAVILONDO: (Cont'g.)

Q. So referring back to the testimony that I read earlier, within the panel's testimony, it describes cyber security investments,

increasing in complexity at a breakneck pace. What does the panel mean by that?

A. (Park) Could you please refer to what page you are looking at again?

Q. Yes, I'm sorry. It's page 39 of the direct testimony and the lines -- the line of the specific sentence that I'm referring to begins on line 19.

A. (Haugen) To answer your question, the cyber security field is ever evolving. Threat actors from nation states are constantly searching for new ways to break into organization's computer systems. And in order to keep up with that pace, we have to follow suit. I'm not sure if that answers your question.

Q. And so I'll just ask based on what you just said Mr. Haugen --.

A. I mean if that's a technical you want, but.

Q. Well, when you say follow suit, do you have any more and -- and I -- and I ask if you feel as though this is getting in a confidential area, please let me know. And we'll -- we'll break, but do you have any clarification or elaboration on

what you mean by follow suit?

A. Well, utilities need to keep investing in technologies, so they'll be able to defend against those ever-escalating threats.

Q. And I think you also, on page 40 of the testimony beginning at line 8, you further identify the criticality of the infrastructure and the increasing temptation that the information that the company handles, and its infrastructure will be for nation states and cyber criminals, is that correct?

A. Yes.

Q. And that will take more investment to address those increasing actors and -- and threats. Is that correct?

A. Yes.

Q. Thank you. Does anyone on the panel have a position with respect to the I.T. needs for other areas other than cyber?

MR. FORST: Objection. That's another vague question.

MR. GAVILONDO: It is, thank you.

A.L.J. LEARY: What time is it?

MR. GAVILONDO: And perhaps it was

because I stopped in midsentence.

BY MR. GAVILONDO: (Cont'g.)

Q. Does anyone on the panel have a position with respect to other information technology areas other than cyber with respect to the need to continue to evolve and invest to address new challenges facing the utilities?

A. (Park) So we acknowledge that these projects are important to -- to the -- to the near future, but we're just trying to also establish an appropriate budget for the company based on -- based on their past history of spending for I.T.

Q. Okay. Thank you.

MR. GAVILONDO: Your Honors, I'd like to introduce a information request into the record. I believe it's been previously circulated to the panel and the parties and the Judges.

A.L.J. COSTELLO: Yes and --

A.L.J. LEARY: Yeah.

A.L.J. COSTELLO: -- that -- that will be marked as Exhibit 619 for identification.

MR. GAVILONDO: Thank you.

A.L.J. LEARY: Yeah.

BY MR. GAVILONDO: (Cont'g.)

Q. Panel, you have before you a document entitled N.G.34. It's an information request propounded by KEDNY and KEDLI to the -- the Staff Information Technology Panel, it's dated -- the date of the request is November 12th. And the date of the response is December 6th. It's been pre marked for identification as Exhibit 619.

And I'd ask whether or not this document -- and I'm sorry, it's a -- it's a three page document, two pages of text and one page of the printout of an Excel spreadsheet. And I'd ask whether or not this document represents a true and accurate -- accurate representation of the staff's response to this information request.

A. (Quackenbush) Yes.

Q. Thank you.

A.L.J. LEARY: Can I just ask, did staff highlight the multifactor authentication or M.F.A. on the attachment? Was that in the original response you highlighted that in the Excel?

THE WITNESS: (Park) Yes. And then we had a note on the bottom with the stars to reference it.

A.L.J. LEARY: Okay. Thank you.

MR. GAVILONDO: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. This information request response the question C, asked whether or not -- I'm sorry. I'll read it. If the answer to A is yes, please provide the rationale for applying a 25.5% reduction to cyber security and technology modernization projects. And the staff's response appears is on page 2. Is that still the staff's response?

A. (Park) Yes, and then we apply that 25.5% to pretty much all the projects.

Q. Thank you.

A.L.J. COSTELLO: I'm sorry. Can I just ask a question about that before you -- you move on? I just -- can you explain to me what -- what do you mean by the I.R. -- it says I.R.S. papers comma which have a high budget tolerance. What do you mean by high budget tolerance? That's on page 2. It's your response to C.

MR. GOODRICH: Panel, you might want to first state what I.R.S. stands for, if you know.

THE WITNESS: (Quackenbush) First I'd just like to say that I.R.S. does stand for Investment Request Paper.

THE WITNESS: (Park) If you look at staff's direct testimony, on page 14, line 8, that's where we start describing the I.R.S. and the different types of sanction papers and the budget tolerances.

A.L.J. LEARY: Okay. I think Judge Costello is asking you in your testimony today, to provide an explanation of high budget tolerance in addition to what's in your direct testimony.

A.L.J. COSTELLO: And also, if you'd just point out where -- where in your direct testimony refers to high budget tolerance?

THE WITNESS: I apologize, we didn't use the term high-budget tolerance, but I believe what we were trying to get at was that, there was high variability of the budget variances for the project, the I.R.S. papers which are the -- which are done in the preliminary phase and -- and the closeout papers which are done after the project is completed.

A.L.J. COSTELLO: Okay. Thank you.

A.L.J. LEARY: So is that plus -- plus 200% or minus 50%? Is that the variance you're referring to?

THE WITNESS: Yes. That's correct.

A.L.J. LEARY: Okay.

A.L.J. COSTELLO: Thank you. Sorry.

MR. GAVILONDO: No, thank you, Judge.

BY MR. GAVILONDO: (Cont'g.)

Q. And -- and if I might just ask a follow-up on that specific answer. You mentioned that the I.R.S. papers have a -- a budget, let's see, I think it's on page 14 of your testimony. It refers to a cost of estimate project to have a variance of plus 200% or minus 50%, is that correct?

A. Yes, that is correct.

Q. Okay. But you also mentioned in your response that closure papers also have a large budget -- I'm sorry, variance is that -- that was -- is that correct? And if so, could you point me to where that is?

A. No. I apologize if you heard it that way. I was just saying based when the I.R.S. papers are -- are created, that's when the final estimated cost can have a plus 200% or minus 50%. But the actual closure paper is when the project closes, so I guess --.

Q. Okay. Thank you. And I'll turn your attention to what's been pre-marked as Exhibit

619. And specifically, the attachment of a listing of projects, would have been identified in the column identified as statuses inflight projects. I believe staff performed an analysis of slipped projects, is that correct?

A. Yes.

Q. And at the bottom of the -- of the sheet that I'm referring to, there's a -- a number that says slipped projects 31. Is that correct?

A. Yes. Based on our -- our analysis, we found 31 slipped projects.

Q. Okay. And the average months delayed was 6.6 months, is that correct?

A. Yes.

Q. Okay. In your analysis, did you factor in any projects that were delivered on time?

A. Would you please clarify what you mean by analysis, on what?

A.L.J. LEARY: First of all --.

MR. GAVILONDO: Staff --?

A.L.J. LEARY: Excuse me.

MR. GAVILONDO: Yes.

A.L.J. LEARY: Is there something in

the testimony that indicates the projects were in fact delivered on time?

MR. GAVILONDO: On the same spreadsheet that we have here, it indicates the dates that projects went into service. The analysis is solely on this spreadsheet, there's nothing in the testimony. This came in after the testimony was submitted.

A.L.J. LEARY: And you're referring to the attachment to Hearing Exhibit 619? Is that --

MR. GAVILONDO: That's correct, Your Honor.

A.L.J. LEARY: -- the spreadsheet? I'm -- I'm going to ask the question again. Does this spreadsheet, attached to hearing Exhibit 619, reflect that there are specific projects that came in on time? Not the date that they, I guess closed, or the in-service date, but how can I tell here that projects are on time?

MR. GAVILONDO: Well, there are a couple of different ways we can tell when we look at the in-service date. There is a column, it says in-service date delayed. And then there are yeses and nos. And then, months delayed, there are a number of

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entries with --

A.L.J. LEARY: So --.

MR. GAVILONDO: -- digits.

A.L.J. LEARY: Got it. So if it -- there's nothing next to it, it means it's on time. If there's no number of months delayed, it means it's on time?

MR. GAVILONDO: If there are no number of months in which it's delayed, it's either on time or on schedule. It -- it's either been delivered already or it's on schedule to be delivered.

A.L.J. LEARY: So those are not delayed?

MR. GAVILONDO: That's correct, Your Honor.

A.L.J. LEARY: Okay. None of these would have been, you know, withdrawn or abandoned or otherwise?

MR. GAVILONDO: The status of these projects at the time that this spreadsheet was developed, all of these projects were still in flight, Your Honor.

A.L.J. LEARY: Okay. Thank you.

MR. GAVILONDO: You're welcome.

BY MR. GAVILONDO: (Cont'g.)

Q. So I'll -- I'll repeat the question. Did the staff's analysis take into account any projects that were not delayed?

A. (Park) So I'm not sure if -- if I'm following because, as you can see, we did take into account all the projects and pretty much where you see the blanks, we took -- where you see the blanks, there is -- it means that it was on time. And then we just took the average of the projects that were delayed and on time. Or that were delayed.

A.L.J. LEARY: I'm sorry. You need to be clear. Did you take into consideration only the projects that were delayed or did you include the projects that were on time? I think that's what counsel's asking you.

THE WITNESS: So we -- we included 31 out of the 64 inflight projects which were -- right, which were the ones that were delayed and then found the average on that.

BY MR. GAVILONDO: (Cont'g.)

Q. Okay. So and just to clarify the average of 6.6 months delayed was based on the total months delayed divided by 31 projects, is that

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correct?

A. I'll have to check that again because I don't have that calculation in front of me.

Q. Okay. All right. Turn your attention to page 45 of the direct testimony of the staff I.T. panel.

A.L.J. COSTELLO: I'm sorry to interject one more time just -- I'm -- can someone on the panel just explain to me and -- if this is the appropriate out -- what the different statuses are, here all of the projects provided their inflight status. Could you explain to me what inflight status means? Just so it's clear for the record.

THE WITNESS: Inflight from what I re -- know is that those are the projects that are currently occurring or have a -- are currently occurring.

A.L.J. COSTELLO: So they're projects that -- that haven't been completed yet, is that -- is that what it stands for?

THE WITNESS: Yes.

A.L.J. COSTELLO: Okay. Thank you.

THE WITNESS: Yeah. So in other words they haven't been closed out yet.

A.L.J. COSTELLO: Okay. Thank you.

MR. GAVILONDO: Thank you. Your Honor, I actually will have to ask some clarifying questions on that line of questioning because -- and -- and well, I'll just asked the panel.

BY MR. GAVILONDO: (Cont'g.)

Q. So panel, on column that is headed, it says per rate case in-service date at the top of the page. Do you see that column?

A.L.J. LEARY: I don't.

MR. GAVILONDO: It's right near the middle of the page, it says per rate case. And that's the first line, and then in-service date immediately be -- below that. And -- and I apologize, the font is microscopic and I -- that's why I had to print mine out on the big page.

A.L.J. LEARY: Let me see yours. We don't have that. That's not reflected on this document. Well, you can see --

MR. GAVILONDO: Right.

A.L.J. LEARY: -- this is a different workbook page, I'm going to bet. But I don't know in Excel.

MR. GAVILONDO: So I will look at the

-- it just says -- I'm going to refer to the document here.

A.L.J. LEARY: Yeah. Does -- did the witness -- the witness panel have --

MR. GAVILONDO: They have, yes, Your Honor.

A.L.J. LEARY: -- what I have or what you have?

THE WITNESS: (Quackenbush) Your Honor, I believe we have what you have.

A.L.J. LEARY: Do you want to see this?

A.L.J. COSTELLO: No, he -- he's going to ask based on what we have.

A.L.J. LEARY: On what we have, all right. Here's my question, the panel authenticated their response to this document --

MR. GAVILONDO: Yes.

A.L.J. LEARY: -- and I'm questioning now whether this is in fact the document that they intend to authenticate based upon what you're showing me that you have in your hand which is an -- 11/17 more -- what I'll call a more complete --.

MR. GAVILONDO: Annotated to help the

--

A.L.J. LEARY: Well --.

MR. GAVILONDO: -- dumb lawyer, yeah.

A.L.J. LEARY: There -- there are columns that you have that we do not have on Hearing Exhibit 619. So does -- does anyone on the panel have your responses? Or counsel for D.P.S. staff have the attachment to 61 -- Hearing Exhibit 619?

MR. FORST: Yes, Your Honor. This is Nick Forst for D.P.S. Staff. I'm looking at the original attachment that was submitted in response to N.G.34 and I can confirm that the headings as they appear on the document that's been passed around to the staff panel and to the other parties is what was submitted with the response.

A.L.J. LEARY: Okay. So you have something that the company must have created, is that right?

MR. GAVILONDO: That's right. For my purposes here on the stand, Your Honor. And I apologize, I referenced the wrong column header, but I will reference the appropriate column header --

A.L.J. LEARY: Okay.

MR. GAVILONDO: -- in the further

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questioning. Thank you.

A.L.J. LEARY: Thanks. I'm glad we caught this.

MR. GAVILONDO: Yes. Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. Okay. So I call your attention to a column that is entitled in-service date. Do you see that, panel?

A. (Park) Yes.

Q. Okay. Thank you. And I apologize for that misdirection there. Now, would you agree with me that some of the in-service dates in this column are prior to the date of this interrogatory response which is November 22nd, 2019? If we go down the column if you could just scan and see whether or not you agree that some of the dates are prior to November of 2019?

A.L.J. LEARY: It should be December 6th, 2019 when the response was provided.

MR. GAVILONDO: Thank you.

THE WITNESS: Yes.

BY MR. GAVILONDO: (Cont'g.)

Q. Okay. Those projects with in-service dates prior to November of 20 -- I'm sorry

December of 2019, were those included in the 31 slipped projects that you've identified below in your analysis? Can I -- I'll withdraw that question because I see that -- that there are some projects. And I'll re -- I'll restate the question, so disregard. Thank you.

Are there some projects with in-service dates in the in-service column that are not delayed projects?

A. Could you repeat the question? Sorry.

Q. Certainly. Maybe I'll ask it a little bit more simply. Are there any projects on this page that were not delayed in their delivery?

A. Yes. As you can see in the in-service date delayed column, I -- I specifically say if they were delayed -- if the -- if the go-live date was delayed or not.

Q. And did you account for those projects that were not delayed when you calculated the average months delayed of 6.6?

A. No. We didn't take the average because they would just be zero.

Q. Okay. Thank you. So now, I'll

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turn your attention to page 45.

MR. GAVILONDO: And Your Honors, I just have about two more minutes of cross-examination.

A.L.J. LEARY: Whatever.

BY MR. GAVILONDO: (Cont'g.)

Q. Page 45 of the panel's direct testimony. And there, the panel states that in referring to the Gas Business Enablement Program and the -- the company's cost recovery proposal in this case. Beginning at line 3, changes include cost related to greater than anticipated complexity, despite foresight knowing it would be a very difficult and complex integration of the legacy interfaces of approximately 21.168 million dollars.

Does the panel have any specific recommendations on what the company should have done differently given that it had foresight of knowing of these complexities what it should have done differently with respect to delivery of the G.B.E. Program?

A. (Fisher) I do not have any recommendations for what the company should have done as I was not a part of the analysis that the company

did when they were forecasting and look -- looking at the changes to what they were, due to the system in forecasting the budget to do that.

Q. Thank you.

MR. GAVILONDO: Your Honor, if I might have a moment?

A.L.J. LEARY: Yes.

MR. GAVILONDO: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. So panel, with respect to the Gas Business Enablement Program, are there portions of the proposed investment that the company is making that the staff deems is imprudent and should not be recovered?

A.L.J. LEARY: Investments in I.T.?

MR. GAVILONDO: Specifically to the Gas Business Enablement Program, Your Honor.

A.L.J. LEARY: Thank you.

THE WITNESS: At this time, I would not be able to say that any of the projects or spending is imprudent.

MR. GAVILONDO: Thank you.

BY MR. GAVILONDO: (Cont'g.)

Q. Do any of the panel members have

any direct experience with implementing or overseeing a large information technology project or program?

A. No.

Q. Thank you.

A.L.J. LEARY: Is that on behalf of the entire panel?

THE WITNESS: Yes, on behalf of the entire panel, no.

BY MR. GAVILONDO: (Cont'g.)

Q. Panel, if the commission were to adopt the proposed slippage adjustment recommended by the Staff I.T. Panel, and the companies were to invest in their capital projects and spend up to their approved budgets including the amounts or in excess of their approved budgets and incur spending that included the amount that was removed from the slippage adjustment or by the slippage adjustment would staff deem that investment by the company imprudent?

Let me restate that question, I apologize. It's late in the day. If the company were to spend in excess of what the rate allowance was because the staff adopted the forecasting adjustment, would staff deem the company's

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investments in excess of the rate allowance as imprudent?

A.L.J. LEARY: You know, I -- I really have to kind of ask you about this for one second. Because prudence determinations are made by this agency in kind of a separate forum. Are you talking about what is done in the -- when -- when staff and the agency initiates a prudence review or -- give me the context of what you're talking about when you refer to imprudent. Is that in the context of what we understand prudently -- prudently incurred or prudently -- prudent expenditures or what do you mean by that? Just for the context here for the record. Is it -- is it that to which you refer?

MR. GAVILONDO: It -- it is Your Honor. It's -- the -- the investments made by the company to provide service for customers and that the commission authorizes re -- a recovery and a return on, yes.

A.L.J. LEARY: And that would -- if there was any question about those that the agency had, they would undertake a kind of formal review of the costs that were incurred, it's my understanding, right?

MR. GAVILONDO: Your Honor, I'll be honest, I haven't been involved in a prudence review in the past. So I don't know the answer.

A.L.J. LEARY: Well, someone else on behalf of the company.

MR. MALONEY: Your Honor, a prudence review can occur in the context of a rate case with respect to past projects that have been placed in service.

A.L.J. LEARY: Right.

MR. MALONEY: But I think the question goes to, with respect to the amount of the slippage adjustment, if the companies actually spend their budget regardless of the fact that the slippage adjustment has been applied, is the staff panel contending that that extra spending is in excessive or imprudent? And if the answer is no, it's no, if it's yes, we might delve into what the basis of that is.

A.L.J. LEARY: Okay.

MR. FORST: Your Honors, I think that the -- the entire context of the question is pretty -- pretty much improper. I mean it's asking the panel to, you know, speak about hypothetical investments

that haven't been made yet that may or may not be prudent that staff would undertake, you know, an analysis of in the future. I mean I just --.

MR. MALONEY: It's asking about the dollars that are reflected in the company's capital expenditures in this rate case.

MR. GOODRICH: But it's also the -- the premise of the question was that there are commission approved budgets and I -- I -- please point us --.

MR. MALONEY: But I -- I didn't say there were --.

MR. GOODRICH: No -- no. Carlos did.

MR. MALONEY: Okay.

MR. GOODRICH: And please point us if there is some place where we've said we recommended that the commission approve a budget. But I believe we are talking merely about a rate allowance for the rate year.

MR. GAVILONDO: You're correct, it is a rate allowance.

A.L.J. LEARY: I understand where you're trying to go, I just don't want the record to be reflecting some anticipated determination that --

to which this panel or the agency will be held. So that's where I'm getting a little bit nervous in terms of the hypothetical. So can we get to a little bit more specific questions and maybe you can get there, but I'm -- I have concerns that testimony that's elicited today somehow you hold it up later and if -- if D.P.S. staff determines to undertake some kind of prudence review about something in the I.T. area.

And I don't -- I'm not ever sure your question is limited to that, but maybe we can get a little tighter to the specifics of what you're asking without a hypothetical because I don't even understand what the hypothetical is.

BY MR. GAVILONDO: (Cont'g.)

Q. Panel, earlier today, there was testimony on the part of the company's I.T. panel indicating a willingness to accept a downward only capital tracker in a 1-year litigated rate case. And I ask whether or not the panel was either in attendance or aware of that testimony on the part of the company's I.T. panel?

A. (Quackenbush) I believe the majority of the panel members were in attendance and

did hear that. But again, we haven't seen the testimony yet.

Q. Okay. Thank you. And with respect to the capital investment plan that the company has put forth in this case, has the staff identified any projects that they believe should be removed or eliminated from the case other than what we described earlier which was the customer information system program which is going to be addressed in a separate panel?

MR. FORST: Objection, that's been asked and answered I think two or three times now.

A.L.J. LEARY: Well, let's give the company a little latitude because I understand, I think, where they want to get, so let's give them a chance. Asked and answered. I know it's getting late in the day, but let's just give the company a bit of latitude, if you would.

MR. GAVILONDO: Thank you, Your Honor.

BY MR. GAVILONDO: (Cont'g.)

Q. So projects that the Staff Panel has proposed to remove from the I.T. investment portfolio are the C.I.S. Program, the C.D.G. Low Income Program and a portion of the G.B.E. Capital

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Investment, is that correct?

A. That's correct.

Q. And there are no other programs that the -- that -- I'm sorry, that the staff panel has proposed to eliminate or otherwise remove from the I.T. capital portfolio?

A. (Park) So we're not removing other projects other than what you just stated, but we're just making an adjustment -- overall an adjustment of what we think is reasonable for the -- what we think the company will be able to deliver in -- what kind of, a reasonable budget.

Q. Okay. Thank you.

A. (Fisher) I'd just like to clarify one thing with respect to that in that I think the way it was worded. We didn't propose to remove the G.B.E. Program, we changed the -- the proposed budget.

Q. Okay. Thank you.

MR. GAVILONDO: I have no further questions, Your Honors.

A.L.J. COSTELLO: Do you need to take a few minutes to see if you need re-direct?

A.L.J. LEARY: I think we need a

little break. Anybody have an objection to that, 10 minutes?

(Off the record 3:26 p.m.)

MR. FORST: Staff has no redirect, Your Honors.

A.L.J. LEARY: Okay. That's it for today.

A.L.J. COSTELLO: That's it. We thank the panel for their testimony, and you're excused.

A.L.J. LEARY: Thanks.

A.L.J. COSTELLO: Thank you. And we'll go off the record and we'll -- off the record.

(Off the record; 3:39 p.m.)

STATE OF NEW YORK

I, KAYLA ALLEN, do hereby certify that the foregoing was reported by me, in the cause, at the time and place, as stated in the caption hereto, at Page 1 hereof; that the foregoing typewritten transcription consisting of pages 1 through 1062, is a true record of all proceedings had at the hearing.

IN WITNESS WHEREOF, I have hereunto subscribed my name, this the 17th day of February, 2020.

Kayla Allen, Reporter